



Integrated Cloud Applications & Platform Services

Oracle Application Express Workshop I

Activity Guide – Volume I
D79653GC30 | D106873

Learn more from Oracle University at education.oracle.com



For Instructor Use Only
This document should not be distributed.

Authors

Sayantani Ghosh
Moitreyee Hazarika

**Technical Contributors
and Reviewers**

Suresh Rajan
Anjani Pothula
David Peake
Chaitanya Koratamaddi
Joe Greenwald

Editors

Adrita Biswas
Aju Kumar
Moushmi Mukherjee

Graphic Editors

Prakash Dharmalingam
Anne Elizabeth
Pushparaj Kundar

Publishers

Pavithran Adka
Veena Narasimhan
Giri Venugopal

3009092019

Copyright © 2019, Oracle and/or its affiliates. All rights reserved.

Disclaimer

This document contains proprietary information and is protected by copyright and other intellectual property laws. You may copy and print this document solely for your own use in an Oracle training course. The document may not be modified or altered in any way. Except where your use constitutes "fair use" under copyright law, you may not use, share, download, upload, copy, print, display, perform, reproduce, publish, license, post, transmit, or distribute this document in whole or in part without the express authorization of Oracle.

The information contained in this document is subject to change without notice. If you find any problems in the document, please report them in writing to: Oracle University, 500 Oracle Parkway, Redwood Shores, California 94065 USA. This document is not warranted to be error-free.

Restricted Rights Notice

If this documentation is delivered to the United States Government or anyone using the documentation on behalf of the United States Government, the following notice is applicable:

U.S. GOVERNMENT RIGHTS

The U.S. Government's rights to use, modify, reproduce, release, perform, display, or disclose these training materials are restricted by the terms of the applicable Oracle license agreement and/or the applicable U.S. Government contract.

Trademark Notice

Oracle and Java are registered trademarks of Oracle and/or its affiliates. Other names may be trademarks of their respective owners.

For Instructor Use Only.
This document should not be distributed.

Table of Contents

Course Practice Environment: Security Credentials	vii
Practices for Lesson 1: Course Overview	9
Practices for Lesson 1: Overview	10
Practices for Lesson 2: Oracle Application Express: Introduction.....	11
Practices for Lesson 2: Overview	12
Practice 2-1: Using Oracle Application Express as a Workspace Administrator	13
Practice 2-2: Using Oracle Application Express as a Developer	14
Practice 2-3: Installing a Sample Dataset	15
Solution 2-1: Using Oracle Application Express as a Workspace Administrator	16
Solution 2-2: Using Oracle Application Express as a Developer	23
Solution 2-3: Installing a Sample Dataset	37
Practices for Lesson 3: Creating a Database Application	41
Practices for Lesson 3: Overview	42
Practice 3-1: Creating Database Applications	43
Practice 3-2: Creating a Database Application from a File	44
Solution 3-1: Creating Database Applications	45
Solution 3-2: Creating a Database Application from a File	48
Practices for Lesson 4: Working with Reports.....	55
Practices for Lesson 4: Overview	56
Practice 4-1: Creating Classic Reports	57
Practice 4-2: Creating a Column Toggle Report.....	59
Practice 4-3: Adding and Modifying a List View Report.....	61
Solution 4-1: Creating Classic Reports	63
Solution 4-2: Creating a Column Toggle Report.....	86
Solution 4-3: Adding and Modifying a List View Report.....	91
Practices for Lesson 5: Working with Interactive Reports	97
Practices for Lesson 5: Overview	98
Practice 5-1: Creating and Manipulating an Interactive Report	99
Practice 5-2: Customizing an Interactive Report	101
Solution 5-1: Creating and Manipulating an Interactive Report	103
Solution 5-2: Customizing an Interactive Report	139
Practices for Lesson 6: Working with Interactive Grids.....	145
Practices for Lesson 6	146
Practice 6-1: Creating an Interactive Grid.....	147
Practice 6-2: Using and Manipulating the Interactive Grid	149

Solution 6-1: Creating an Interactive Grid	151
Solution 6-2: Using and Manipulating the Interactive Grid	163
Practices for Lesson 7: Working with Pages and Regions	169
Practices for Lesson 7: Overview	170
Practice 7-1: Creating and Modifying Pages and Regions	171
Practice 7-2: Working with Global Pages	173
Solution 7-1: Creating and Modifying Pages and Regions	175
Solution 7-2: Working with Global Pages	197
Practices for Lesson 8: Managing Forms	203
Practices for Lesson 8: Overview	204
Practice 8-1: Creating a Form.....	205
Practice 8-2: Creating a List View with Form.....	207
Practice 8-3: Creating a Master Detail Form (Side By Side)	210
Practice 8-4: Creating a Master Detail Form (Stacked).....	212
Practice 8-5: Creating a Master Detail Form (Drill-Down)	214
Solution 8-1: Creating a Form.....	216
Solution 8-2: Creating a List View with Form.....	227
Solution 8-3: Creating a Master Detail Form (Side by Side).....	234
Solution 8-4: Creating a Master Detail Form (Stacked).....	246
Solution 8-5: Creating a Master Detail Form (Drill Down)	252
Practices for Lesson 9: Adding Items and Buttons	259
Practices for Lesson 9: Overview	260
Practice 9-1: Adding Items and Buttons	261
Practice 9-2: Manipulating Items	263
Solution 9-1: Adding Items and Buttons	265
Solution 9-2: Manipulating Items	289
Practices for Lesson 10: Understanding Session State.....	319
Practices for Lesson 10: Overview	320
Practice 10-1: Understanding Session State	321
Solution 10-1: Understanding Session State	323
Practices for Lesson 11: Including Page Processing.....	339
Practices for Lesson 11: Overview	340
Practice 11-1: Creating and Manipulating Computations, Processes, and Validations.....	341
Solution 11-1: Creating and Manipulating Computations, Processes, and Validations.....	343
Practices for Lesson 12: Using Dynamic Actions and Plug-Ins.....	379
Practices for Lesson 12: Overview	380
Practice 12-1: Using Dynamic Actions on Your Page	381
Practice 12-2: Importing and Using Plug-Ins on Your Page	383
Solution 12-1: Using Dynamic Actions on Your Page	384

Solution 12-2: Importing and Using Plug-Ins on Your Page	417
Practices for Lesson 13: Validating and Debugging Your Application	433
Practices for Lesson 13: Overview	434
Practice 13-1: Using the Advisor.....	435
Practice 13-2: Modifying the Attribute Dictionary.....	436
Solution 13-1: Using the Advisor	438
Solution 13-2: Modifying the Attribute Dictionary.....	445
Practices for Lesson 14: Adding Shared Components That Aid Navigation	469
Practices for Lesson 14: Overview	470
Practice 14-1: Creating Lists and List Regions.....	471
Practice 14-2: Creating and Editing a Navigation Menu	473
Practice 14-3: Adding Navigation Bar Entries.....	475
Practice 14-4: Adding Breadcrumbs to an Existing Page	477
Solution 14-1: Creating Lists and List Regions.....	480
Solution 14-2: Creating and Editing a Navigation Menu	499
Solution 14-3: Adding Navigation Bar Entries.....	510
Solution 14-4: Adding Breadcrumbs to an Existing Page	526
Practices for Lesson 15: Working with Themes, Templates, and Files	537
Practices for Lesson 15: Overview	538
Practice 15-1: Creating a Theme from Scratch	539
Practice 15-2: Copying, Editing, and Switching a Theme.....	541
Practice 15-3: Editing Templates	543
Practice 15-4: Uploading and Applying a CSS	545
Solution 15-1: Creating a Theme from Scratch	547
Solution 15-2: Copying, Editing, and Switching a Theme.....	550
Solution 15-3: Editing Templates	560
Solution 15-4: Uploading and Applying a CSS	567
Practices for Lesson 16: Implementing Security	573
Practices for Lesson 16: Overview	574
Practice 16-1: Creating an Authentication Scheme	575
Practice 16-2: Restricting Users by Using Access Control	576
Practice 16-3: Enabling Session State Protection	578
Solution 16-1: Creating an Authentication Scheme	580
Solution 16-2: Restricting Users by Using Access Control	587
Solution 16-3: Enabling Session State Protection	622
Practices for Lesson 17: Managing Application Navigation	637
Practices for Lesson 17: Overview	638
Practice 17-1: Building a Hierarchical List with Images	639
Practice 17-2: Building a Database-Driven Report	642

Practice 17-3: Building a Site Map	644
Practice 17-4: Enforcing Authorization in Site Map	647
Solution 17-1: Building a Hierarchical List with Images	650
Solution 17-2: Building a Database-Driven Report	668
Solution 17-3: Building a Site Map	679
Solution 17-4: Enforcing Authorization in Site Map	701
Practices for Lesson 18: Extending Your Application	711
Practices for Lesson 18: Overview	712
Practice 18-1: Creating Data Load Wizard Pages	713
Practice 18-2: Adding an Upload and Download Page	716
Solution 18-1: Creating Data Load Wizard Pages	719
Solution 18-2: Adding an Upload and Download Page	739
Practices for Lesson 19: Creating and Editing Charts	759
Practices for Lesson 19: Overview	760
Practice 19-1: Creating and Modifying Charts	761
Practice 19-2: Creating Pie and Doughnut Charts	764
Practice 19-3: Creating Combination Chart	766
Solution 19-1: Creating and Modifying Bar Charts	769
Solution 19-2: Creating a Pie Chart and Converting it to a Doughnut Chart	778
Solution 19-3: Creating Combination Chart	786
Practices for Lesson 20: Adding Calendars and Trees	803
Practices for Lesson 20: Overview	804
Practice 20-1: Creating a Calendar	805
Practice 20-2: Creating a Tree	807
Solution 20-1: Creating a Calendar	810
Solution 20-2: Creating a Tree	822
Practices for Lesson 21: Managing Application Feedback	833
Practices for Lesson 21: Overview	834
Practice 21-1: Adding and Monitoring Feedback in Your Application	835
Solution 21-1: Adding and Monitoring Feedback in Your Application	838

Course Practice Environment: Security Credentials

For OS usernames and passwords, see the following:

- If you are attending a classroom-based or live virtual class, ask your instructor or LVC producer for OS credential information.
- If you are using a self-study format, refer to the communication that you received from Oracle University for this course.

For product-specific credentials used in this course, see the following table:

Product-Specific Credentials		
Product/Application	Username	Password
Workspace Administrator User	apex_admin	<u>Initial login:</u> apex <u>Changed To:</u> welcome (will be using this all through the course)
Workspace Developer User	apex_dev	apex
Workspace Developer User	brad.knight	apex
Workspace Administrator User	susie.parker	apex
Workspace End User	john.bell	apex

Note: The credentials mentioned above are for the users of the **APEX** Workspace. The first user (that is, `apex_admin`) is already created. You will be creating all the other users during the practices.

Practices for Lesson 1: Course Overview

For Instructor Use Only.
This document should not be distributed.

Practices for Lesson 1: Overview

Overview

- There are no practices for Lesson 1.
- The OEHR schema is preinstalled in your workspace.
- The files used in the practices are available in the `/home/oracle/labs/labs` directory.
- The solution files for the practices are available in the `/home/oracle/labs/solutions` directory.
- Practice 3-2 is an optional practice and does not have a solution script.

For Instructor Use Only.
This document should not be distributed.

Practices for Lesson 2:
Oracle Application Express:
Introduction

Practices for Lesson 2: Overview

Overview

There are three practices for this lesson. In these practices, you sign in and use Oracle Application Express as a Workspace Administrator and then as a Developer.

For Instructor Use Only.
This document should not be distributed.

Practice 2-1: Using Oracle Application Express as a Workspace Administrator

Overview

As a new user, you are given access to Oracle APEX as a Workspace Administrator. To develop GlobalMart Management Tool (GMT), you need a Developer user. Therefore, you sign in to Oracle APEX as a Workspace Administrator and create a new user, Developer.

Assumptions

None

Tasks

1. Sign in to Oracle Application Express with the following details:
 - Workspace: APEX
 - Username: apex_admin
 - Password: **<mentioned in the password document>**
2. Create a developer user with the following details:
 - Username: apex_dev
 - Password: **<mentioned in the password document>**
 - Email: apexdev@oracle.com

For Instructor Use Only.
This document should not be distributed.

Practice 2-2: Using Oracle Application Express as a Developer

Overview

By now, you have created the `apex` user as a developer. Before getting into developing the GMT application, you should familiarize yourself with using Oracle APEX as a developer. To achieve this, in this practice, you:

- Sign in to Oracle Application Express as a developer user
- Run the sample database application
- Install a packaged application and use it
- Remove the packaged application from the workspace

Assumptions

You have completed Practice 2-1.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Sign in to Oracle Application Express with the following details:
 - Workspace: `APEX`
 - Username: `apex_dev`
 - Password: *<mentioned in the password document>*
2. Install and Run the **Sample Database Application**, which is already available in your workspace.
3. Install and use a packaged application. You install and use **Customer Tracker** available under packaged applications in your workspace.
4. Remove the **Customer Tracker** packaged application from your workspace.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon to create the `apex` user.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

Practice 2-3: Installing a Sample Dataset

Overview

Oracle Application Express includes four sample datasets. Each dataset includes sufficient data to enable you to build applications with pages containing different types of components.

With a few clicks of the mouse you can install, refresh, replace, or remove sample datasets in one of the schemas associated with your workspace.

Assumptions

You have completed Practice 2-1.

Tasks

1. Sign in to Oracle Application Express with the following credentials:
 - Workspace: `APEX`
 - Username: `apex_dev`
 - Password: **<mentioned in the password document>**
2. Install the sample dataset `HR Data` from SQL Workshop Utilities in the `OEHR` schema.

Catch up

If you haven't completed **Practice 2-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon to create the `apex_dev` user.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

For Instructor Use Only.
This document should not be distributed.

Solution 2-1: Using Oracle Application Express as a Workspace Administrator

Overview

In this practice solution, steps are provided to create a new developer user by logging in as a Workspace Administrator.

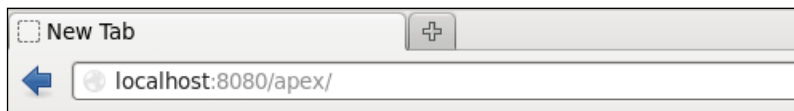
Steps

1. Sign in to Oracle Application Express with the following details:

- Workspace: Enter `apex`
- Username: Enter `apex_admin`
- Password: <mentioned in the password document>

- a. Open a browser and enter the following URL in the address bar:

`http://localhost:8080/apex/`

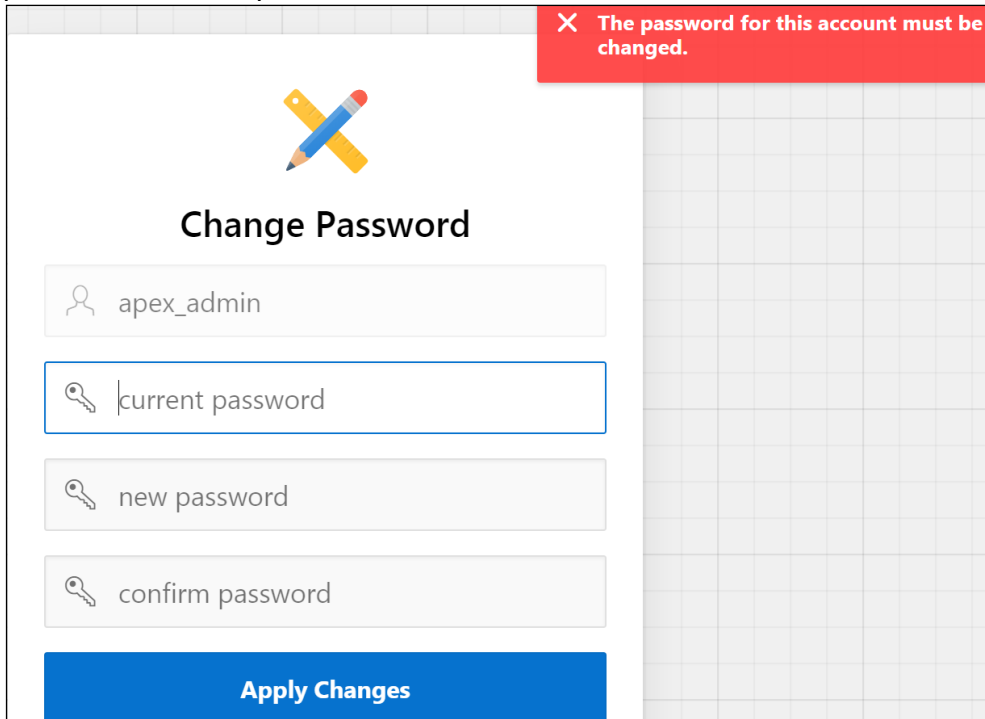


- b. On the Sign in page, enter `apex` for **Workspace**, `apex_admin` for **Username**, and the password (as mentioned in the password document). Click **Sign In**.

A screenshot of the Oracle Application Express Sign In page. At the top is the Oracle logo (two crossed pencils) and the text 'Oracle Application Express'. Below this are three input fields: the first contains 'apex' with a database icon and a green checkmark; the second contains 'apex_admin' with a person icon and a green checkmark; the third contains masked characters '....' with a key icon and a green checkmark. Below these fields is a checkbox labeled 'Remember workspace and username' with a question mark icon. At the bottom is a large blue button labeled 'Sign In' with a hand cursor icon pointing at it.

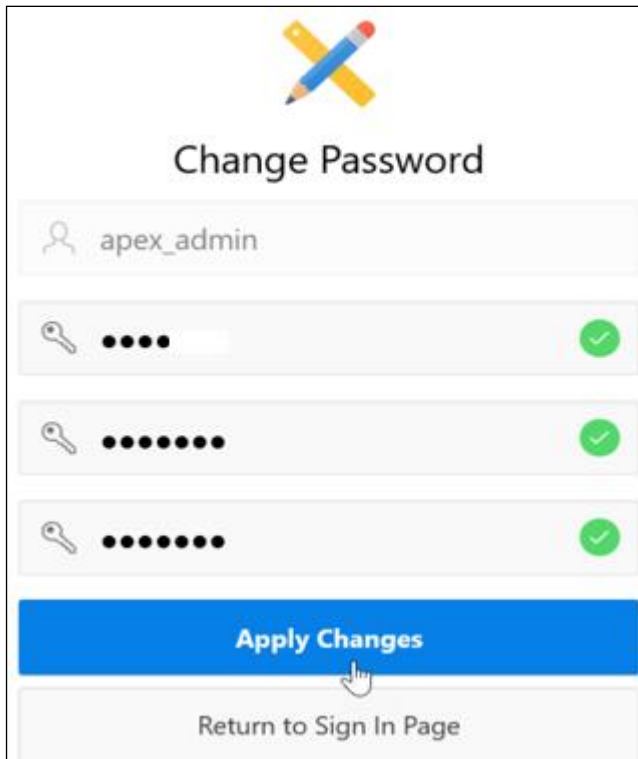
For Instructor Use Only.
This document should not be distributed.

- c. On your first login, you will be prompted to change your password. Enter the current password and new password.



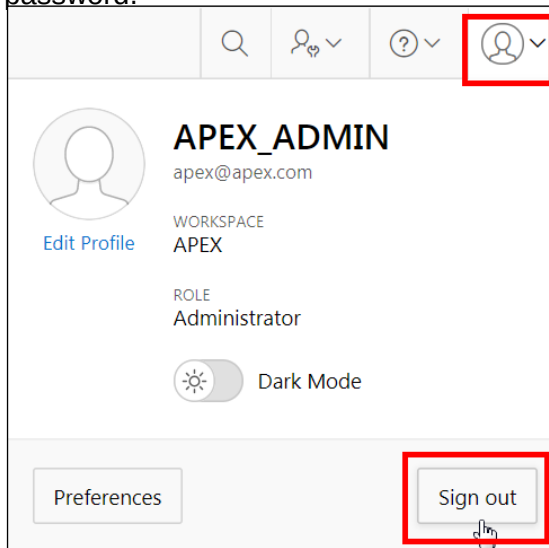
The screenshot shows a web interface for changing a password. At the top right, a red banner with a white 'X' icon contains the text: "The password for this account must be changed." Below this, the title "Change Password" is centered. Underneath the title is a pencil icon. The form contains four input fields: a username field with "apex_admin", a "current password" field, a "new password" field, and a "confirm password" field. Each password field has a key icon on the left. At the bottom of the form is a blue button labeled "Apply Changes".

- d. Click **Apply Changes**.

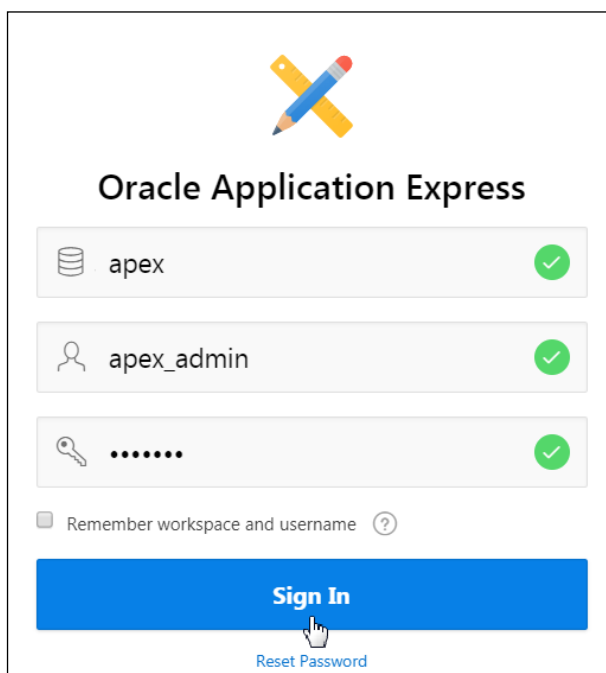
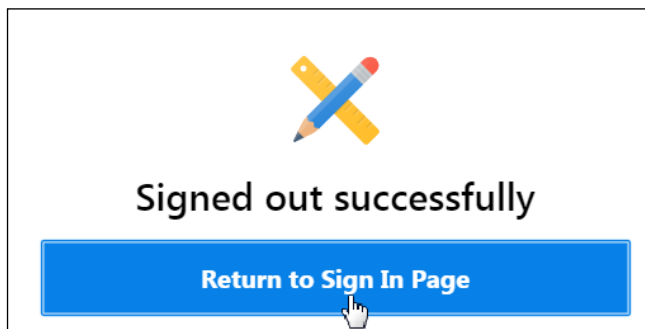


The screenshot shows the same "Change Password" form after the "Apply Changes" button has been clicked. The password fields now show masked characters (dots) and each has a green checkmark icon on the right, indicating successful validation. The "Apply Changes" button is still present, and a hand cursor is shown clicking it. Below the button is a link that says "Return to Sign In Page".

- e. You have successfully logged in to the Oracle Application Express workspace as a workspace administrator. You can sign out of the workspace and sign in with your new password.



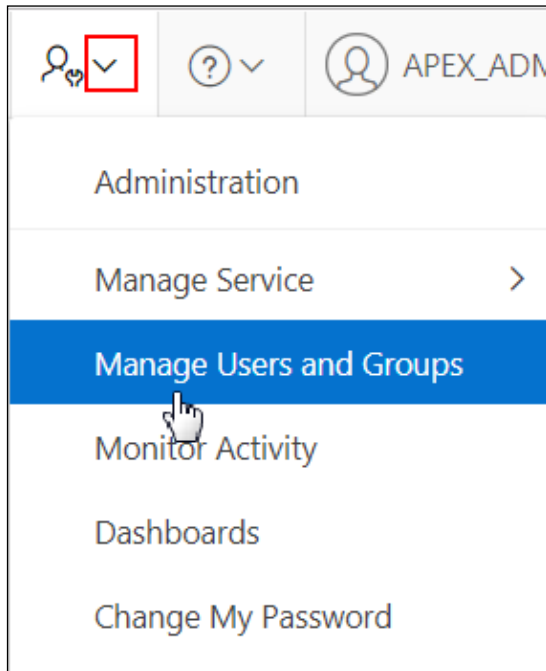
Click the **Return to Sign In Page** button and sign in again.



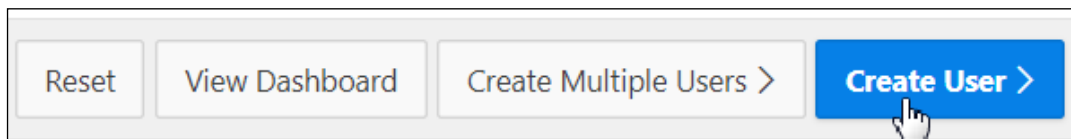
2. Create a developer user with the following details:

- Username: Enter apex_dev
- Password: <mentioned in the password document>
- Email: Enter apexdev@oracle.com

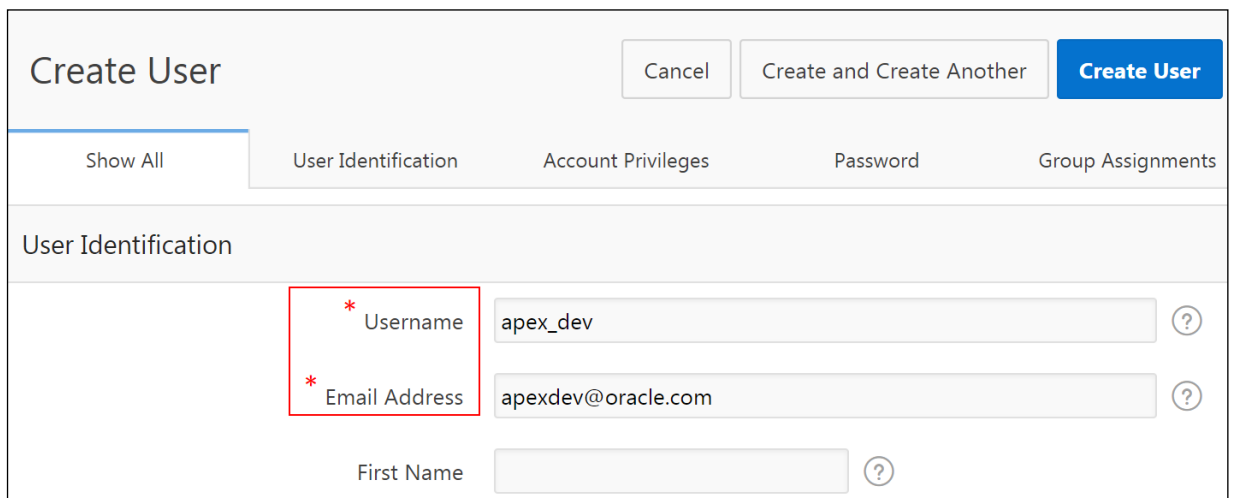
a. Click the **Administration** icon  present in the top-right corner of the page and select **Manage Users and Groups** from the drop-down menu.



b. Click **Create User >**.



c. Enter apex_dev in the **Username** field and enter apexdev@oracle.com in the **Email Address** field.

A screenshot of the Oracle APEX 'Create User' form. The form has a title 'Create User' and three buttons: 'Cancel', 'Create and Create Another', and 'Create User'. Below the title is a tabbed interface with tabs for 'Show All', 'User Identification', 'Account Privileges', 'Password', and 'Group Assignments'. The 'User Identification' tab is selected. The form contains three input fields: 'Username' (containing 'apex_dev'), 'Email Address' (containing 'apexdev@oracle.com'), and 'First Name' (empty). The 'Username' and 'Email Address' fields are highlighted with a red box.

- d. Scroll down and set the **Default Schema** to **OEHR**. Remove text from the **Accessible Schemas** field to make it null. Ensure that **User is a workspace administrator** is set to **No** and **User is a developer** is set to **Yes**.

Account Privileges

Default Schema: OEHR

Accessible Schemas (null for all):

User is a workspace administrator: ☐ Yes ☒ No

User is a developer: ☒ Yes ☐ No

- e. In the Password section, set the password in the **Password** field. Enter the same password in the **Confirm Password** field. In **Require Change of Password on First Use**, select **No**.

Password

* Password:

* Confirm Password:

Require Change of Password on First Use: No

- f. In the Group Assignments section, click >> to assign the OAuth2 Client Developer, RESTful Services, and SQL Developer roles to the user.

Group Assignments

Group Assignments:

>>

OAuth2 Client Developer
RESTful Services
SQL Developer

- g. Go to the top of the page and click **Create User**. After the user is created, the message *User created* appears.

Create User

Cancel Create and Create Another Create User

Group Assignments

Group Assignments:

>>

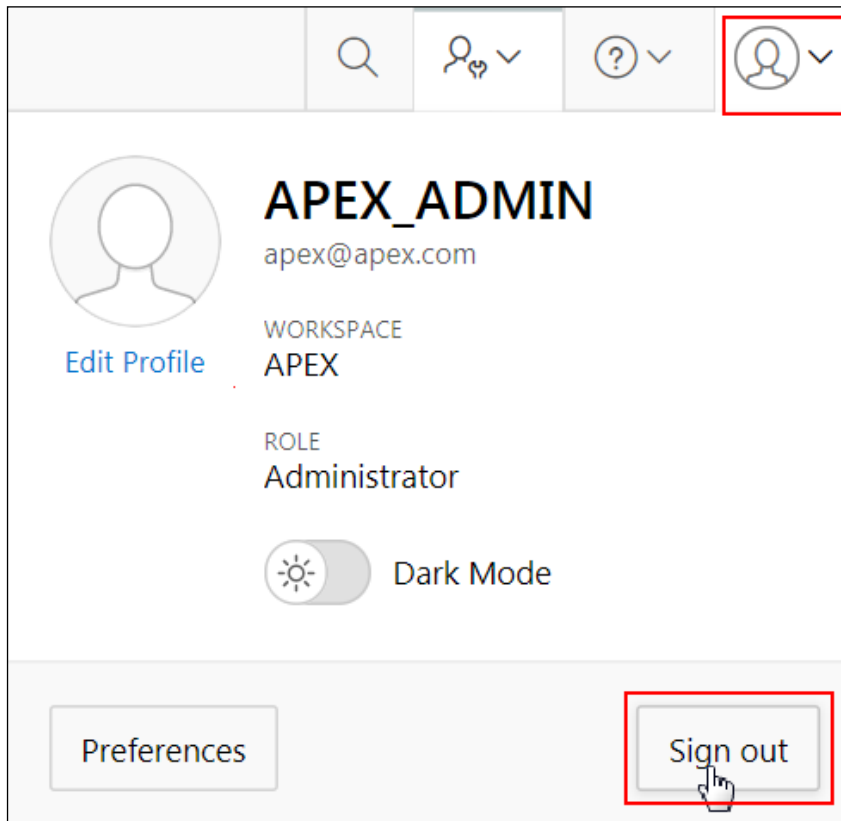
OAuth2 Client Developer
RESTful Services
SQL Developer

Manage Users and Groups

✓ User created.

For Instructor Use Only.
This document should not be distributed.

- h. Click the **Account Menu** icon  present in the top-right corner of the page and click **Sign out**.



You have successfully created a developer user by signing in to the Oracle Application Express workspace as a workspace administrator.

Solution 2-2: Using Oracle Application Express as a Developer

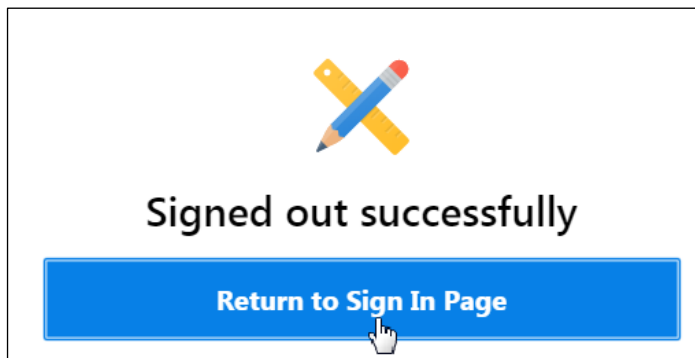
Overview

In this practice solution, steps are provided to:

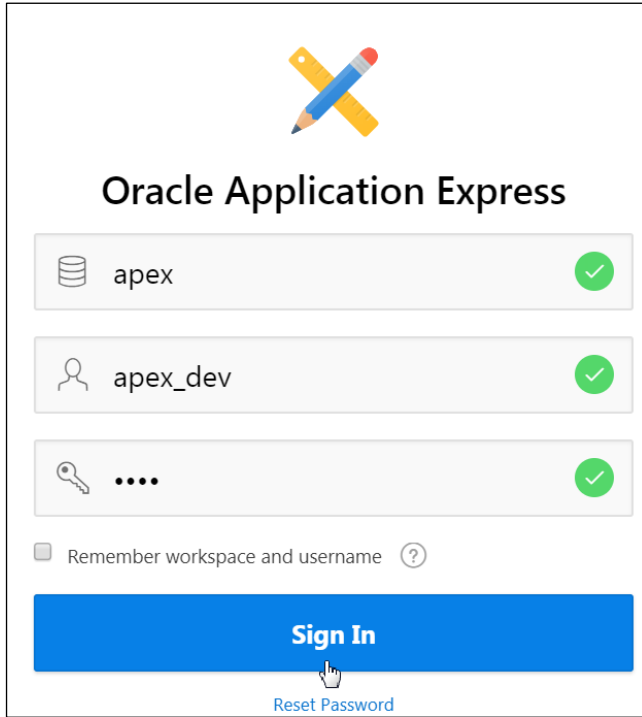
- Sign in to Oracle Application Express as a developer user
- Run the sample database application
- Install a packaged application and use it
- Remove the packaged application from the workspace

Steps

1. Sign in to Oracle Application Express with the following details:
 - **Workspace:** Enter `APEX`
 - **Username:** Enter `apex_dev`
 - **Password:** *<mentioned in the password document>*
- a. Click the **Return to Sign In Page** button.

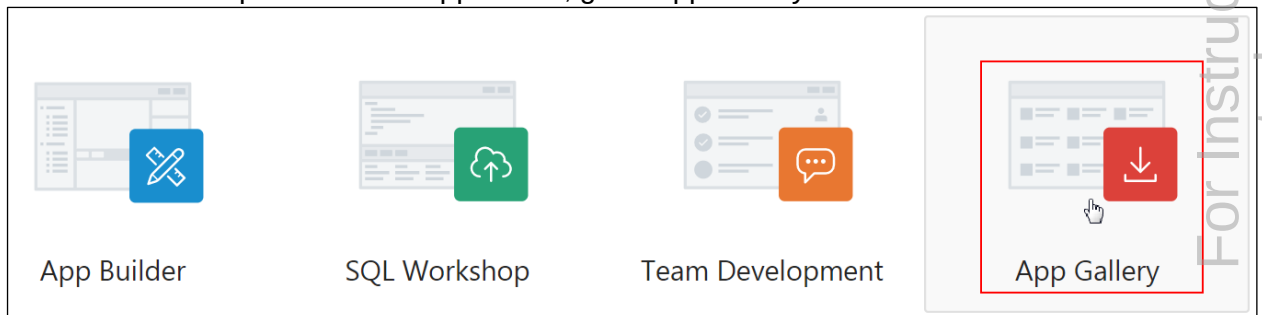


- b. Enter `apex` for **Workspace** and enter the `apex_dev` user credentials. Click **Sign In**.

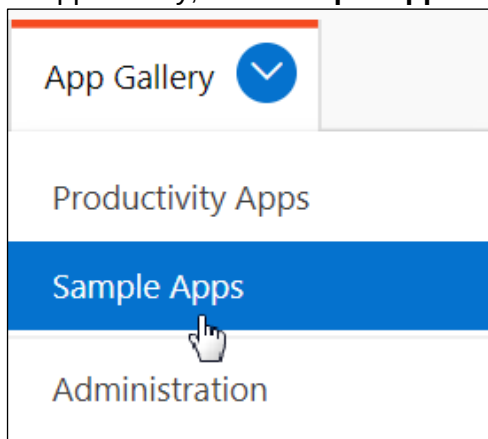


The screenshot shows the Oracle Application Express Sign In interface. At the top is the Oracle APEX logo (two crossed pencils). Below it is the title "Oracle Application Express". There are three input fields: the first contains "apex" with a database icon and a green checkmark; the second contains "apex_dev" with a user icon and a green checkmark; the third contains masked characters "...." with a key icon and a green checkmark. Below these fields is a checkbox labeled "Remember workspace and username" with a question mark icon. At the bottom is a large blue "Sign In" button with a hand cursor pointing to it, and a link "Reset Password" below it.

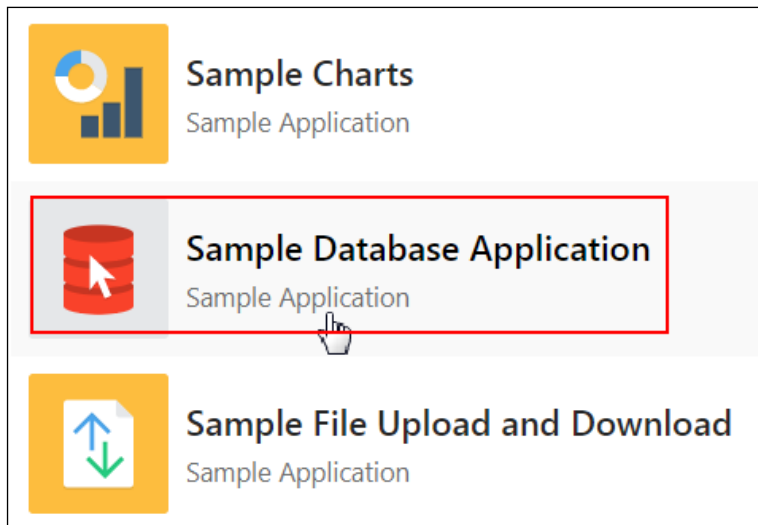
2. **Install and Run the Sample Database Application**, which is already available in your workspace.
- a. To install the Sample Database Application, go to App Gallery.



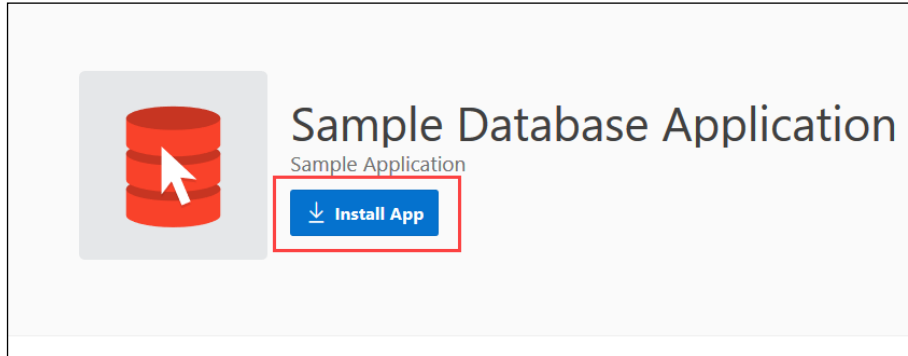
- b. In App Gallery, click **Sample Apps**.



- c. Under Sample, click **Sample Database Application**.



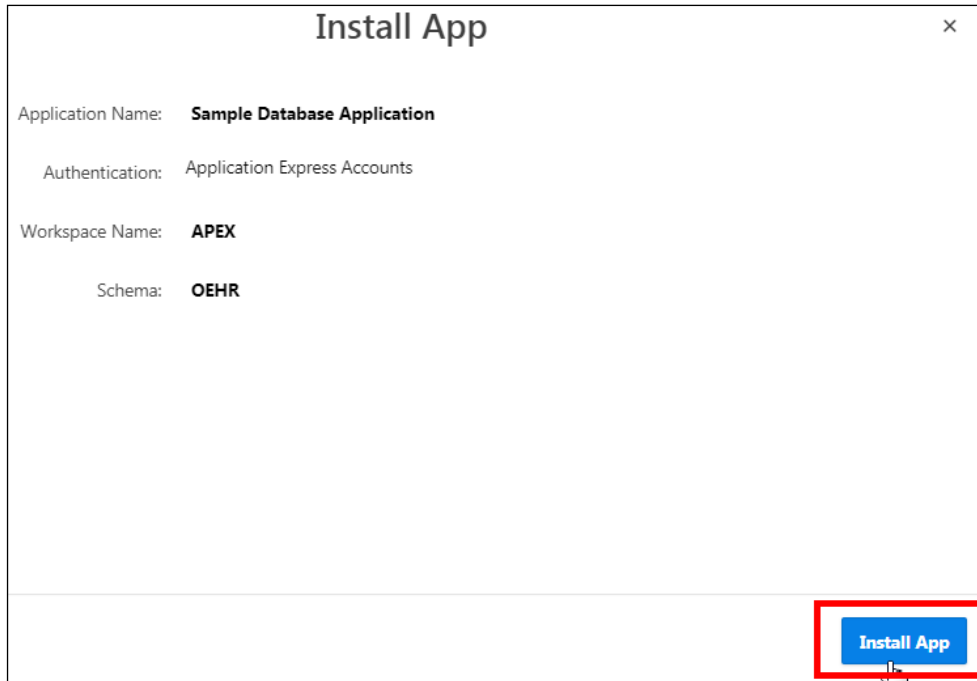
- d. Click **Install App**.



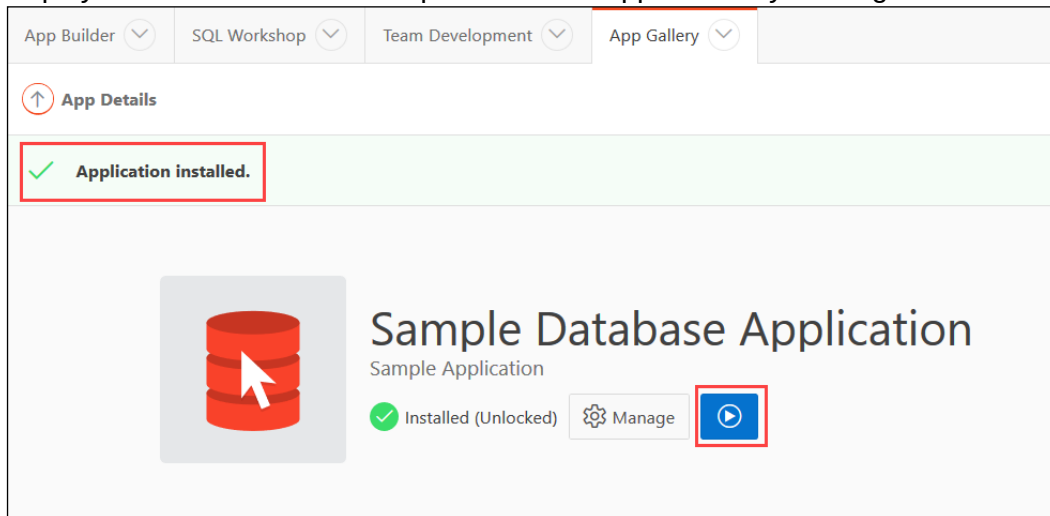
- e. In the Install App Wizard, ensure that, under **Authentication**, **Application Express Account** is selected. Click **Next >**.

The screenshot shows the 'Install App' wizard. The title is 'Install App'. Below it, the 'Application Name' is 'Sample Database Application'. The 'Authentication' tab is selected and highlighted with a red box. The dropdown menu for authentication is open, showing 'Application Express Accounts' as the selected option. Below this, the 'Workspace Name' is 'APEX' and the 'Schema' is 'OEHR'.

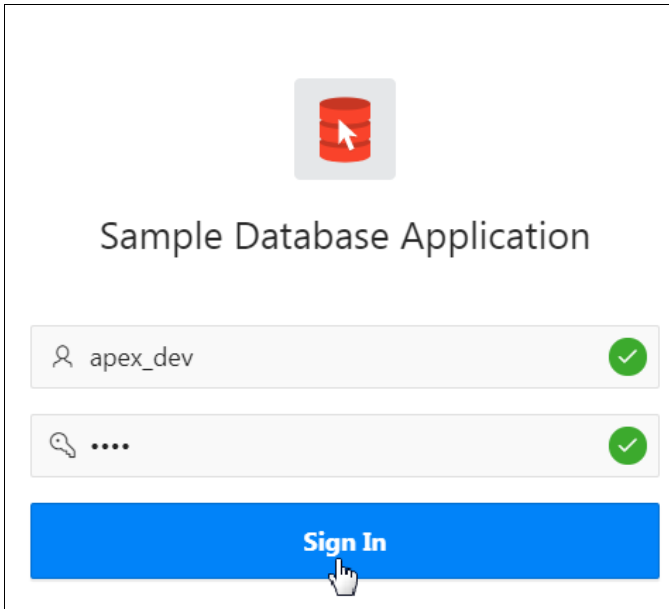
- f. Click **Install App**. The installation process begins.



- g. After the installation is complete, the following message “*Application installed*” is displayed. You can run the Sample Database Application by clicking the Run icon.

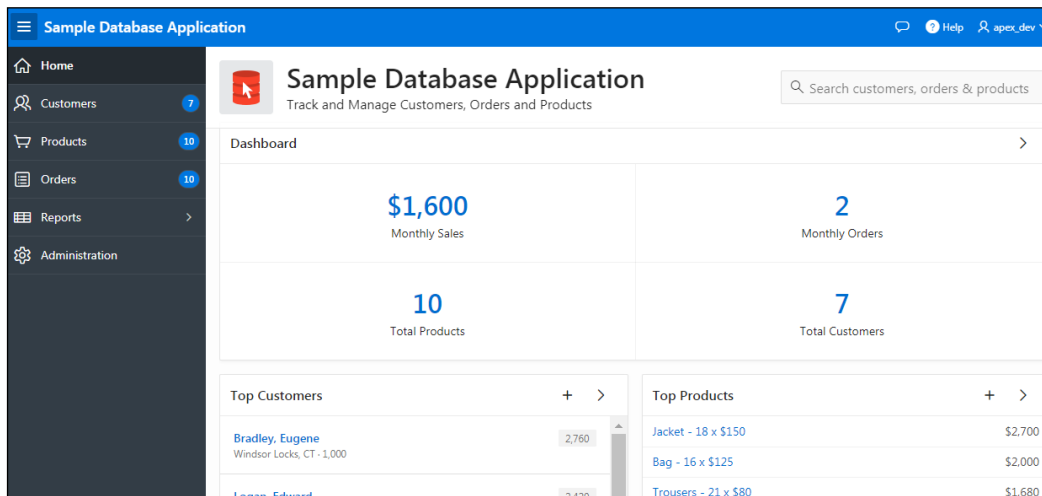


- h. You are prompted to enter your login credentials to sign in to the application. Enter your Application Express Developer Username and Password and click **Sign In**.



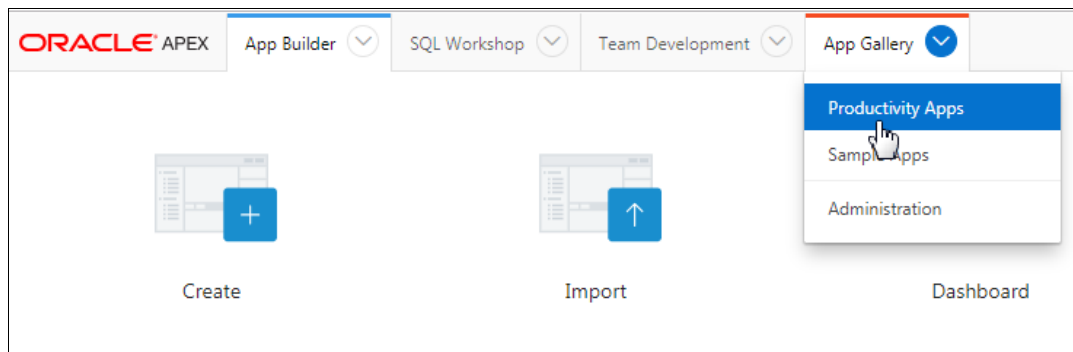
The image shows the login screen for the 'Sample Database Application'. At the top is a red database icon. Below it is the title 'Sample Database Application'. There are two input fields: the first contains 'apex_dev' with a green checkmark to its right, and the second contains four dots with a green checkmark to its right. At the bottom is a large blue button labeled 'Sign In' with a mouse cursor pointing at it.

- i. The Sample Database Application opens. Click the various tabs and understand what the application does.

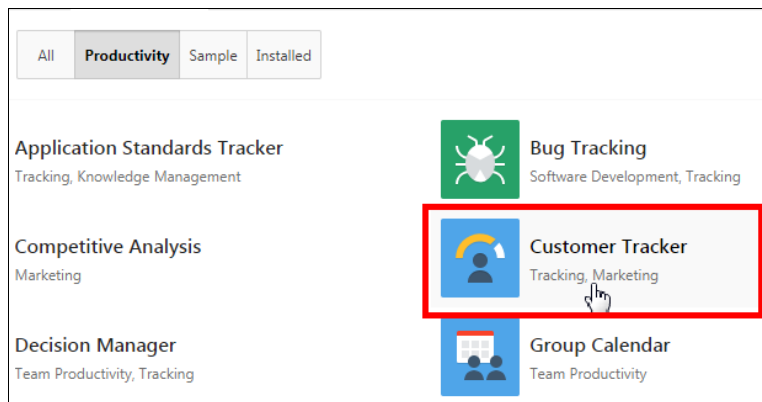


After you have explored the application, close the browser. You have successfully run the sample database application.

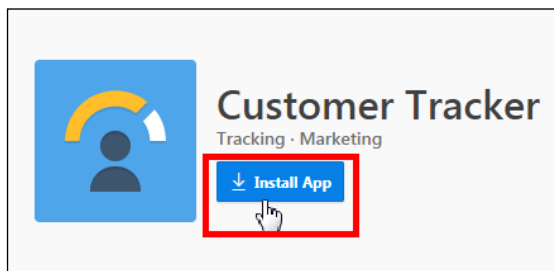
3. Install and use a packaged application. You install and use **Customer Tracker** available under packaged applications in your workspace.
- a. Click the arrow next to **App Gallery** in the navigation bar and select **Productivity Apps**.



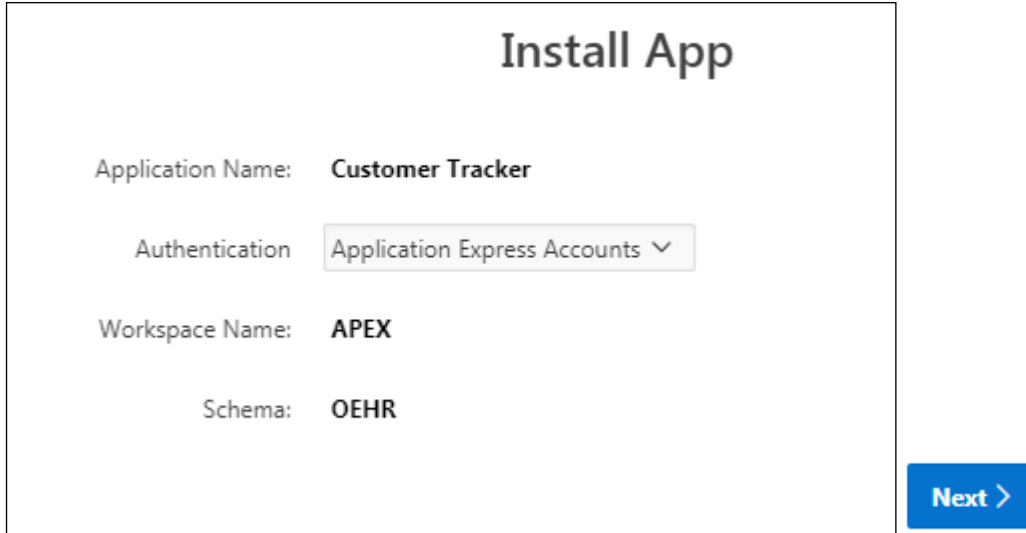
- b. You find a set of packaged applications that are available for you to install. In this activity, you install the Customer Tracker application. Click **Customer Tracker**.



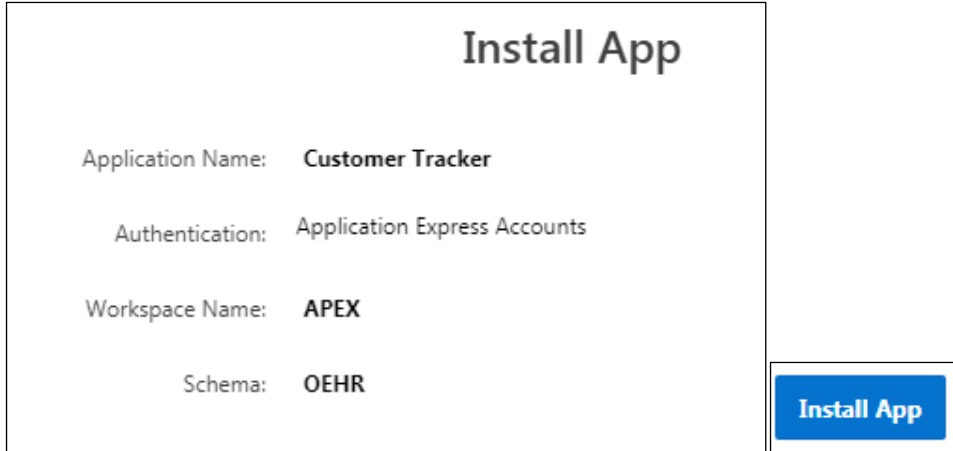
- c. A description about the application is available. Read the description and **click Install Application**.



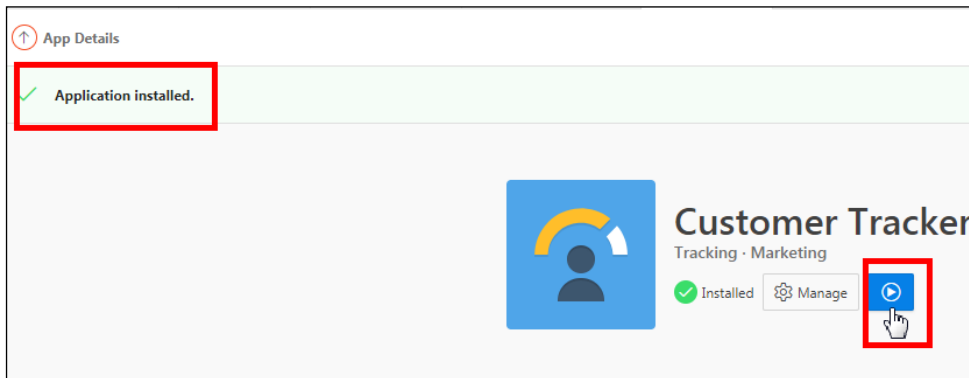
- d. Retain the default authentication and click **Next >**.



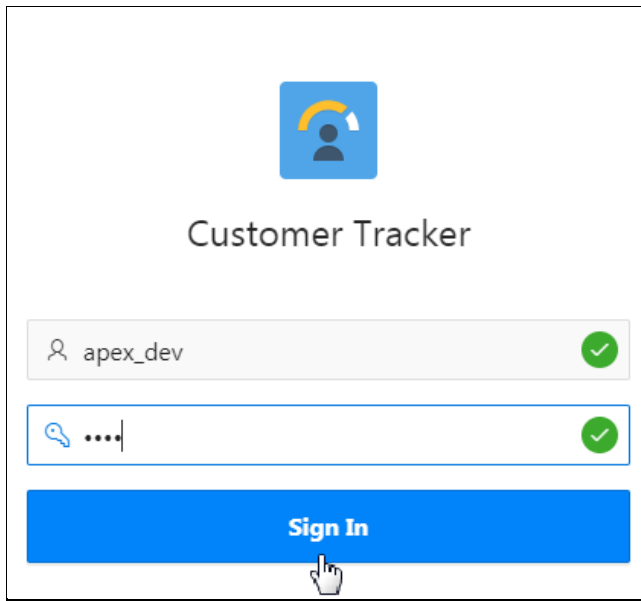
- e. Click **Install App**.



- f. After the application is installed successfully, click the **Run** icon .



- g. Enter your login credentials to sign in to the application (if prompted) and click **Sign In**.



The image shows a login interface for an application titled "Customer Tracker". At the top, there is a blue square icon with a white person silhouette and a yellow arc above it. Below the icon, the text "Customer Tracker" is displayed. There are two input fields: the first contains the text "apex_dev" and has a green checkmark to its right; the second contains four dots "...." and also has a green checkmark to its right. Below these fields is a large blue button with the text "Sign In" in white. A mouse cursor is pointing at the "Sign In" button.

- h. You are taken to the Welcome page of the application. You must do the first-time setup:
- Under Access Control, click **Contributor access for any authenticated user**.
 - Under Users, click **Add User** to add a new user for Customer Tracker. In the screenshot below, you can see that apex_dev is added as an Administrator. In this practice, apex_member is added as a Contributor. To add more users, click **Add User**.

Customer Tracker

Welcome to Customer Tracker

Please take a moment to complete this first time setup.

Access Control

Access Control enables you to control the level of access users have to the application. Your user account is automatically configured as an Administrator.

1

Access Control Level

- ☐ Reader access for any authenticated user
- ☒ **Contributor access for any authenticated user**
- ☐ Restricted to users defined in access control list

Users

2

apex_dev
Administrator

[+ Add User](#)

New User Details

When Access Control is enabled, Administrators have the ability to restrict access to certain application features, for authenticated users. Customer Tracker supports the following 3 access levels; Reader, Contributor and Administrator. **Reader's** have read-only access and can also view reports. **Contributor's** can create, edit, delete and view reports. **Administrator's**, in addition to Contributor's capability, can also perform Customer Tracker administration, including configuration of access control, managing application look-up data and installing or uninstalling sample data.

* Username

* Access Level ☐ Administrator ☒ **Contributor** ☐ Reader

[Cancel](#) [Add and Add Another](#) [Add User](#)

Customer Tracker

Welcome to Customer Tracker

Please take a moment to complete this first time setup.

1

Access Control

Access Control enables you to control the level of access users have to the application. Your user account is automatically configured as an Administrator.

Access Control Level

☒ Reader access for any authenticated user

☐ Contributor access for any authenticated user

☐ Restricted to users defined in access control list

2

Users

apex_dev	Administrator
apex_member	Contributor

[Add User](#)

- iii. Under Sample Data, click **Yes – Load Sample Data**.

3

Sample Data

Loading Sample Data will enable you to explore the functionality of the app with pre-populated sample data.

☒ **Yes - Load Sample Data**

☐ No - Do Not Load Sample Data

- iv. Under Application Options, select Yes to enable the following options: Competitors, Contacts, Feedback, Geography, and Partners.

4

Application Options

Additional options are available, to view and set these navigate to the build option console under administration.

Feature	Enabled
Competitors This feature allows the tracking of competitors, and which customers they are competing against you on.	☒ Yes ☐ No
Contacts Use this build option to enable or disable tracking of customer contacts.	☒ Yes ☐ No
Feedback This feature allows end-users to submit feedback about the application that can be emailed and viewed by application administrators.	☒ Yes ☐ No
Geography Including will allow reporting and management of geography by customer. Excluding will not show the geography attribute for customers.	☒ Yes ☐ No
Partners This feature allows the tracking of partners, and which customers they are working with you on.	☒ Yes ☐ No

- v. Click **Complete Set Up**.

4

Competitors This feature allows the tracking of competitors, and which customers they are competing against you on.	Yes No
Contacts Use this build option to enable or disable tracking of customer contacts.	Yes No
Feedback This feature allows end-users to submit feedback about the application that can be emailed and viewed by application administrators.	Yes No
Geography Including will allow reporting and management of geography by customer. Excluding will not show the geography attribute for customers.	Yes No
Partners This feature allows the tracking of partners, and which customers they are working with you on.	Yes No

Complete Set Up ✓

- i. After the setup is completed, you are signed in to the application.

Customer Tracker

Customers 19
Categories
Geographies
Referenceability
Products
Dashboard
Products 5
Partners
Competitors
Contacts 28

Customer Tracker
Track and Manage Customers

Access Control has been enabled, defined users have been added, and application settings have been successfully updated.

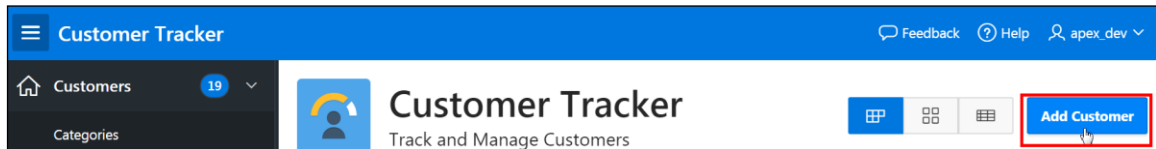
Search Customers

Product: - All -
Geography: - All -
Publicly Referenceable: - All -
Reference Types (any):
☐ Analyst Interview
☐ Available for Calls

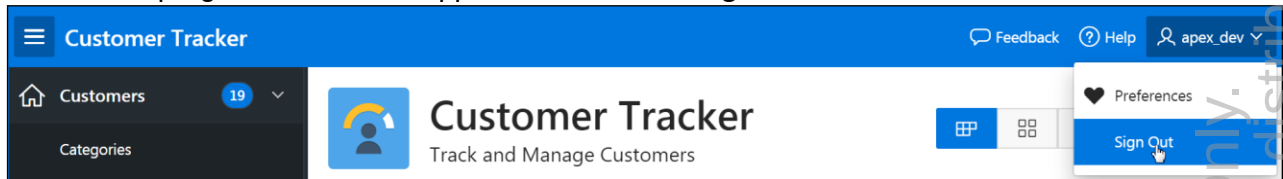
Illumina Biotech IB Specialize in validation, Life Sciences, North A...	Cyphria CY Manufacture drugs primarily for Life Sciences, North A...	Mogul Mashups MM Produce new age marketing Communications, EMEA
Bankgra... BA Specialize in financial	Sakoro Speakers SS Manufacture high-fidelity speaker	Synergy Sales Solutions SS Sell financial

j. Explore the Customer Tracker application. You can do the following:

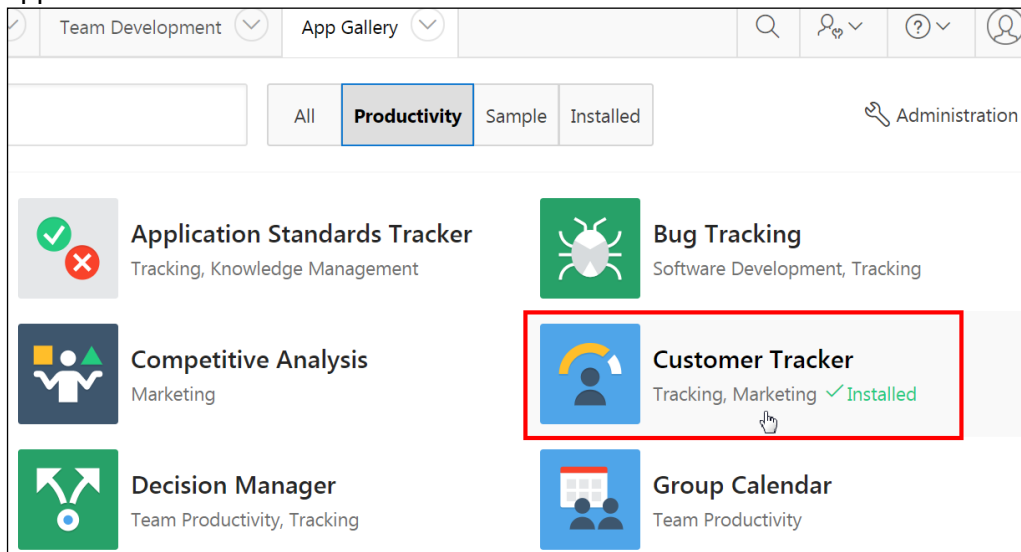
- Click **Add Customer** to add new customers.
- Click  to view the page in Card view.
- Click  to view the page in Logo view.
- Click  to view the page in Report view.



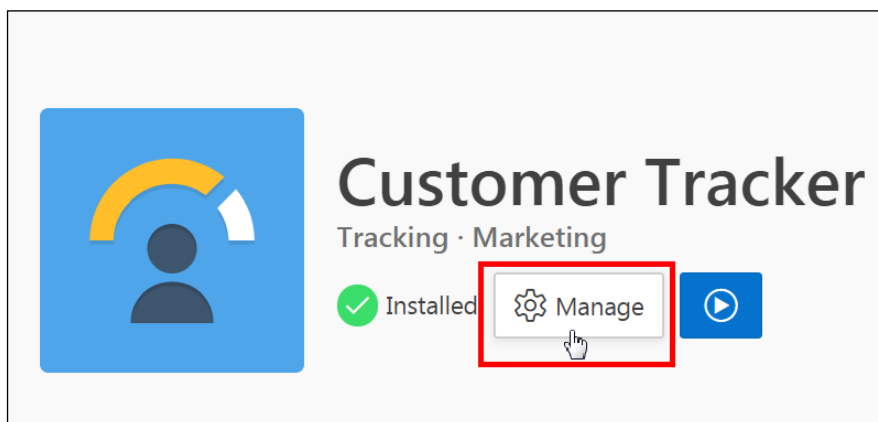
k. Go to the top-right corner of the application and click **Sign Out**. Close the browser.



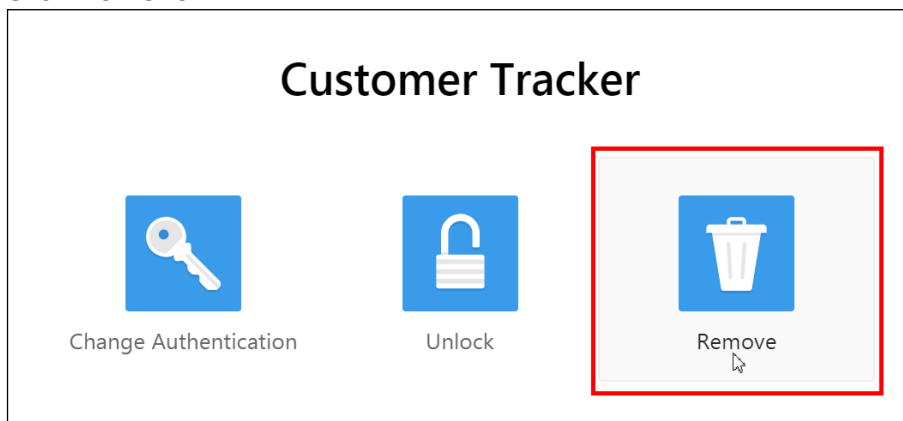
4. To remove the **Customer Tracker** packaged application from your workspace:
- a. Go to App Gallery and then click **Productivity**. Click the **Customer Tracker** application.



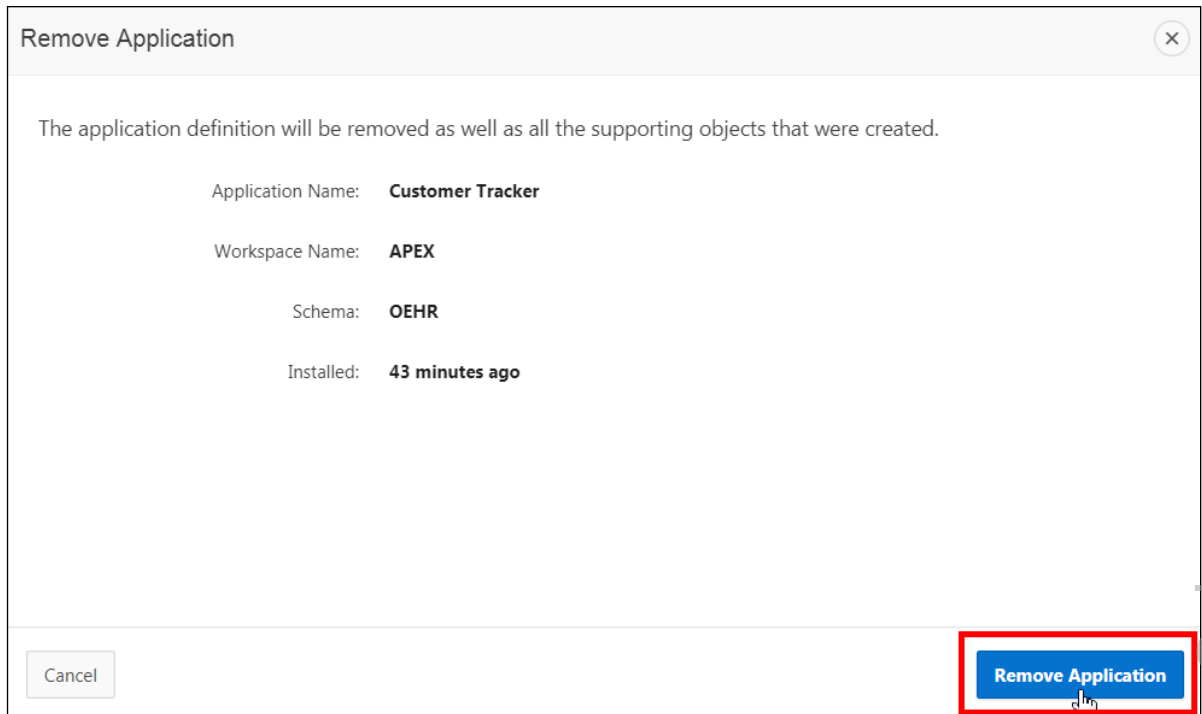
- b. On the Customer Tracker page, click **Manage**.



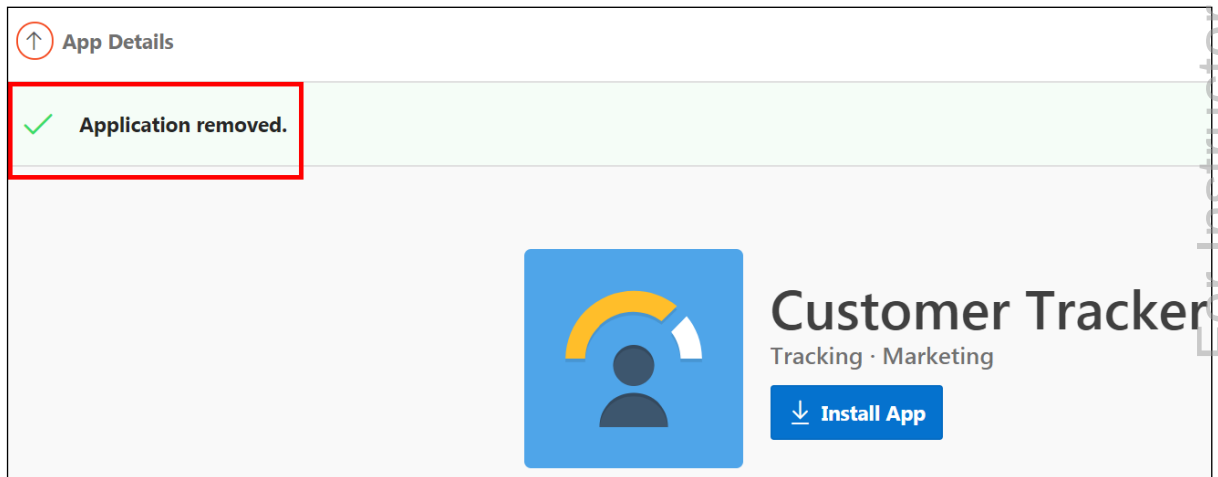
- c. Click **Remove**.



- d. Click **Remove Application** to confirm.



Wait until the application is removed completely. After the application is removed, the message "Application removed" is displayed.



Solution 2-3: Installing a Sample Dataset

Overview

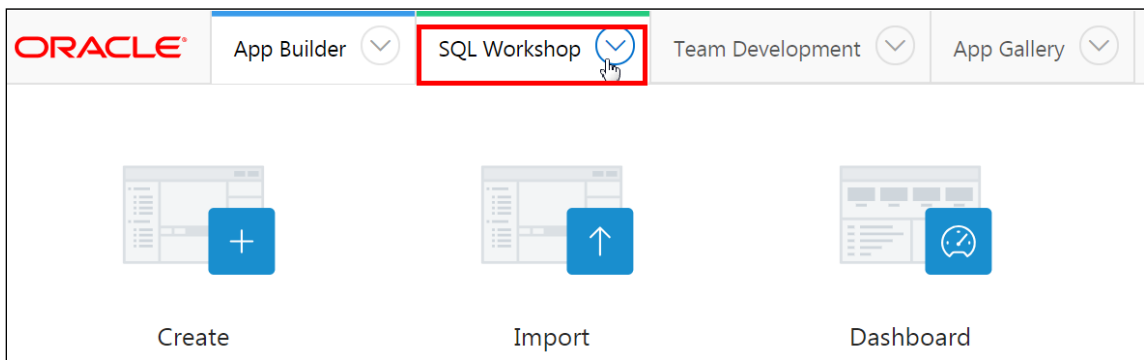
In this practice solution, steps are provided to:

- Sign in to Oracle Application Express as a developer user.
- Install the sample dataset `HR Data` in the `OEHR` schema.

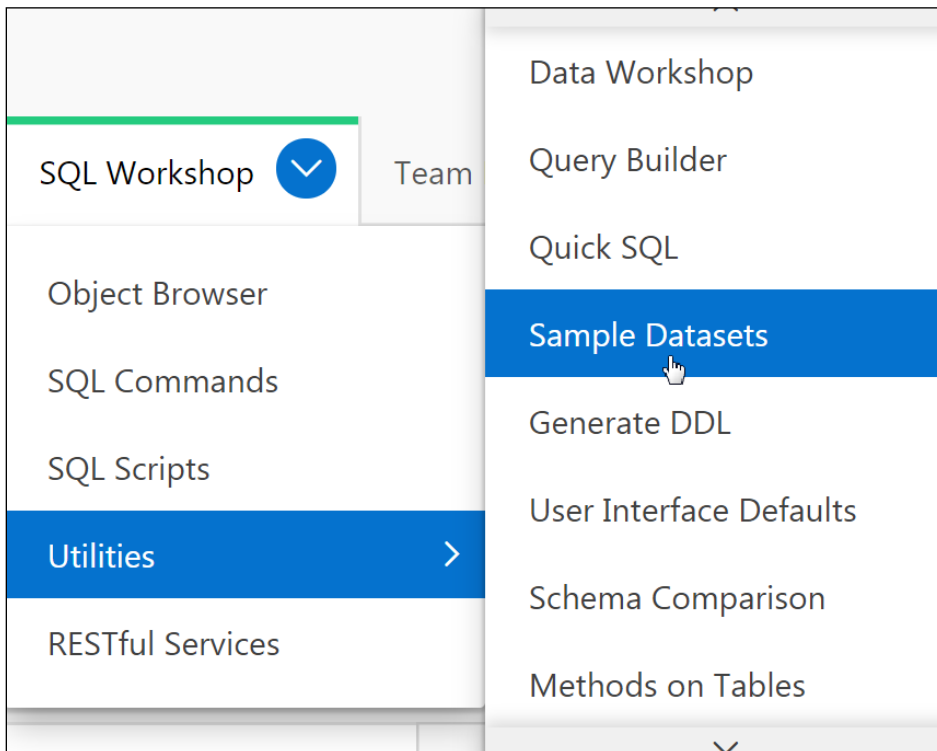
Steps

To install the sample dataset:

1. On the Oracle Application Express Workspace home page, click **SQL Workshop**.



2. Select **Utilities** and click **Sample Datasets**.



For Instructor Use Only.
This document should not be distributed.

3. On the Sample Datasets page, find the dataset **HR Data** and click **Install**. If the sample dataset is already installed, click **Update**.

Sample Datasets			
Action	Name	Languages	Description
Install	Project Data	English	A collection of projects, milestones, tasks, and more. This dataset includes master-detail-detail relationships and a useful view for charting.
Install	EMP / DEPT	English, Chinese, Czech, French, German, Japanese, Korean, Polish, Russian, Spanish	The generic EMP and DEPT tables.
Install	HR Data	English	The generic HR tables commonly used by Oracle Education.
Install	Tasks Spreadsheet	English	A single table with unnormalized data containing tasks, dates, status, assigned to, cost, and budget.

4. On the Manage Sample Datasets page, select options for:
 - a. **Language:** Select *English*
Schema: Select your schema *OEHR*.
 - b. Click **Next**. The Load Sample Dataset page opens, displaying the database object that will be installed.

Manage Sample Dataset

Dataset

SAMPLE_DS.OEHR

Description

SAMPLE_DS.OEHR.DESC

Change History

Installed during Application Express installation.

Dataset Last Updated

08-APR-2019

Table Prefix

OEHR

Install Dataset

Language

English

Schema

OEHR

APEX

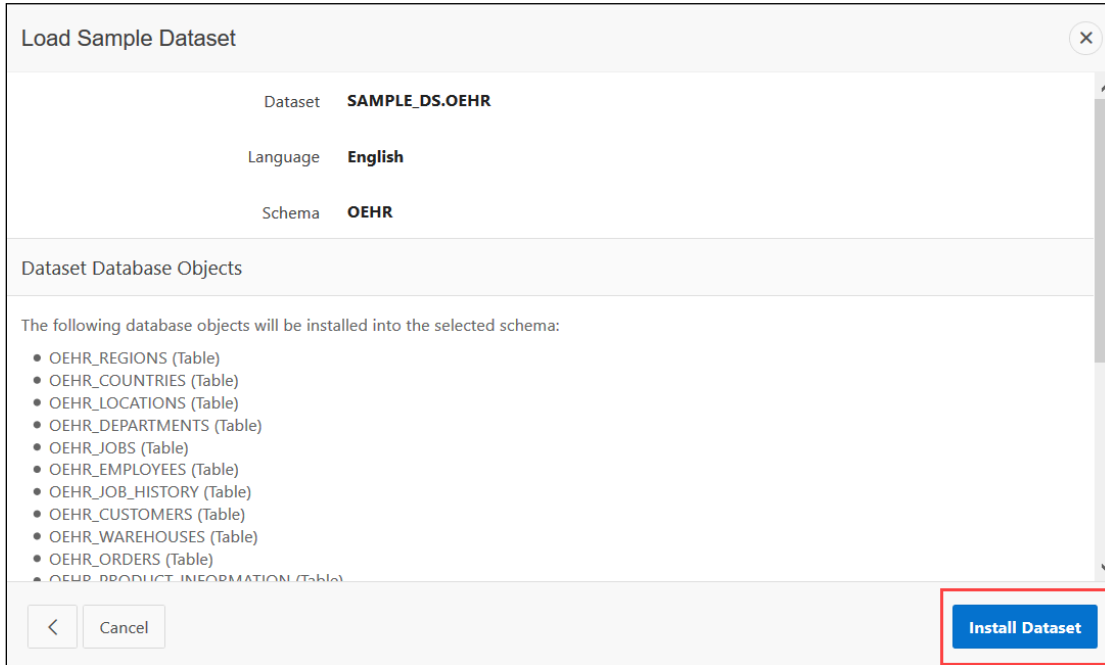
OEHR

PTS

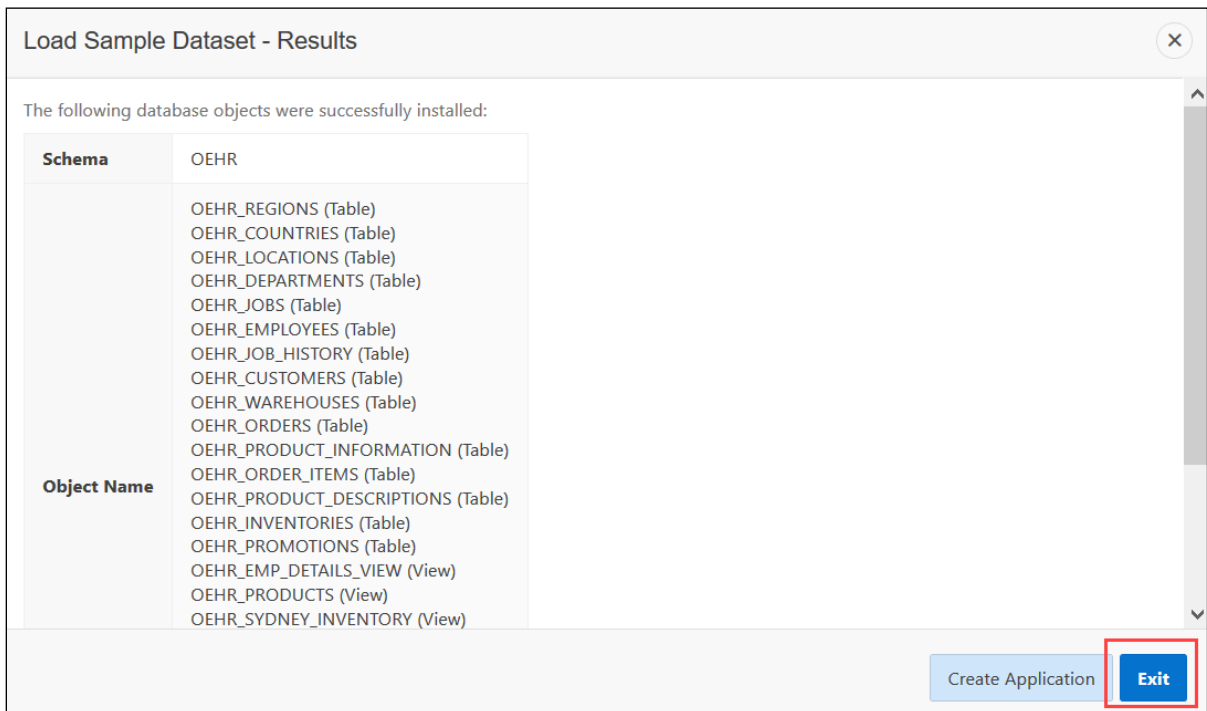
Cancel

Next

5. Click **Install Dataset**.



6. The Load Sample Dataset - Results page appears. At this point, you can create an application or exit. Click **Exit**.



The Sample Dataset page displays again. This completes the task of loading Project Data sample dataset.

Note: The Action column now displays Update instead of Install.

Tip: You can click **Update** to refresh a sample dataset or remove it.

**Practices for Lesson 3:
Creating a Database
Application**

Practices for Lesson 3: Overview

Overview

In these practices, you create a database application. You also create a database application by using a spreadsheet.

For Instructor Use Only.
This document should not be distributed.

Practice 3-1: Creating Database Applications

Overview

In this practice, you create GlobalMart Management Tool (GMT).

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a database application named **GlobalMart Management Tool**. The **Application ID** entered here is **111**; In the rest of the practices in this course, you will enhance and complete this application by creating pages, regions, processes, and so on.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 3-1**.

Practice 3-2: Creating a Database Application from a File

Overview

In this practice, you create a database application from a file.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a database application from a file. You create an application named **Tasks** based on the `tasks.txt` file available in the `/home/oracle/labs/labs/` folder. The file has tab-delimited information, and the first row contains the column names. Name the table, where the data from the file is uploaded, **Tasks**.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 3-2**.

If you haven't completed **Practice 3-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_03_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

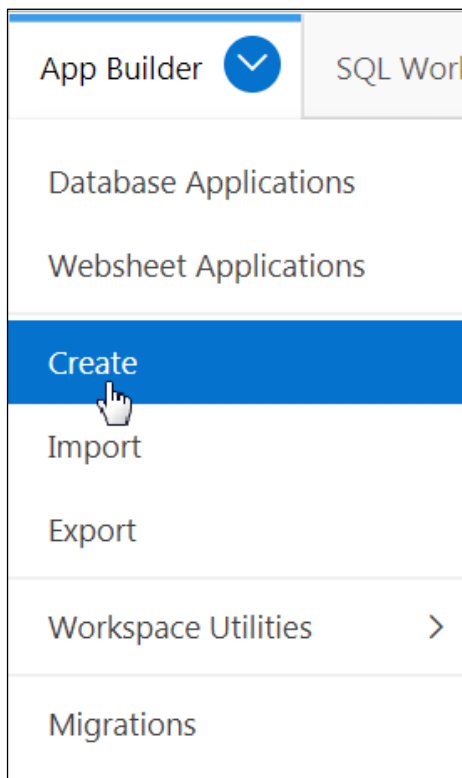
Solution 3-1: Creating Database Applications

Overview

In this practice solution, steps are provided for creating the GlobalMart Management Tool application.

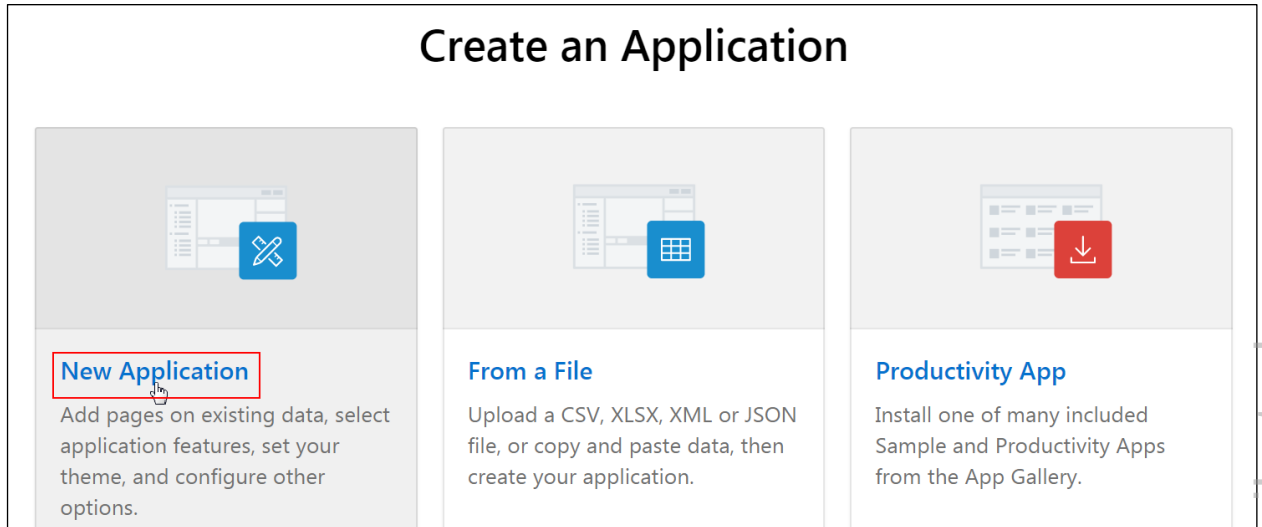
Steps

1. Create a database application named **GlobalMart Management Tool**. In the rest of the practices in this course, you will enhance and complete this application by creating pages, regions, processes, and so on.
 - a. Select the **App Builder** pull-down menu and click **Create**.



For Instructor Use Only.
This document should not be distributed.

- b. In the Create Application Wizard, click **New Application**.



- c. In the *Create an Application* Wizard, enter the following:

- **Name:** Enter GlobalMart Management Tool.
- Click **Set Appearance**  to customize the appearance of the application.

The screenshot shows the 'Create an Application' wizard with the following fields and options:

- Name**: GlobalMart Management Tool
- Appearance**: Vita, Side Menu
- Pages**:  **Add Page**

- d. Scroll down the Create App Wizard page and enter the following under **Settings**:

- In the **Application ID** field, enter 111.
- In the **Schema** field, select OEHR.
 - **Note:** Throughout the course, the application ID remains 111.

- Click **Create Application**.

Settings ?

Application ID: 111

Schema: OEHR

Authentication: Application Express Accounts

Language: English (en)

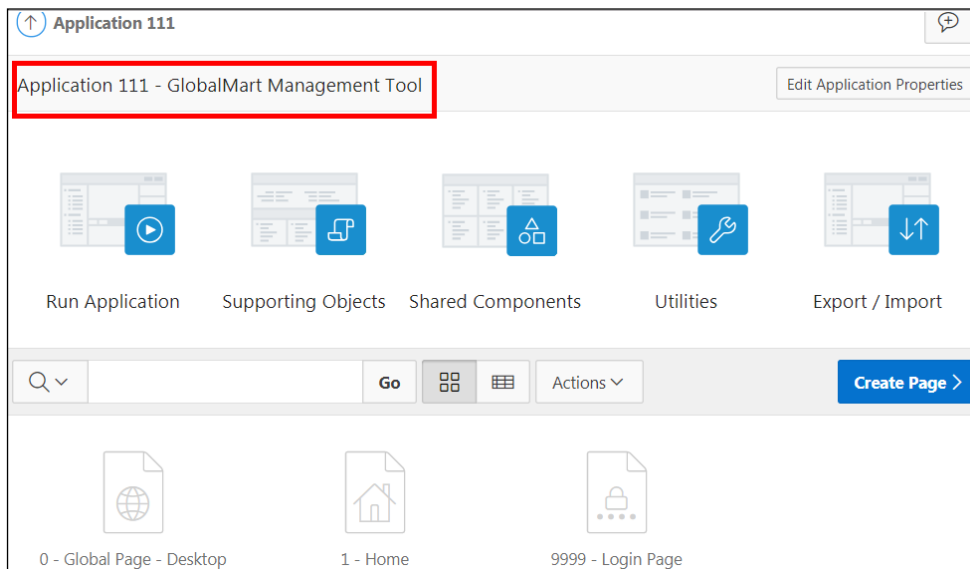
Advanced Settings

User Interface Defaults

Cancel

Create Application

- e. The *GlobalMart Management Tool* application is created. By default, the following three pages are created:
- Global Page – 0 – Global Page – Desktop
 - Home page – 1 - Home
 - Login page – 9999 – Login page



Solution 3-2: Creating a Database Application from a File

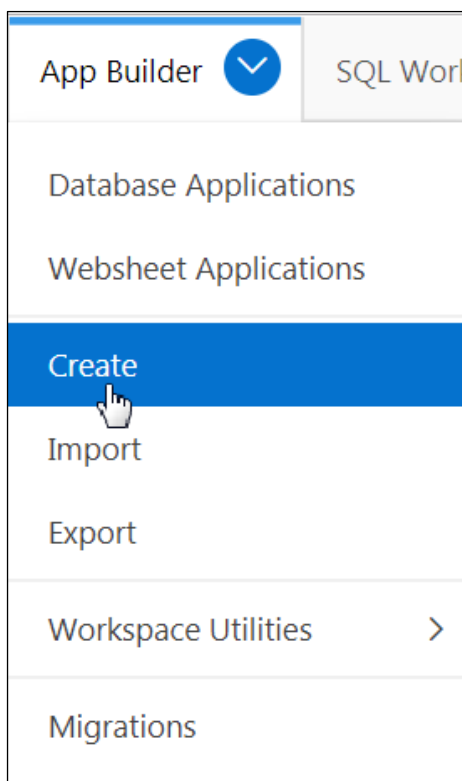
Overview

In this practice solution, steps are provided for creating a database application from a file.

Note: Practice 3-2 is an optional practice and does not have a solution script.

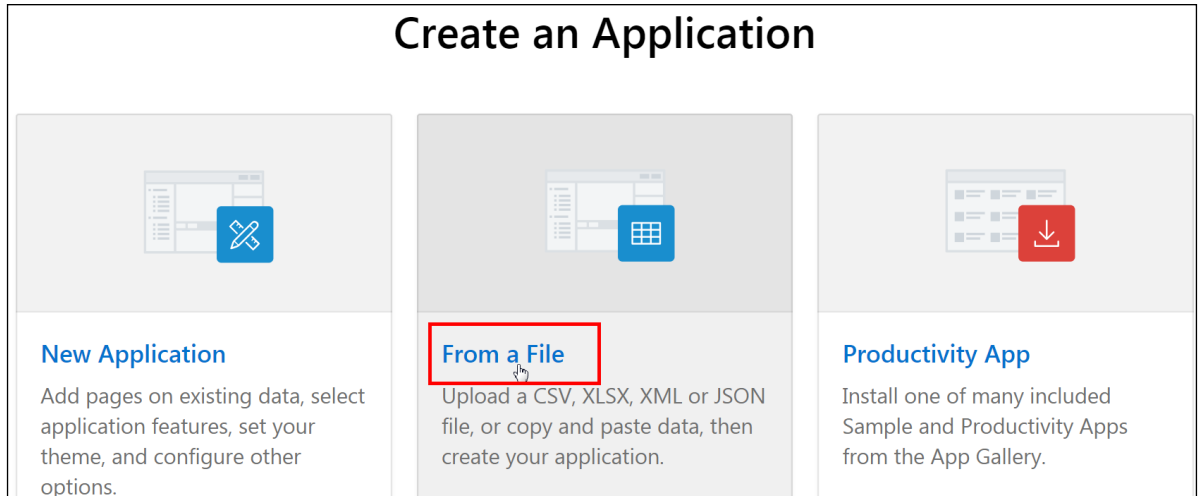
Steps

1. Create a database application from a file. You create an application named **Tasks** based on the `tasks.txt` file available in the `/home/oracle/labs/labs/` folder. The file has tab-delimited information and the first row contains the column names. Name the table, where the data from the file is uploaded, **Tasks**.
 - a. Select the **App Builder** pull-down menu and click **Create**.



For Instructor Use Only.
This document should not be distributed.

- b. In the Create Application Wizard, click **From a File**.



For Instructor Use Only.
This document should not be distributed.

- c. The Load Data page opens. You can either upload a file or copy and paste a file. Here we select **Upload a File** tab. You can either drag and drop file or select the Choose File option. Here we click **Choose File**.

Load Data

Upload a File

Copy and Paste

Drag and drop file here or

Choose File

Supported formats csv, xlsx, txt, xml, json

- d. Select **tasks.txt** file from the `/home/oracle/labs/labs/` folder. The Load Data page opens with a preview. Click **Load Data**.

Load Data

tasks.txt

Settings

Column Headers ?

☒ First line contains headers

Column Delimiter ?

,

;

|

#

tab

Enclosed By ?

None

"

'

File Encoding ?

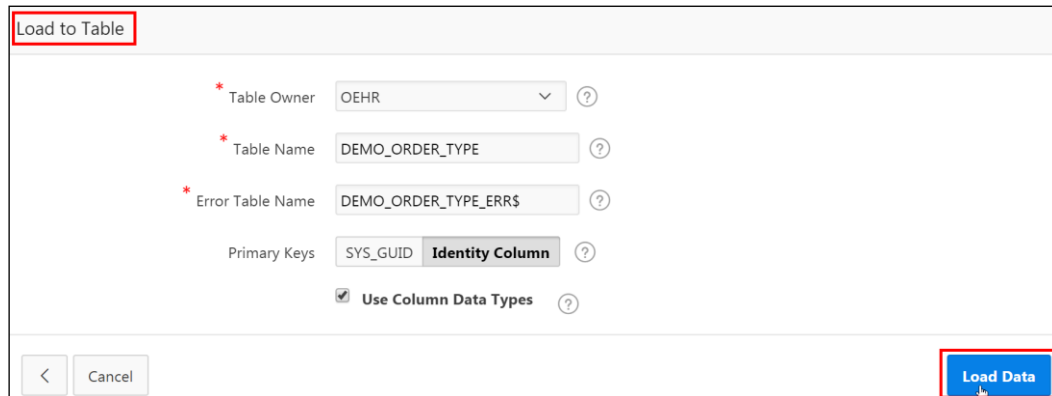
Unicode UTF-8

The Load Data page opens with a preview of rows and columns.

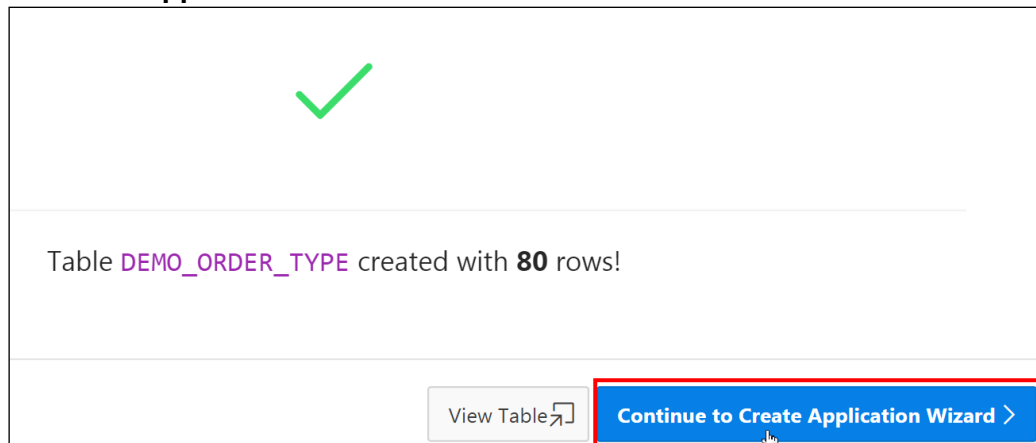
Preview								
Parsed first 81 rows to sample the column types. The preview below only displays the first 8 columns and 5 rows. To view the full preview, configure data load settings, and set which columns to load, please click Configure button. Configure								
1	PROJECT	TASK_NAME	START_DATE	END_DATE	STATUS	ASSIGNED_TO	COST	BUDGET
2	Maintain Support Systems	HR software upgrades	1-Jan-10	27-Feb-10	Closed	Pam King	8000	7000
3	Maintain Support Systems	Apply Billing System updates	1-Jan-10	28-Feb-10	closed	Russ Sanders	5000	7000

e. Enter the following details under Load to Table and click **Load Data**:

- Table Owner: Select OEHR
- In the **Table Name** field, enter DEMO_ORDER_TYPE.
- The **Error Table Name** field gets populated by default as DEMO_ORDER_TYPE_ERR\$.



You get the message DEMO_ORDER_TYPE created with 80 rows. Click **Continue to Create Application Wizard**.



2. The Create Application page appears. You can now start creating your application. In the Create App Wizard, ensure that Application Name is **TASKS**.

View Blueprint | Load Blueprint

Create an Application

Name: Tasks

Appearance: Vita, Side Menu

Pages ?

+ Add Page

3. Click **Create Application**.

Settings ?

Application ID: 109

Schema: OEHR

Authentication: Application Express Accounts

Language: English (en)

Advanced Settings

User Interface Defaults

Cancel | **Create Application**

4. To view the application, click **Run Application**.

Application 109 - Tasks

Edit Application Properties

Run Application

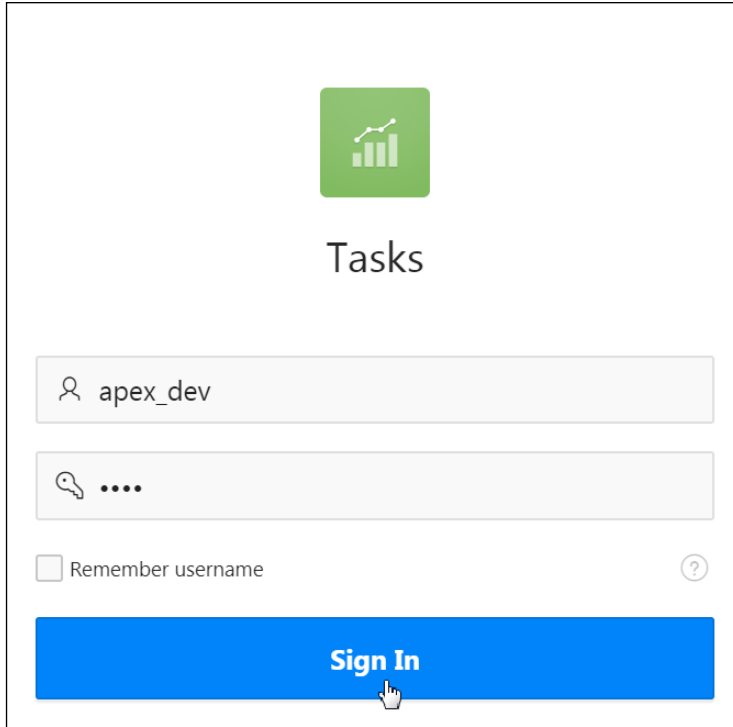
Supporting Objects | Shared Components | Utilities | Export / Import

Go | Actions

Create Page >

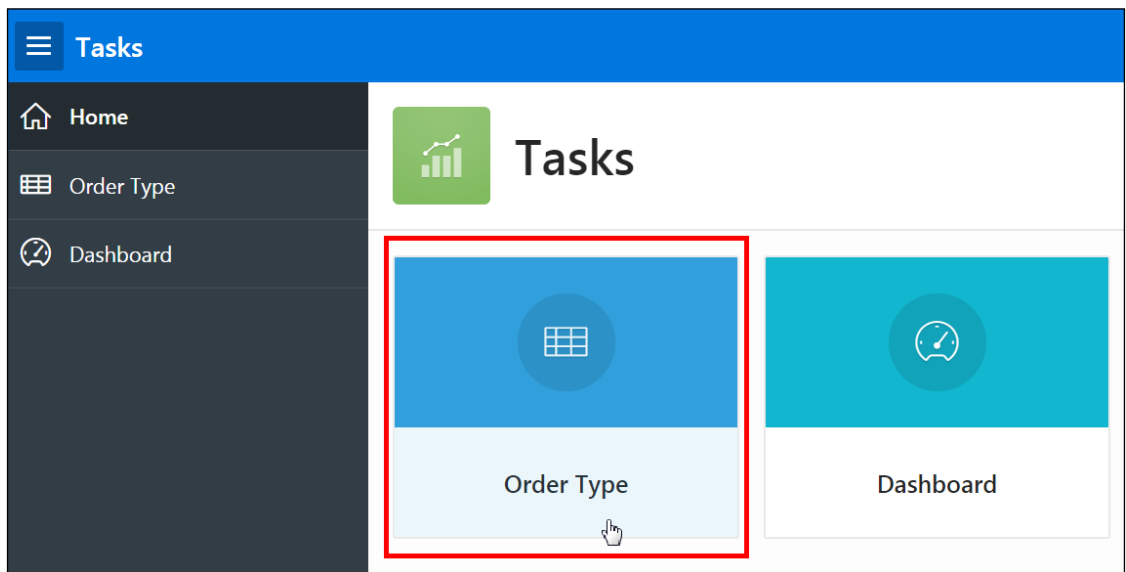
0 - Global Page - Desktop | 1 - Home | 2 - Order Type | 3 - Order Type | 4 - Dashboard | 9999 - Login Page

- a. Enter the developer user credentials and click **Sign in**.



The sign-in form features a green square icon with a white bar chart and an upward arrow. Below the icon is the title "Tasks". The form includes a username field with a person icon and the text "apex_dev", a password field with a key icon and four dots, a "Remember username" checkbox, and a "Sign In" button. A question mark icon is located to the right of the checkbox. A mouse cursor is pointing at the "Sign In" button.

- b. The home page of the **TASKS** application is displayed. Click the **Order Type**. It displays an interactive report by using the data from the file (*tasks.txt*). Click **Home** on the Developer toolbar to return to the development environment.



Tasks

Home

Order Type

Dashboard

Order Type

Q

Go

Actions

Reset

Create

	Project	Task Name	Start Date	End Date	Status	Assigned To	Cost	Budget
	1000	-	-	-	-	-	-	-
	APEX Environment Configuration	Run installation	28-FEB-10	01-MAR-10	Pending	John Watson	100	100
	APEX Environment Configuration	Integrate authentication schemes	01-MAR-10	05-MAR-10	Pending	Scott Spencer	250	500

Home

Application 109

Edit Page 2

Session

View Debug

Debug

Page Info

Quick Edit

Theme Roller

For Instructor Use Only.
This document should not be distributed.

Practices for Lesson 4: Working with Reports

For Instructor Use Only.
This document should not be distributed.

Practices for Lesson 4: Overview

Overview

In these practices, you create Classic reports and two reports optimized for Smartphones, a Column Toggle report and a List View report.

For Instructor Use Only.
This document should not be distributed.

Practice 4-1: Creating Classic Reports

Overview

In this practice, you create a:

- Classic report for GMT application (called Top Tier Salary) by using a SQL query
- Classic report (called Customer Address List) for GMT application by selecting a table

Assumptions

You have completed the previous practices.

Note: If you have not completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. On a new page in the GMT application, create a Classic report by using a SQL query. Name the new page and the report **Top Tier Salary**. The report should display the last name, email, and salary of employees who earn between \$5000 and \$12000 from the `OEHR_EMPLOYEES` table.
2. On a new page in the GMT application, create a Classic report by using the source type as Table. Name the page and the report **Customer Address List**. The report should display the address information of the customers, such as the first and last names, address, city, and state from the `OEHR_CUSTOMERS` table.
3. Modify the Customer Address List report with the following requirements:
 - Rename the report region Customer Address List. Then change the report headings for `CUST_FIRST_NAME` and `CUST_LAST_NAME` to `First Name` and `Last Name`, respectively.
 - Change the Pagination to **Row Ranges X to Y of Z (with pagination)**.
 - Review the query definition and add the `COUNTRIES.COUNTRY_NAME` column to the report.

Hint: You will need to create a join.

Catch Up

If you have not completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 4-1**.

If you have not completed **Practice 3-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **App Builder** and select **Import**.
- c. For Import File, click the button and browse to the `sol_03_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next**.
- d. Click **Next**.
- e. For Install As Application, select **Reuse Application ID 111 From Export File** and click **Install Application**.
- f. Click **Replace Application** (if prompted).
- g. Wait for the application to successfully install.

Practice 4-2: Creating a Column Toggle Report

Overview

In this practice, you create a new page containing a Column Toggle report for the GMT application. Column Toggle reports feature a responsive report optimized for mobile applications and Smartphones. You will deselect a few columns to hide them in the report. Then you will select those columns again to show them in the report.

Assumptions

You have completed the previous practices.

Note: If you have not completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. On a new page in the GMT application, create a **Column Toggle Report**. Enter `Product Information` as the Page Name. Change the Page Number to 202. The report should select all the columns from the `OEHR_PRODUCT_INFORMATION` table.
 - After the page is created, run the page to view the Product Information report with all the columns from `OEHR_PRODUCT_INFORMATION` displayed. Check out the responsive nature of the report.
 - Using the Columns option in the report, hide the columns Product Description, Category ID, and Supplier ID from the report.

Catch Up

If you have not completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 4-2**.

If you have not completed **Practice 4-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **App Builder** and select **Import**.
- c. For Import File, click the button and browse to the `sol_04_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next**.
- d. Click **Next**.
- e. For Install As Application, select **Reuse Application ID 111 From Export File** and click **Install Application**.
- f. Click **Replace Application** (if prompted).
- g. Wait for the application to successfully install.

For Instructor Use Only.
This document should not be distributed.

Practice 4-3: Adding and Modifying a List View Report

Overview

In this practice, you create a list of employees by creating a List View report. List View reports feature a responsive design to display data and provide easy navigation on Smartphones.

Assumptions

You have completed the previous practices.

Note: If you have not completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. On a new page in the GMT application, create a **List View report**. Enter `Employee List` as the Page Name. Change the Page Number to **201**. The list view should select all the columns from the `OEHR_EMPLOYEES` table and should display the last names as the text column. The list view should also be enabled to search the last name column with the Search Type **Like & Ignore Case**.

Catch Up

If you have not completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 4-3**.

If you have not completed **Practice 4-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **App Builder** and select **Import**.
- c. For Import File, click the button and browse to the `sol_04_02.sql` file in the `/home/oracle/labs/solutions` directory and then click **Next**.
- d. Click **Next**.
- e. For Install As Application, select **Reuse Application ID 111 From Export File** and click **Install Application**.
- f. Click **Replace Application** (if prompted).

- g. Wait for the application to successfully install.

For Instructor Use Only.
This document should not be distributed.

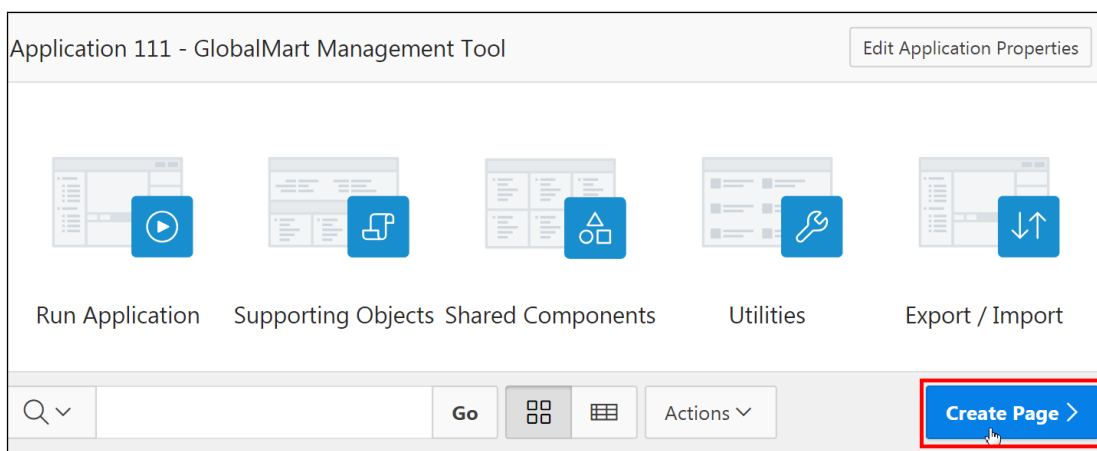
Solution 4-1: Creating Classic Reports

Overview

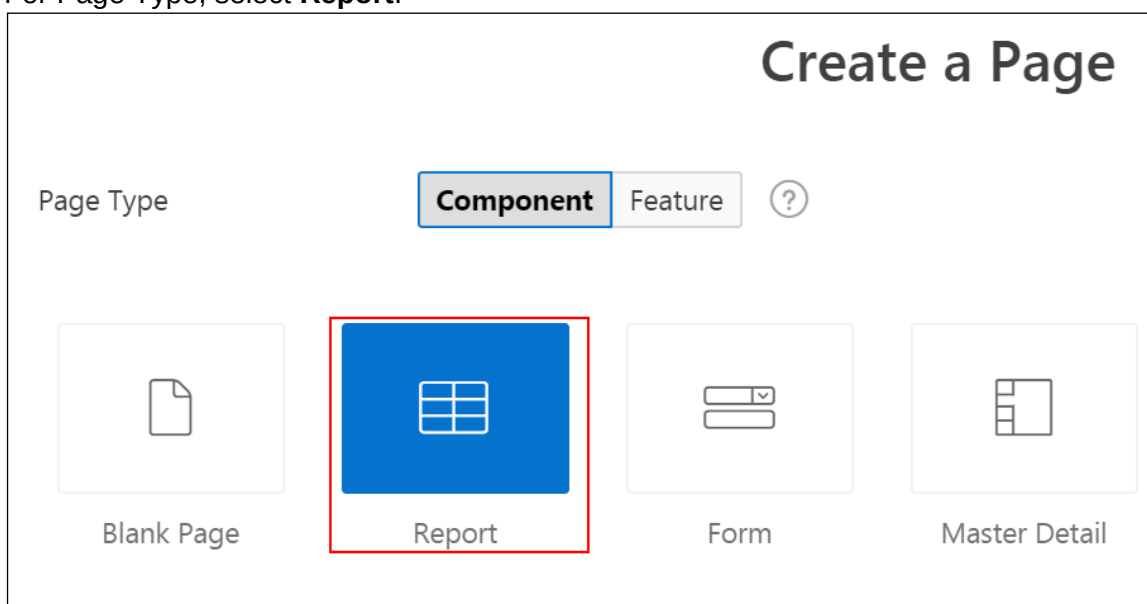
In this practice solution, you create Classic reports for the GlobalMart Management Tool (GMT) application.

Steps

1. On a new page in the GMT application, create a Classic report by using a SQL query. Name the new page and the report **Top Tier Salary**. The report should display the last name, email, and salary of employees who earn between \$5000 and \$12000 from the `OEHR_EMPLOYEES` table.
 - a. On the Workspace home page, click **App Builder**, select the **GlobalMart Management Tool** application, and click **Create Page**.



- b. For Page Type, select **Report**.



- c. Select **Classic Report**.

Create Page


Interactive Grid


Classic Report


Report with Form


List View

- d. Update the following Page Attributes:

- **Page Number:** Enter 2.
- **Page Name:** Enter Top Tier Salary.

Create Classic Report

Page Attributes

Type **Classic Report**

* Page Number ?

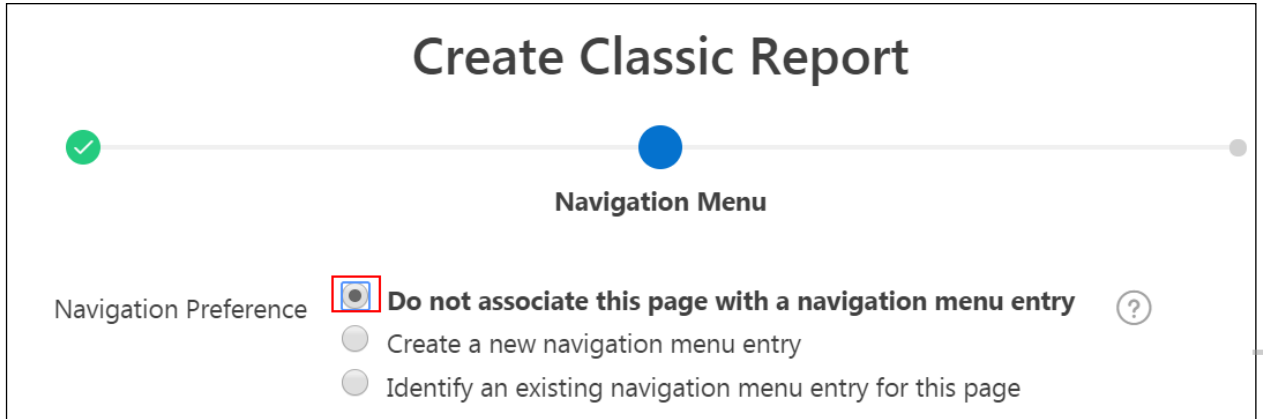
* Page Name ?

Page Mode **Normal** Modal Dialog ?

Breadcrumb ?

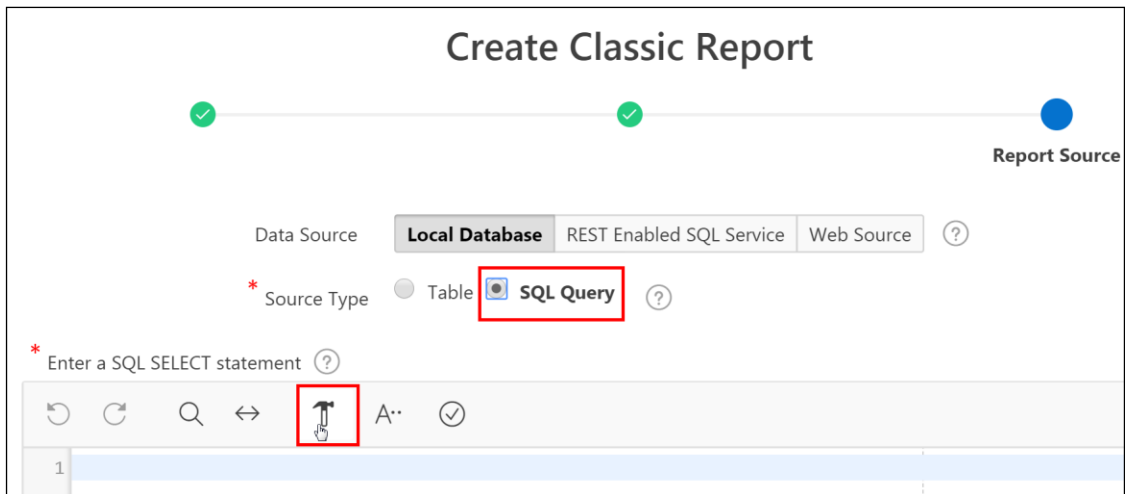
- e. Click **Next >**.

- f. For Navigation Menu, accept the default and click **Next**.

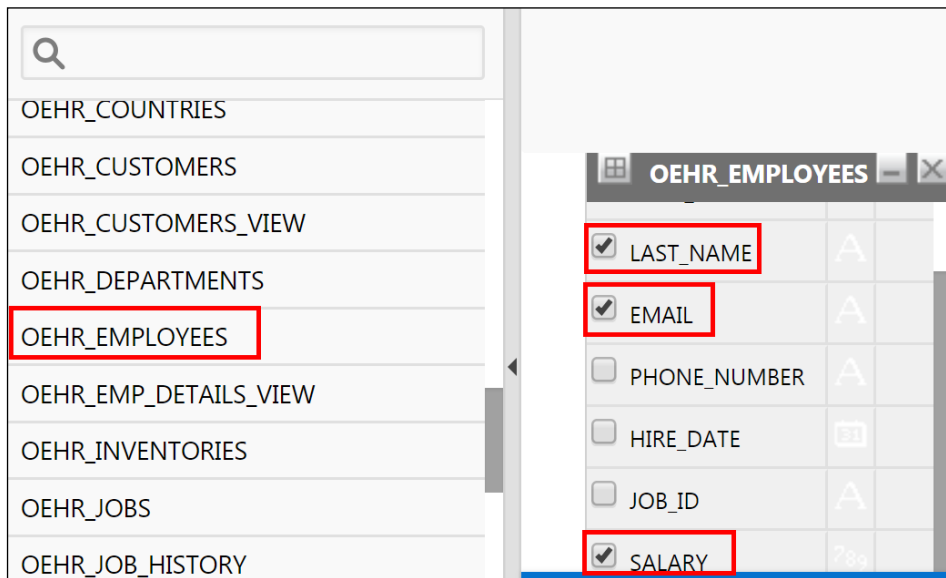


- g. For **Source Type**, select **SQL Query**.

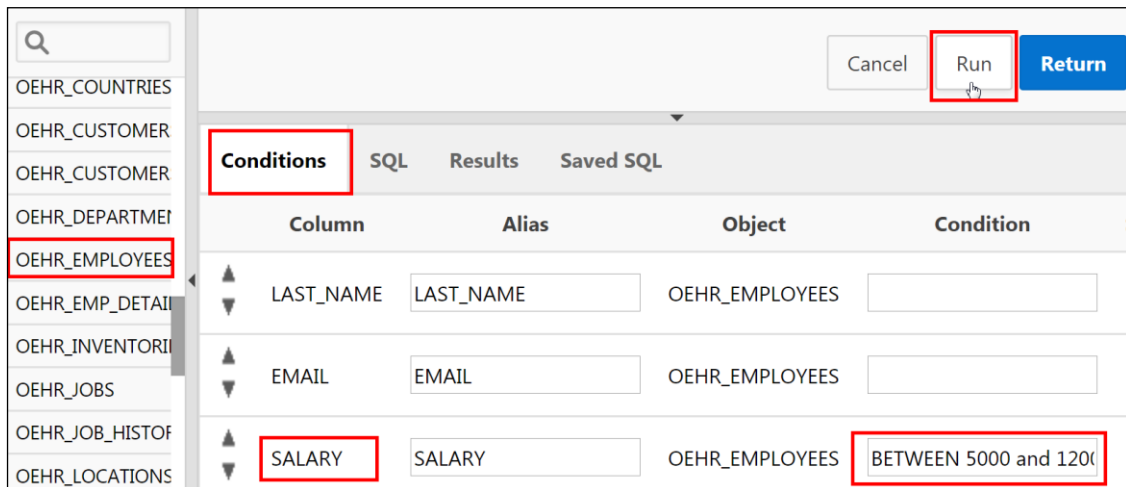
- h. Create the query using Query Builder. Click the **Query Builder** icon .



- i. In the Object Selection pane (left pane), select `OEHR_EMPLOYEES`.
- j. Select the check boxes to select the following columns:
- `LAST_NAME`
 - `EMAIL`
 - `SALARY`

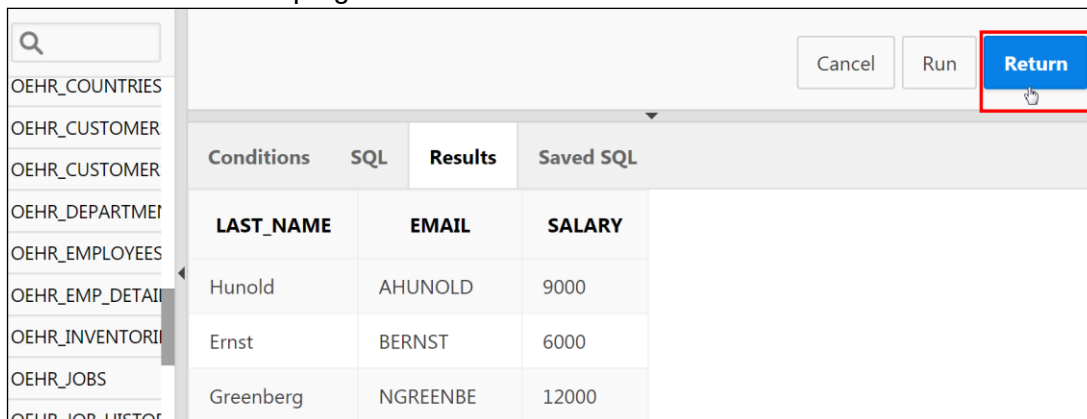


On the Conditions tab, for the SALARY > Condition, enter BETWEEN 5000 and 12000 and click **Run** in the top-right corner.



The results of the query are displayed on the Results tab at the bottom of the screen.

- k. Click **Return** on the top right of the screen.



- l. The query appears in the **Enter a SQL SELECT** statement field. Click **Create**.

Create Classic Report

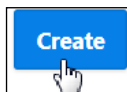
Report Source

Data Source: **Local Database** REST Enabled SQL Service Web Source

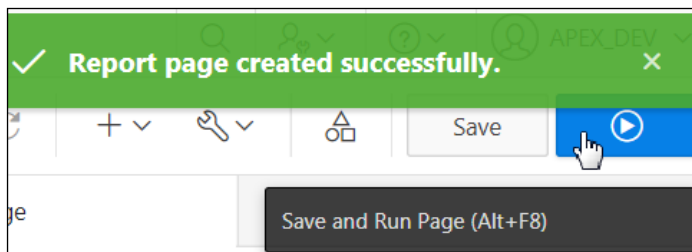
* Source Type: ☐ Table ☒ **SQL Query**

* Enter a SQL SELECT statement

```
1 select OEHR_EMPLOYEES.LAST_NAME as LAST_NAME,  
2 OEHR_EMPLOYEES.EMAIL as EMAIL,  
3 OEHR_EMPLOYEES.SALARY as SALARY  
4 from OEHR_EMPLOYEES OEHR_EMPLOYEES  
5 where OEHR_EMPLOYEES.SALARY BETWEEN 5000 and 12000
```



- m. Page Designer appears, displaying the new page. Click the **Save and Run Page**.



If prompted, enter your developer credentials.

GlobalMart Management Tool

apex_dev

....

☐ Remember username

Sign In

- n. Review the report. When finished, click **Application 111** on the Runtime Developer toolbar (at the bottom of the page) to return to your application home page.

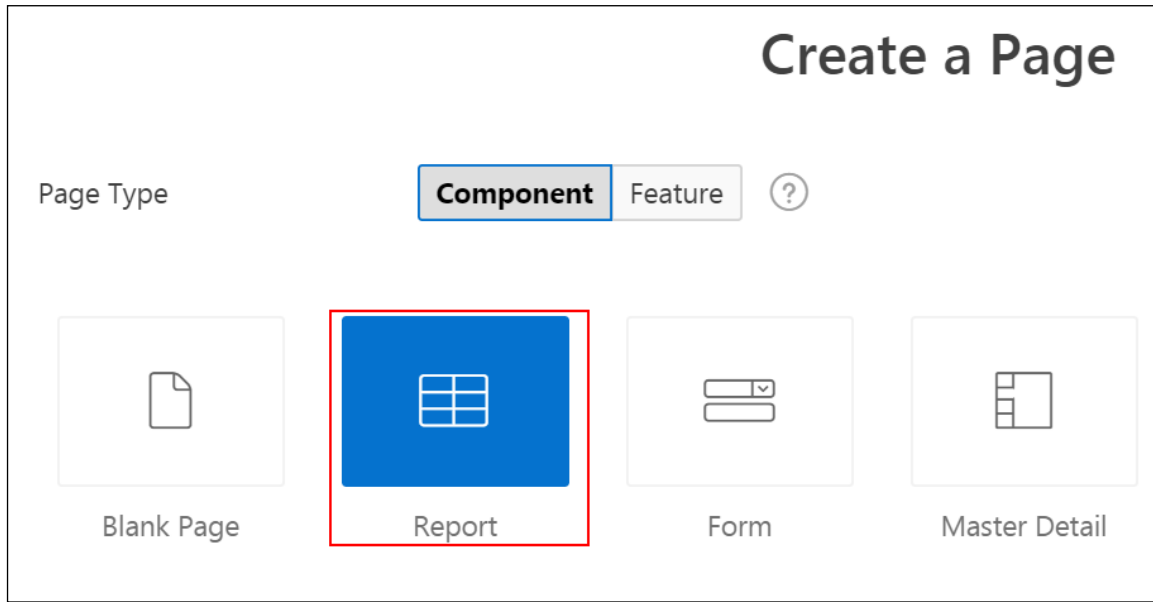
The screenshot shows the GlobalMart Management Tool interface. On the left is a dark sidebar with a 'Home' link. The main area displays 'Report 1' with a table of employee data. The bottom toolbar contains several icons, with 'Application 111' highlighted by a red box.

Last Name ↑	Email	Salary
Abel	EABEL	11000
Ande	SANDE	6400
Baer	HBAER	10000
Banda	ABANDA	6200
Bates	EBATES	7300
Bernstein	DBERNSTE	9500
Bloom	HBLOOM	10000

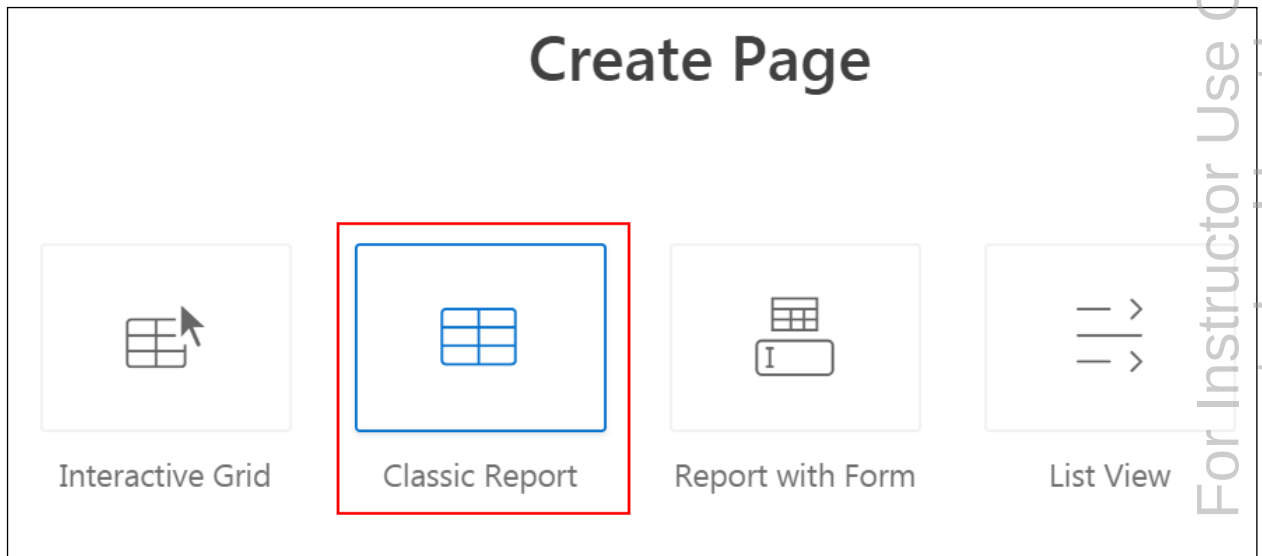
2. Create another classic report on a new page, but this time select the source type, Table. Name the page and the report Customer Address List. The report should display the address information of the customers, such as the first and last names, address, city, and state from the OEHR_CUSTOMERS table.
- a. On the Application home page, click **Create Page**.

The screenshot shows the Application 111 - GlobalMart Management Tool home page. It features a header with the application name and an 'Edit Application Properties' link. Below the header are five icons representing different functions: Run Application, Supporting Objects, Shared Components, Utilities, and Export / Import. At the bottom, there is a search bar, a 'Go' button, a grid icon, a list icon, an 'Actions' dropdown, and a 'Create Page >' button highlighted with a red box.

- b. Select **Report**.



- c. Select **Classic Report**.



- d. Update the following attributes:

- Page Number: Enter 3.
- Page Name: Enter `Customer Address List`.
- Click **Next >**.

Create Classic Report

●
●
●

Page Attributes

Type **Classic Report**

* Page Number ?

* Page Name ?

Page Mode **Normal** Modal Dialog ?

Breadcrumb ?

- e. For Navigation Menu, accept the default and click **Next**.

Create Classic Report

✓
●
●

Navigation Menu

Navigation Preference ☒ **Do not associate this page with a navigation menu entry** ?

☐ Create a new navigation menu entry

☐ Identify an existing navigation menu entry for this page

- f. For Source Type, select **Table**.

- g. For Table / View Name, click the icon  to open a Search dialog box.

Create Classic Report

Progress bar: [Green Checkmark] [Green Checkmark] [Blue Circle] **Report Source**

Data Source: **Local Database** REST Enabled SQL Service Web Source ?

* Source Type: ☒ **Table** ☐ SQL Query ?

* Table / View Owner: OEHR ?

* Table / View Name:  ?

- h. Select OEHR_CUSTOMERS (table) .

Search Dialog

Search

OEHR_BOMBAY_INVENTORY (view)
OEHR_COUNTRIES (table)
OEHR_CUSTOMERS (table)
OEHR_CUSTOMERS_VIEW (view)
OEHR_DEPARTMENTS (table)

- i. Click the **Columns** region to expand it.

Create Classic Report

Progress: [Step 1: ✓] [Step 2: ✓] [Step 3: Report Source]

Data Source: **Local Database** | REST Enabled SQL Service | Web Source (?)

* Source Type: ☒ **Table** | ☐ SQL Query (?)

* Table / View Owner: OEHR (?)

* Table / View Name: OEHR_CUSTOMERS (table) [Menu Icon] (?)

> **Lookup Columns**

Columns [Hand Icon]

- j. By default, the wizard selects all the columns. Move all the columns to the left by clicking the **double-left arrow (<<)** labeled **Remove All**.

Create Classic Report

Progress: [Step 1: ✓] [Step 2: ✓] [Step 3: Report Source]

* Source Type: ☒ **Table** | ☐ SQL Query (?)

* Table / View Owner: OEHR (?)

* Table / View Name: OEHR_CUSTOMERS (table) [Menu Icon] (?)

> **Lookup Columns**

Columns [Hand Icon]

* Select Columns (?)

	[Left Arrow]	CUSTOMER_ID (Number)	[Up Arrow]
	[Right Arrow]	CUST_FIRST_NAME (Varchar2)	[Up Arrow]
	[Right Arrow]	CUST_LAST_NAME (Varchar2)	[Up Arrow]
	[Right Arrow]	STREET_ADDRESS (Varchar2)	[Down Arrow]
	[Right Arrow]	POSTAL_CODE (Varchar2)	[Down Arrow]
	[Right Arrow]	CITY (Varchar2)	[Down Arrow]
	[Right Arrow]	STATE_PROVINCE (Varchar2)	[Down Arrow]
	[Right Arrow]	COUNTRY_ID (Char)	[Down Arrow]

[Double-Left Arrow] [Hand Icon]

Create Classic Report

✓ ✓ ● **Report Source**

* Source Type ☒ **Table** ☐ SQL Query ?

* Table / View Owner ?

* Table / View Name ⋮ ?

> **Lookup Columns**

▼ **Columns**

* Select Columns ?

CUSTOMER_ID (Number)
CUST_FIRST_NAME (Varchar2)
CUST_LAST_NAME (Varchar2)
STREET_ADDRESS (Varchar2)
POSTAL_CODE (Varchar2)
CITY (Varchar2)
STATE_PROVINCE (Varchar2)
COUNTRY_ID (Char)

⌂
>>
>
<
<<

- k. Deselect all the columns.
- l. Select the columns to include in the report. Press and hold the **Ctrl** key and click the following items:
 - CUST_FIRST_NAME
 - CUST_LAST_NAME
 - STREET_ADDRESS
 - CITY
 - STATE_PROVINCE

Create Classic Report

Report Source

* Source Type ☒ **Table** ☐ SQL Query ?

* Table / View Owner OEHR ?

* Table / View Name OEHR_CUSTOMERS (table) ⋮ ?

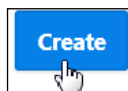
> **Lookup Columns**

▼ **Columns**

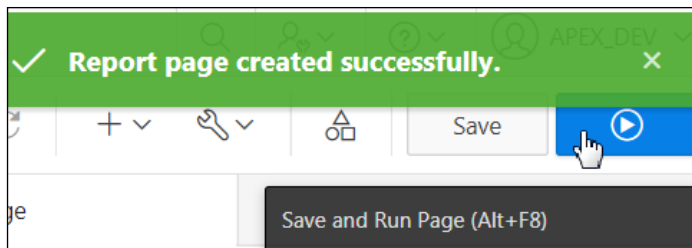
* Select Columns ?

CUSTOMER_ID (Number) POSTAL_CODE (Varchar2) COUNTRY_ID (Char) PHONE_NUMBER (Varchar2) NLS_LANGUAGE (Varchar2) NLS_TERRITORY (Varchar2) CREDIT_LIMIT (Number) CUST_EMAIL (Varchar2)	⌵ >> << < >	CUST_FIRST_NAME (Varchar2) CUST_LAST_NAME (Varchar2) STREET_ADDRESS (Varchar2) CITY (Varchar2) STATE_PROVINCE (Varchar2)
--	--	---

n. Click **Create**.



o. Page Designer appears. Click the **Save and Run Page**.



- p. Review the report. When finished, click **Edit Page 3** on the Runtime Developer toolbar to return to Page Designer.

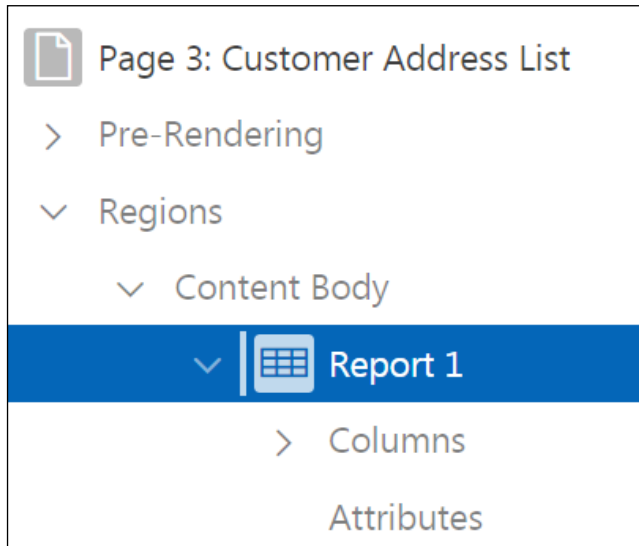
The screenshot displays the GlobalMart Management Tool interface. On the left is a dark sidebar with a 'Home' link. The main area shows 'Report 1' with a table of customer data. At the bottom is a toolbar with various icons. The 'Edit Page 3' icon, which shows a document with a pencil, is highlighted with a red rectangle.

Cust First Name ↑	Cust Last Name	Street Address
Ajay	Sen	220 Penn Ave # 300
Ajay	Andrews	223 4Th Ave # 1100
Alain	Dreyfuss	Harmoniegasse 3
Alain	Barkin	Sonnenberg 4
Alain	Siegel	Alfred E. Neumann-Weg 3
Alan	Minnelli	Dr. Herbert Bitto Str 23

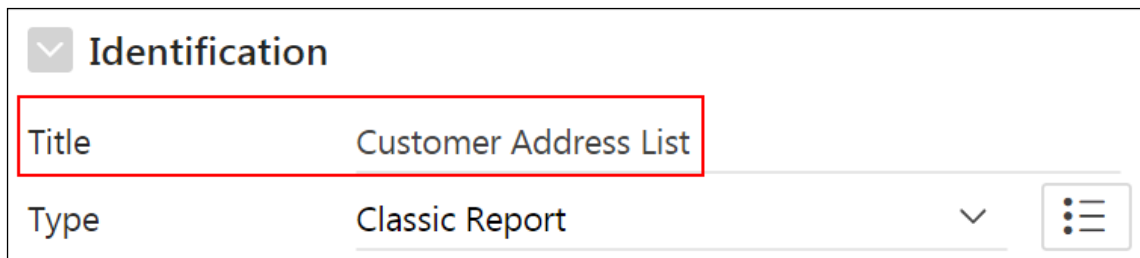
Toolbar icons: Home, Application 111, **Edit Page 3**, Session, View Debug, Debug, Page Info, Quick Edit, Theme Roller.

For Instructor Use Only.
This document should not be distributed.

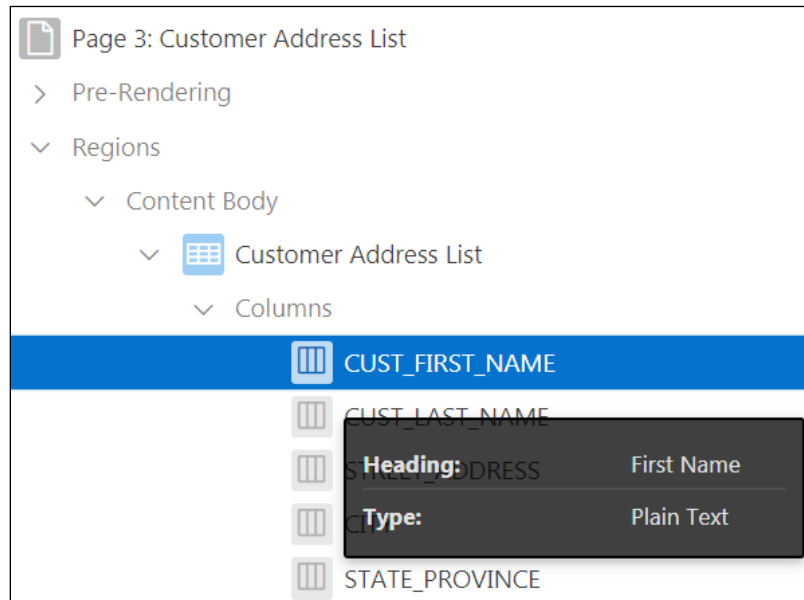
3. Modify the Customer Address List report with the following requirements:
- Change the report region from the default Report 1 to Customer Address List.
 - Change the report headings for CUST_FIRST_NAME and CUST_LAST_NAME to First Name and Last Name, respectively.
- a. On the Rendering tab (the left pane), under **Regions > Content Body**, select **Report 1**.



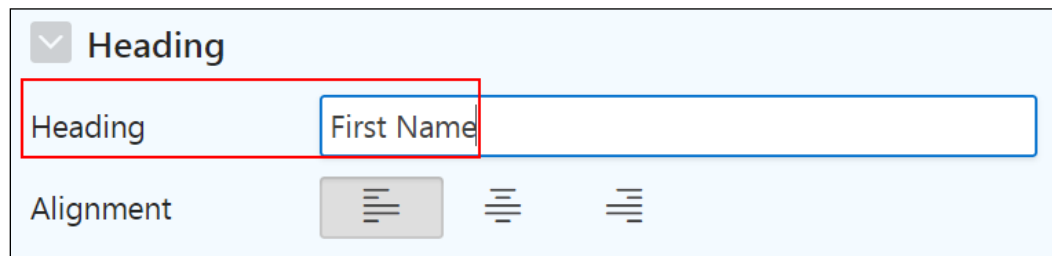
- b. In the Property Editor (the right pane), under Identification > Title, enter Customer Address List.



- c. Change the heading for the CUST_FIRST_NAME column.
 - On the Rendering tab (the left pane), under **Regions > Content Body > Customer Address List**, expand **Columns** and select CUST_FIRST_NAME.

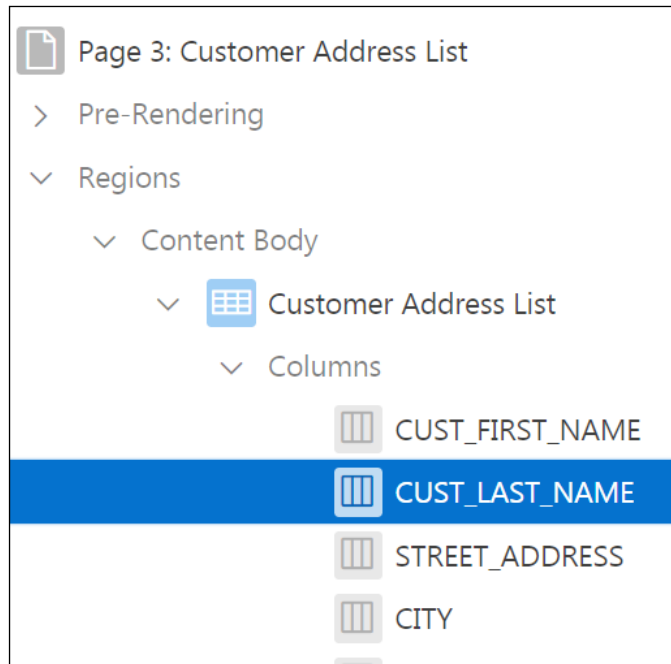


- In the Property Editor (right pane), change **Heading > Heading** to First Name.

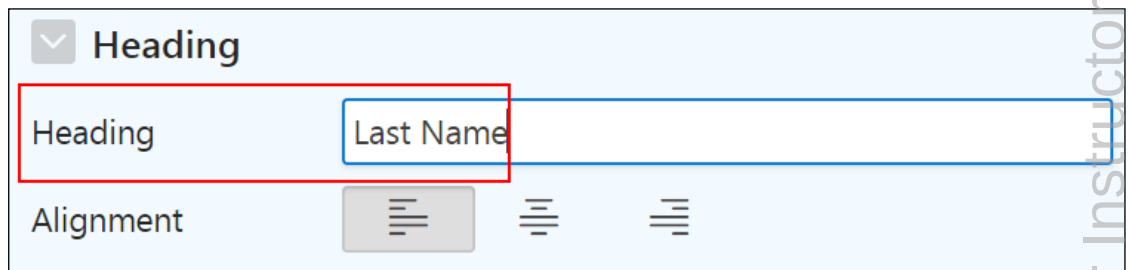


d. Change the heading for the CUST_LAST_NAME column.

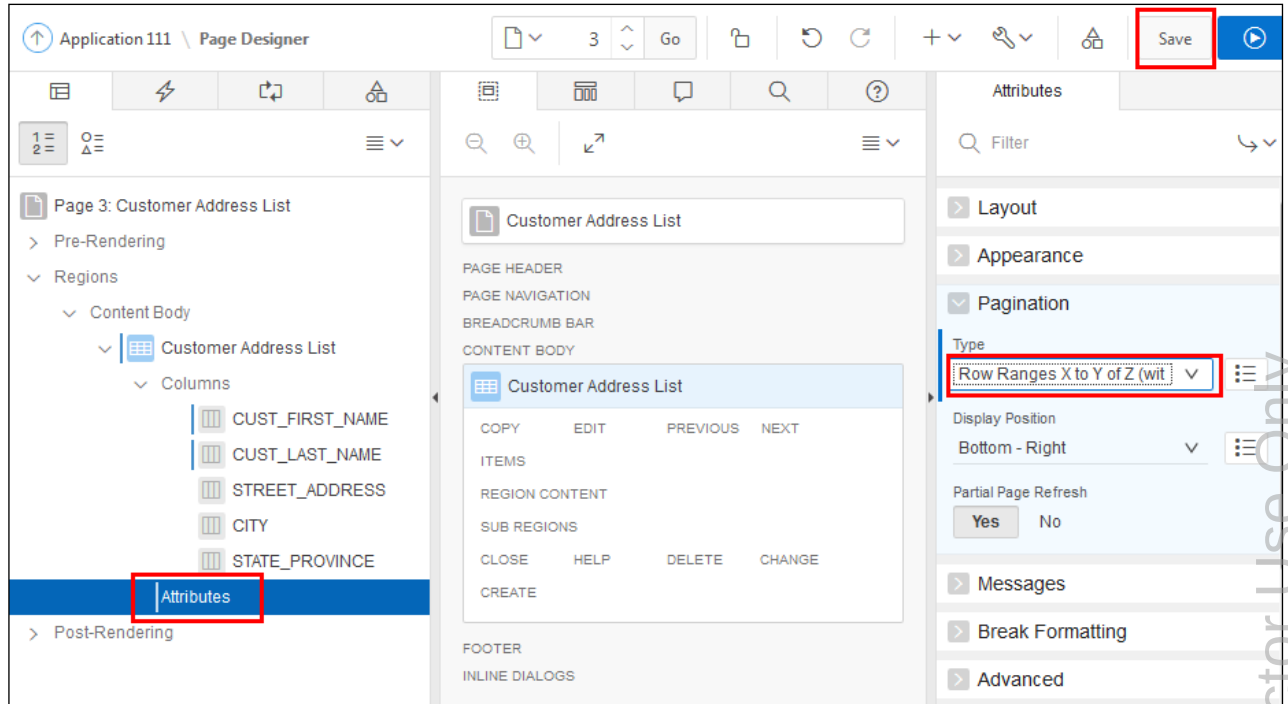
- On the Rendering tab (left pane), under **Regions > Content Body > Customer Address List > Columns**, select CUST_LAST_NAME.



- In the Property Editor, change **Heading > Heading** to Last Name.



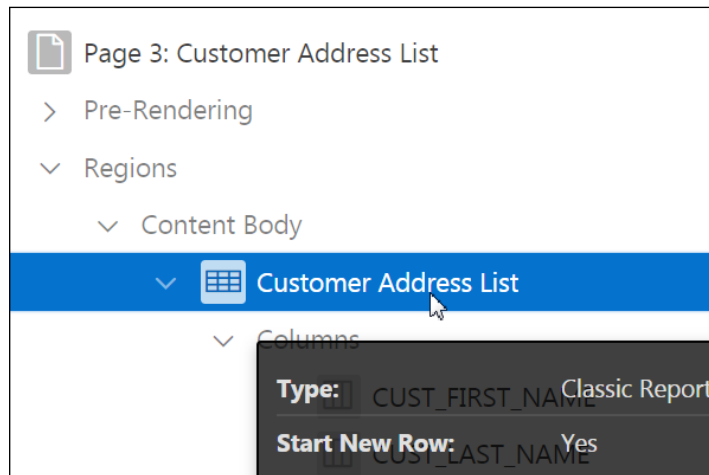
4. Change the Pagination to Row Ranges X to Y of Z (with pagination).
 - a. On the Rendering tab, under **Regions > Content Body > Customer Address List**, select **Attributes**.
 - b. In the Property Editor, under the **Pagination > Type**, select **Row Ranges X to Y of Z (with pagination)**.
 - c. Click the **Save** button (top-right corner).



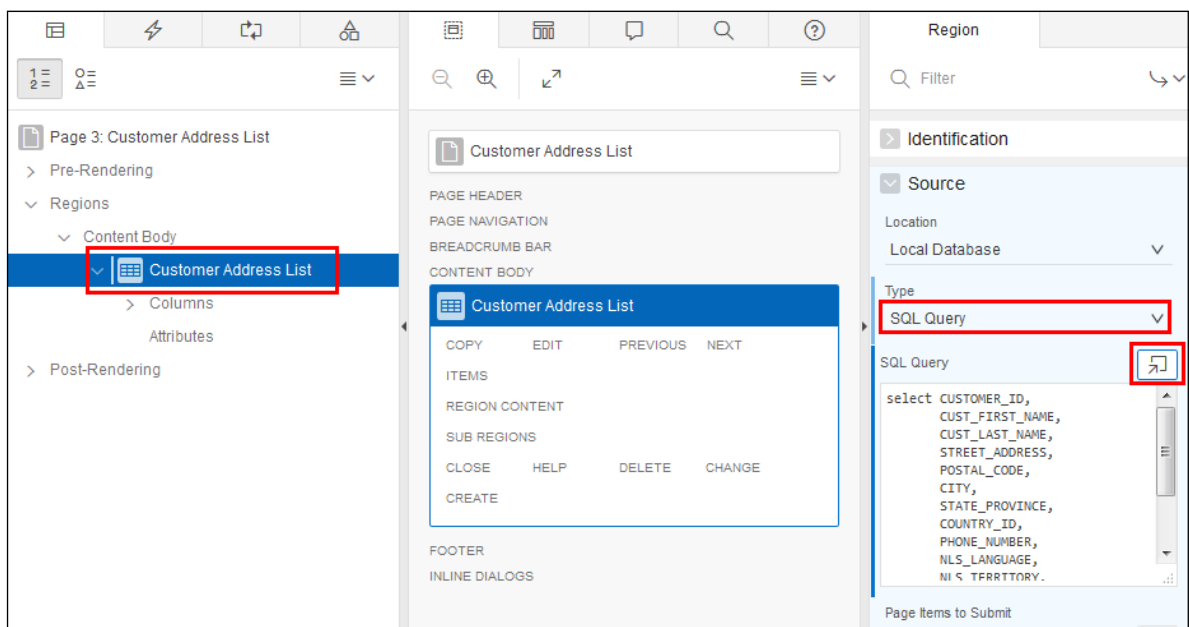
5. Review the query definition and add the `OEHR_COUNTRIES.COUNTRY_NAME` column to the report.

Hint: You will need to create a join.

- a. On the Rendering tab, under **Regions > Content Body**, select **Customer Address List**.

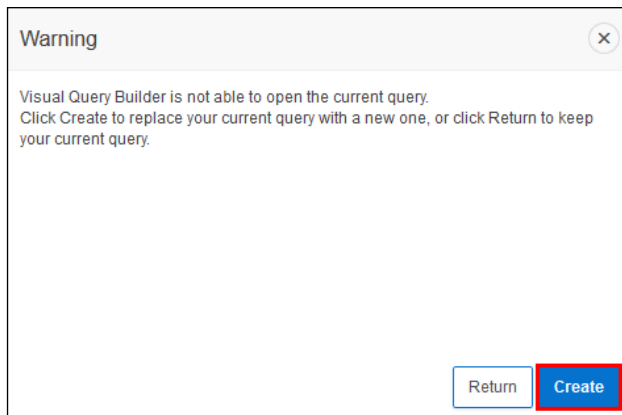
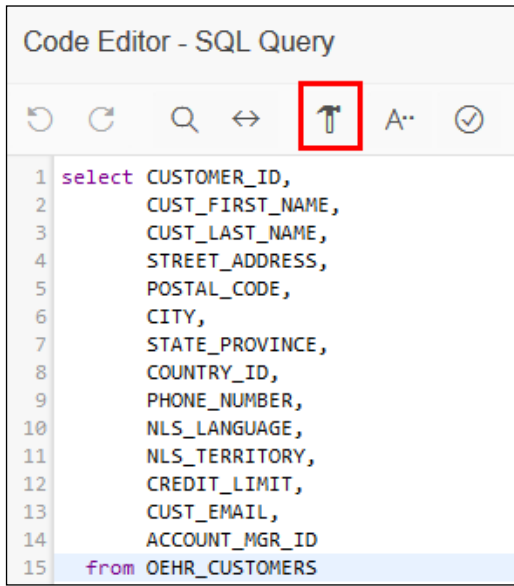


- b. In the Property Editor, under Source > Type, select **SQL Query**. A SQL Query attribute and field appear.
- c. Modify the query definition using Query Builder. Under **Source > SQL Query**, click the **Code Editor: SQL Query** icon .



For Instructor Use Only. This document should not be distributed.

- d. Click the **Query Builder** icon . A warning is displayed. Click **Create** to proceed.



- e. On the Object Selection pane (on the left side of the page), select OEHR_CUSTOMERS and OEHR_COUNTRIES.
- f. In the OEHR_CUSTOMERS table, select the CUST_FIRST_NAME, CUST_LAST_NAME, STREET_ADDRESS, CITY, and STATE_PROVINCE columns.

- g. In the OEHR_COUNTRIES table, select the COUNTRY_NAME column.

The screenshot shows the Oracle SQL Developer interface with two tables, OEHR_CUSTOMERS and OEHR_COUNTRIES, displayed in a grid. The OEHR_CUSTOMERS table has columns: CUSTOMER_ID (789), CUST_FIRST_NAME (A), CUST_LAST_NAME (A), STREET_ADDRESS (A), POSTAL_CODE (A), CITY (A), and STATE_PROVINCE (A). The OEHR_COUNTRIES table has columns: COUNTRY_ID (A), COUNTRY_NAME (A), and REGION_ID (789). The columns CUST_FIRST_NAME, CUST_LAST_NAME, STREET_ADDRESS, CITY, STATE_PROVINCE, and COUNTRY_NAME are selected, indicated by checkmarks in the selection column. The join column field to the right of the COUNTRY_ID entry in the OEHR_COUNTRIES table is highlighted with a red box.

Table	Column	Length	Selected	Join Column
OEHR_CUSTOMERS	CUSTOMER_ID	789	☐	
	CUST_FIRST_NAME	A	☒	
	CUST_LAST_NAME	A	☒	
	STREET_ADDRESS	A	☒	
	POSTAL_CODE	A	☐	
	CITY	A	☒	
	STATE_PROVINCE	A	☒	
OEHR_COUNTRIES	COUNTRY_ID	A	☐	
	COUNTRY_NAME	A	☒	
	REGION_ID	789	☐	

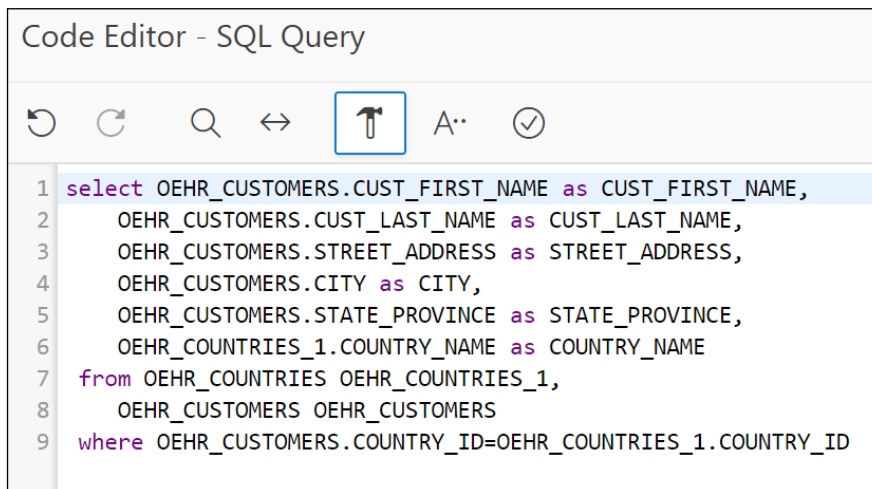
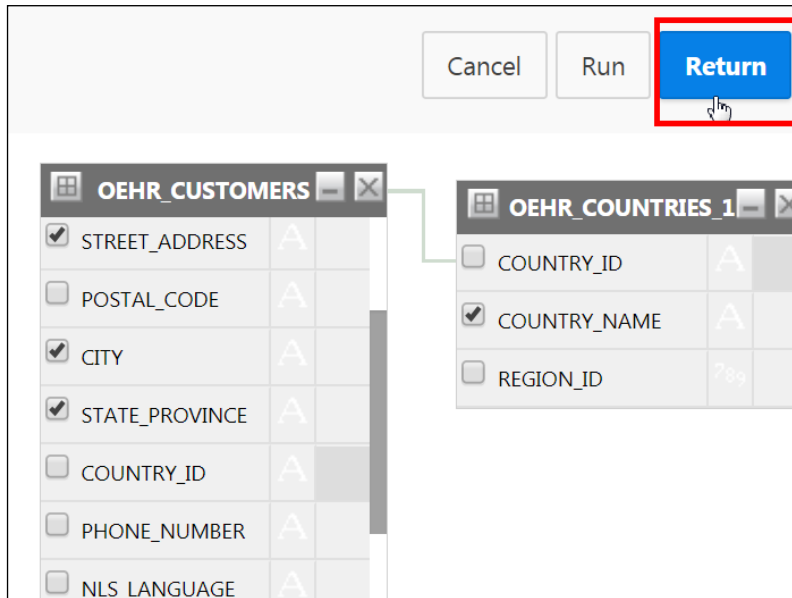
Conditions SQL Results Saved SQL

- h. Join the tables by clicking the **join column field** to the right of the COUNTRY_ID entry in both the tables.

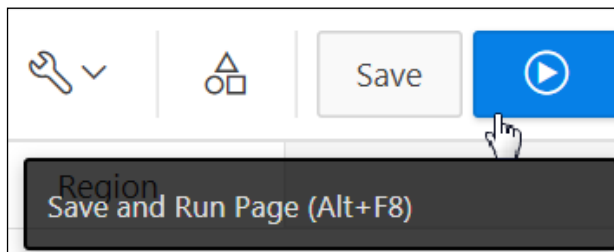
The screenshot shows the Oracle SQL Developer interface with the same two tables, OEHR_CUSTOMERS and OEHR_COUNTRIES, displayed in a grid. The OEHR_CUSTOMERS table has columns: STATE_PROVINCE (A), COUNTRY_ID (A), PHONE_NUMBER (A), NLS_LANGUAGE (A), NLS_TERRITORY (A), CREDIT_LIMIT (789), and CUST_EMAIL (A). The OEHR_COUNTRIES table has columns: COUNTRY_ID (A), COUNTRY_NAME (A), and REGION_ID (789). The columns STATE_PROVINCE, COUNTRY_ID, and COUNTRY_NAME are selected, indicated by checkmarks in the selection column. The join column field to the right of the COUNTRY_ID entry in the OEHR_CUSTOMERS table is highlighted with a red box. The join column field to the right of the COUNTRY_ID entry in the OEHR_COUNTRIES table is also highlighted with a red box. A green line connects the two highlighted join column fields, indicating the join operation.

Table	Column	Length	Selected	Join Column
OEHR_CUSTOMERS	STATE_PROVINCE	A	☒	
	COUNTRY_ID	A	☐	
	PHONE_NUMBER	A	☐	
	NLS_LANGUAGE	A	☐	
	NLS_TERRITORY	A	☐	
	CREDIT_LIMIT	789	☐	
	CUST_EMAIL	A	☐	
OEHR_COUNTRIES	COUNTRY_ID	A	☐	
	COUNTRY_NAME	A	☒	
	REGION_ID	789	☐	

- i. Click **Run** and then **Return** in the top right of the screen.



- j. Click **OK** to commit the new query.
- k. The SQL updates in the SQL Query field in the Property Editor. Click the **Save and Run Page**.



- I. Note the changes that you made to the report. Click **Edit Page 3** on the Runtime Developer toolbar to return to Page Designer.

The screenshot displays the GlobalMart Management Tool interface. On the left is a dark sidebar with a 'Home' link. The main area shows a 'Customer Address List' report. The report has five columns: 'First Name', 'Last Name', 'Street Address', 'City', and 'State Province'. The first two columns are highlighted with red boxes. Below the report is a toolbar with various icons. The 'Edit Page 3' icon, which shows a document with a pencil, is highlighted with a red box. Other icons include 'Home', 'Application 111', 'Session', 'View Debug', 'Debug', 'Page Info', 'Quick Edit', and 'Theme Roller'.

First Name	Last Name	Street Address	City	State Province
Ajay	Sen	220 Penn Ave # 300	Scranton	PA
Ajay	Andrews	223 4Th Ave # 1100	Pittsburgh	PA
Alain	Dreyfuss	Harmoniegasse 3	Baden-Daettwil	AG
Alain	Barkin	Sonnenberg 4	Baden-Daettwil	AG
Alain	Siegel	Alfred E. Neumann-Weg 3	Baden-Daettwil	AG

For Instructor Use Only.
This document should not be distributed.

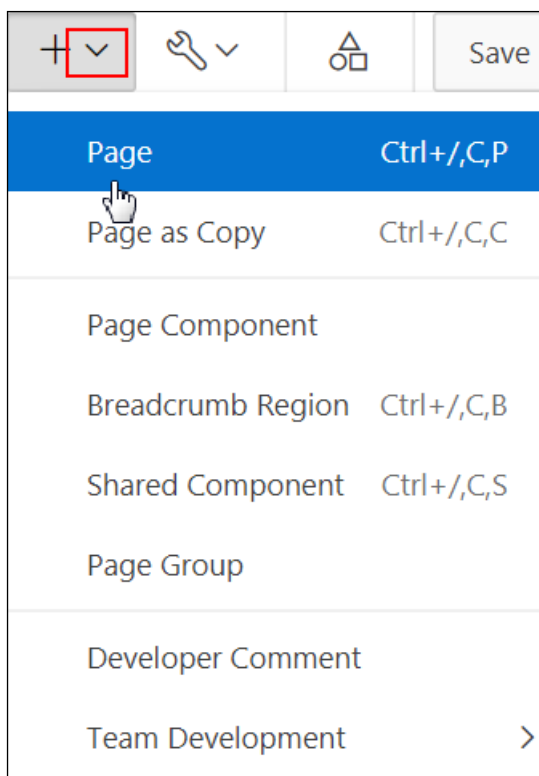
Solution 4-2: Creating a Column Toggle Report

Overview

In this practice solution, you create a new page containing a Column Toggle report for your GlobalMart Management Tool (GMT) application. Column Toggle reports feature a responsive report optimized for mobile applications and Smartphones. By default, column toggle reports are created with all columns set to the same priority. However, the developer can edit the report column attributes and rank columns by importance.

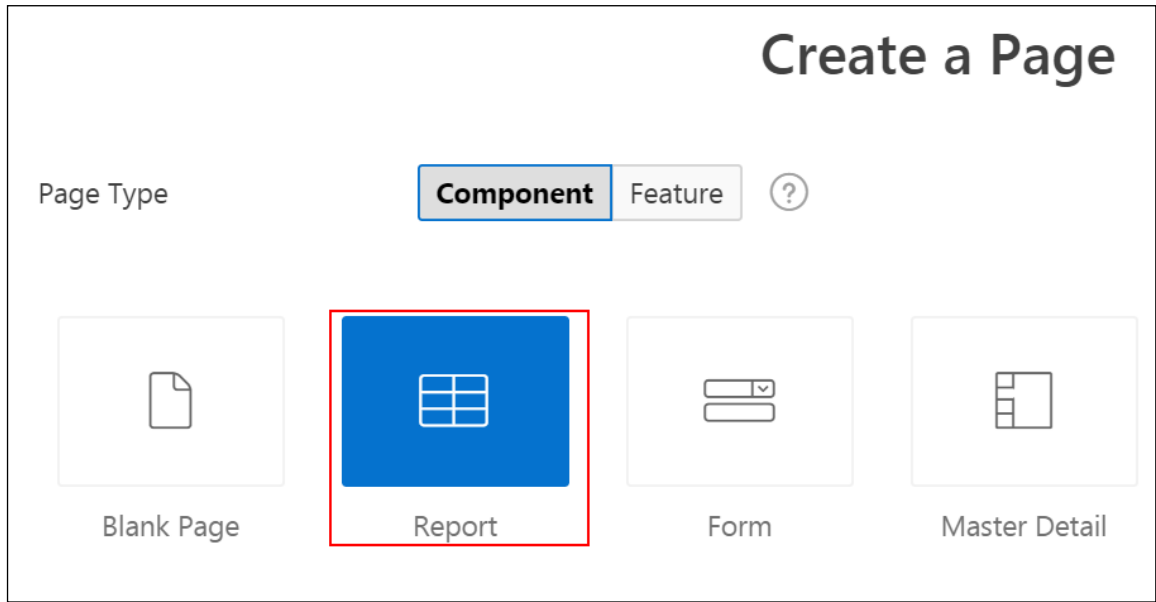
Steps

1. Create a new page in the GMT application containing a Column Toggle report. Enter Product Information as the Page Name. Change Page Number to 202. The report should select all the columns from the `OEHR_PRODUCT_INFORMATION` table.
 - a. Return to Page Designer (already open in another window). From the Page Designer toolbar, select Create  and then Page.

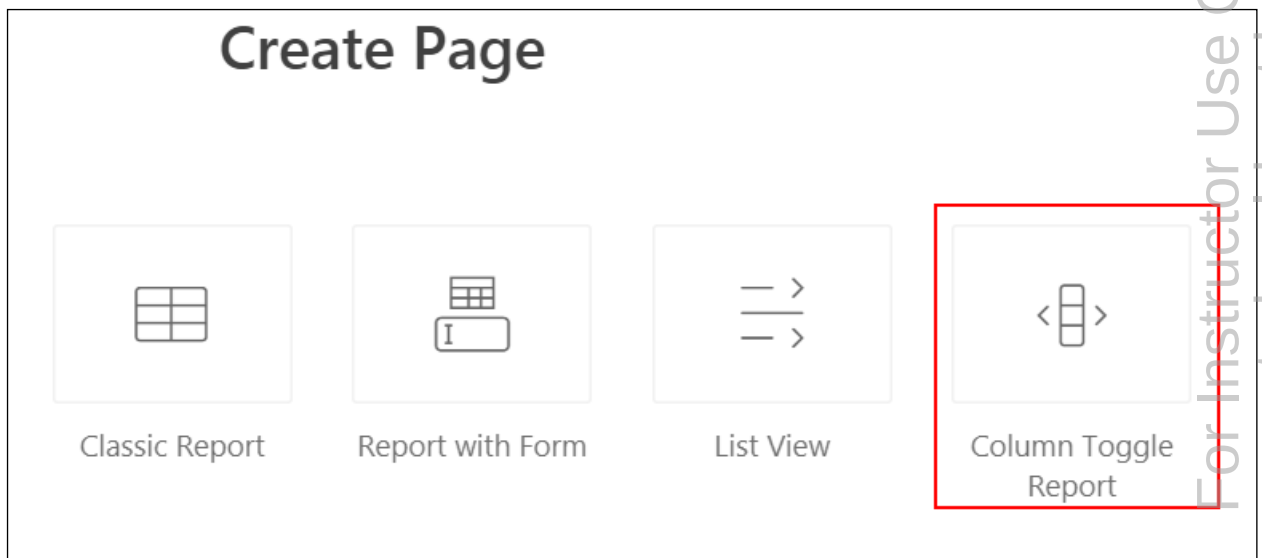


For Instructor Use Only.
This document should not be distributed.

- b. Click **Report**.



- c. Click **Column Toggle Report**.



- d. Change Page Number to 202. For Page Name, enter Product Information.

The screenshot shows the 'Create Page' dialog box with the 'Page Attributes' tab selected. A progress bar at the top has four steps: the first is completed (green checkmark), the second is active (blue circle), and the third and fourth are pending (grey circles). The 'Region Type' is set to 'Column Toggle Report'. The 'Page Number' is 202 and the 'Page Name' is 'Product Information'. The 'Page Mode' is 'Normal'. The 'Page Group' is '- Select Page Group -'. The 'Breadcrumb' is '- do not use breadcrumbs on page -'. The 'Page Number' and 'Page Name' fields are highlighted with a red box.

Create Page

Page Attributes

Region Type **Column Toggle Report**

* Page Number 202 ?

* Page Name Product Information ?

* Page Mode **Normal** Modal Dialog ?

Page Group - Select Page Group - ?

Breadcrumb - do not use breadcrumbs on page - ?

- e. Click **Next >**.
For Navigation menu, accept the defaults and click **Next >**.

The screenshot shows the 'Create Page' dialog box with the 'Navigation Menu' tab selected. The progress bar at the top has four steps: the first and second are completed (green checkmarks), the third is active (blue circle), and the fourth is pending (grey circle). The 'Navigation Preference' is 'Do not associate this page with a navigation menu entry'. The other two options are 'Create a new navigation menu entry' and 'Identify an existing navigation menu entry for this page'. The 'Do not associate this page with a navigation menu entry' option is highlighted with a red box.

Create Page

Navigation Menu

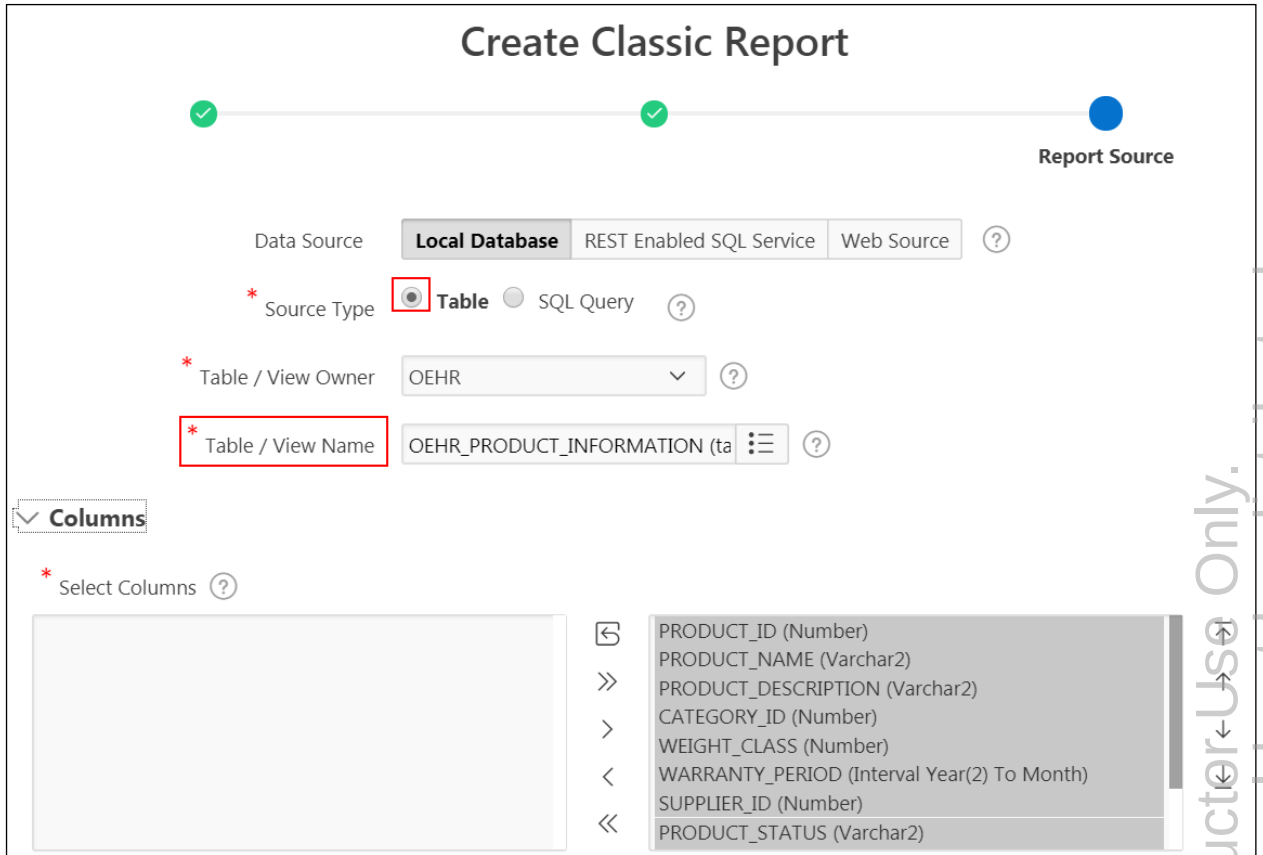
Navigation Preference ☒ **Do not associate this page with a navigation menu entry** ?

☐ Create a new navigation menu entry

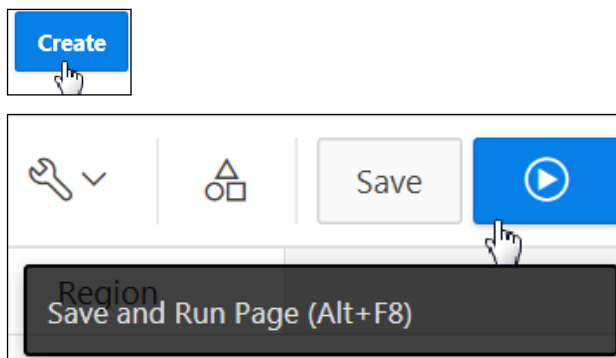
☐ Identify an existing navigation menu entry for this page

- f. For Source Type, click **Table**.

- g. From Table / View Name, click the  icon to open a search dialog box and select OEHR_PRODUCT_INFORMATION (table) . By default, all columns are selected.



- h. Click **Create**. Click the **Save and Run Page** to view the report.



Note: If prompted, enter your developer credentials.

All the columns from PRODUCT_INFORMATION table appear.

- i. Try out the responsive nature of the report. Slowly reduce the size of the browser horizontally. Notice that the column starts disappearing from the right side (that is, starting from the Catalog URL field) as the size of the browser is being reduced.
2. Using the Columns option in the report, hide the columns Product Description, Category ID, and Supplier ID from the report.

- a. On the Product Information Page, click **Columns** and then click the following columns to deselect them:

- Product Description
- Category Id
- Supplier Id

Product Id	Product Name	Weight Class	Warranty Period	Product Status	List Price	Min Price	Catalog Url
2453	Inkjet C/4	3	+000000002-00	orderable	195	174	http://www.supp-102090.com/cat/p2453.html
1797	Inkjet C/8/HQ	3	+000000002-00	orderable	349	288	http://www.supp-102094.com/cat/p1797.html
2459	LaserPro 1200/8/BW	5	+000000003-00	under development	699	568	http://www.supp-102099.com/cat/p2459.html
3127	LaserPro 600/6/BW	4	+000000003-00	orderable	498	444	http://www.supp-102087.com/cat/p3127.html
2254	HD 10GB /I	2	+000000002-00	obsolete	453	371	http://www.supp-102071.com/cat/hw/p2254.html

- b. Click anywhere on the page to close the Columns list. Notice that the columns that you deselected are hidden in the report.

Product Id	Product Name	Weight Class	Warranty Period	Product Status	List Price	Min Price	Catalog Url
1772	HD 9.1GB @10000	3	+000000005-00	orderable	456	393	http://www.supp-102070.com/cat/hw/p1772.html
2414	HD 9.1GB @10000 /I	3	+000000005-00	orderable	454	399	http://www.supp-102098.com/cat/hw/p2414.html
2415	HD 9.1GB @7200	3	+000000005-00	orderable	359	309	http://www.supp-102063.com/cat/hw/p2415.html
2395	32MB Cache /M	1	+000000000-06	orderable	123	109	http://www.supp-102093.com/cat/hw/p2395.html
1755	32MB Cache /NM	1	+000000000-06	orderable	121	99	http://www.supp-102076.com/cat/hw/p1755.html

- c. The column toggle report is complete. Click **Application 111** on the Runtime Developer toolbar to return to the Application home page.



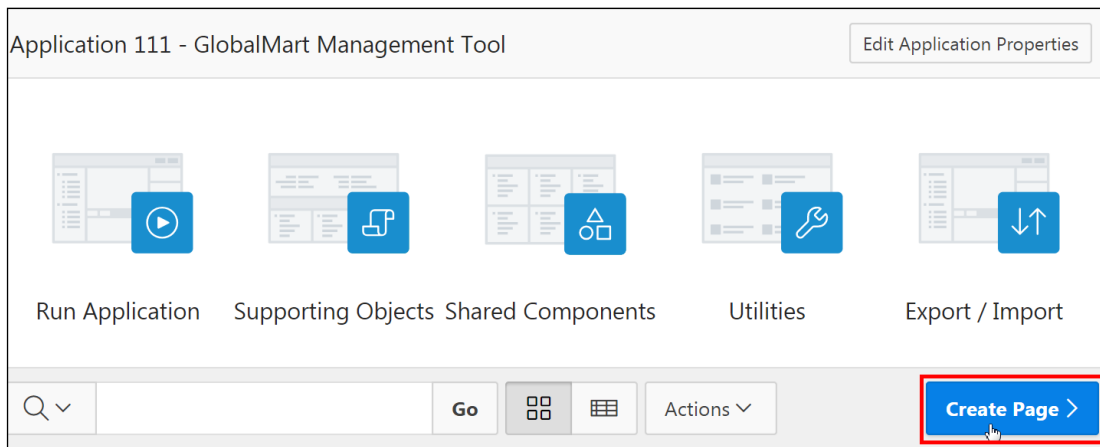
Solution 4-3: Adding and Modifying a List View Report

Overview

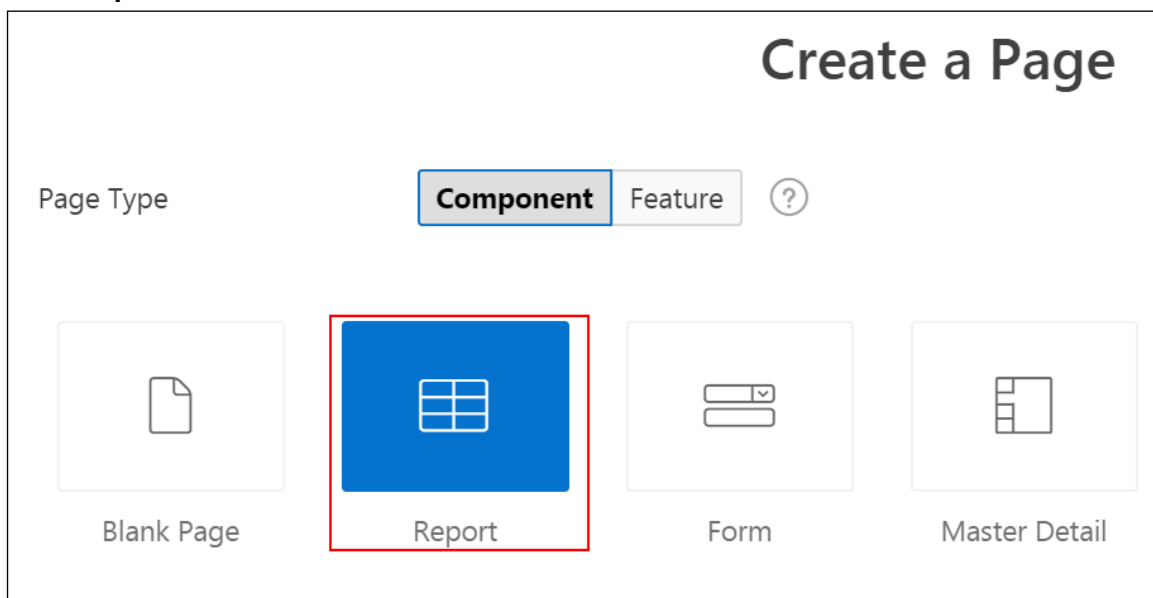
In this practice solution, you add and modify a List View report in your GlobalMart Management Tool (GMT) application. List View reports feature a responsive design to display data and provide easy navigation on Smartphones.

Steps

1. On a new page in the GMT application, create a List View report. Enter Employee List as the Page Name. Change Page Number to 201.
 - a. Navigate to the Application home page and click **Create Page**.



- b. Click **Report**.



- c. Click **List View**.

Create Page


Interactive Grid


Classic Report


Report with Form


List View

- d. Change Page Number to 201 and in Page Name enter `Employee List`.

Create Page

✓

●

●

●

Page Attributes

Region Type **List View**

* Page Number

201 ?

* Page Name

Employee List ?

* Page Mode

Normal Modal Dialog ?

Page Group

- Select Page Group - ?

Breadcrumb

- do not use breadcrumbs on page - ?

- e. Click **Next >**.

- f. For Navigation Menu, accept the defaults and click **Next >**.

The screenshot shows the 'Create Page' wizard at the 'Navigation Menu' step. A progress bar at the top has four steps: the first two are marked with green checkmarks, the third (Navigation Menu) is marked with a blue circle, and the fourth is marked with a grey circle. Below the progress bar, the 'Navigation Preference' section has two radio buttons. The first radio button, labeled 'Do not associate this page with a navigation menu entry', is selected and highlighted with a red box. The second radio button is labeled 'Create a new navigation menu entry' and 'Identify an existing navigation menu entry for this page'. A help icon (?) is visible to the right of the first radio button.

- g. For Source Type, select **SQL Query**.

- h. Under Enter Region Source, enter the following SQL `SELECT` statement (in the `/home/oracle/labs/labs/lab_04_03_01.txt` file):

```
SELECT * from OEHR_EMPLOYEES
```

The screenshot shows the 'Create Page' wizard at the 'Source' step. The progress bar at the top has four steps: the first three are marked with green checkmarks, and the fourth (Source) is marked with a blue circle. Below the progress bar, the 'Data Source' dropdown is set to 'Local Database'. The 'Source Type' section has two radio buttons: 'Table' and 'SQL Query'. The 'SQL Query' radio button is selected and highlighted with a red box. Below the radio buttons, the 'Enter Region Source:' label is followed by a help icon (?). A text area below contains the SQL statement '1 SELECT * from OEHR_EMPLOYEES', which is highlighted with a red box. The text area has a toolbar with icons for undo, redo, search, undo/redo, insert, and a checkmark.

- i. Click **Next >**.
2. The list view should select all the columns from the `OEHR_EMPLOYEES` table and should display the last names as the text column. The list view should also be enabled to search the last name column with the Search Type Like & Ignore Case.
- a. Under Features, select **Enable Search** and **Inset List**. For **Text Column**, select `LAST_NAME` to display the last name as the text column. For **Search Type**, select **Server: Like & Ignore Case**.

Create Page

✓
✓
✓
✓
●

Settings

Features

- ☐ Advanced Formatting ?
- ☐ Show Image
- ☐ Show List Divider
- ☐ Has Split Button
- ☒ **Enable Search**
- ☐ Is Nested List View
- ☒ **Inset List**

Text Column LAST_NAME ?

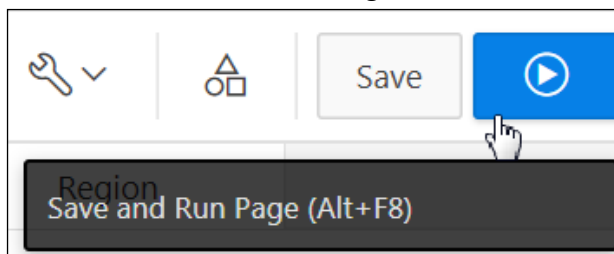
Supplemental Information Column - Select - ?

Counter Column - Select - ?

Link Target ?

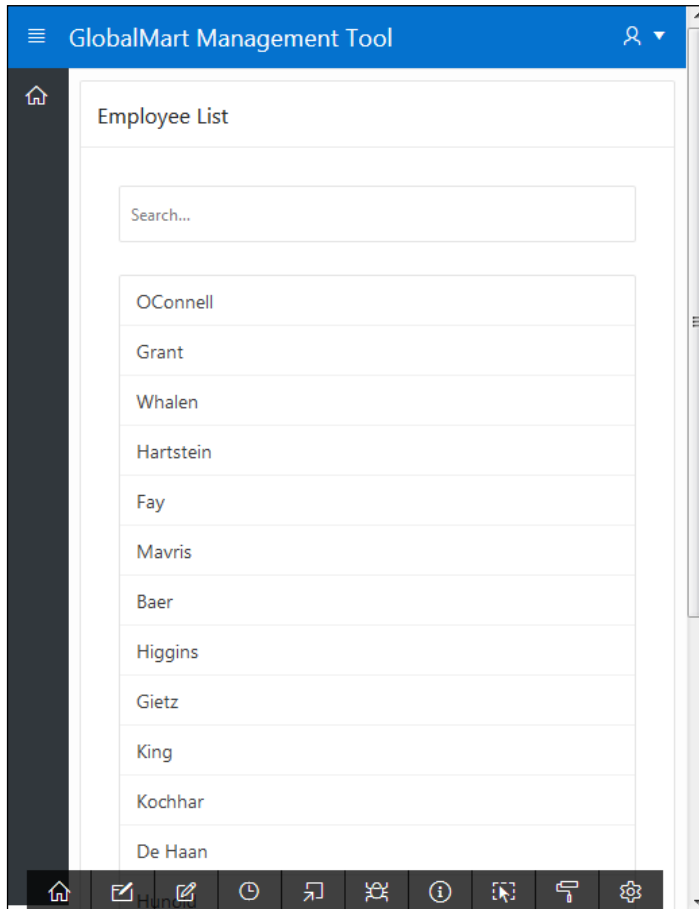
Search Type Server: Like & Ignore Case ?

- b. Click **Create**. The Page Designer opens, displaying Page 201 - Employee List.
- c. Click the **Save and Run Page**.

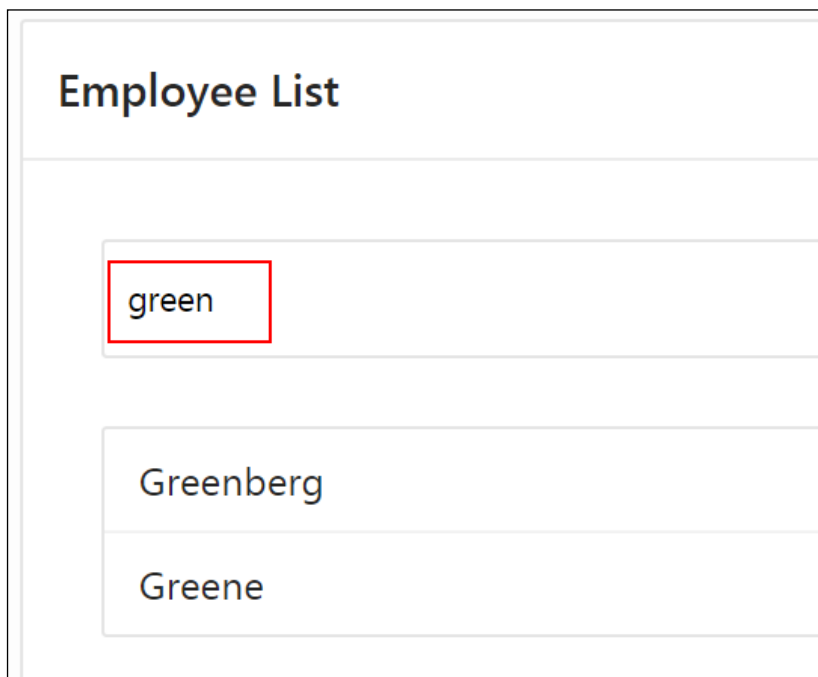


Note: If prompted, enter your developer workspace credentials.

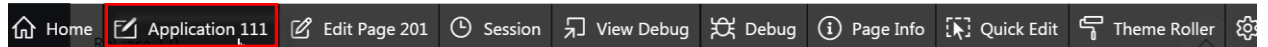
- d. Review the List View report.



- e. Enter `green` in the search bar and press the **Enter** key to display the names containing “green.” Note the search results.



- f. Clear the search field and press **Enter** to reset the report.
- g. This List View report is complete. Click **Application 111** on the Runtime Developer toolbar to return to the Application home page.



For Instructor Use Only.
This document should not be distributed.

**Practices for Lesson 5:
Working with Interactive
Reports**

Practices for Lesson 5: Overview

Overview

In these practices, you create, manipulate, and customize an interactive report to generate separate private, public, and default reports.

For Instructor Use Only.
This document should not be distributed.

Practice 5-1: Creating and Manipulating an Interactive Report

Overview

In this practice, you create two interactive reports—Customer Report and Employee Commissions for GMT application. These two reports provide flexible customization for users.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create an interactive report to display all the columns from the `OEHR_CUSTOMERS` table on a new page in the GMT application. Name the page **Customers**. Perform the following customizations to the report:
 - Create a **filter** on the city of Philadelphia.
 - Create a **highlight** where the credit limit is greater than 2500.
 - **Hide** columns to display only the Customer ID, Customer Name, Address, Phone Number, and Credit Limit.
 - Delete the filter that you created first.
 - **Sort** the rows in ascending order by State.
 - Change the pagination to display all rows.
 - Create a **Group By** report to display the number of customers in a particular province or state. Change the **pagination** to display all rows.
 - Create a **Pivot** report to display the total credit limit of customers for each account manager.
 - Save the report as a private report called **My Report**.
 - **Reset** the Primary Report to its original state.
 - Create a new computation. The computation should add .15 to a commission of .1 and .05 to a commission of .15. Use the following CASE statement to perform the computation (located in `/home/oracle/labs/labs/lab_05_02_02.txt`).
2. Create an interactive report on a new page in the GMT application to display details (`employee_id`, `first_name`, `last_name`, `manager_id`, and `commission_pct`) from the `OEHR_EMPLOYEES` table about who earns a commission. Name the page and the report **Employee Commissions**. From the Create Report Wizard, create a breadcrumb for the page and select the Home page as the parent page. After creating the report, make the following customizations to the report:
 - Management proposes to increase the commission of employees whose existing commission percentage is less than or equal to 0.15. Create a computed column called New Commission, to display the new commission. The new commission scheme includes:
 - Employees with a commission percentage of 0.1 will get an additional commission of 0.15.

- Employees with a commission percentage of 0.15 will get an additional commission of 0.05.
- Create a **control break** on Manager ID.
- Display the maximum commission earned by employees under each manager.
- **Download** the HTML report.
- Save all the customizations that you have made to this report so that it is available to other users, but is not the default report. Name the saved report **Commission Proposal**.
- Reset the Primary Report.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 5-1**.

If you have not completed **Practice 4-3**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **App Builder** and select **Import**.
- c. For Import File, click the button and browse to the `sol_04_03.sql` file in the `/home/oracle/labs/solutions` directory and click **Next**.
- d. Click **Next**.
- e. For Install As Application, select **Reuse Application ID 111 From Export File** and click **Install Application**.
- f. Click **Replace Application** (if prompted).
- g. Wait for the application to successfully install.

Practice 5-2: Customizing an Interactive Report

Overview

By now, you have created two reports (Customer Report and Employee Commissions) with features for customization at the users' end. In this practice, you customize those two reports using all the features provided by Oracle Application Express.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Make the following customizations to the Employee Commissions report:
 - Hide `EMPLOYEE_ID` and change `MANAGER_ID` to be the `FIRST_NAME` and `LAST_NAME` of the manager.
 - Remove the Highlight option from the Actions menu.
 - Run the report and confirm the changes that you made.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- a. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- b. Click the **Upload >** button.
- c. Browse and upload the `sol_02_01.sql` file to the `/home/oracle/labs/solutions` directory.
- d. Run the script by clicking the **Run** icon.
- e. Click **Run Now**.
- f. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 5-2**.

If you have not completed **Practice 5-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_dev` user.
- b. Click the arrow next to **App Builder** and select **Import**.
- c. For Import File, click the button and browse to the `sol_05_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next**.
- d. Click **Next**.
- e. For Install As Application, select **Reuse Application ID 111 From Export File** and click **Install Application**.

- f. Click **Replace Application** (if prompted).
- g. Wait for the application to successfully install.

For Instructor Use Only.
This document should not be distributed.

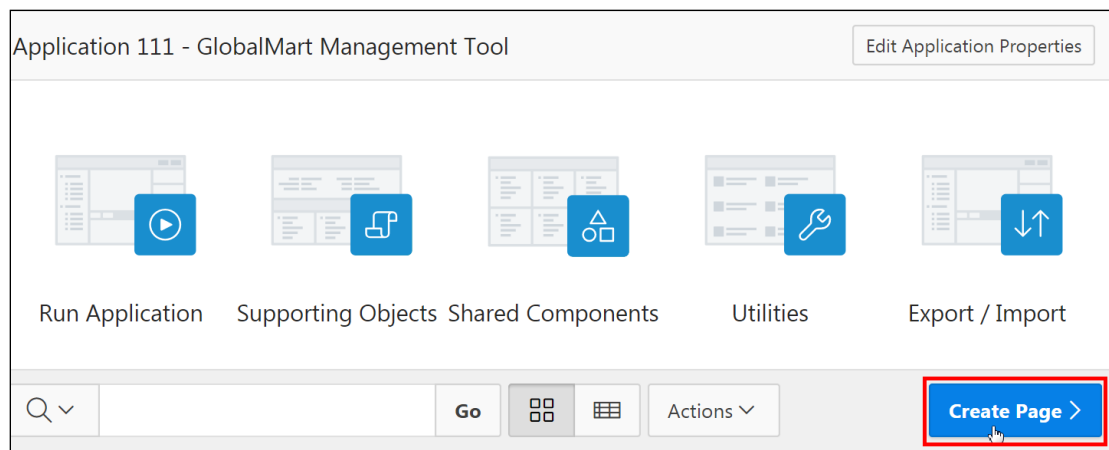
Solution 5-1: Creating and Manipulating an Interactive Report

Overview

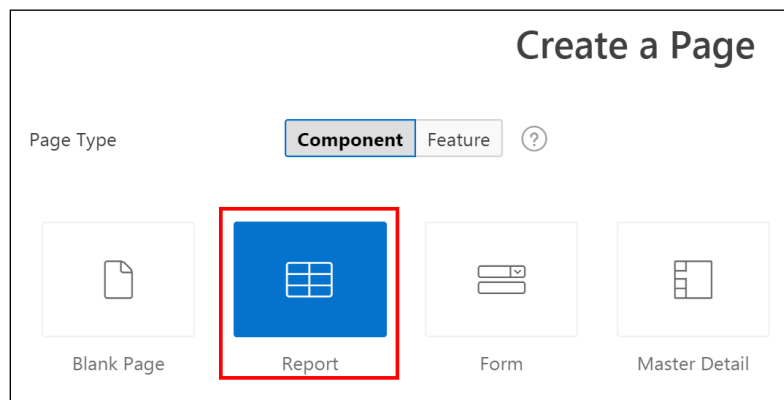
In this practice solution, steps are provided to create and manipulate the interactive report for the GMT application.

Steps

1. Create an interactive report to display all the columns from the OEHR_CUSTOMERS table on a new page in the GMT application. Name the page Customers and the region Customer Report.
 - a. Navigate to the GMT application home page and click **Create Page**.



- b. Select **Report**.

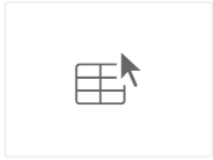


For Instructor Use Only.
This document should not be distributed.

- c. Select **Interactive Report**.

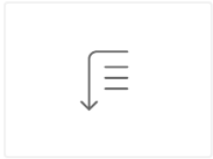
Create Page


Interactive Report


Interactive Grid


Classic Report


Report with Form


Reflow Report

- a. In Report Page Number, enter 4; in Report Page Name, enter `Customers`. Click **Next**.

Create Interactive Report

Page Attributes

Type **Interactive Report**

* Page Number

?

* Page Name

?

Page Mode **Normal** Modal Dialog ?

Breadcrumb - do not use breadcrumbs on page - v ?

- b. In Navigation Preference, select **Create a new navigation menu entry**. Notice that Customers is auto-populated in the New Navigation Menu Entry field. Click **Next**.

The screenshot shows the 'Create Interactive Report' wizard at the 'Navigation Menu' step. A progress bar at the top has three dots; the first is green with a checkmark, and the second is blue. Below the progress bar, the 'Navigation Preference' section has three radio button options: 'Do not associate this page with a navigation menu entry', 'Create a new navigation menu entry' (which is selected and highlighted with a red box), and 'Identify an existing navigation menu entry for this page'. Below this, the 'New Navigation Menu Entry' field is a text box containing the word 'Customers'. The 'Parent Navigation Menu Entry' is a dropdown menu showing '- No parent selected -' and 'Home'. Both fields have a question mark icon to their right.

Create Interactive Report

Navigation Menu

Navigation Preference

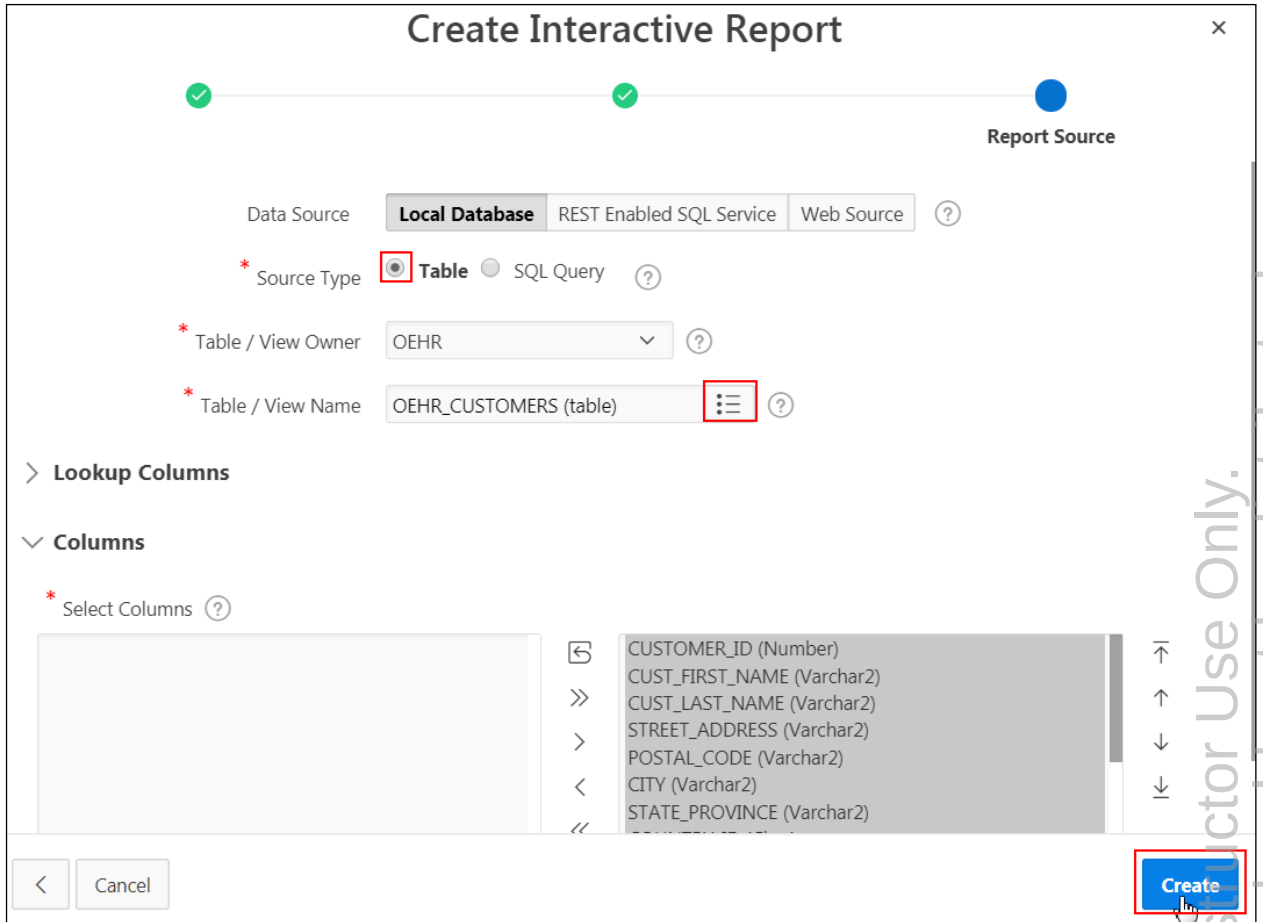
- ☐ Do not associate this page with a navigation menu entry ?
- ☒ **Create a new navigation menu entry**
- ☐ Identify an existing navigation menu entry for this page

* New Navigation Menu Entry Customers ?

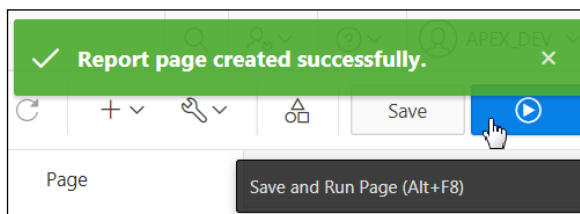
Parent Navigation Menu Entry - No parent selected -
Home ?

For Instructor Use Only.
This document should not be distributed.

- c. From Table / View Name, click the  icon to search for a table. In the Search dialog box, find and select OEHR_CUSTOMERS (table) . Click **Create**.



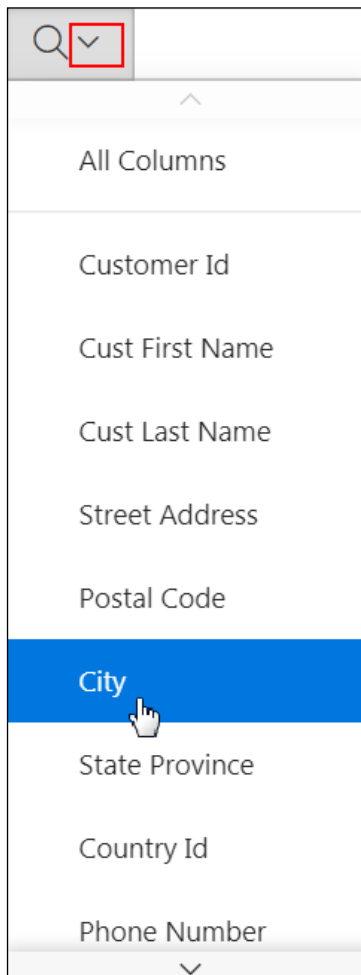
- d. The report is created. To view the report, click the **Save and Run Page** icon .



Note: If prompted, enter your developer user credentials.
The interactive report is displayed.

Q Go Actions									
Customer Id	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province	Country Id	Phone Number	Nls Language
218	Bryan	Dvrrie	3376 Perrysville Ave	15214	Pittsburgh	PA	US	+1 412 123 4740	us
219	Ajay	Sen	220 Penn Ave # 300	18503	Scranton	PA	US	+1 717 123 4741	us
220	Carol	Jordan	135 S 18Th St # 1	19103	Philadelphia	PA	US	+1 215 123 4743	us

2. Create a filter on the city of Philadelphia.
 - a. Click the **magnifying glass** icon labeled Select columns to search and select **City** from the Column Selector. Note that the selected search term appears as placeholder text in the field (Search: City). If you did not select a column, the search criteria you enter in the next step would search all columns.



- b. Enter `phil` in the search field and click **Go**. Note that the % wildcard is implied and is not entered.

A filter is applied, and only the records that meet the search criteria are displayed.

Q

▼

Go

Actions ▼

▼

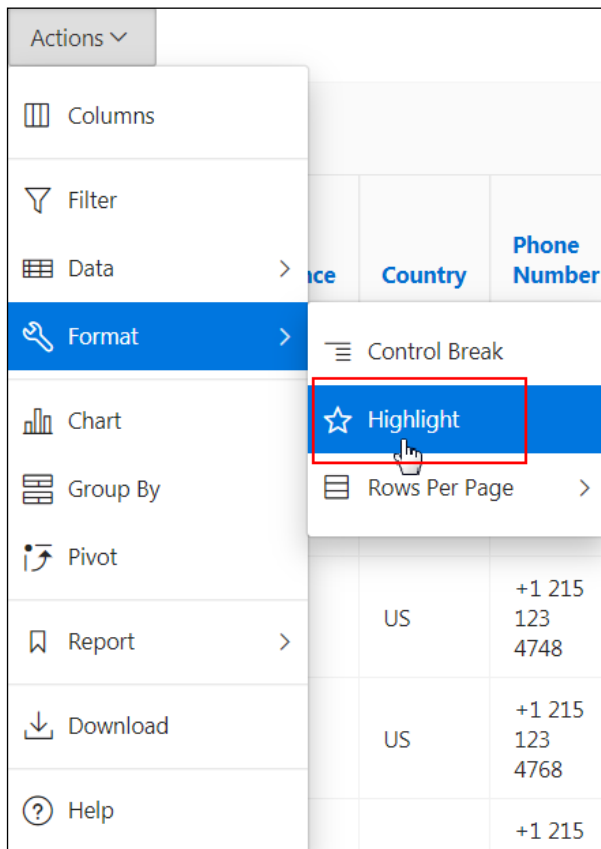
City contains 'phil'

×

Customer Id	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province	Country Id
220	Carol	Jordan	135 S 18Th St # 1	19103	Philadelphia	PA	US
222	Cary	Stockwell	7708 City Ave	19151	Philadelphia	PA	US
224	Clara	Krige	101 E Olney Ave	19120	Philadelphia	PA	US
229	Ian	Chapman	601 Market St	19106	Philadelphia	PA	US
230	Danny	Wright	5565 Baynton St	19144	Philadelphia	PA	US

For Instructor Use Only.
This document should not be distributed.

3. Create a highlight where the credit limit is greater than 2500.
 - a. To highlight only those customers whose credit limit is greater than 2500, click **Actions** and select **Format > Highlight**.



- b. In Name, enter `Good Credit Rating`; in Background Color, select **[yellow]**; in Column, select **Credit Limit**; in Operator, select **>** (greater than), and in Expression, enter `2500`. Click **Apply**.

×

Name

Good Credit Rating

Sequence

10

Enabled

Yes ▼

Highlight Type

Row ▼

Background Color

#FFFF99 ^

[yellow]

[green]

[blue]

[orange]

[red]

Text Color

^

[yellow]

[green]

[blue]

[orange]

[red]

Highlight Condition

Column	Operator	Expression
Credit Limit ▼	> ▼	2500 ▼

Cancel

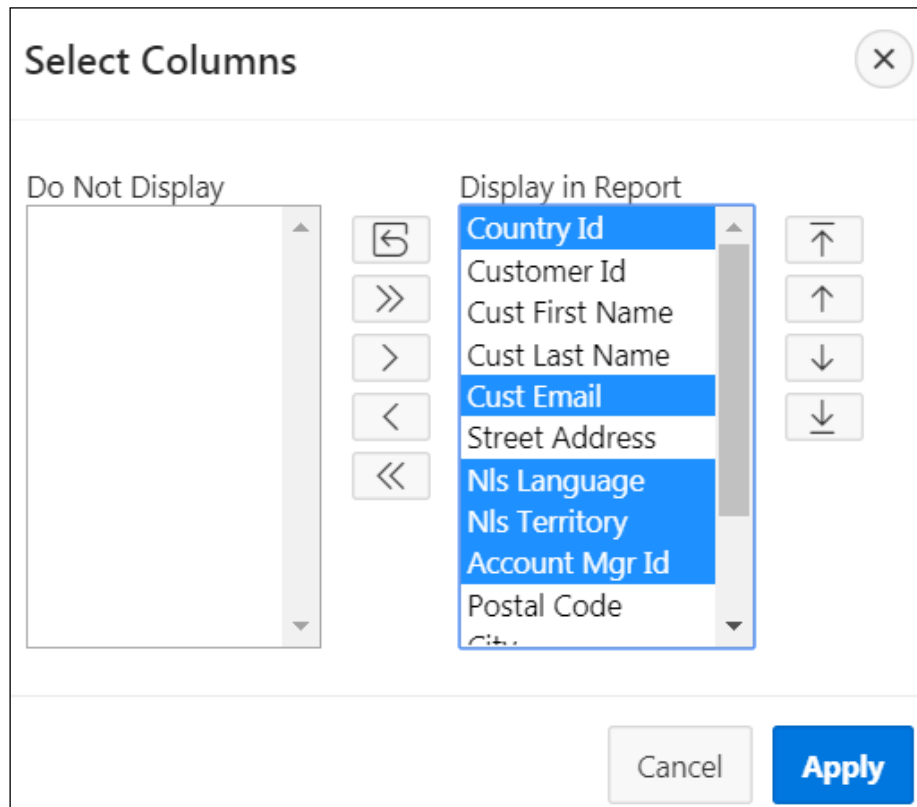
Apply

The records that meet the criteria are highlighted. (Scroll down to see them.)

Go Actions							
☒ x ☒ x							
Customer Id	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province	Country Id
220	Carol	Jordan	135 S 18Th St # 1	19103	Philadelphia	PA	US
222	Cary	Stockwell	7708 City Ave	19151	Philadelphia	PA	US
224	Clara	Krige	101 E Olney Ave	19120	Philadelphia	PA	US
229	Ian	Chapman	601 Market St	19106	Philadelphia	PA	US
230	Danny	Wright	5565 Baynton St	19144	Philadelphia	PA	US
264	George	Adjani	1136 Arch St	19107	Philadelphia	PA	US
265	Irene	Laughton	6Th And Master St	19122	Philadelphia	PA	US
280	Rob	MacLaine	5344 Haverford Ave	19139	Philadelphia	PA	US
281	Don	Barkin	6959 Tulip St	19135	Philadelphia	PA	US

For Instructor Use Only.
This document should not be distributed.

4. Display only the Customer ID, Customer Name, Address, Phone Number, and Credit Limit by hiding the other columns in the report.
 - a. Click **Actions > Columns**.
 - b. Ctrl + select **only** the following columns:
 - Country ID
 - Cust Email
 - Nls Language
 - Nls Territory
 - Account Manager ID



- c. Click the < icon (**Remove**) to move them to the Do Not Display area. Note that you can still perform other actions against these columns even though they are not displayed in the report.

Select Columns

Do Not Display

Display in Report

Country Id
Customer Id
Cust First Name
Cust Last Name
Cust Email
Street Address
Nls Language
Nls Territory
Account Mgr Id
Postal Code

Cancel Apply

Select Columns

Do Not Display

Country Id
Cust Email
Nls Language
Nls Territory
Account Mgr Id

Display in Report

Customer Id
Cust First Name
Cust Last Name
Street Address
Postal Code
City
State Province
Phone Number
Credit Limit

Cancel Apply

- d. Click **Apply**.

Only the columns that were under the **Display in Report** area are displayed.

Q

Go

Actions

▼

☒

City contains 'phil'

☒

Good Credit Rating

×

×

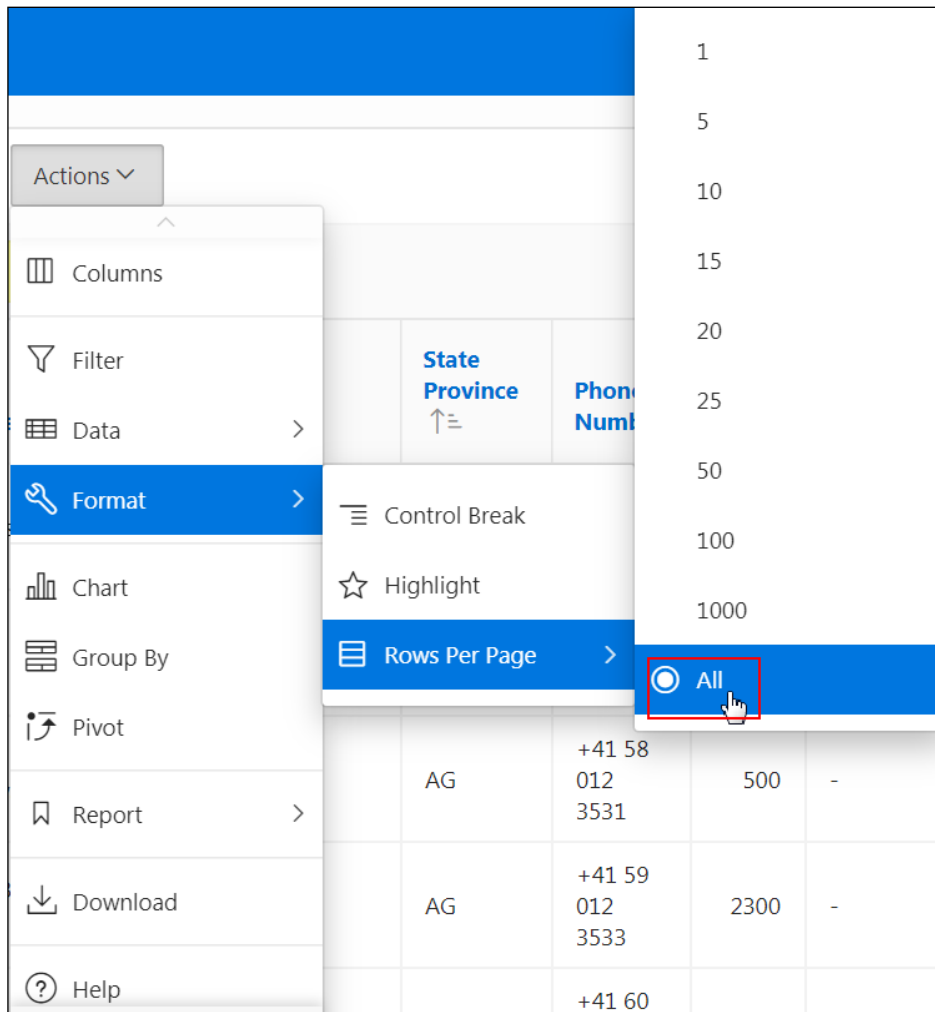
Customer Id	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province	Phone Number	Credit Limit
220	Carol	Jordan	135 S 18Th St # 1	19103	Philadelphia	PA	+1 215 123 4743	2300
222	Cary	Stockwell	7708 City Ave	19151	Philadelphia	PA	+1 215 123 4745	2300
224	Clara	Krige	101 E Olney Ave	19120	Philadelphia	PA	+1 215 123 4748	2300
229	Ian	Chapman	601 Market St	19106	Philadelphia	PA	+1 215 123 4768	2400
230	Danny	Wright	5565 Baynton St	19144	Philadelphia	PA	+1 215 123 4771	2400
264	George	Adjani	1136 Arch St	19107	Philadelphia	PA	+1 215 123 4702	3600
265	Irene	Laughton	6Th And Master St	19122	Philadelphia	PA	+1 215 123 4749	3600
280	Rob	MacLaine	5344 Haverford Ave	19139	Philadelphia	PA	+1 215 123 4709	5000
281	Don	Barkin	6959 Tulip St	19135	Philadelphia	PA	+1 215 123 4751	5000

5. Delete the “phil” filter.

- a. Click the **X** icon (**Remove Filter**) for the City contains 'phil' filter.

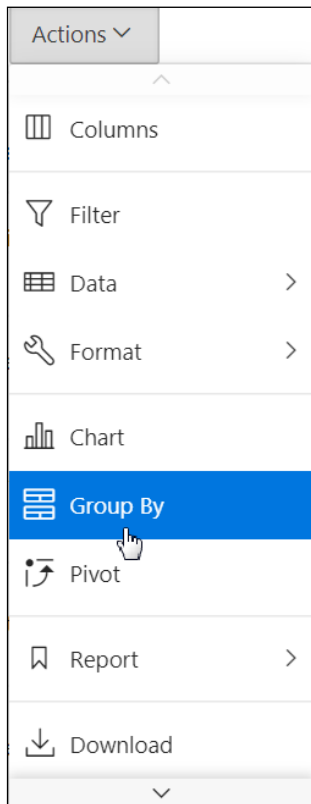
Go Actions						
☐ ☒ ☆ Good Credit Rating x						
Customer Id	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province
825	Alain	Dreyfuss	Harmoniegasse 3	3413	Baden-Daettwil	AG
853	Amrish	Palin	1540 Stripes Crt	3413	Baden-Daettwil	AG
827	Alain	Siegel	Alfred E. Neumann-Weg 3	3413	Baden-Daettwil	AG
828	Alan	Minnelli	Dr. Herbert Bitto Str 23	3413	Baden-Daettwil	AG
829	Alan	Hunter	Taefernstr 4	3413	Baden-Daettwil	AG
830	Albert	Dutt	Kreuzritterplatz 5	3413	Baden-Daettwil	AG
831	Albert	Bel Geddes	Helebardenweg 5	3413	Baden-Daettwil	AG

7. Change the pagination to display all rows.
 - a. Because the pagination is set to default (50), you do not see all the rows. To view all the rows, click **Actions > Format > Rows Per Page > All**.

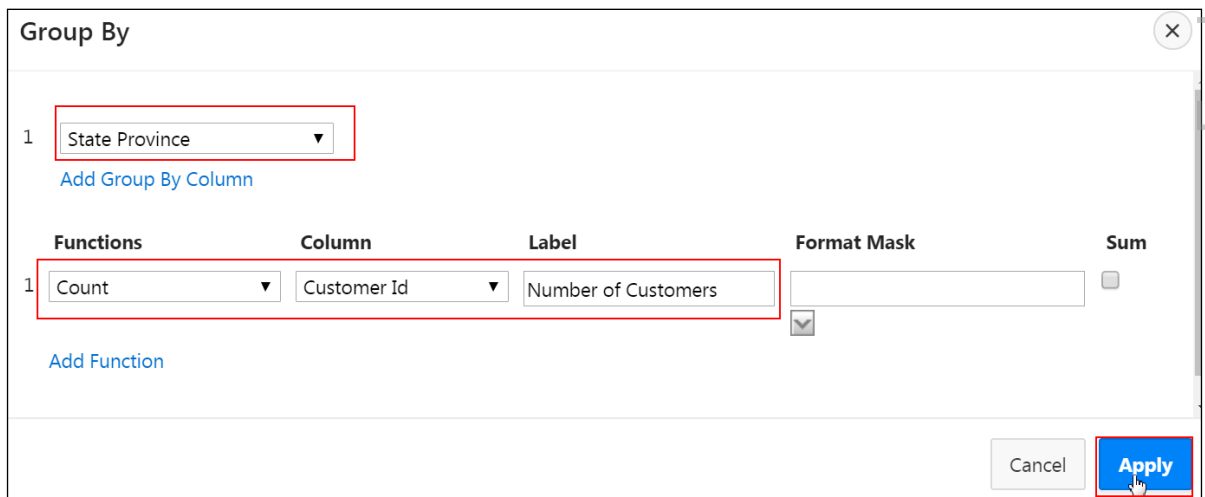


8. Create a Group By report to display the number of customers in a particular province or state.

a. To create a Group By report, click **Actions** and select **Group By**.



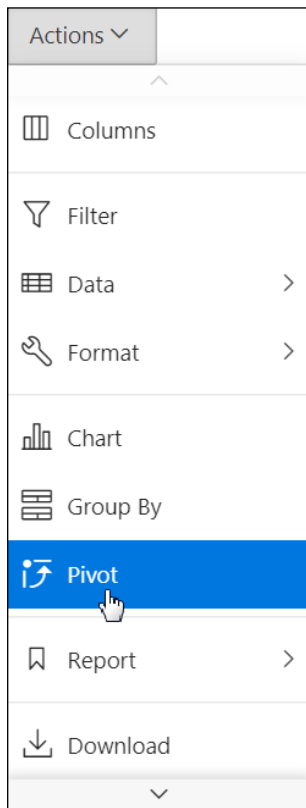
b. Click **Select Group By Column** and select **State Province**. In Functions, select **Count**; in Column, select **Customer Id**; and in Label, enter **Number of Customers**. Click **Apply**.



For Instructor Use Only.
This document should not be distributed.

- c. The state provinces and the number of customers per state appear. To return to the report view, click the **View Report** icon .

9. Create a Pivot report to display the total credit limit of customers for each account manager to see how much potential debt you may hold.
- a. To create a Pivot, click **Actions > Pivot**.



- b. In the Pivot dialog box:
- In **Pivot Columns**, select **Account Manager Id**
 - In **Row Columns**, select **Customer Id**
 - In **Functions**, select **Sum**
 - In **Column**, select **Credit Limit**
 - In **Label**, enter `Total Credit Limit`
 - In **Sum**, select the **check box** to enable

Pivot

Pivot Columns

1 Account Mgr Id

Add Pivot Column

Row Columns

1 Customer Id

Add Row Column

Functions	Column	Label	Format Mask	Sum
1 Sum	Credit Limit	Total Credit Limit		☒

Add Function

Cancel

Apply

c. Click **Apply**.

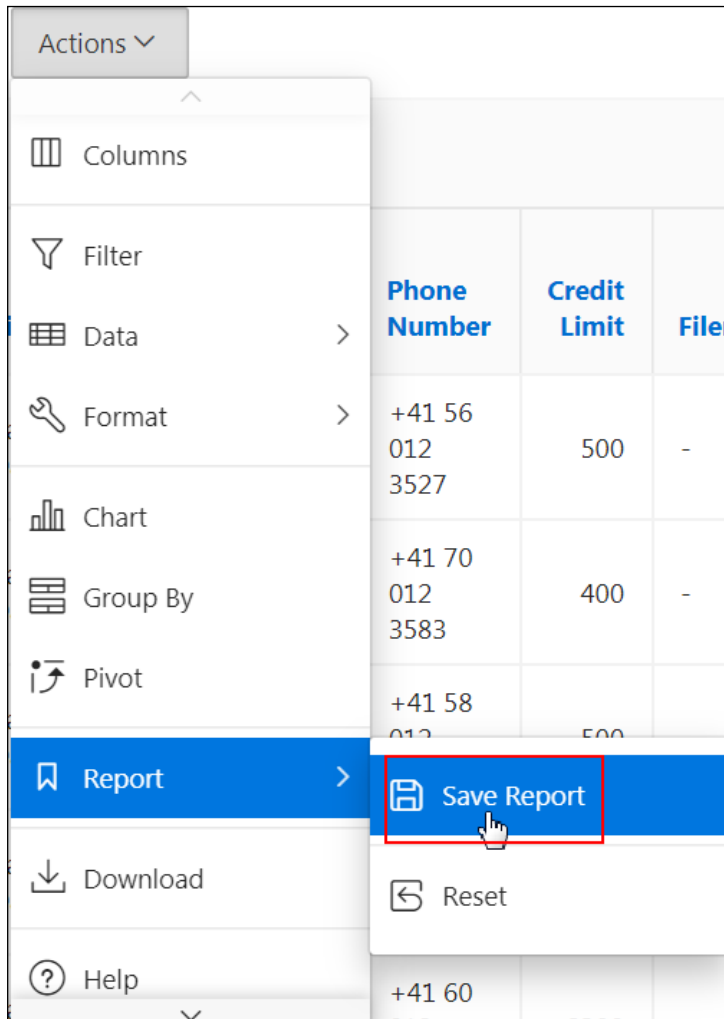
The credit limits for each customer per manager are displayed with an aggregation row at the bottom with the total amount of credit each manager has allocated.

	145	147	148	149
Customer Id	Total Credit Limit	Total Credit Limit	Total Credit Limit	Total Credit Limit
833	-	3,500	-	-
840	-	1,400	-	-
934	-	-	600	-
919	-	-	3,500	-
920	-	-	3,500	-
239	2,400	-	-	-
448	-	500	-	-
450	-	1,500	-	-
451	-	700	-	-
475	-	-	5,000	-
487	-	-	1,900	-
771	-	400	-	-
	151,400	136,100	157,500	159,400

d. To return to the report view, click the **View Report** icon .

10. Save the report as a private report that is available only to you.

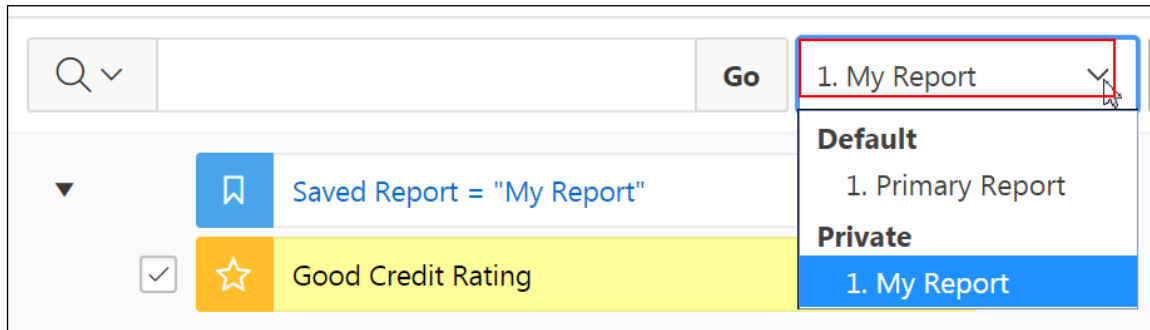
a. Click **Actions > Report > Save Report**.



b. In Name, enter My Report. Click **Apply**.

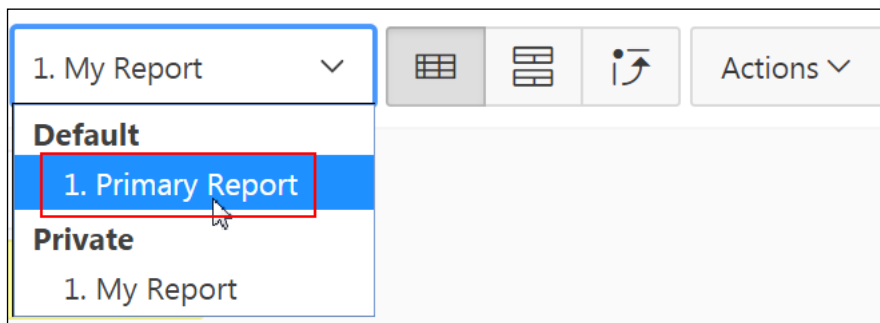
The screenshot shows the 'Save Report' dialog box. It has a title bar with a close button. Inside, there is a 'Save' section with a dropdown menu set to 'As Named Report'. Below this, there is a 'Name' field containing 'My Report', which is highlighted with a red box. There is also a 'Description' field. At the bottom right, there are 'Cancel' and 'Apply' buttons, with the 'Apply' button highlighted by a red box and a mouse cursor. A note '(Only displayed for developers)' is visible next to the dropdown menu.

- c. The report is saved. You can click the Reports drop-down list to see the list of reports.

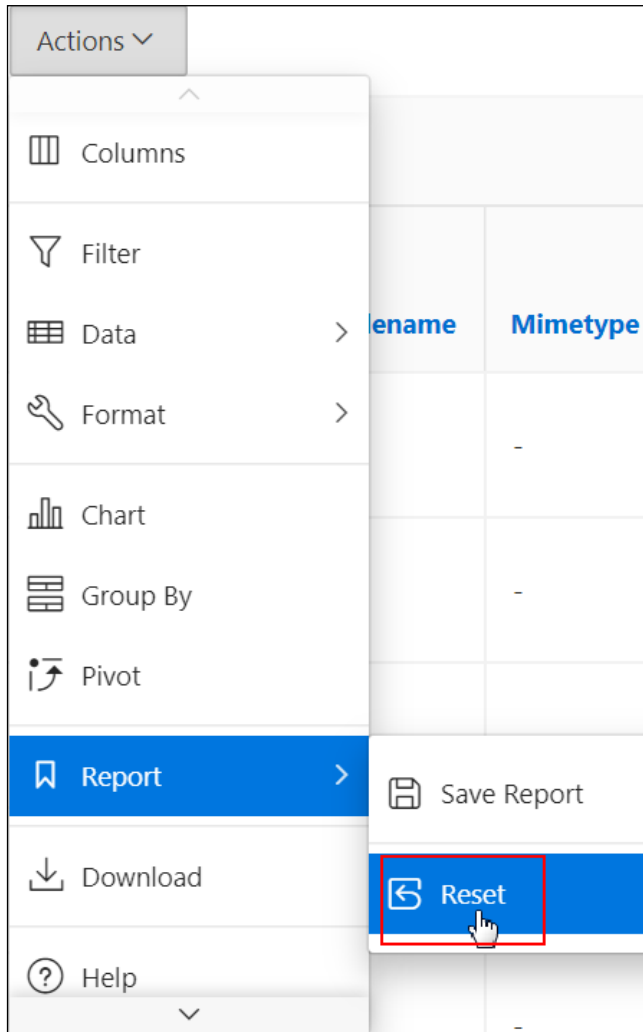


11. Reset the Primary Report to its original state.

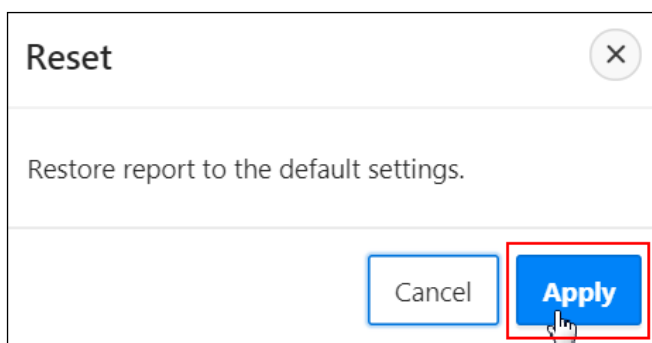
- a. Click the **Reports** drop-down menu (**1. My Report**) and select **1. Primary Report**.



- b. To return to the default settings, click **Actions > Report > Reset**.



- c. Click **Apply**.



You see the original report without customizations.

- d. Click the **Application 111** link at the bottom of the page to return to the Application home page.

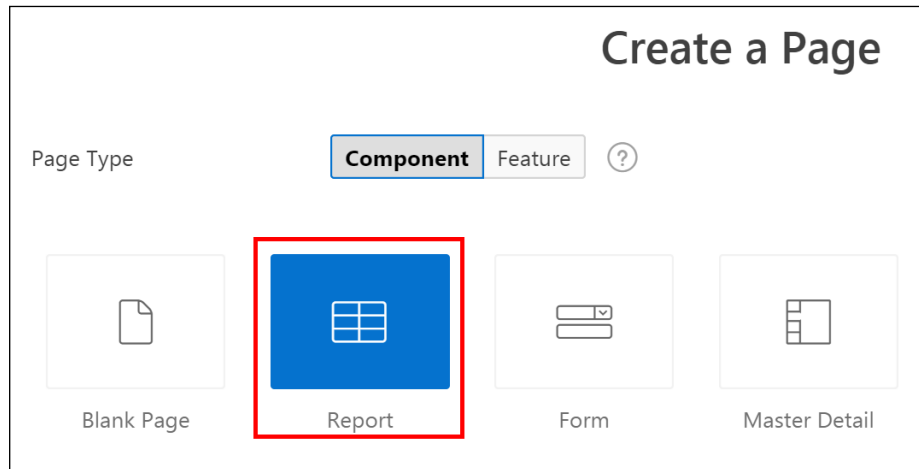
The screenshot displays the GlobalMart Management Tool interface. On the left is a dark sidebar with a menu containing 'Home' and 'Customers'. The main content area features a table with customer information. Above the table is a search bar with a magnifying glass icon, a 'Go' button, a dropdown menu set to '1. Primary Report', and an 'Actions' button. The table has columns for Customer Id, Cust First Name, Cust Last Name, Street Address, Postal Code, City, State Province, and Country Id. It contains three rows of data. At the bottom, a dark navigation bar includes links for Home, Application 111 (highlighted with a red box), Edit Page 4, Session, View Debug, Debug, Page Info, Quick Edit, and Theme Roller.

Customer Id	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province	Country Id
218	Bryan	Dvrrie	3376 Perrysville Ave	15214	Pittsburgh	PA	US
219	Ajay	Sen	220 Penn Ave # 300	18503	Scranton	PA	US
220	Carol	Jordan	135 S 18Th St # 1	19103	Philadelphia	PA	US

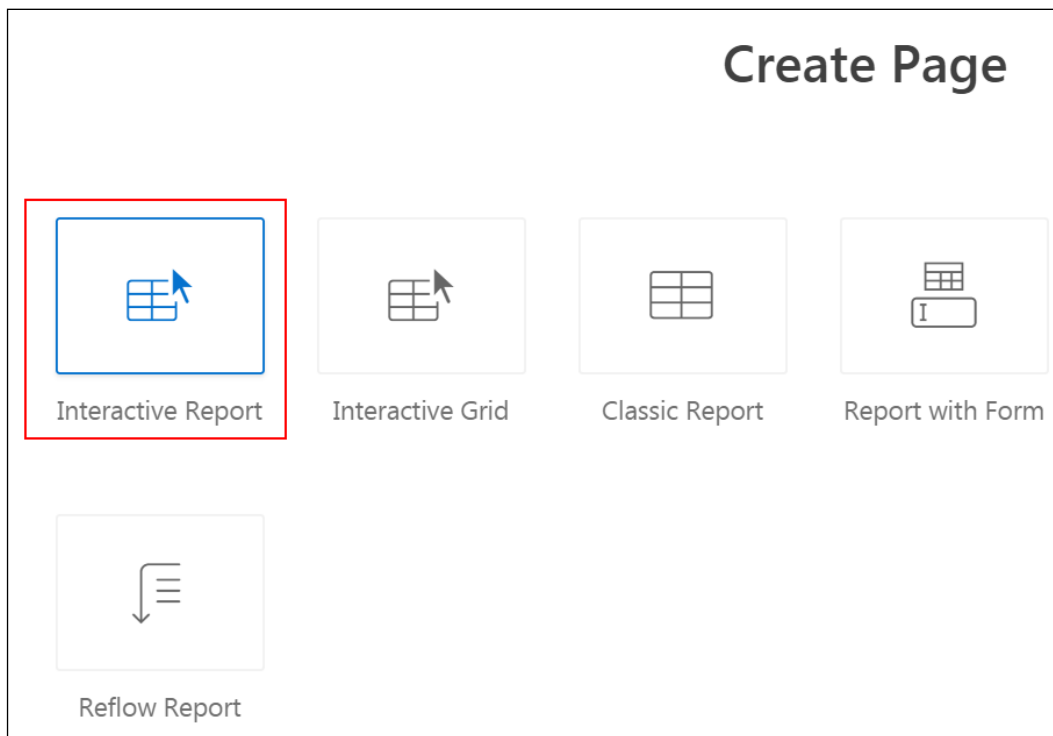
For Instructor Use Only.
This document should not be distributed.

12. Create a stand-alone interactive report page with breadcrumbs in the GMT application named Employee Commissions. This interactive report displays details from the OEHR_EMPLOYEES table about who earns a commission.

- a. Navigate to the GMT application home page and click **Create Page**.
- b. Select **Report**.



- c. Select **Interactive Report**.



- a. In Page Number, enter 5; in Page Name, enter `Employee Commission`; in Breadcrumb, select **Breadcrumb**; and in Parent Entry, select **Home (Page 1)**. Click **Next**.

Create Interactive Report

Page Attributes

Type	Interactive Report
* Page Number	5 ?
* Page Name	Employee Commission ?
Page Mode	Normal Modal Dialog ?
Breadcrumb	Breadcrumb v ?
Parent Entry	Home (Page 1) v
Entry Name	Employee Commission ?

- b. In Navigation Preference, accept the default settings and click **Next**.
- c. In Source Type, select **SQL Query**. Enter the following SQL `SELECT` statement. You can copy and paste the SQL statement from `/home/oracle/labs/labs/lab_05_01_01.txt`. Alternatively, you can use the Query Builder button to create the report query.

```
SELECT employee_id, first_name, last_name, manager_id,
commission_pct
FROM OEHR_EMPLOYEES
WHERE commission_pct IS NOT NULL;
```

Create Interactive Report

Report Source

Data Source: Local Database REST Enabled SQL Service Web Source

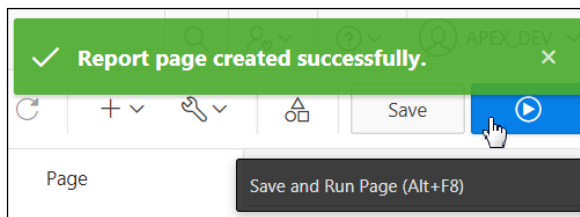
Source Type: Table **SQL Query**

* Enter a SQL SELECT statement

```

1 SELECT employee_id, first_name, last_name, manager_id, commission_pct
2 FROM OEHR_EMPLOYEES
3 WHERE commission_pct IS NOT NULL;
4
    
```

- d. Click **Create**. The report is created.
- e. To view the report, click the **Save and Run Page** icon.



Note: If prompted, enter your developer user login credentials.
Your report is displayed.

Home \

Employee Commission

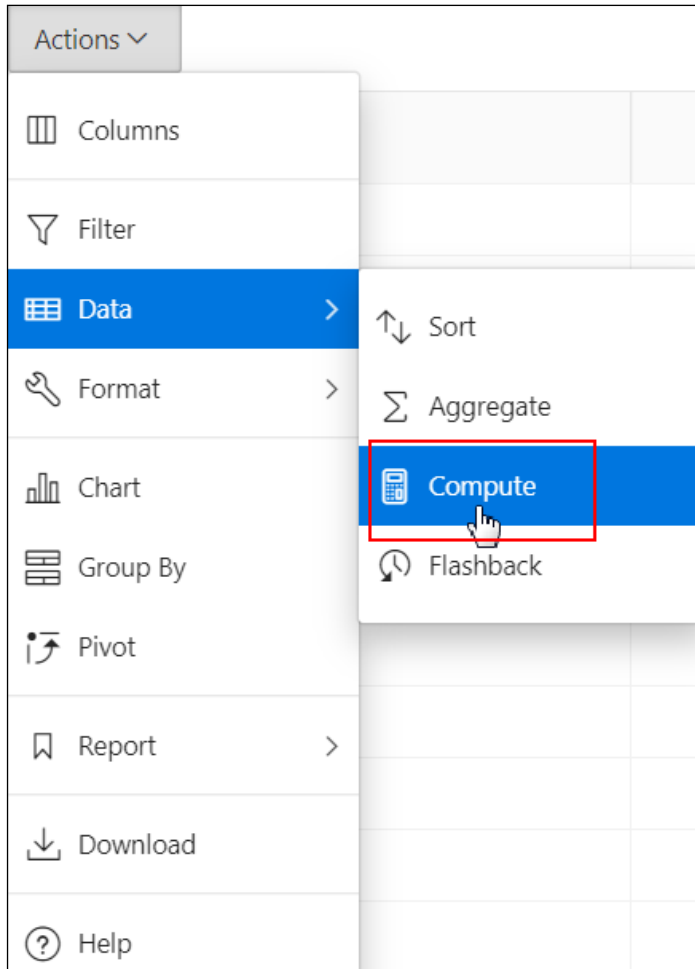
Employee Id	First Name	Last Name	Manager Id	Commission Pct
145	John	Russell	100	.4
146	Karen	Partners	100	.3
147	Alberto	Errazuriz	100	.3
148	Gerald	Cambrault	100	.3

Home
Application 111
Edit Page 5
Session
View Debug
Debug
Page Info
Quick Edit
Theme Roller

13. Management proposes to increase the commission of employees whose existing commission percentage is less than or equal to 0.15. Create a computed column called New Commission, to display the new commission. The new commission scheme includes:

- Employees with a commission percentage of 0.1 will get an additional commission of 0.15.
- Employees with a commission percentage of 0.15 will get an additional commission of 0.05.

a. To create a new computation, click **Actions > Data > Compute**.



b. For Column Label, enter **New Commission**.

c. The computation should add .15 to a commission of .1 and .05 to a commission of .15. For Computation Expression, enter the CASE statement (located in /home/oracle/labs/labs/lab_05_01_02.txt):

```
CASE E WHEN 0.1 THEN E+0.15 WHEN 0.15 THEN E+0.05 ELSE E END
```

You can use the columns list, functions list, and keypad to enter the code in the computation area or paste the previous CASE statement into the computation area. When you select Commission Pct from Columns, notice that the letter corresponding to the column appears in the computation area. This is the column alias.

After entering the code, click **Apply**.

Compute

Computation

- New Computation - ▼

Column Label

New Commission

Format Mask

Computation Expression

CASE E WHEN 0.1 THEN E+0.15 WHEN 0.15 THEN E+0.05 ELSE E END

Columns

A. Employee Id

B. First Name

C. Last Name

D. Manager Id

E. Commission Pct

Keypad

() ' ||

7 8 9 -

4 5 6 +

1 2 3 *

0 . /

space ,

Functions / Operators

!=

<

<=

=

>

>=

ABS

Create a computation using column aliases. Examples:

(B+C)*100

INITCAP(B)||', '||INITCAP(C)

CASE WHEN A = 10 THEN B + C ELSE B END

ROUND(C / 1000000)

Cancel

Apply

The computed column appears in the report.

Employee Id	First Name	Last Name	Manager Id	Commission Pct	New Commission
146	Karen	Partners	100	.3	.3
147	Alberto	Errazuriz	100	.3	.3
148	Gerald	Cambrault	100	.3	.3
149	Eleni	Zlotkey	100	.2	.2
150	Peter	Tucker	145	.3	.3
151	David	Bernstein	145	.25	.25

14. Create a control break on Manager ID.

- Click the **Manager Id** column heading and click the **Control Break** icon. (To more fully configure a control break, you can click **Actions > Format > Control Break**.)

Employee Id	First Name	Last Name	Manager Id	Commission Pct
145	John	Russell	100	
146	Karen	Partners	100	
147	Alberto	Errazuriz	100	
148	Gerald	Cambrault	100	
149	Eleni	Zlotkey	100	
150	Peter	Tucker	145	
151	David	Bernstein	145	
152	Peter	Hall	145	

The column now appears before each break point in the report.

Home \

Employee Commission

Actions

Manager Id

Manager Id : 100

Employee Id	First Name	Last Name	Commission Pct	New Commission
147	Alberto	Errazuriz	.3	.3
146	Karen	Partners	.3	.3
145	John	Russell	.4	.4

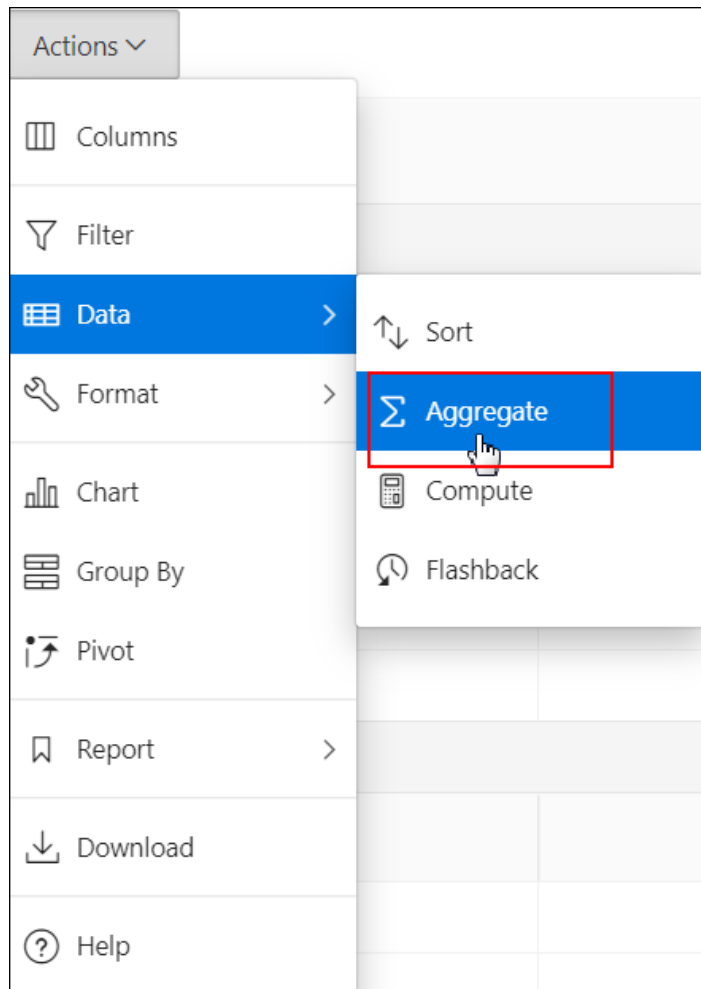
Manager Id : 145

Employee Id	First Name	Last Name	Commission Pct	New Commission
152	Peter	Hall	.25	.25
151	David	Bernstein	.25	.25
150	Peter	Tucker	.3	.3
153	Christopher	Olsen	.2	.2
155	Oliver	Tuvault	.15	.2

Home Application 111 Edit Page 5 Session View Debug Debug Page Info Quick Edit Theme Roller

For Instructor Use Only.
This document should not be distributed.

15. Display the maximum commission earned by employees under each manager.
- a. To find out the maximum commission earned by employees under each manager, click **Actions > Data > Aggregate**.



- b. In Function, select **Maximum**, and in Column, select ****New Commission** (a computed column from the column list).

Aggregate

Aggregation: - New Aggregation - ▼

Function: Maximum ▼

Column: **New Commission ▼

Displayed:

- Manager Id
- Employee Id
- Commission Pct
- **New Commission**

Apply

- c. Click **Apply**.

The report includes the maximum commission earned by employees under each manager.

Home \

Employee Commission

Actions ▼

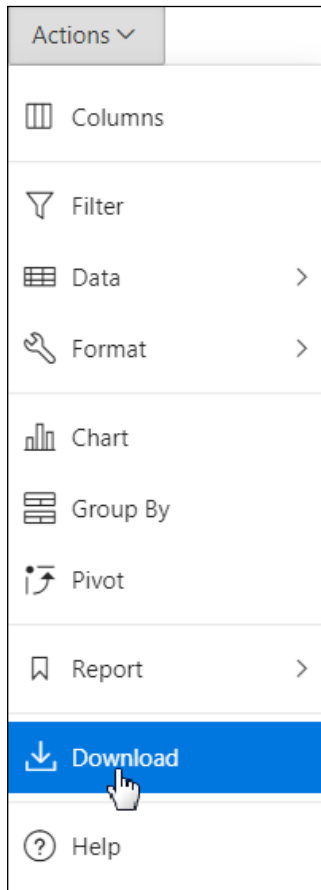
▼ ☒ Manager Id

Manager Id : 100

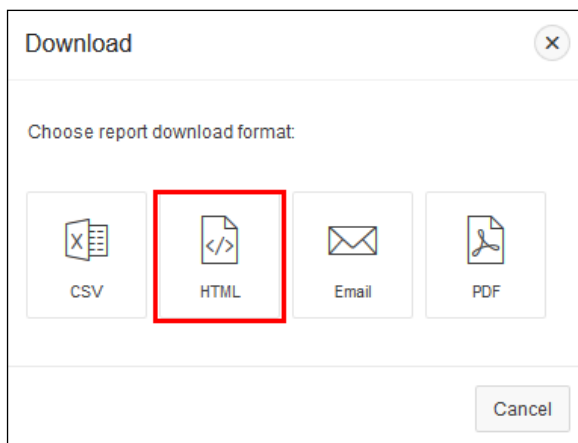
Employee Id	First Name	Last Name	Commission Pct	New Commission
147	Alberto	Errazuriz	.3	.3
146	Karen	Partners	.3	.3
145	John	Russell	.4	.4
148	Gerald	Cambrault	.3	.3
149	Eleni	Zlotkey	.2	.2
				Maximum: .4

16. Download the HTML report.

a. To download a report, click **Actions** and select **Download**.



b. Click **HTML**.

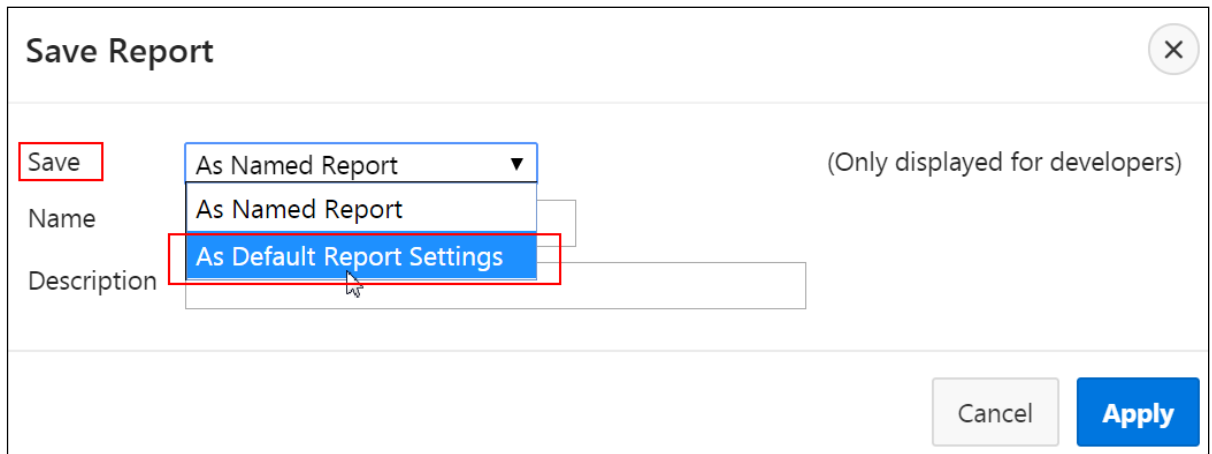


- c. Open the file to view the report in your web browser.

		Search			
Manager Id	Employee Id	First Name	Last Name	Commission Pct	New Commission
100	147	Alberto	Errazuriz	.3	.3
100	146	Karen	Partners	.3	.3
100	145	John	Russell	.4	.4
100	148	Gerald	Cambrault	.3	.3
100	149	Eleni	Zlotkey	.2	.2
145	152	Peter	Hall	.25	.25
145	151	David	Bernstein	.25	.25
145	150	Peter	Tucker	.3	.3
145	153	Christopher	Olsen	.2	.2
145	155	Oliver	Tuvault	.15	.2
145	154	Nanette	Cambrault	.2	.2
146	156	Janette	King	.35	.35
146	157	Patrick	Sully	.35	.35
146	158	Allan	McEwen	.35	.35
146	160	Louise	Doran	.3	.3

- d. Close the HTML report.
e. Click **Cancel** to close the Download window.

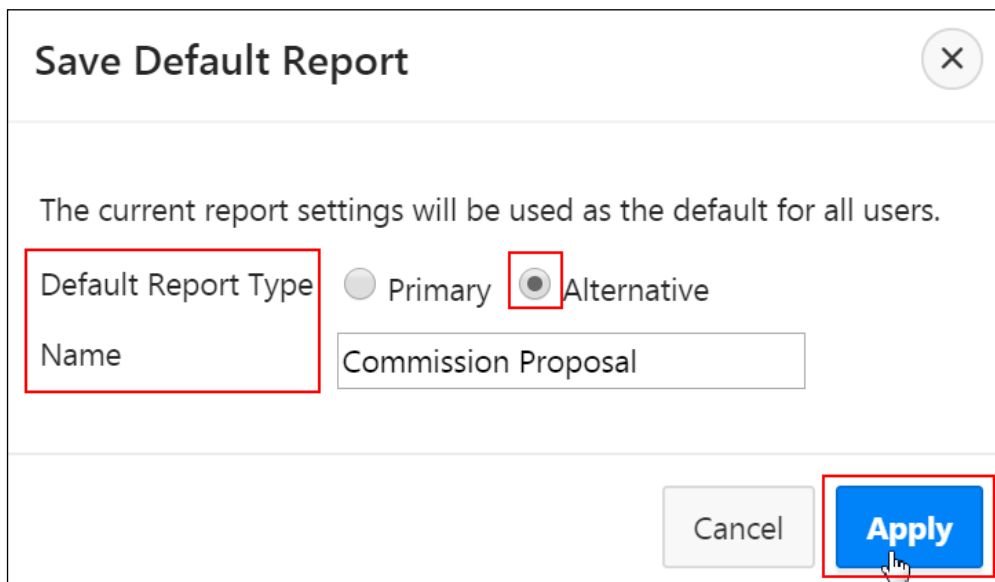
17. Save this as a public report. Name the saved report **Commission Proposal**.
- Click **Actions > Report > Save Report**.
 - In the Save drop-down list, select **As Default Report Settings**.



The 'Save Report' dialog box is shown. It has a title bar with a close button (X). The main area contains a 'Save' button (highlighted with a red box), a dropdown menu (highlighted with a blue box) showing 'As Named Report' with a downward arrow, and a text input field for 'Name' containing 'As Named Report'. Below the dropdown is a 'Description' label and a text input field containing 'As Default Report Settings' (highlighted with a blue box). To the right of the dropdown is the text '(Only displayed for developers)'. At the bottom right are 'Cancel' and 'Apply' buttons.

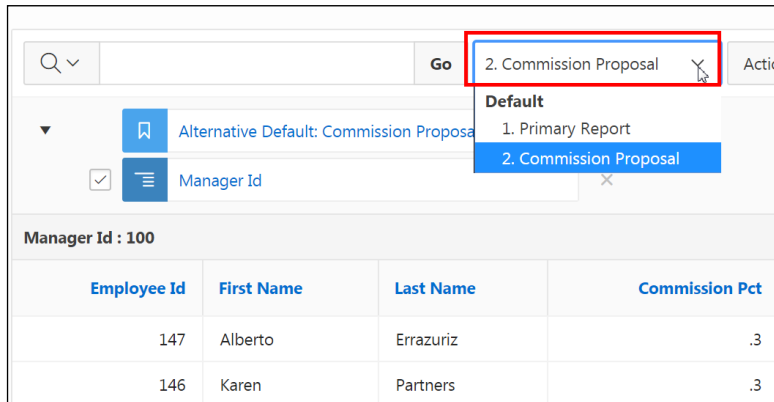
The Save Report dialog changes to **Save Default Report**.

- In Default Report Type, select **Alternative**, and in Name, enter **Commission Proposal**. Click **Apply**.



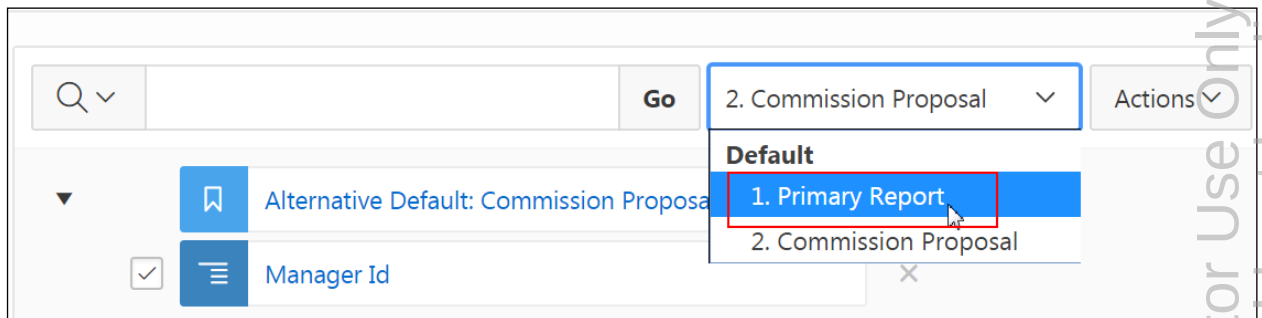
The 'Save Default Report' dialog box is shown. It has a title bar with a close button (X). The main area contains the text 'The current report settings will be used as the default for all users.' Below this is a 'Default Report Type' label (highlighted with a red box) and two radio buttons: 'Primary' and 'Alternative' (the 'Alternative' radio button is selected and highlighted with a red box). Below the radio buttons is a 'Name' label (highlighted with a red box) and a text input field containing 'Commission Proposal'. At the bottom right are 'Cancel' and 'Apply' buttons (the 'Apply' button is highlighted with a red box and a mouse cursor is clicking it).

The alternative report now appears in the Reports drop-down list.



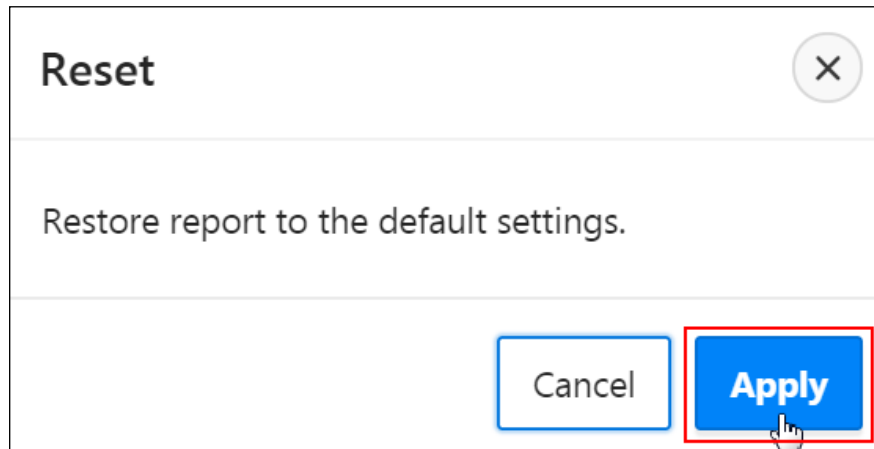
18. Reset the Primary Report.

a. Click the Reports drop-down list and select **1. Primary Report**.



b. To reset this primary report to its original settings, click **Actions > Report > Reset**.

c. Click **Apply**.



- d. The primary report is reset to its original settings. Click **Edit Page 5** on the Runtime Developer toolbar at the bottom of the page to return to Page Designer.

Home \

Employee Commission

Q ▾ Go 1. Primary Report ▾ Actions ▾

Employee Id	First Name	Last Name	Manager Id	Commission Pct
145	John	Russell	100	.4
146	Karen	Partners	100	.3
147	Alberto	Errazuriz	100	.3
148	Gerald	Cambrault	100	.3
149	Eleni	Zlotkey	100	.2
150	Peter	Tucker	145	.3

Home Application 111 Edit Page 5 Session View Debug Debug Page Info Quick Edit Theme Roller

For Instructor Use Only.
This document should not be distributed.

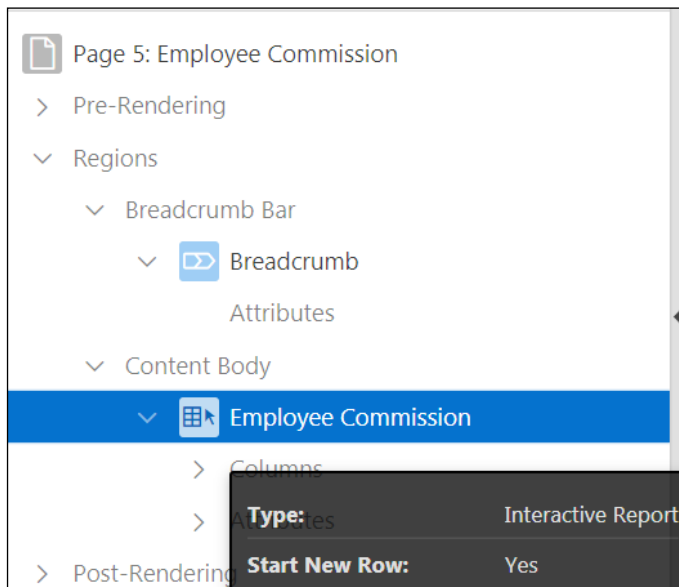
Solution 5-2: Customizing an Interactive Report

Overview

In this practice solution, steps are provided to customize the interactive report that you created for the GMT application.

Steps

1. Make the following customizations to the Employee Commissions report:
 - Hide `EMPLOYEE_ID`.
 - Change `MANAGER_ID` to be the `FIRST_NAME` and `LAST_NAME` of the manager.
- a. Navigate to the Page Designer view of the Employee Commission page.
- b. On the Rendering tab (left pane), expand **Regions > Content Body** and select **Employee Commission**.



For Instructor Use Only.
This document should not be distributed.

- c. In the Property Editor (right pane), in the Source section, change the SQL Query field value to reflect the following query (located in /home/oracle/labs/labs/lab_05_02_01.txt):

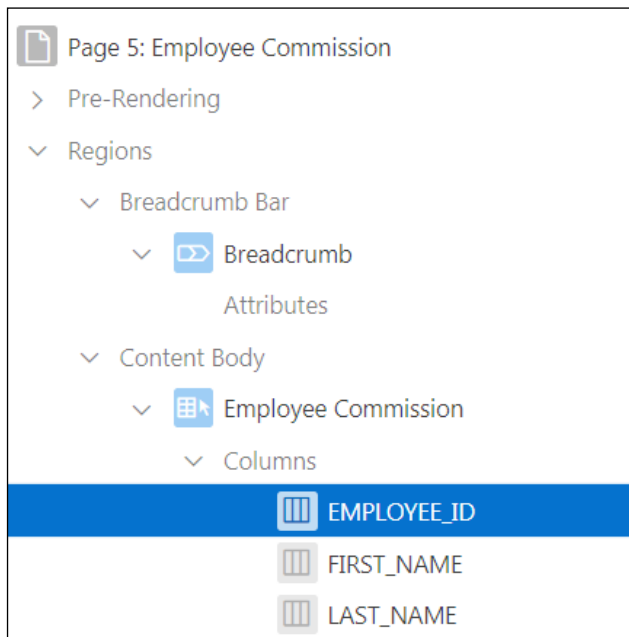
```
SELECT e.employee_id,  
e.first_name,  
e.last_name,  
(SELECT e1.first_name||' '||e1.last_name from  
OEHR_EMPLOYEES e1 where e1.employee_id =  
e.manager_id) as  
"Manager",  
e.commission_pct  
FROM OEHR_EMPLOYEES e  
where e.commission_pct IS NOT NULL;
```

The screenshot shows the 'Source' section of a Property Editor. It has a dropdown menu for 'Location' set to 'Local Database' and a dropdown menu for 'Type' set to 'SQL Query'. Below these is a text area labeled 'SQL Query' containing the following SQL code:

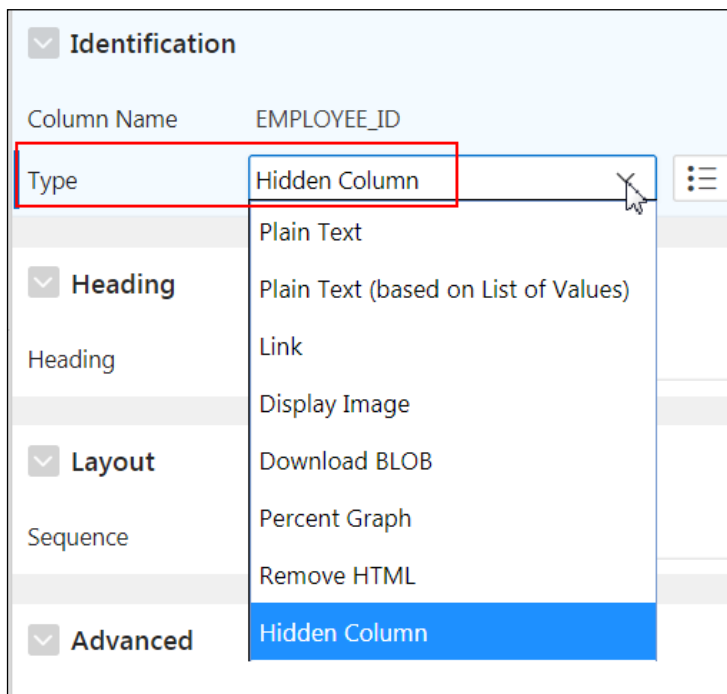
```
OEHR_EMPLOYEES e1 where e1.employee_id =  
e.manager_id) as  
"Manager",  
e.commission_pct  
FROM OEHR_EMPLOYEES e  
where e.commission_pct IS NOT NULL;
```

The text area has a red border and a blue border. There is a small icon in the top right corner of the text area.

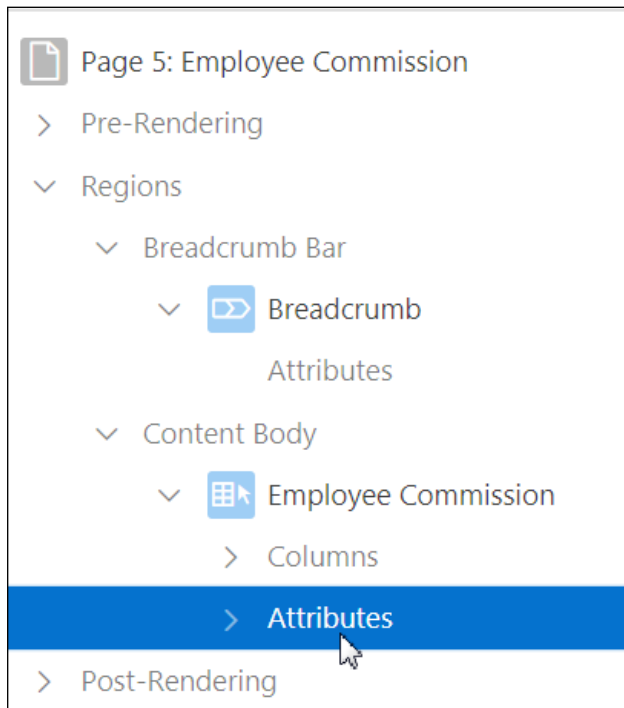
- d. On the Rendering tab (left pane), expand **Regions > Content Body > Employee Commission > Columns** and select `EMPLOYEE_ID`.



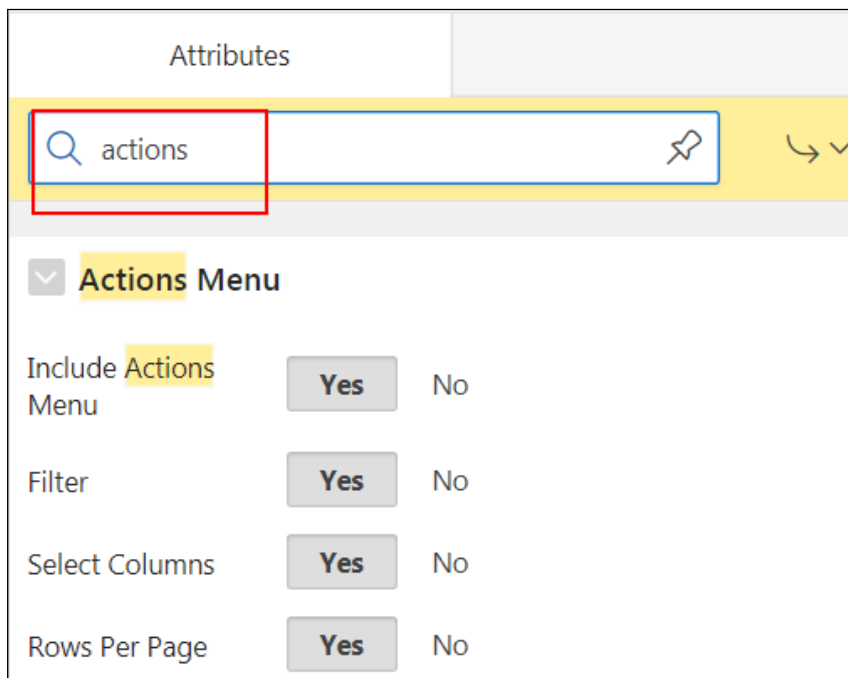
- e. In the Property Editor (right pane), under Identification, click the **Type** drop-down list and select **Hidden Column**.



2. Remove the Highlight option from the Actions menu.
 - a. On the Rendering tab (left pane), expand **Regions > Content Body > Employee Commission** and select **Attributes**.



- b. In the Property Editor (right pane), scroll down to the Actions Menu section. Alternatively, enter `actions` in the Filter field.

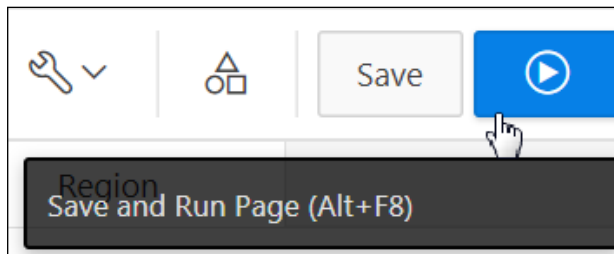


- c. For Highlight, select **No** to remove it from the Actions Menu. (If you used the filter, clear the Filter field to redisplay all options.)

The screenshot shows the 'Attributes' panel with a search bar containing 'actions'. Below the search bar is a section titled 'Actions Menu' with a dropdown arrow. The 'Actions Menu' section contains several options, each with 'Yes' and 'No' buttons. The 'Highlight' option is highlighted with a red box, and its 'No' button is also highlighted with a red box. The other options are 'Include Actions Menu', 'Filter', 'Select Columns', 'Rows Per Page', 'Sort', 'Control Break', 'Compute', and 'Aggregate', all of which have their 'Yes' buttons selected.

Attribute	Yes	No
Include Actions Menu	<input checked="" type="button" value="Yes"/>	No
Filter	<input checked="" type="button" value="Yes"/>	No
Select Columns	<input checked="" type="button" value="Yes"/>	No
Rows Per Page	<input checked="" type="button" value="Yes"/>	No
Sort	<input checked="" type="button" value="Yes"/>	No
Control Break	<input checked="" type="button" value="Yes"/>	No
Highlight	Yes	<input checked="" type="button" value="No"/>
Compute	<input checked="" type="button" value="Yes"/>	No
Aggregate	<input checked="" type="button" value="Yes"/>	No

- d. Click the **Save and Run Page**.

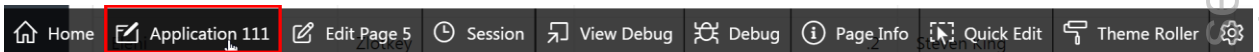


The report appears. Note that Employee ID is excluded from the report. The Manager column displays the First Name and Last Name of the manager.

First Name	Last Name	Commission Pct	Manager
John	Russell	.4	Steven King
Karen	Partners	.3	Steven King
Alberto	Errazuriz	.3	Steven King
Gerald	Cambrault	.3	Steven King

Additionally, the Highlight option is removed in the **Actions > Format** menu.

- e. The interactive report creation is complete. Click **Application 111** on the Runtime Developer Toolbar at the bottom of the page to return to the Application home page.



For Instructor Use Only.
This document should not be distributed.

**Practices for Lesson 6:
Working with Interactive
Grids**

Practices for Lesson 6

Overview

In these practices, you will create a page with an interactive grid, customize it in the application, then convert it to an editable interactive grid, and customize it further.

For Instructor Use Only.
This document should not be distributed.

Practice 6-1: Creating an Interactive Grid

Overview

In this practice, you create a report on `OEHR_PRODUCT_INFORMATION` for the GMT application. This report uses an interactive grid for end users to update the data more easily.

Assumptions

You have completed the previous practices.

Note: If you have not completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. On a new page in the GMT application, create an interactive grid to display columns from the `OEHR_PRODUCT_INFORMATION` table. Name the new page and report `Product Information Details` and number it page 6.
2. Run the app and customize the report.
 - a. Sort the report by Product Id.
 - b. Hide the Category Id column.
 - c. Break the report by Product Status. (**Hint:** Apply a control break.)
 - d. Save the report as the Primary report.
 - e. Create a Pie chart that counts the number of items in each weight class.
 - f. Save your customizations as a private report named `My Report`.

Catch Up

If you have not completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 6-1**.

If you have not completed **Practice 5-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_dev` user.
- b. Click the arrow next to **App Builder** and select **Import**.

- c. For Import File, click the button and browse to the `sol_05_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next**.
- d. Click **Next**.
- e. For Install As Application, select **Reuse Application ID 111 From Export File** and click **Install Application**.
- f. Click **Replace Application** (if prompted).
- g. Wait for the application to successfully install.

Practice 6-2: Using and Manipulating the Interactive Grid

Overview

In this practice, you convert the interactive grid to an editable interactive grid and explore the data entry capabilities.

Assumptions

You have completed the previous practices.

Note: If you have not completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Convert the interactive grid to an editable interactive grid in Page Designer.
2. In the app run time, customize the report further:
 - Add a blank new row to the report.
 - Duplicate the blank row.
 - Attempt to save the report.
 - Delete the blank rows.
3. In Page Designer, lock all displayed columns but one: Product Description. (**Hint:** Lock a column by making it Display Only.)

Catch Up

If you have not completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 6-2**.

If you have not completed **Practice 6-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **App Builder** and select **Import**.
- c. For Import File, click the button and browse to the `sol_06_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next**.

- d. Click **Next**.
- e. For Install As Application, select **Reuse Application ID 111 From Export File** and click **Install Application**.
- f. Click **Replace Application** (if prompted).
- g. Wait for the application to successfully install.

For Instructor Use Only.
This document should not be distributed.

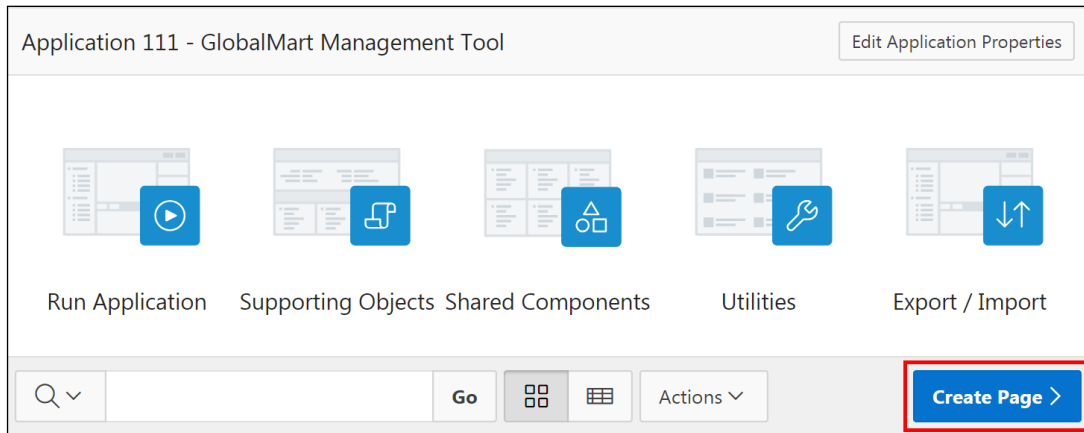
Solution 6-1: Creating an Interactive Grid

Overview

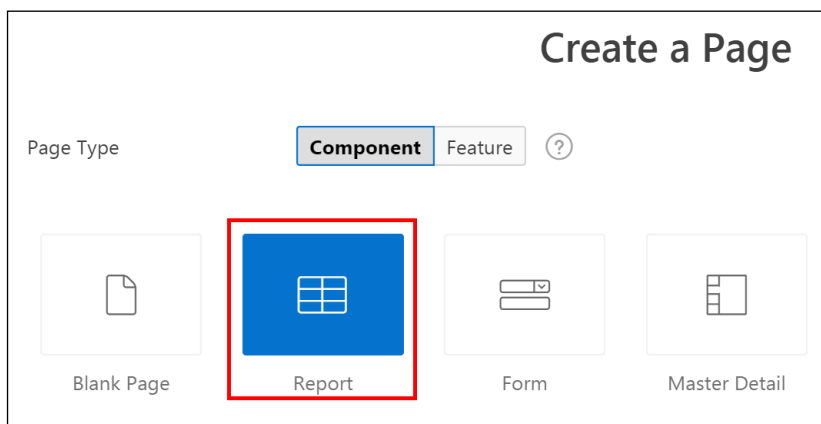
In this practice solution, steps are provided to create the interactive grid for the GMT application.

Steps

1. Create an interactive grid to display columns from the `OEHR_PRODUCT_INFORMATION` table.
 - a. Navigate to the GMT application home page and click **Create Page**.

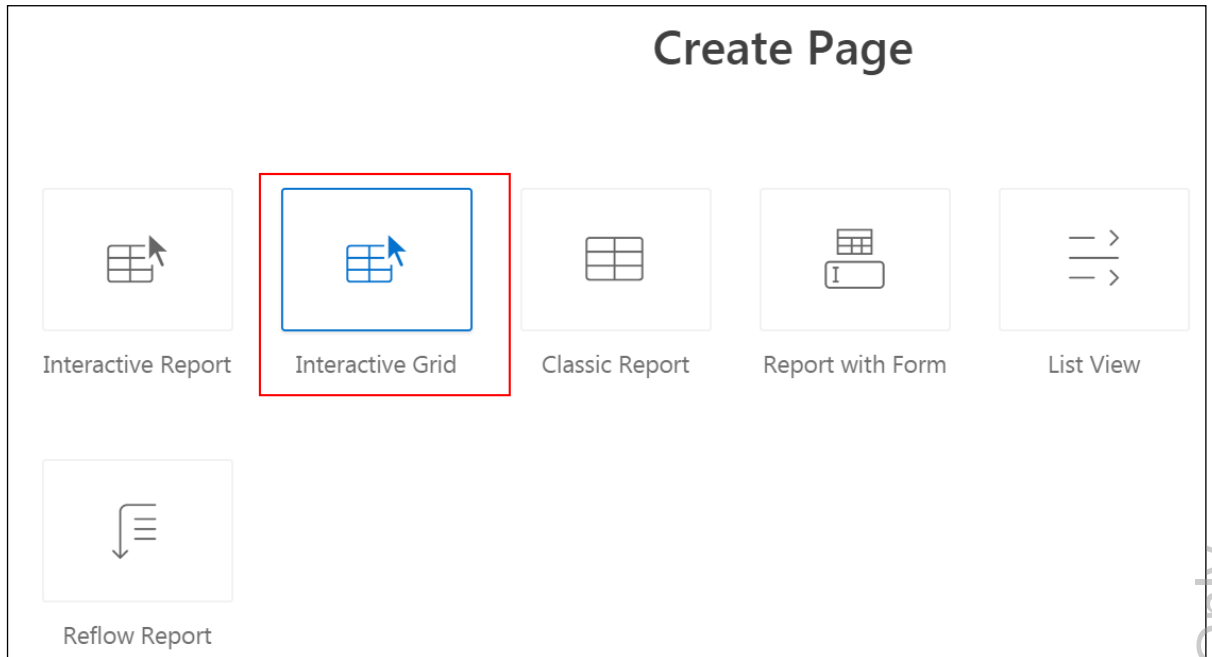


- b. Select **Report**.



For Instructor Use Only.
This document should not be distributed.

- c. Select **Interactive Grid**.



- d. For Page Number, enter 6. For Page Name, enter Product Information Details.

The screenshot shows the 'Create Interactive Grid' dialog box with the following fields:

- Type: **Interactive Grid**
- Page Number: 6
- Page Name: Product Information Details
- Page Mode: **Normal** (Modal Dialog)
- Breadcrumb: - do not use breadcrumbs on page -

- e. Click **Next**.

- f. For **Navigation Preference**, accept the default and click **Next**.

Create Interactive Grid

Navigation Preference ☒ Do not associate this page with a navigation menu entry ?

☐ Create a new navigation menu entry

☐ Identify an existing navigation menu entry for this page

- g. For **Source Type**, select **Table**.

- h. For **Table / View Name**, click the  icon to search for a table. In the Search dialog box, find and select OEHR_PRODUCT_INFORMATION (table) and click **Create**.

Create Interactive Grid

Report Source

Editing Enabled Yes No ?

* Source Type ☒ Table ☐ SQL Query ?

* Table / View Owner OEHR ?

* Table / View Name OEHR_PRODUCT_INFORMATION (ta  ?

Columns

* Select Columns ?

PRODUCT_ID (Number)
 PRODUCT_NAME (Varchar2)
 PRODUCT_DESCRIPTION (Varchar2)
 CATEGORY_ID (Number)
 WEIGHT_CLASS (Number)
 WARRANTY_PERIOD (Interval Year(2) To Month)
 SUPPLIER_ID (Number)
 PRODUCT_STATUS (Varchar2)

< Cancel Create

- i. The report is created. To view the report, click the **Save and Run Page**.

✓ Report page created successfully.

Save

Save and Run Page (Alt+F8)

Note: If prompted, enter your developer user credentials. The Interactive Grid is displayed.

Product Information Details							
Search: All Text Columns Go Actions							
Product Id	Product ...	Product ...	Category...	Weight ...	Warrant...	Supplier Id	Product ...
2257	HD 8GB /I	8GB capa...	13	1	+03-00	102061	orderable
3400	HD 8GB /SE	8GB capa...	13	2	+03-00	102063	orderable
3355	HD 8GB /SI	8GB SCSI ...	13	1	+02-00	102050	orderable
1772	HD 9.1GB...	Hard disk ...	13	3	+05-00	102070	orderable
2414	HD 9.1GB...	9.1 GB SC...	13	3	+05-00	102098	orderable

2. Explore the interactive grid interface. Interactive grids share many of the same features as interactive reports.
 - a. Sort by **Product Id**.
 - i. Place the cursor over the **Product Id** column heading.
 - ii. On the right side of the column name, two arrow icons have now appeared. Click the **up arrow** icon (**Sort Ascending**).

Product Information Details										
Search: All Text Columns Go Actions Reset										
Product Id	Product ...	Product ...	Categor...	Weight ...	Warrant...	Supplier Id	Product ...	List Price	Min Price	Catalog Url
205	Sort Ascending	Standard ...	16	1	+01-00	102099	planned	8	6	http://ww...
2211	Wrist Pad	A foam st...	16	1	+01-00	102072	orderable	4	3	http://ww...
2944	Wrist Pad...	Wrist Pad...	16	1	+01-00	102063	under de...	3	2	http://ww...
1742	CD-ROM ...	CD drive, ...	17	1	+00-06	102052	orderable	101	81	http://ww...
2402	CD-ROM ...	600 MB e...	17	2	+00-09	102052	orderable	127	113	http://ww...
2403	CD-ROM ...	600 MB i...	17	2	+01-00	102052	orderable	117	103	http://ww...
1761	CD-ROM ...	600 MB l...	17	2	+01-00	102052	under de...	134	119	http://ww...

The interactive grid after sorting the **Product ID** column.

Product Information Details										
Search: All Text Columns Go		Actions		Reset						
Product Id	Produ...	Produ...	Categ...	Weig...	Warra...	Suppli...	Produ...	List Pr...	Min Pr...	Catalog Url
1726	LCD M...	Liquid ...	11	3	+00-03	102067	under ...	259	208	http://www...
1729	Chemi...	Cleani...	39	2	+05-00	103094	ordera...	80	66	http://www...
1733	PS 220...	220V P...	19	2	+00-09	102080	ordera...	89	76	http://www...
1734	Cable ...	10 ft R...	19	2	+01-00	102055	ordera...	6	5	http://www...

- b. Hide the **Category Id** column.
 - i. Click the column heading **Category Id** to open the Column Heading menu.
 - ii. Click the Hide icon.

cription	Category	Weight Class	Warranty Period
color (wi...		3	+02-00
color, 8 ...		3	+02-00
black and...		5	+03-00
ck and w...		4	+03-00

The column is hidden from display.

- c. Apply a Control Break to organize the report by **Product Status**.
 - i. Click the column heading **Product Status** to open the Column Heading menu.
 - ii. Click the Control Break icon.

Supplier Id	Product Status	List Price
102090	☰	195
102094	Σ	349
102099	✱	699
102087	☰x	498
102071	Filter...	453
102071	under development	489
102051	obsolete	436
102069	orderable	399
	planned	
	under development	
	obsolete	

The Control Break is applied. A Control Break item also appears above the report ("Product Status").

Q

▼

Search: All Text Columns

Go

Actions ▼

▼

☒

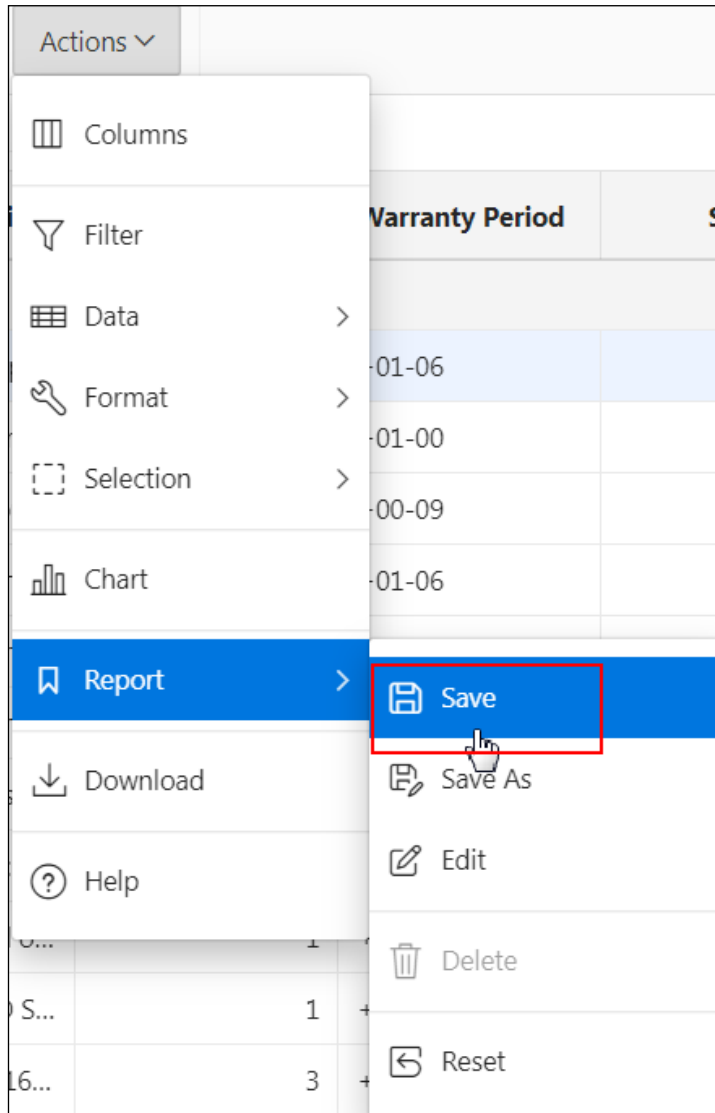
Product Status

×

Product Id	Product ...	Product ...	Weight C...	Warrant...	Supplier Id
▼ Product Status: obsolete					
2252	Monitor 2...	Monitor 2...	5	+01-06	102079
1782	Compact ...	400 chara...	4	+01-06	102088
3073	HD 6GB /I	6GB capa...	2	+05-00	102072
2253	HD 10GB ...	10GB cap...	2	+03-00	102069
▼ Product Status: orderable					
3060	Monitor 1...	CRT Moni...	4	+00-06	102081
2243	Monitor 1...	Monitor 1...	4	+00-06	102060
3057	Monitor 1...	CRT Moni...	4	+00-06	102055
3061	Monitor 1...	CRT Moni...	5	+00-09	102094
2245	Monitor 1...	Monitor 1...	5	+00-09	102053

- d. Save this as the new Primary report.

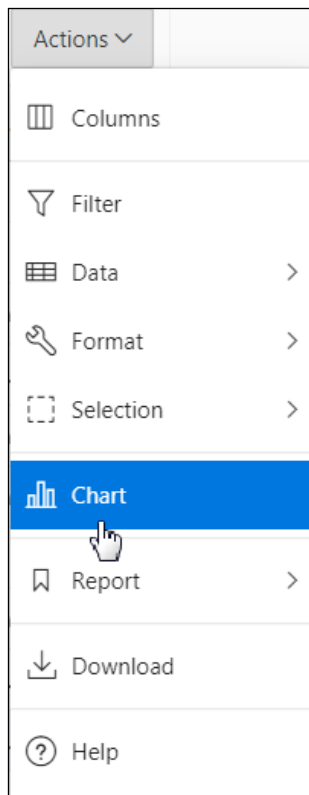
- i. Click **Actions > Report > Save**.



This is now the default report for all users.



- e. Create a chart to see the proportion of products by weight class.
- i. Click **Actions > Chart**.



- ii. For **Type**, select **Pie**.
- iii. Update the following fields:
- **Label:** Select **Weight Class**
 - **Value:** Select **Weight Class**
 - **Aggregation:** Select **Count**

Chart

Type

Area

Bar

Bubble

Donut

Funnel

Line

Line with Area

Pie

Polar

Radar

Range

Scatter

Stock

Label

Weight Class

Value

Weight Class

Aggregation

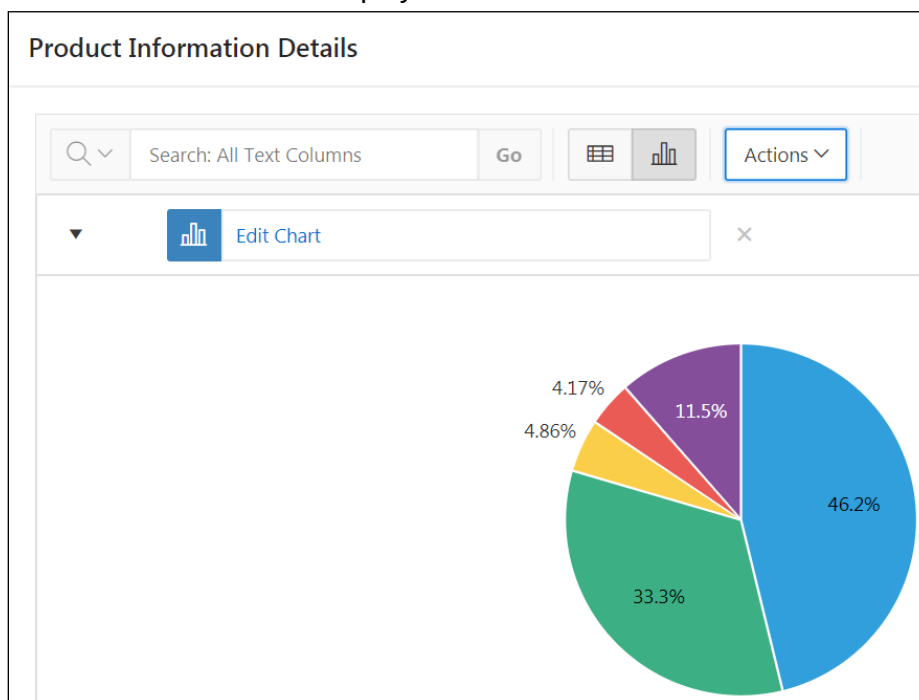
Count

Help

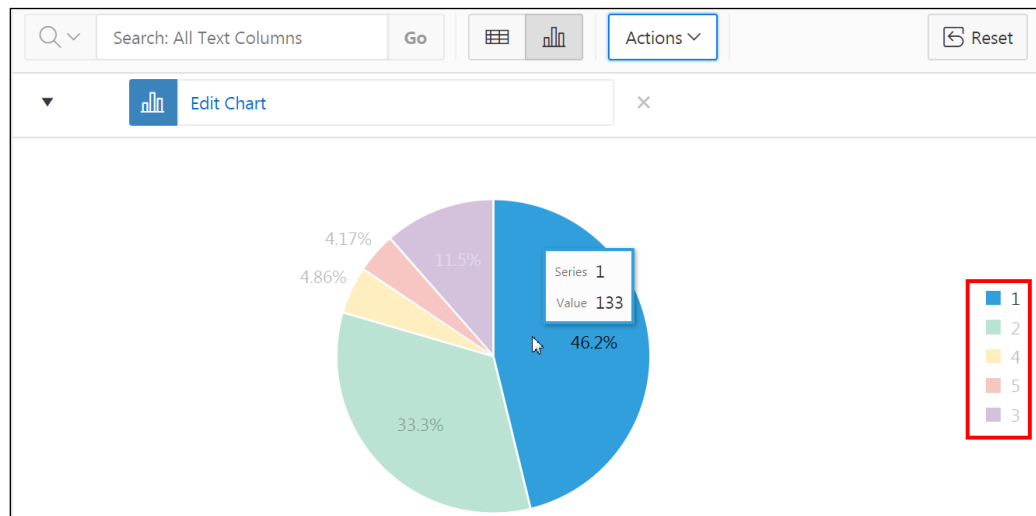
Cancel

Save

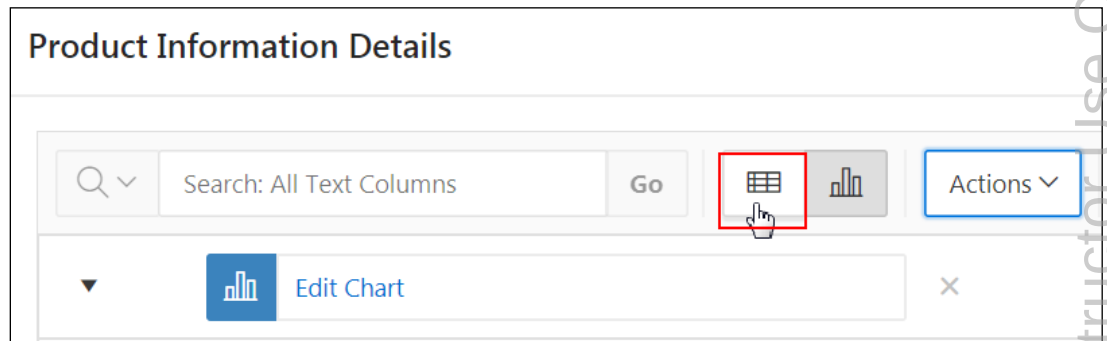
- iv. Click **Save**. The chart is displayed.



- v. Move the mouse over different parts of the chart and the legend to see how the interactive grid responds.



- vi. When finished, click the Grid icon  near the **Actions** menu button.



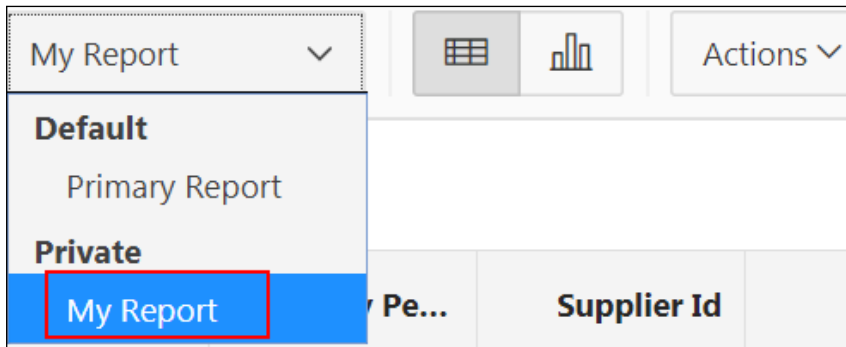
- f. Save this as a private report.

- i. Click **Actions > Report > Save As**.
- ii. For **Name**, enter `My Report` and click **Save**.

The screenshot shows a dialog box titled "Report - Save As". It has a close button (X) in the top right corner. Below the title bar, there is a "Type" dropdown menu currently showing "Private". Below that is a "Name" field containing the text "My Report". At the bottom right of the dialog, there are three buttons: "Help", "Cancel", and "Save". The "Save" button is highlighted with a red rectangular box.

You get the message *"Private report saved"*.

Note: A drop-down menu appears with a list of available reports for the interactive grid.



- iii. On the Runtime Developer toolbar at the bottom of the screen, click **Edit Page 6** to return to Page Designer.



Solution 6-2: Using and Manipulating the Interactive Grid

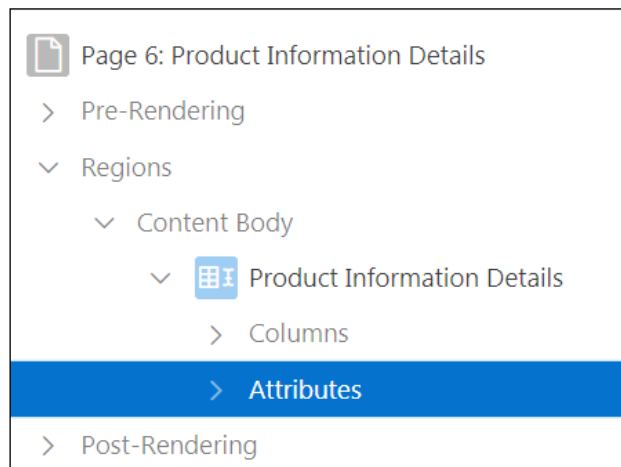
Overview

In this practice solution, steps are provided to convert the interactive grid for the GMT application to an editable interactive grid and customize it further in the application and in Page Designer.

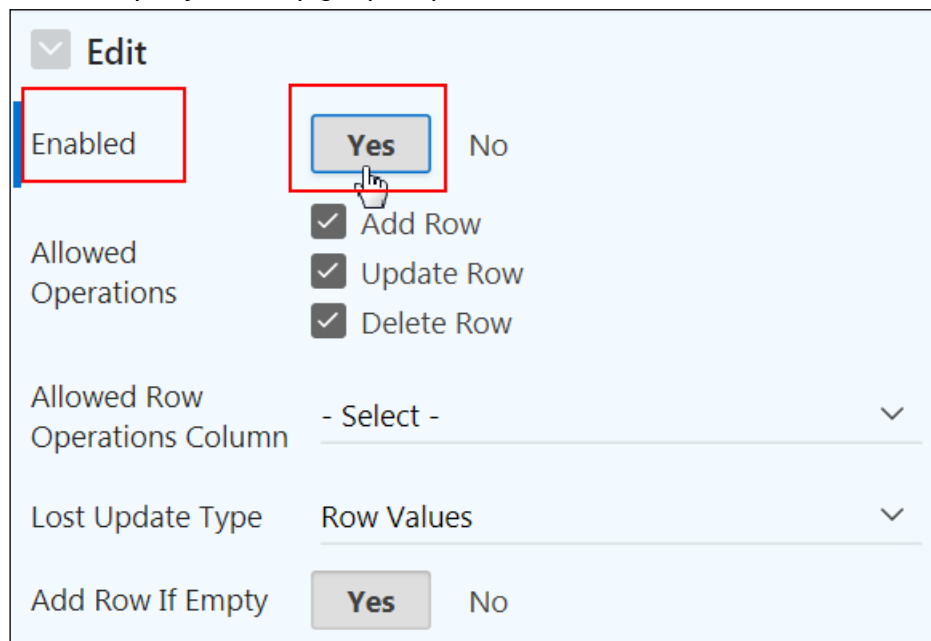
Steps

1. Convert the interactive grid to an editable interactive grid in Page Designer and run the page again.
 - a. Return to Page Designer and make the interactive grid region editable.

- On the Rendering tab (left pane), under **Regions > Content Body > Product Information Details**, select **Attributes**.

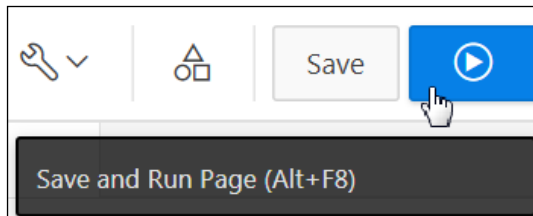


- In the Property Editor (right pane), under the Edit section, for **Enabled** select **Yes**.



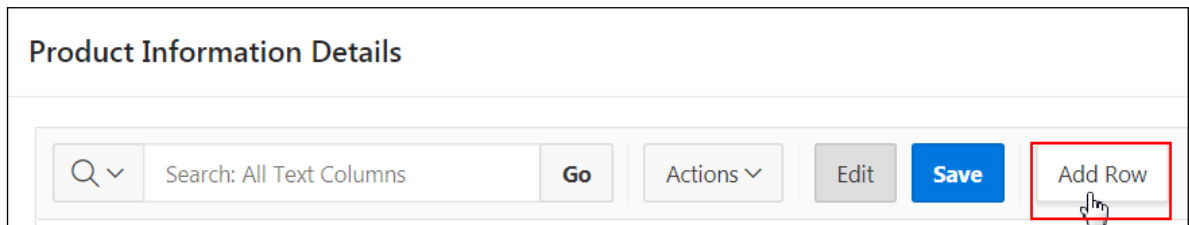
The interactive grid is now an editable interactive grid.

- Click the **Save and Run Page**.



Note: If prompted, enter your developer user credentials.

- b. Note the greatly expanded toolset at the top of the interactive grid. Editable interactive grids enable users to add to and modify the table directly without a form.
 - i. Near the Actions menu, click the **Add Row** button.

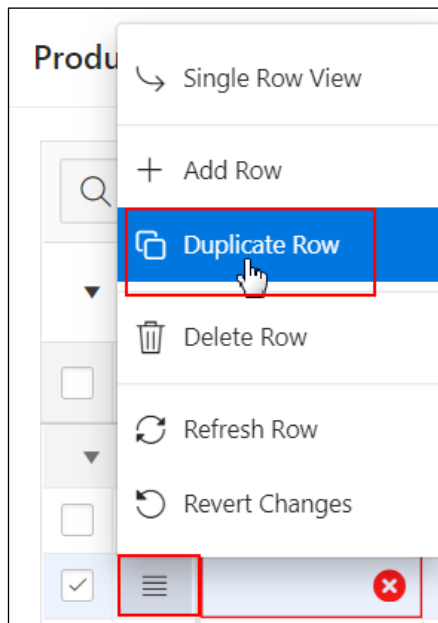


A blank row appears. The two left-most columns of an editable interactive grid are:

- The Selection column
- The Actions column

The top-most icons (located in the same row as the column headings) are for batch actions, such as Select All or Select None. The icons below the column heading row are for individual rows.

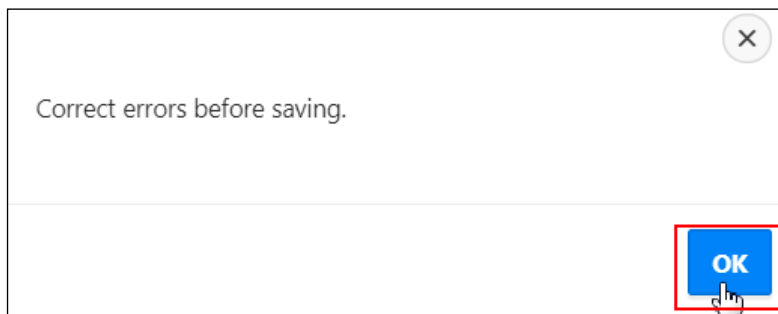
- ii. Click the **Row Actions** menu for the blank row.
- iii. Click **Duplicate Row**.



A second blank row appears. Note the error markings that appear.

☐		Product Id ↑	Product Name	Product Descr...	Weight Class
▼		Product Status: obsolete			
☐		1782	Compact 400/...	400 characters ...	4
☐					
☒					
☐		2049	MB - S300	PC type mother...	2
☐		2152	Router - DTMF4	DTMF 4 port ro...	3
☐		2252	Monitor 21/HR...	Monitor 21 inc...	5
☐		2253	HD 10GB @540...	10GB capacity ...	2

- iv. Try clicking the **Save** button at the top of the report. You see an error. The table does not change until the report saves, and Application Express prevents you from saving in the event of an error. Click **OK**.



- v. Select the check box for each blank row to select both, click the **Selection Actions** icon at the top of the column, and click **Delete Rows**.

Product Information Details

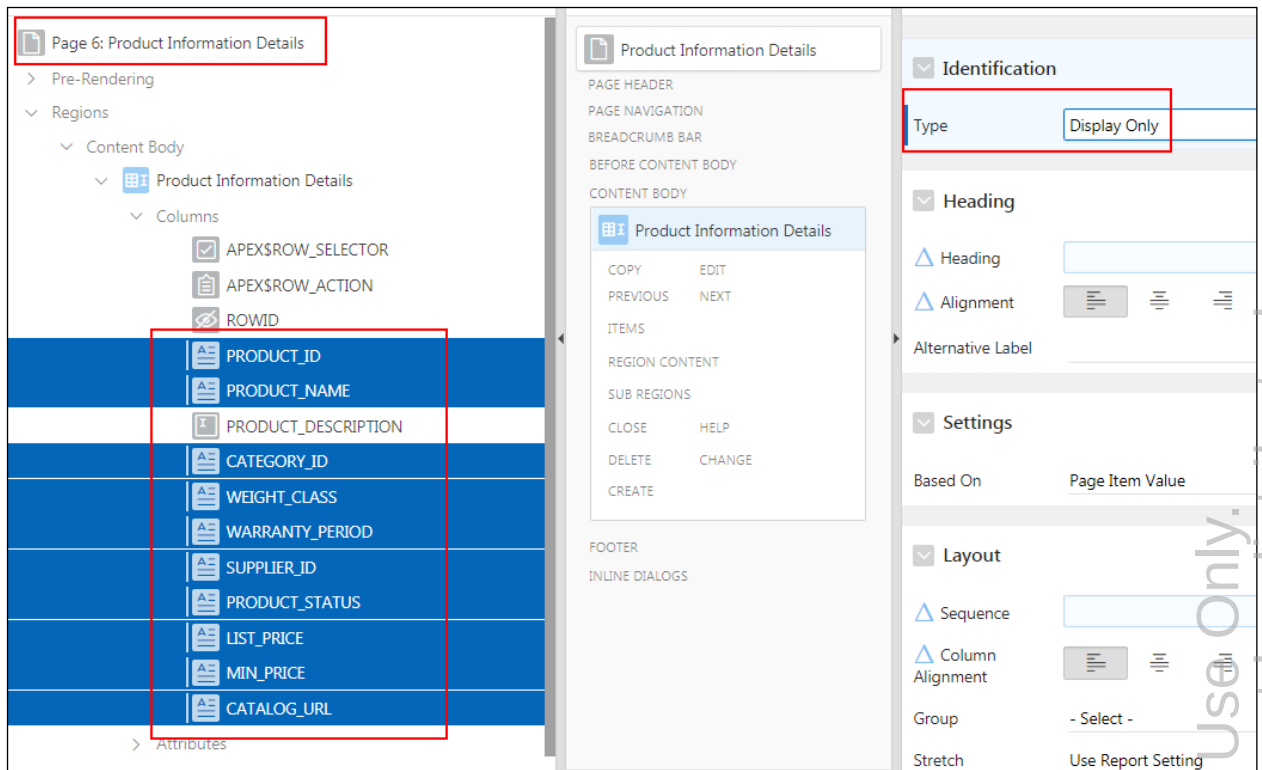
Search: All Text Columns Go My Report Actions Edit Save Add

☐	Product Id	Product Name	Product Des...	Weight Class	W...	S...	Product Status	Li...
☐			Inkjet Printer, ...	3	+0...	10...	orderable	195
☒								
☒								
☐			Inkjet printer, ...	3	+0...	10...	orderable	349
☐			Professional b...	5	+0...	10...	under development	699
☐			Standard blac...	4	+0...	10...	orderable	498
☐			10GB capacity...	2	+0...	10...	obsolete	453
☐			10GB Remova...	1	+0...	10...	obsolete	489
☐			10GB hard dis...	1	+0...	10...	obsolete	436

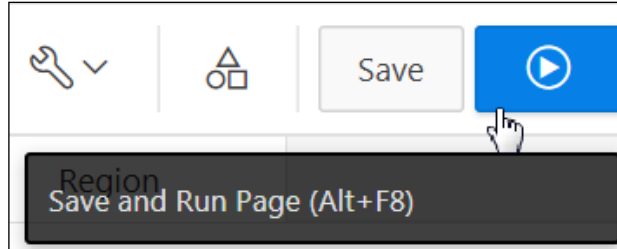
- c. You want to keep most of the information in this report intact, so set columns to **Display Only** to prevent users from updating. Lock columns in the report so that only Product Description is editable.
 - i. Click **Edit Page 6** on the Runtime Developer toolbar at the bottom of the page to return to Page Designer.
 - ii. On the Rendering tab, expand the **Columns** node.
 - iii. Ctrl-click the following items: (Do NOT select all columns.)

- PRODUCT_ID
- PRODUCT_NAME
- CATEGORY_ID
- WEIGHT_CLASS
- WARRANTY_PERIOD
- SUPPLIER_ID
- PRODUCT_STATUS
- LIST_PRICE
- MIN_PRICE
- CATALOG_URL

- iv. In the Property Editor (right pane), under the **Identification** section, for **Type** select **Display Only**.



- v. Click the **Save and Run Page**.



- d. The Product Description column is now the only editable column in the report. Try double-clicking within the Product Description column to see what happens and then click other entries in different columns too.

Product Information Details						
Q Search: All Text Columns Go My Report ▼ Table Icon Bar Chart Icon Actions ▼ Edit						
▼ ☑ Product Status ×						
☐	☰	Product Id ↑	Product Name	Product Descripti...	Weight Class	Warranty Period
Product Status: obsolete						
☑	☰	1782	Compact 400/DQ	400 characters per second high-speed draft printer. Dimensions (HxWxD): ▲ ▼		+01-06
☐	☰	2049	MB - S300			+01-00
☐	☰	2152	Router - DTMF4			+00-09
☐	☰	2252	Monitor 21/HR/M		5	+01-06
☐	☰	2253	HD 10GB @5400 /...		2	+03-00

- e. The interactive grid is complete. Click **Application 111** on the Runtime Developer toolbar to return to the Application home page.



**Practices for Lesson 7:
Working with Pages and
Regions**

Practices for Lesson 7: Overview

Overview

In these practices, you work with pages and regions and update the Global Page for the application.

For Instructor Use Only.
This document should not be distributed.

Practice 7-1: Creating and Modifying Pages and Regions

Overview

In this practice, you create a report region, create a subregion, and modify the attributes of the report. You create a sidebar region and add footer to the Customer Details page.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a blank page called **Products** with a region called **Products List**. Set the page number as 7. Create a new tab called **Products** by using the existing tab set for this page.
2. Create a new report region called **Products List** on the Products page. Use a Classic Report and display all the rows in the `OEHR_PRODUCTS_INFORMATION` table. Run the report after it is created.
3. Modify the report so that it is a subregion in the Products region and displays only the Product ID, Product Name, Category ID, Product Status, and List Price columns in the report. Also, you want to display only 10 rows at a time and show the pagination in a select list. After you have performed these tasks, run the report.
4. Create a **Hint** sidebar region on the right side of the Customer Details page that includes the following HTML code (located in `/home/oracle/labs/labs/lab_07_01_01.txt`):

```
Use this page to enter and <br> maintain customer information.
```

5. Add a footer to the Customer Details region on the Customer Details page so that the following message (located in `/home/oracle/labs/labs/lab_07_01_02.txt`) appears at the bottom of the region when a user enters this page:

```
<i>The record created or modified in this form is reflected in  
the Customer Report.</i>
```

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 7-1**.

If you haven't completed **Practice 6-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_06_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

For Instructor Use Only.
This document should not be distributed.

Practice 7-2: Working with Global Pages

Overview

In this practice, you view the Global page in your application and then add a Static HTML subregion to the Global Desktop Page.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Navigate to Global desktop page in the GlobalMart Management Tool application and review it in Application Properties.
2. Add a Static HTML region to the Global Desktop page. Call the region as **In the News** on the Global page with the following code (located in `/home/oracle/labs/labs/lab_07_02.txt`):

```
<html><body>
<p><strong>News and Events</strong></p>
<p>Visit us at <a href="http://www.oracle.com">
www.oracle.com</a></p>
</body></html>
```

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 7-2**.

If you haven't completed **Practice 7-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.

- d. Browse and open the `sol_07_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

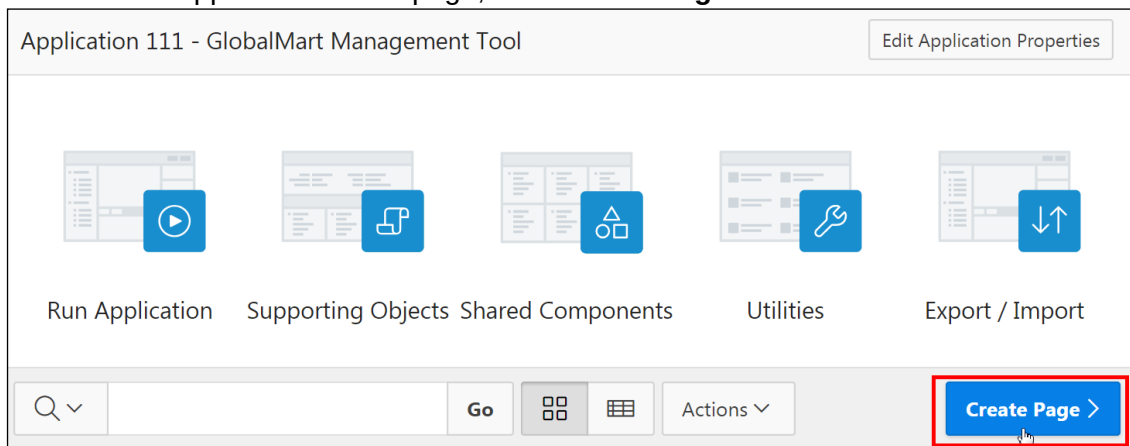
Solution 7-1: Creating and Modifying Pages and Regions

Overview

In this practice solution, steps are provided for creating and modifying pages and regions in the GMT application.

Steps

1. Create a blank page called **Products** with a region called **Products List**. Set the page number as 7. Create a new tab called **Products** by using the existing tab set for this page.
 - a. On the GMT Application Home page, click **Create Page**.



For Instructor Use Only.
This document should not be distributed.

- b. Select **Blank Page**.

Create a Page

Page Type: **Component** Feature ?

Blank Page Report Form Master Detail

Dashboard Calendar Tree Wizard

- c. For Page Attributes:

- **Page Number:** Enter 7.
- **Name:** Enter `Products`.
- **Optional Static Content Regions:** Expand and enter `Products` for **Region 1**.

Create a Blank Page

Page Attributes

* Page Number: 7 ?

* Name: Products ?

* Page Mode: **Normal** Modal Dialog Non-Modal Dialog ?

Page Group: - Select Page Group - ?

Breadcrumb: - don't use breadcrumbs on page - ?

☒ **Optional Static Content Regions**

Region 1: Products ?

2: ?

- d. Click **Next >**.

e. For Navigation Menu:

- **Navigation Preference:** Select **Create a new navigation menu entry**.
- **New Navigation Menu Entry:** Ensure that the text field is set to **Products**.
- **Parent Navigation Menu Entry:** Ensure that **- No parent selected -** is selected.

The screenshot shows the 'Create a Blank Page' wizard with the 'Navigation Menu' step highlighted by a blue dot on the progress bar. The 'Navigation Preference' section has three radio buttons: 'Do not associate this page with a navigation menu entry' (unselected), 'Create a new navigation menu entry' (selected and highlighted with a red box), and 'Identify an existing navigation menu entry for this page' (unselected). Below this, a red box highlights the 'New Navigation Menu Entry' and 'Parent Navigation Menu Entry' labels. The 'New Navigation Menu Entry' text field contains the word 'Products'. The 'Parent Navigation Menu Entry' dropdown menu is open, showing options: '- No parent selected -' (highlighted), 'Home', and 'Customers'. A vertical watermark on the right side of the image reads 'For Instructor Use Only. This document should not be distributed.'

f. Click **Next >**.

- g. Click **Finish**.

Create a Blank Page

✓

✓

Confirm

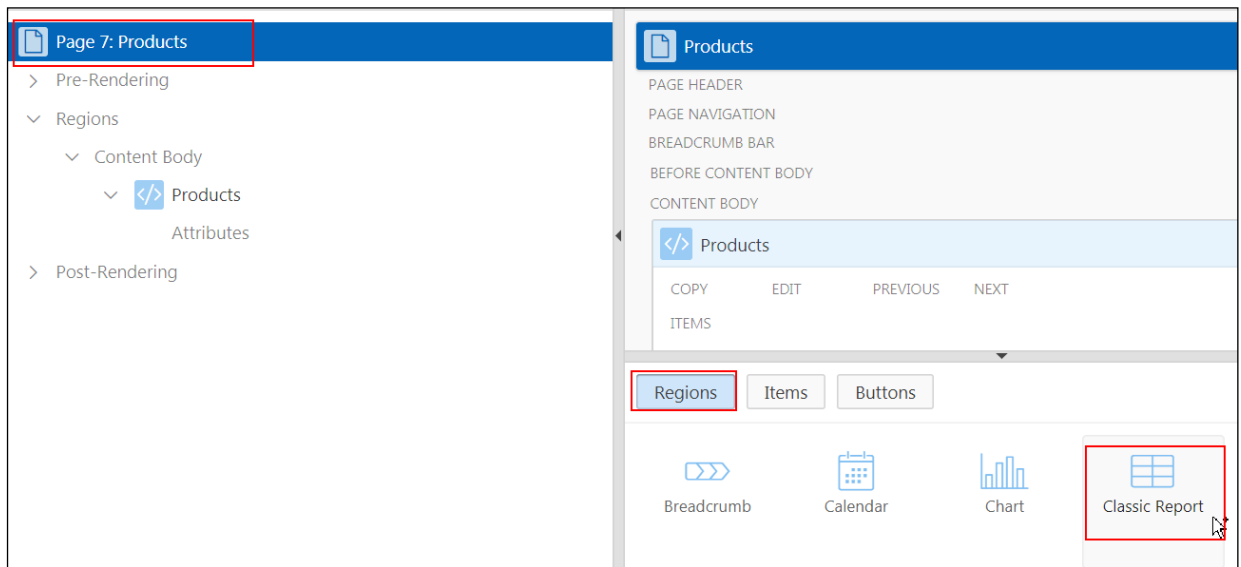
You have requested to create a page with the following attributes. Please confirm your selections.

Application	111
Page	7
Page Name	Products
Page Title	Products

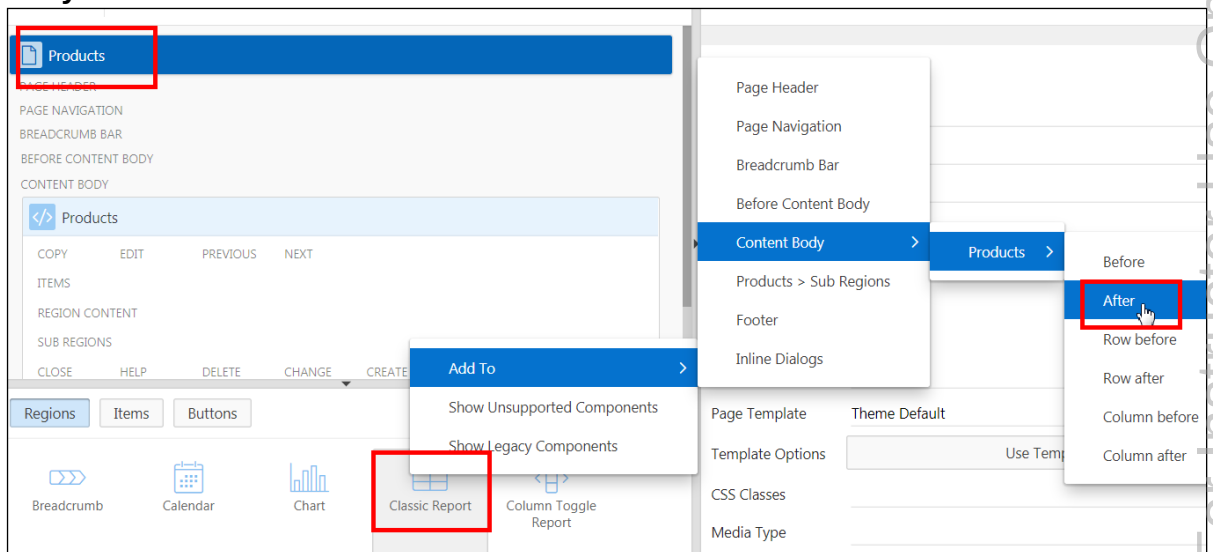
< Cancel

Finish

2. Create a new report region called **Products List** on the Products page. Use a Classic Report and display all the rows in the `OEHR_PRODUCTS_INFORMATION` table. Run the report after it is created.
 - a. In the Gallery at the bottom of the central pane in Page Designer, select the **Regions** tab and find **Classic Report**.



- b. Right-click **Classic Report** to display the context menu and select **Add To > Content Body > Products > After**.

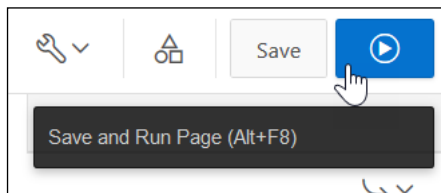


- c. Edit the appropriate attributes in the Property Editor (the right pane):
- **Identification > Title:** Enter `Products List`.
 - **Source > Type:** Select **Table/View** (if not selected by default).

- **Source > Table Name:** Select `OEHR_PRODUCT_INFORMATION` by clicking the  icon.

▼ Identification	
Title	Products List
Type	Classic Report ▼ 
▼ Source	
Location	Local Database ▼
Type	Table / View ▼
Table Owner	Parsing Schema ▼
Table Name	OEHR_PRODUCT_INFORMATION 
Include ROWID Column	Yes No

- Click **Save**.
- Test your page by clicking **Save and Run Page**.



- f. The report is displayed. (**Note:** If you are prompted to sign in, the Home page is displayed. Click the **Products** tab to view the report.) Click the **Edit Page 7** link on the Runtime Developer toolbar.

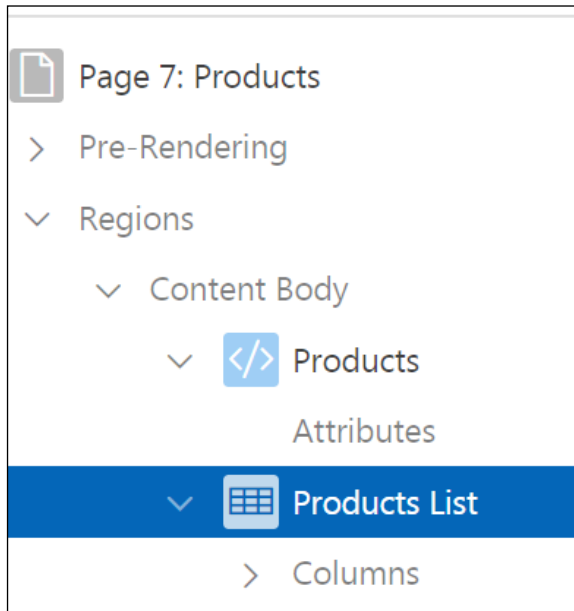
Products

Products List

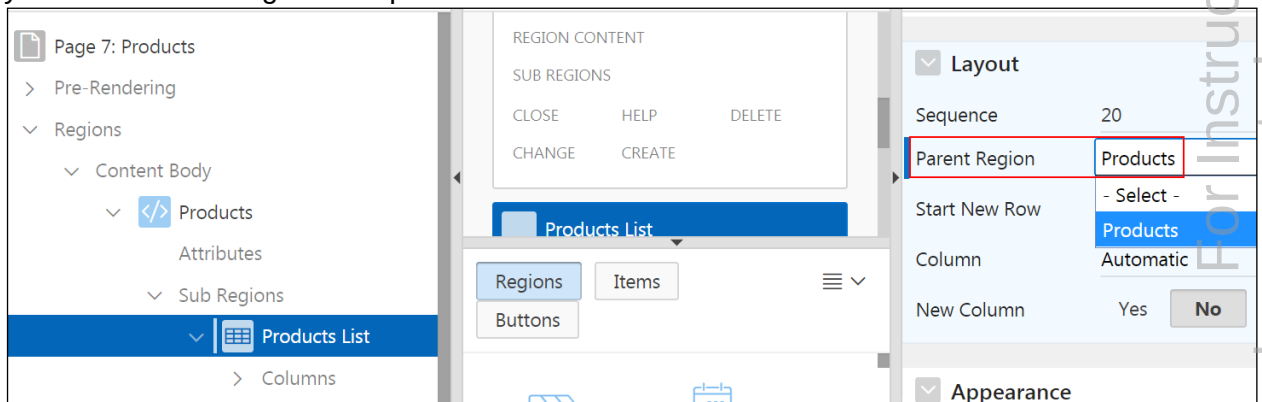
Product Id ↑≡	Product Name	Product Description	Category Id	Weight Class	Warranty Period
1726	LCD Monitor 11/PM	Liquid Cristal Display 11 inch passive monitor. The virtually-flat, high-resolution screen delivers outstanding image quality with reduced glare.	11	3	+000000000-03
1729	Chemicals - RCP	Cleaning Chemicals - 3500 roller clean pads	39	2	+000000005-00
1733	PS 220V /UK	220V Power supply type - United Kingdom	19	2	+000000000-09
1734	Cable RS232 10/AM	10 ft RS232 cable with M/M and 9M/25M adapters	19	2	+000000001-00

Home Application 111 Edit Page 7 Session View Debug Debug Page Info Quick Edit Theme Roller

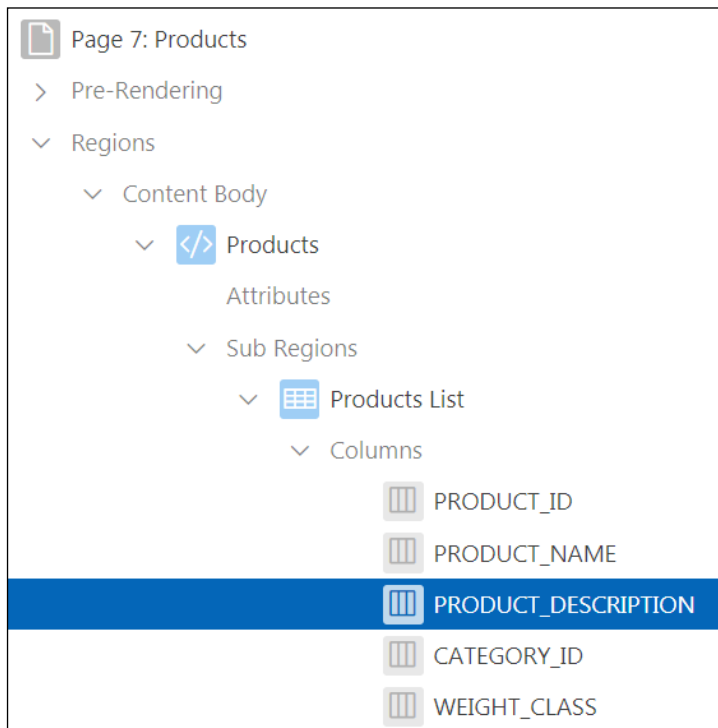
3. Modify the **Products List** report so that it becomes a subregion under the Products region and only displays Product ID, Product Name, Category ID, Product Status, and List Price columns in the report. Also, you want to display only 10 rows at a time and show the pagination in a select list. After you have performed these tasks, run the report.
 - a. On the Rendering tab, expand **Regions > Content Body** and select **Product List**.



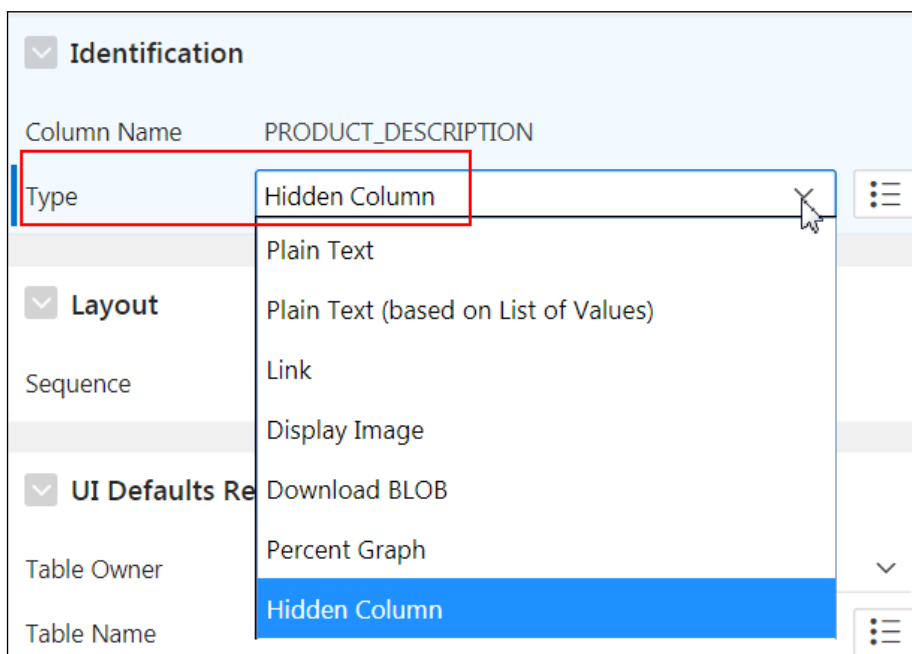
- b. In the Property Editor (the right pane), **Layout > Parent Region**, select **Products**. Notice that the Products List region becomes a subregion of Products. Alternatively, you can use the drag-and-drop feature.



- c. On the Rendering tab, expand **Regions > Content Body > Products > Sub Regions > Products List > Columns >** and select `PRODUCT_DESCRIPTION`.

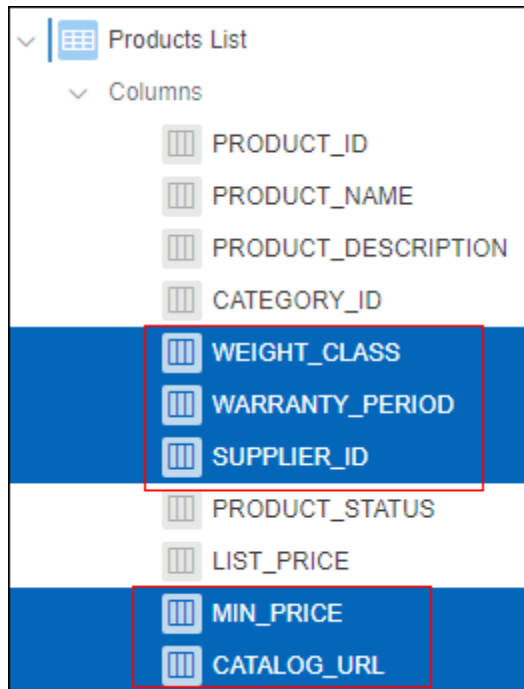


- d. In the Property Editor (the right pane), **Identification > Type**, select **Hidden Column**.

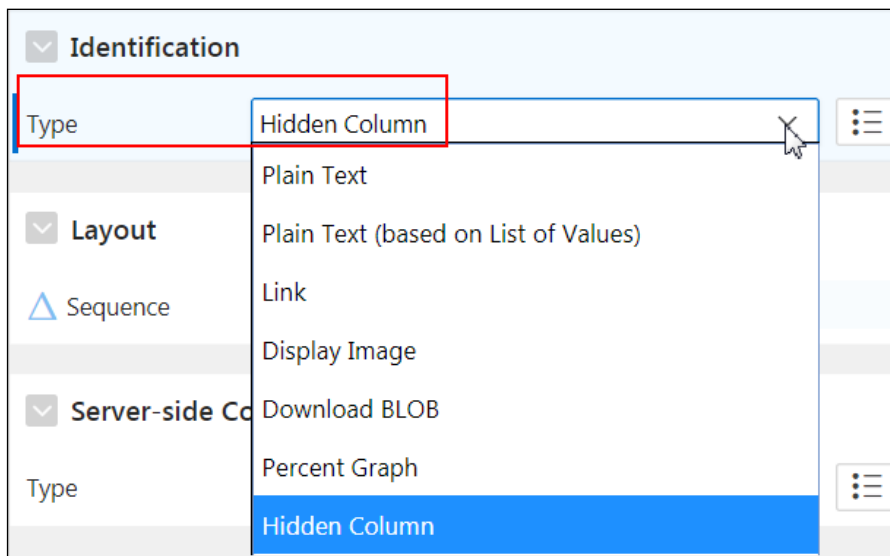


e. On the Rendering tab, press Ctrl and select the following columns:

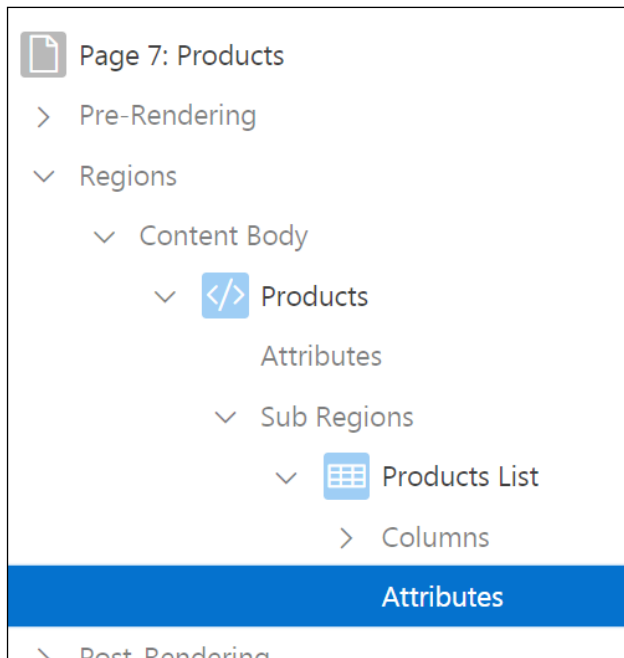
- WEIGHT_CLASS
- WARRANTY_PERIOD
- SUPPLIER_ID
- MIN_PRICE
- CATALOG_URL



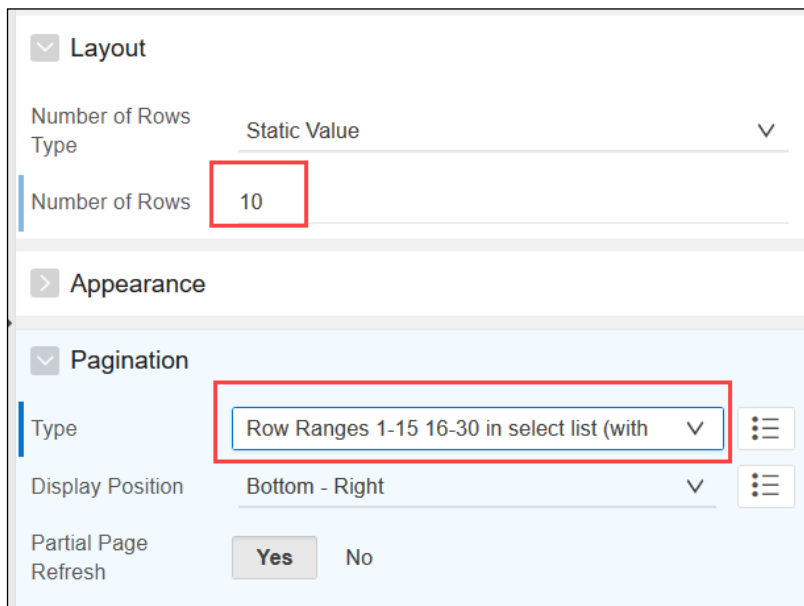
f. In the Property Editor (the right pane), **Identification** > **Type**, select **Hidden Column**.



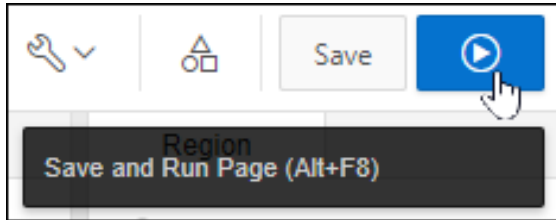
- g. On the Rendering tab, expand **Regions > Content Body > Products > Sub Regions > Products List >** and select **Attributes**.



- h. In the Property Editor (the right pane):
- 1) **Layout > Number of Rows:** Enter 10.
 - 2) **Pagination > Type:** Select **Row Ranges 1-15 16-30 in select list (with pagination)** from the drop-down list.



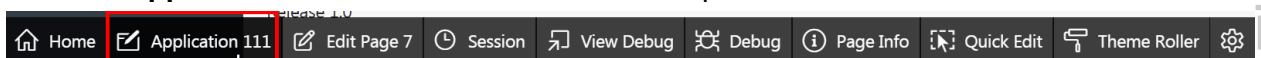
- i. Click **Save** and then **Save and Run Page**.



- j. Notice that the **Products List** region is now displayed in the Products parent region, and only the columns that you specified appear. Also, notice the pagination is now in a select list.

Products				
Products List				
Product Id ↑	Product Name	Category Id	Product Status	List Price
1726	LCD Monitor 11/PM	11	under development	259
1729	Chemicals - RCP	39	orderable	80
1733	PS 220V /UK	19	orderable	89
1734	Cable RS232 10/AM	19	orderable	6
1737	Cable SCSI 10/FW/ADS	19	orderable	8
1738	PS 110V /US	19	orderable	86
1739	SDRAM - 128 MB	14	orderable	299
1740	TD 12GB/DAT	17	orderable	134
1742	CD-ROM 500/16x	17	orderable	101
1743	HD 18.2GB @10000 /E	13	planned	800
row(s) 1 - 10 of 292 ▼				Next ►

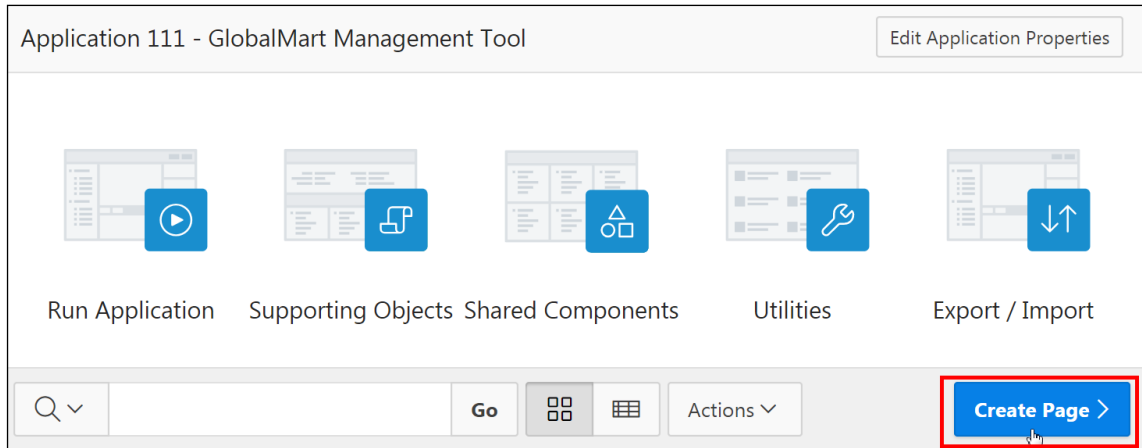
Click the **Application 111** link on the Runtime Developer toolbar.



4. Create a page (**Page 8 – Update Customer**) and modify it to create a **Hint** sidebar region on the right side of the Update Customer page that includes the following HTML code (located in /home/oracle/labs/labs/lab_07_01_01.txt):

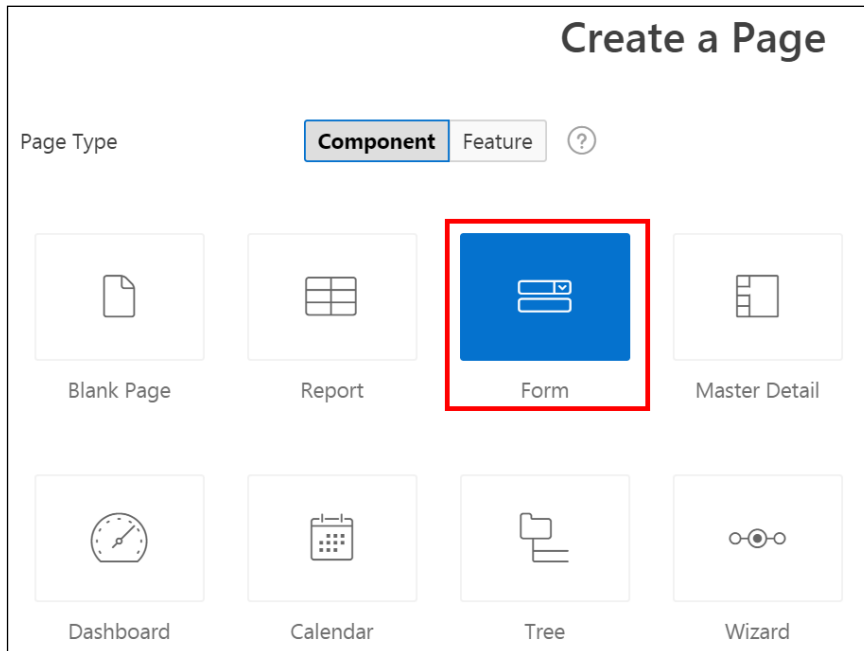
```
Use this page to enter and <br> maintain customer information.
```

- a. You will first create a Form Page. Go to your **Application 111** and click **Create Page**.

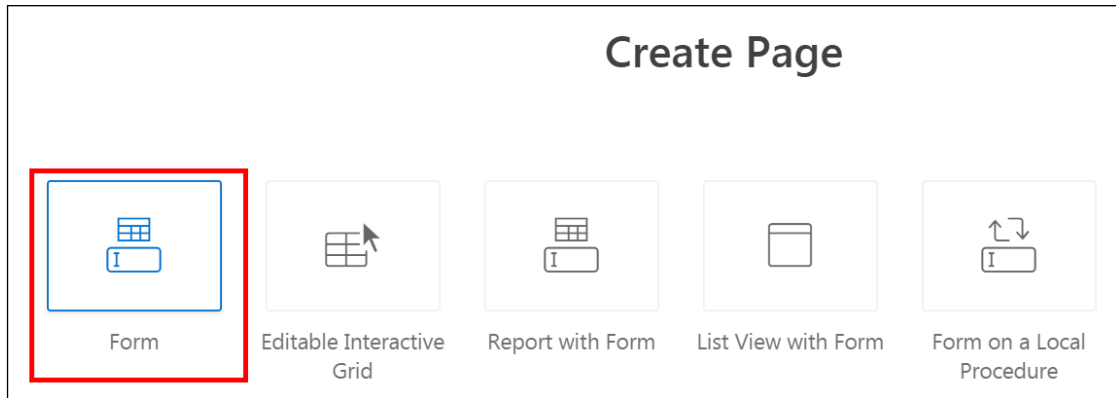


- b. Click **Form** and click **Next >**.

Note: In this example, we create a simple Form and add a **Hint** sidebar region. In the lesson titled “Managing Forms,” you will be learning more about creating different types of Forms.



c. Click **Form**.



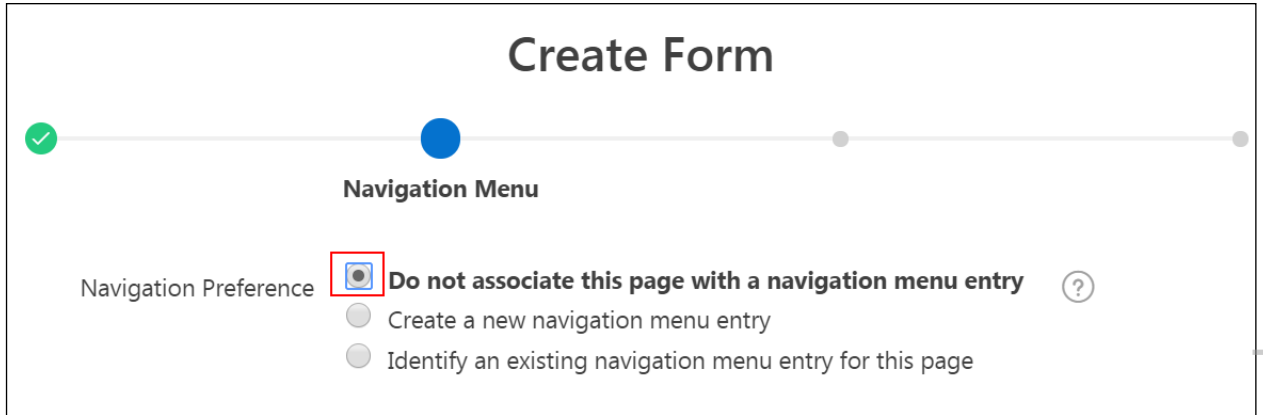
d. In the Create Form on a Table page, enter the following details:

- **Page Number:** Enter 8
- **Page Name:** Update Customers
- **Branch here on Submit:** Select 4
- **Cancel and Go to Page:** Select 4
- Click **Next>**.

The 'Create Form' dialog box shows the following details:

- Page Attributes**
- Page Number:** 8
- Page Name:** Update Customers
- Page Mode:** Normal
- Page Group:** - Select Page Group -
- Branch Here on Submit:** 4
- Cancel and Go To Page:** 4
- Breadcrumb:** - do not add breadcrumb region to page -

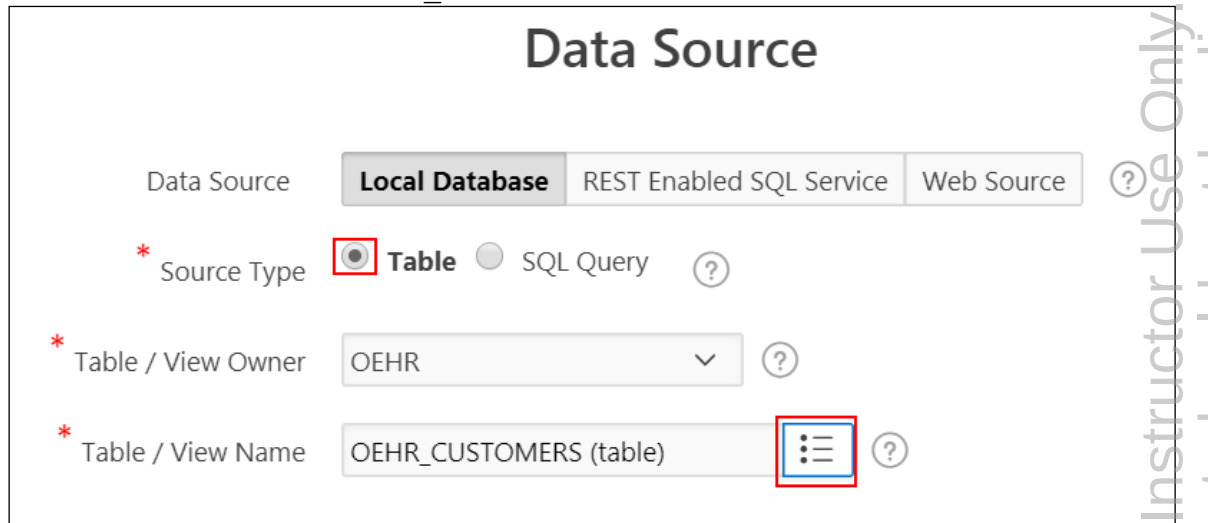
- e. Keep the default navigation settings and click **Next>**.



The screenshot shows the 'Create Form' wizard, specifically the 'Navigation Menu' step. A progress bar at the top has four dots; the first is green with a checkmark, and the second is blue. Below the title, there is a 'Navigation Preference' section. It features a radio button icon (a square with a circle inside) that is highlighted with a red box. To its right is the text 'Do not associate this page with a navigation menu entry' followed by a help icon (?). Below this are two other radio buttons: 'Create a new navigation menu entry' and 'Identify an existing navigation menu entry for this page'.

- f. To define the table source, select the following:

- **Source Type:** Select **Table**
- **Table/View Name:** Select `OEHR_CUSTOMERS` (table)

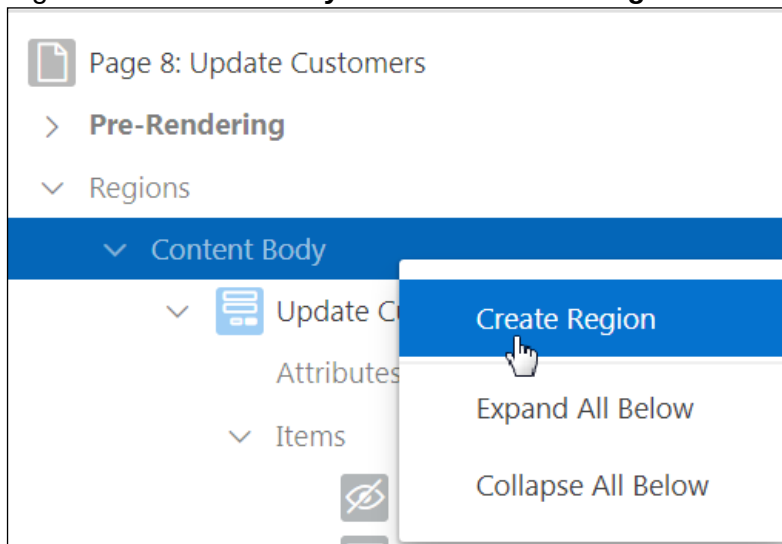


The screenshot shows the 'Data Source' configuration screen. At the top, there are three tabs: 'Local Database' (selected), 'REST Enabled SQL Service', and 'Web Source'. Below the tabs, the 'Source Type' is set to 'Table' (indicated by a radio button icon highlighted with a red box). The 'Table / View Owner' is set to 'OEHR' in a dropdown menu. The 'Table / View Name' is set to 'OEHR_CUSTOMERS (table)' in a text field, with a menu icon (three horizontal lines) highlighted by a red box. There are help icons (?) next to the 'Source Type', 'Table / View Owner', and 'Table / View Name' fields.

- Click **Next>**.

- Select **Select Primary Key Column(s)** for Primary Key Type. For **Primary Key Column**, select **CUSTOMER_ID (Number)**.

- Click **Create**.
- g. The Update Customers form page is created and opens in Page Designer view. In the bottom of the central pane of Page Designer, click the **Regions** tab.
- h. Right-click **Content Body** and select **Create Region**.



- i. In the Property Editor (the right pane):
 - **Identification > Title:** Enter **Hint**.
 - **Identification > Type:** Select **Static Content** (if not selected by default).
 - **Source > Text:** Enter the following (located in `/home/oracle/labs/labs/lab_07_01_01.txt`) in the **Text** field.

Use this page to enter and
 maintain customer information.

Identification

Title

Hint

Type

Static Content

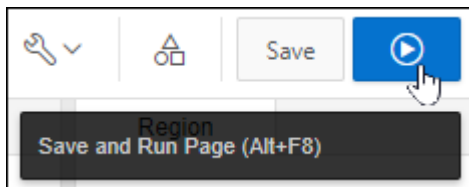
Source

Text

Use this page to enter and

 maintain customer
information.

- j. Click **Save** and then **Save and Run Page**.



- k. Note that although the Hint region is created, it is not displayed as a sidebar region on the right side of the page. To display it as a side bar region, you need to change the page template so that a sidebar is provided. Click the **Edit Page 8** link on the Runtime Developer toolbar.

Customers

Products

Cust Email

Account Mgr Id

Cancel

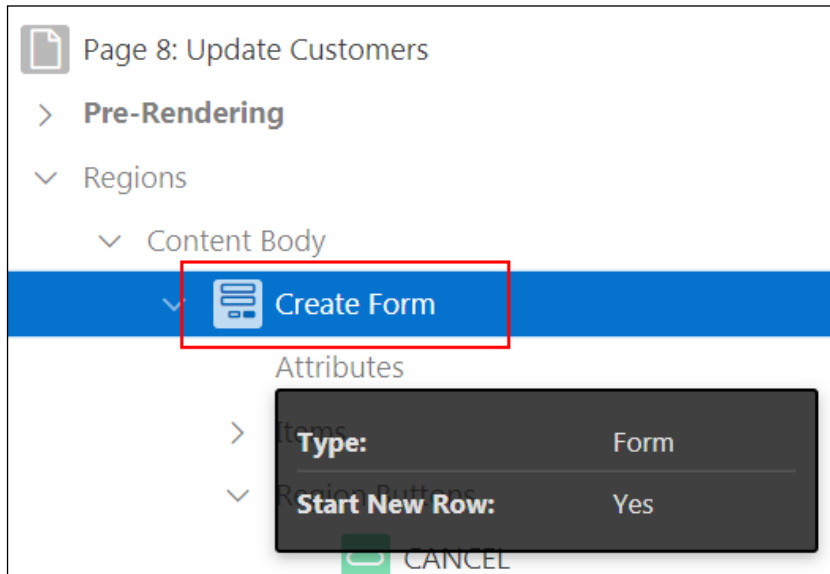
Create

Hint

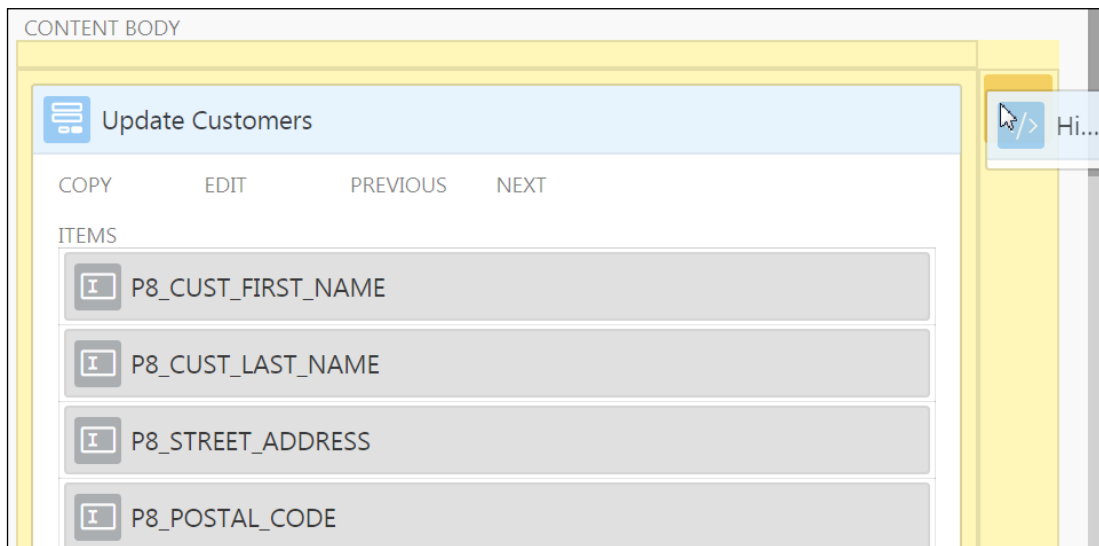
Use this page to enter and
maintain customer information.

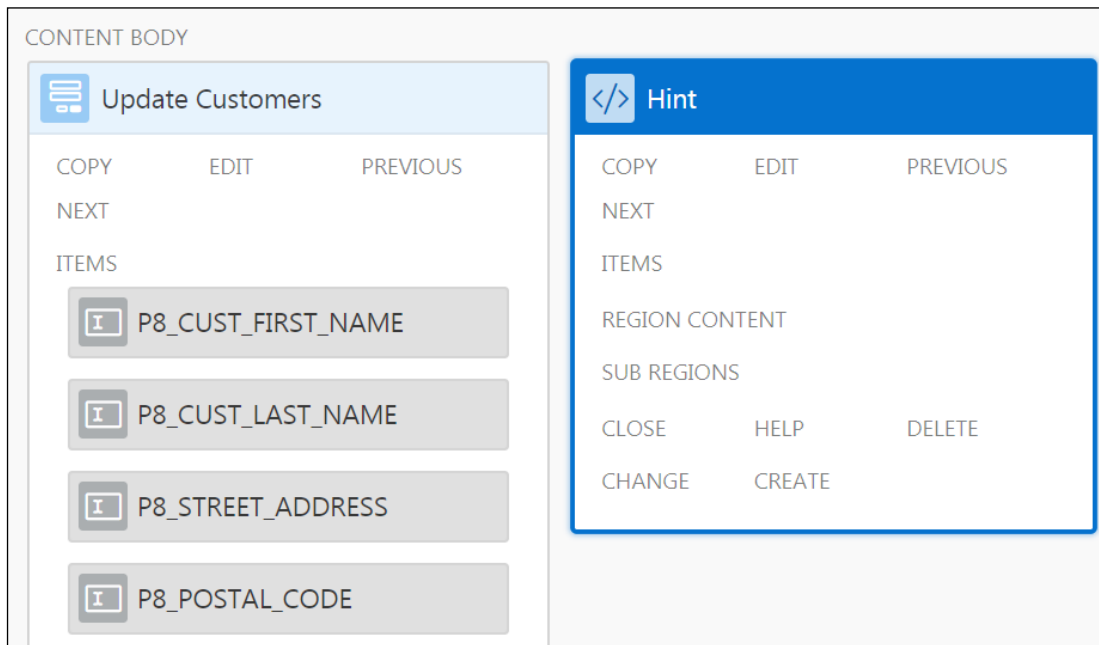
Home Application 111 Edit Page 8 Session View Debug Debug Page Info Quick Edit Theme Roller

- l. Select **Create Form**, and then in the Property Editor, **Identification > Title**: Enter Update Customers.

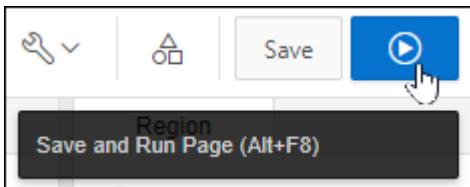


- m. Drag the **Hint** region to the right of the **Update Customers** region.

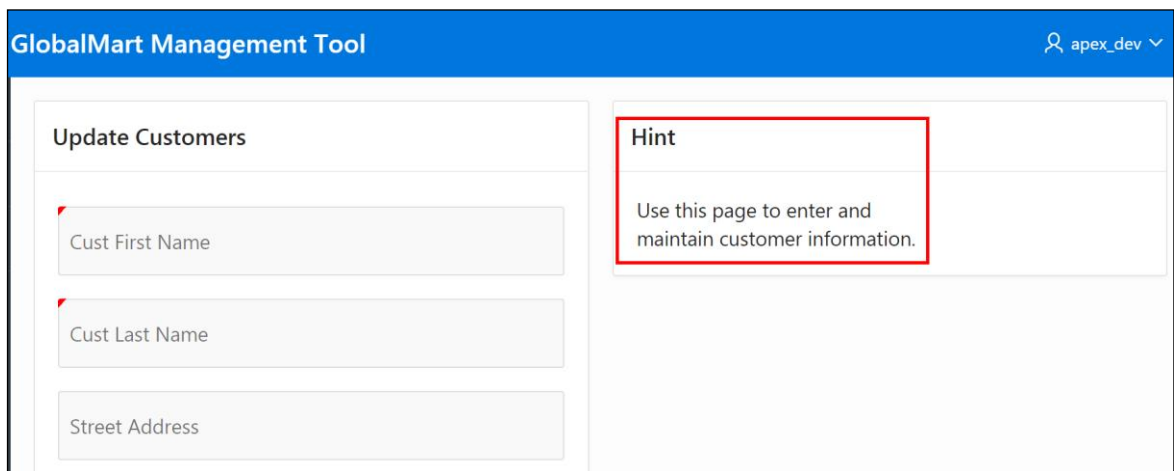




- n. Run the page again and you see the Hint region on the right. Click **Save**. Run the page by clicking **Save and Run Page**.



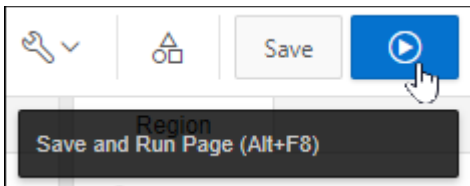
- o. Modify the Hint region such that it takes only two column span. Click the **Edit Page 8** link on the Runtime Developer toolbar.



- p. Ensure that the **Hint** region is selected either on the Rendering tab or on the Layout tab (Central pane). In the Property Editor, under **Layout > Column Span**, select **2**.

The screenshot shows the 'Layout' tab in the Property Editor. The 'Column Span' property is highlighted with a red box and set to '2'. Other properties visible include 'Sequence' (20), 'Parent Region' (- Select -), 'Position' (Content Body), 'Start New Row' (Yes/No), 'Column' (Automatic), 'New Column' (Yes/No), and 'Column CSS Classes'.

- q. Click **Save** and then **Save and Run Page**.



- r. Note that the Hint region now displays on the right side of the page, occupying lesser space than before. Click the **Edit Page 8** link on the Runtime Developer toolbar.

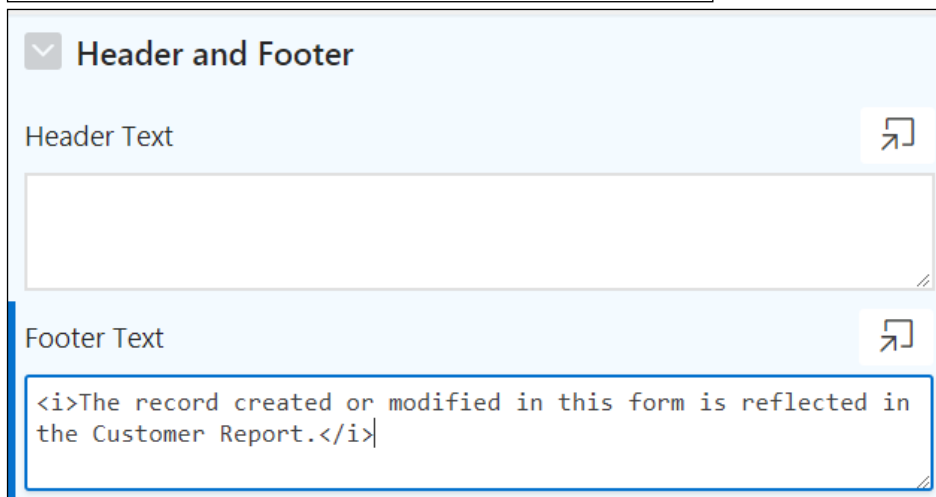
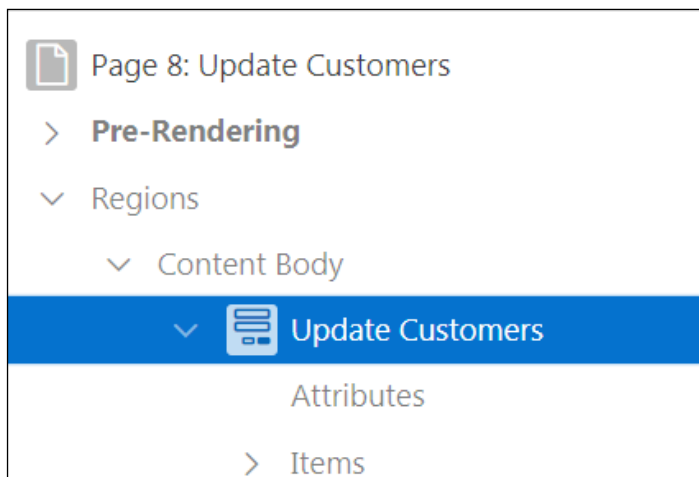
The screenshot shows the 'Update Customers' page in the Runtime Developer. The 'Hint' region is visible on the right side, displaying the text: 'Use this page to enter and maintain customer information.' The 'Edit Page 8' link in the Runtime Developer toolbar is highlighted with a red box.

5. Add a footer to the Update Customer region on the Update Customer page so that the following message (located in /home/oracle/labs/labs/lab_07_01_02.txt) appears at the bottom of the region when a user enters this page:

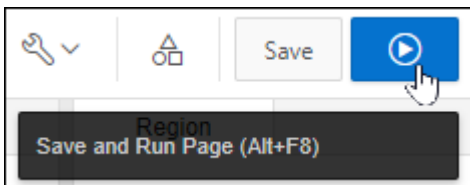
```
<i>The record created or modified in this form is reflected in  
the Customer Report.</i>
```

- On the Rendering tab, select **Regions > Content Body > Update Customers**.
- In the Property Editor (the right pane): Copy the footer text (located in the /home/oracle/labs/labs/lab_07_01_02.txt) and paste it into the **Header Footer > Footer Text**.

```
<i>The record created or modified in this form is reflected in  
the Customer Report.</i>
```



- Click **Save** and then **Save and Run Page**.



- d. The footer text is displayed. Click the **Application 111** link on the Runtime Developer toolbar.

Credit Limit

Cust Email

Account Mgr Id

The record created or modified in this form is reflected in the Customer Report.

Cancel Create

Home Application 111 Edit Page 8 Session View Debug Debug Page Info Quick Edit Theme Roller

For Instructor Use Only
This document should not be distributed.

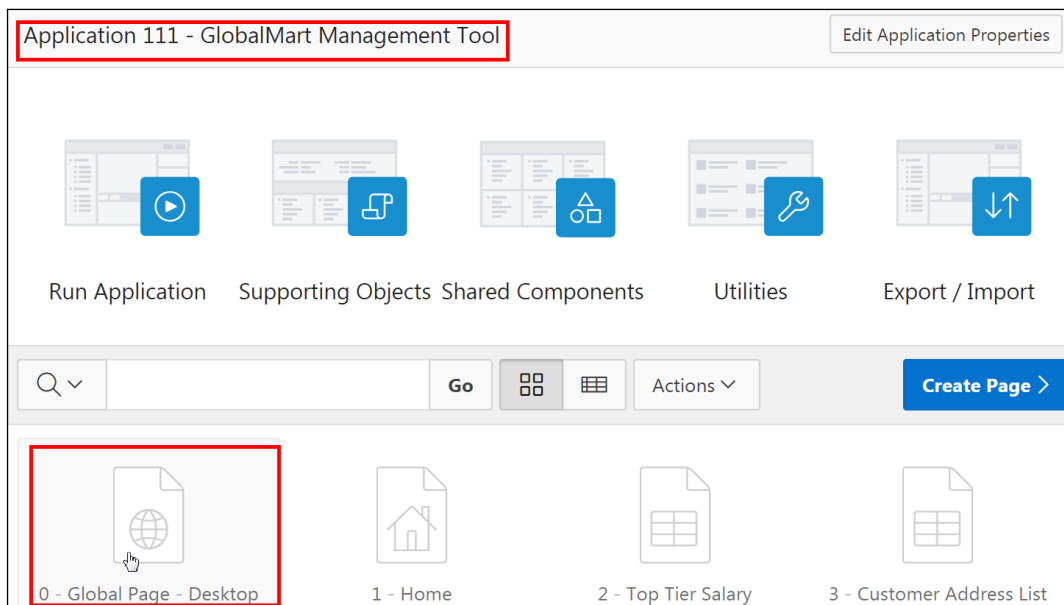
Solution 7-2: Working with Global Pages

Overview

In this practice solution, steps are provided to update the Global page in the GMT application.

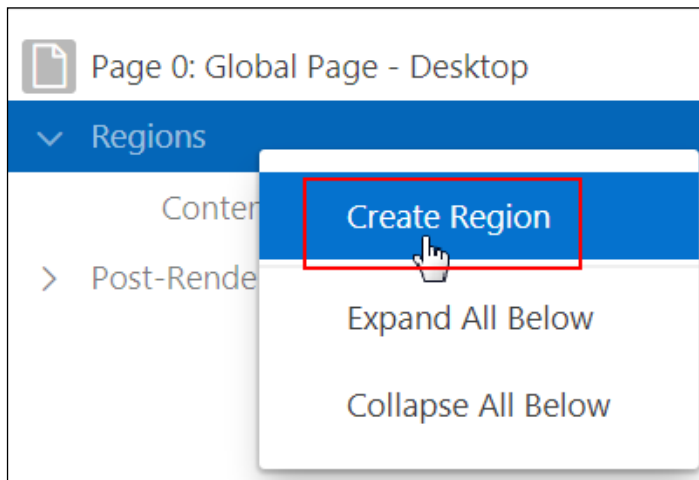
Steps

1. View the **0 – Global Page – Desktop** page in the GlobalMart Management Tool application and review it in Application Properties.
 - a. Navigate to the **GlobalMart Management Tool** application home page and click **0-Global Page**.



- b. The Global Page - Desktop opens in Page Designer.
2. Add a Static HTML region to the Global Desktop page. Name the region as **In the News** on the Global page with the following code (located in `/home/oracle/labs/labs/lab_07_02.txt`):
 - a. On the Rendering tab, right-click the **Regions** and click **Create Region**.

For Instructor Use Only.
This document should not be distributed.



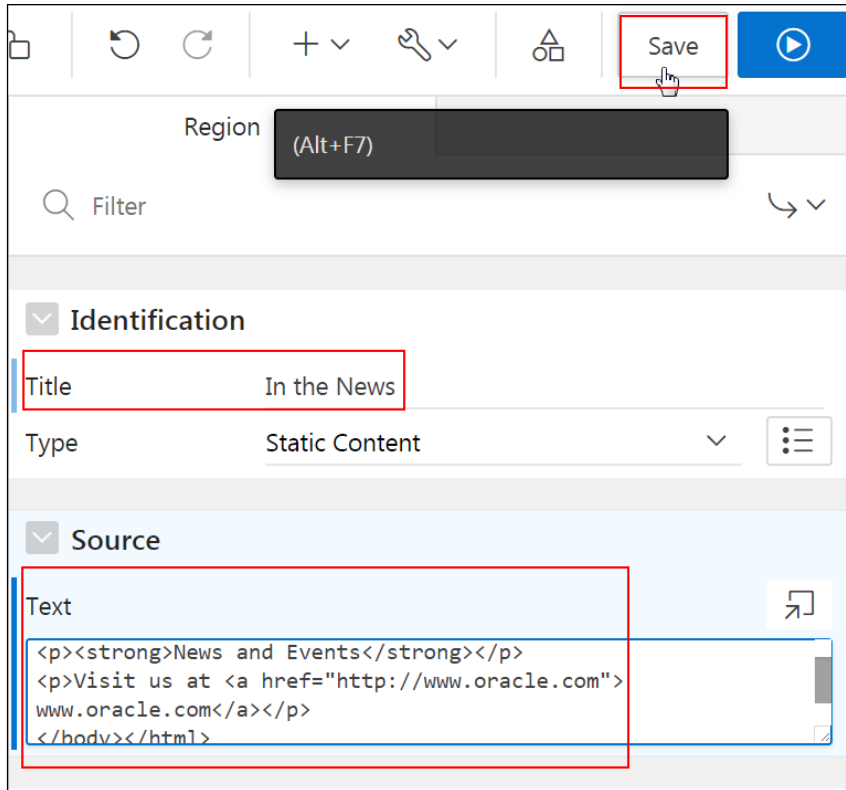
b. In the Property Editor (the right pane):

- 1) **Identification > Title:** Enter In the News.
- 2) **Source > Text:** (Copy content from the
/home/oracle/labs/labs/lab_07_02.txt.)

```
<html><body>
<p><strong>News and Events</strong></p>
<p>Visit us at <a href="http://www.oracle.com">
www.oracle.com</a></p>
</body></html>
```

3) **Layout > Position:** Select **Before Content Body**

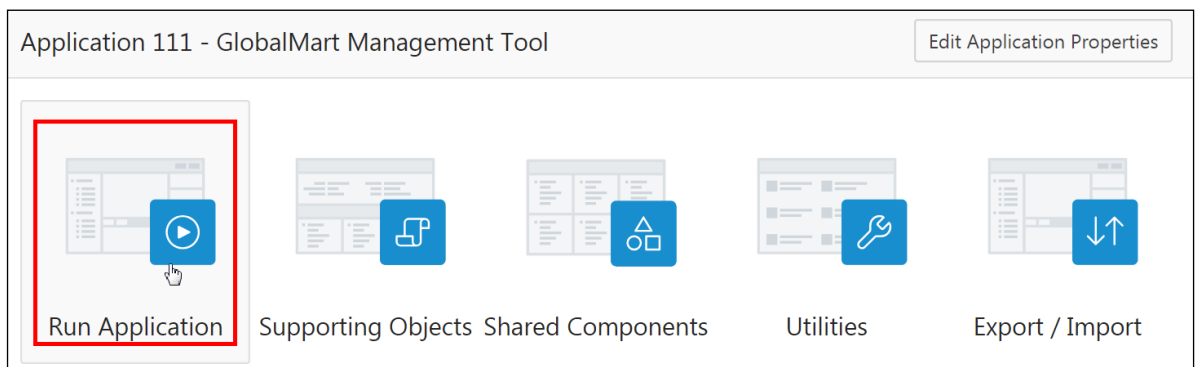
4) Click **Save**.



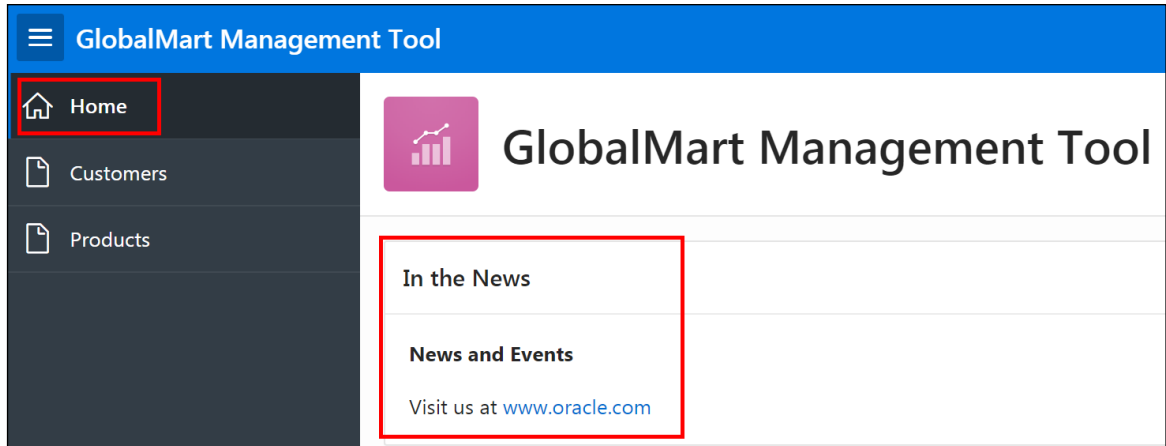
c. Click the **Application 111** breadcrumb.



d. Click **Run Application**.



The home page is displayed with the **In the News** region.



For Instructor Use Only.
This document should not be distributed.

- e. Click the **Customers** tab. You will notice that the **In the News** region appears on this page too.

The screenshot shows the GlobalMart Management Tool interface. The left sidebar has a menu with 'Home', 'Customers', and 'Products'. The 'Customers' tab is selected and highlighted with a red box. The main content area has a header with 'In the News' and 'News and Events', followed by a link 'Visit us at www.oracle.com'. Below this is a search bar with a 'Go' button and a dropdown menu showing '1. Primary Report'. A table with customer data is displayed below the search bar.

Customer Id	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province
218	Bryan	Dvrrie	3386 Perrysville	15214	Pittsburgh	PA

- f. Click the **Products** tab. The region appears here as well.

The screenshot shows the GlobalMart Management Tool interface with the 'Products' tab selected and highlighted with a red box. The main content area has a header with 'In the News' and 'News and Events', followed by a link 'Visit us at www.oracle.com'. Below this is a section titled 'Products' containing a 'Products List' table.

Product Id ↑	Product Name	Category Id	Product Status	List Price
1726	LCD Monitor 11/PM	11	under development	259

- g. Click the **Application 111** link on the Runtime Developer toolbar.

The screenshot shows the Runtime Developer toolbar. The 'Application 111' link is highlighted with a red box. Other icons in the toolbar include Home, Edit Page 7, Session, View Debug, Debug, Page Info, Quick Edit, Theme Roller, and a settings gear icon.

Practices for Lesson 8: Managing Forms

For Instructor Use Only.
This document should not be distributed.

Practices for Lesson 8: Overview

Overview

In these practices, you create and use different types of forms for GMT application, namely, Form on a Table, Report with List View on a Table, and Master Detail Form – Side By Side, Drill-down, and Stacked.

For Instructor Use Only.
This document should not be distributed.

Practice 8-1: Creating a Form

Overview

In this practice, you create a form and then link the form to the Customers report that was created previously.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a form based on the `CUSTOMERS` table. Enter 9 as the page number. The form page and region should be titled **Customer Details** and contain the customer's First Name, Last Name, Email, and Manager ID fields. The following are the other requirements:
 - Ensure that the primary key has the value `CUSTOMER_ID`.
 - The Create and Cancel buttons are required.
 - Branch to the Customers page for both: "After Page Submit and Processing" and "When Cancel Button Pressed."
 - Run the form for verification.
2. Link the form to the report. When the Edit icon in the report is clicked, the user should be redirected to the form and the details populated in the form.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 8-1**.

If you haven't completed **Practice 7-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_dev` user.
- b. Click the arrow next to **Application Builder** and select **Import**.

- c. Click the **Browse** button.
- d. Browse and open the `sol_07_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

Practice 8-2: Creating a List View with Form

Overview

In this practice, you create a list view with form.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a list view with a form based on the `Customers` table. Enter 10 as the report page number and 11 as the form page number. The report page should be titled **Customers List View** and contain the Customer's First Name, Last Name, Email, and Manager ID fields. The form page should be titled **Form on Customers** and contain the Customer's First Name, Last Name, Email, and Manager ID fields. The following are the other requirements:
 - Ensure that the primary key has the value `Customer_ID`.
 - The Create, Cancel, Delete, and Apply Changes buttons are required.
 - Run the form for verification.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 8-2**.

If you haven't completed **Practice 8-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_dev` user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_08_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.

- e. Click **Next** >.

For Instructor Use Only.
This document should not be distributed.

- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

For Instructor Use Only.
This document should not be distributed.

Practice 8-3: Creating a Master Detail Form (Side By Side)

Overview

In this practice, you create a form that is based on two tables (Master table and Detail table) and make some modifications to it.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. On a new page called **Master Detail Form**, create a Master Detail Form (List of Orders page 12) based on the `ORDERS` and `ORDER_ITEMS` tables.
Note: Two additional pages are created while creating the Master Detail Form (Side by Side). You cannot choose the page numbers of these two pages. In this activity, the page numbers are 13 (Oehr Orders) and 14 (Oehr Order Items). The page numbers may vary with different application IDs.
 - Include a report page called **List of Orders** on the master table.
 - Display the master table and detail table forms on a single page.
 - Exclude the `ORDER_ID` column in the `ORDER_ITEMS` table.
 - Include navigation based on the `ORDER_ID` column.
2. On the Master Detail Form page, change the region name for the master region to **Order Master** and change the region name for the detail region to **Order Details**.
3. On the Master Detail page, move the **Order Status** item to appear before the **Customer ID** in the Order Master region.
4. On the **List of Orders (master report)** page, do not show the `PROMOTION_ID` column. Change the heading of **Sales Rep ID** to **Sales Rep**.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 8-3**.

If you haven't completed **Practice 8-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_08_02.sql` file in the `/home/oracle/labs/solutions` directory, and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

For Instructor Use Only.
This document should not be distributed.

Practice 8-4: Creating a Master Detail Form (Stacked)

Overview

In this practice, you create a form that is based on two tables (Master table and Detail table).

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. On a new page called **Master Detail Form**, create a Master Detail Form based on the `ORDERS` and `ORDER_ITEMS` tables and specify the page number as 15.
 - Include a report page called **Orders List** on the master table.
 - Display the master table and detail table forms on a single page.
 - Include navigation based on the `ORDER_ID` column.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 8-4**.

If you haven't completed **Practice 8-3**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_08_03.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

Practice 8-5: Creating a Master Detail Form (Drill-Down)

Overview

In this practice, you create a form that is based on two tables (Master table and Detail table).

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. On a new page called **Master Detail Form**, create a Master Detail Form based on the `CUSTOMERS` and `ORDERS` tables and specify the page numbers as 20 and 21, respectively.
 - Include a report page called **Customer Master Information** on the master table.
 - Display the master table and detail table forms on a single page.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 8-5**.

If you haven't completed **Practice 8-4**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_08_04.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

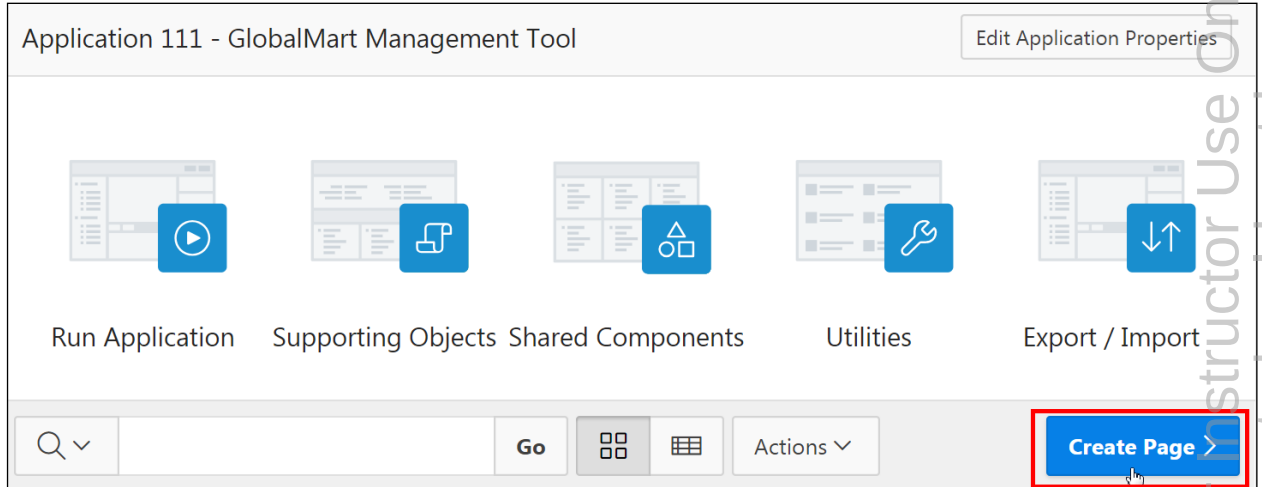
Solution 8-1: Creating a Form

Overview

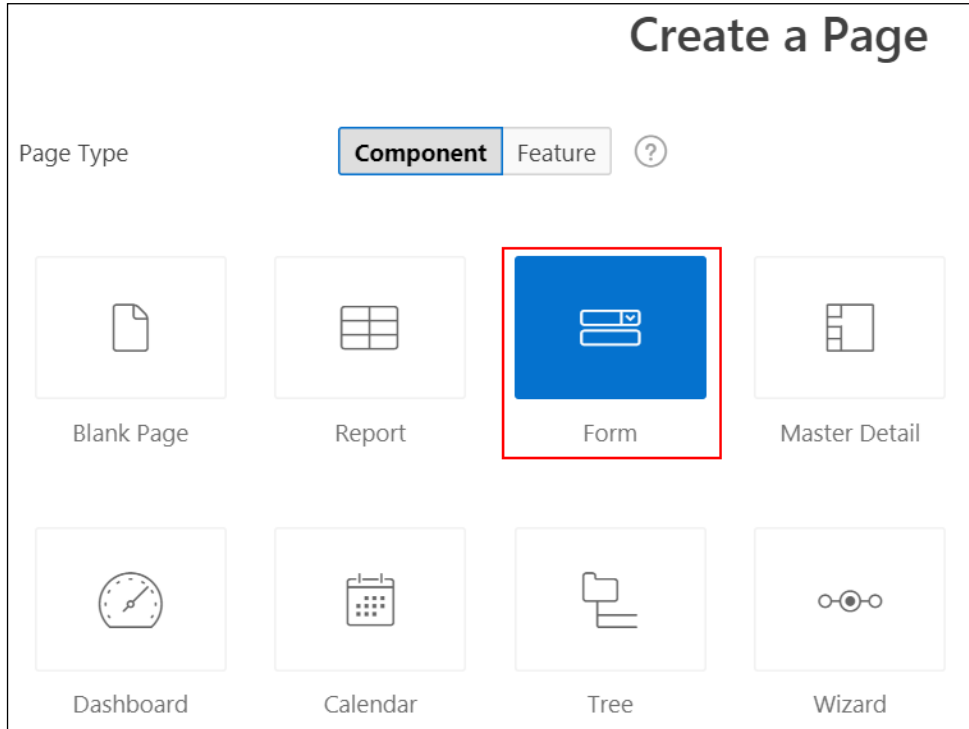
In this practice solution, steps are provided for creating a form for the GMT application.

Steps

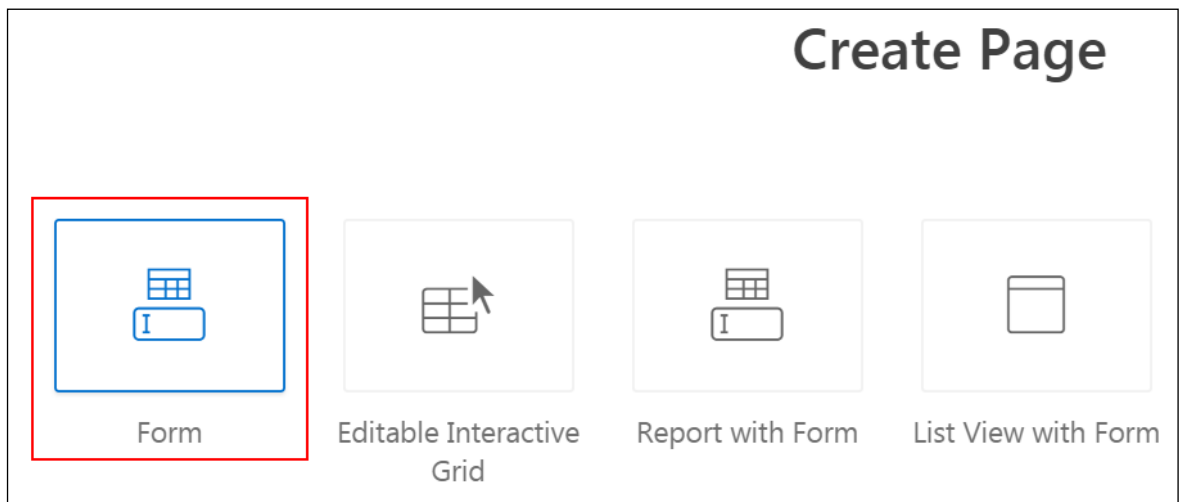
1. Create a form based on the `OEHR_CUSTOMERS` table. Enter 9 as the page number. The form page and region should be titled **Customer Details** and contain the customer's First Name, Last Name, Email, and Manager ID fields. The following are the other requirements:
 - Ensure that the primary key has the value `CUSTOMER_ID`.
 - The Create and Cancel buttons are required.
 - Branch to the Customers page for both: "After Page Submit and Processing" and "When Cancel Button Pressed."
 - Run the form for verification.
- a. On the GMT Application Home page, click **Create Page**.



- b. Select the **Form** option.



- c. Click **Next >**.
d. Select **Form**.



- e. Click **Next**.
- f. Enter the Page Attributes:
 - **Page Number:** Enter 9
 - **Page Name:** Enter Customer Details
 - **Branch Here on Submit:** Select 4 (Customer Report) page.
 - **Cancel and Go To Page:** Select 4 (Customer Report) page.

The screenshot shows the 'Create Form' wizard at the 'Page Attributes' step. A progress bar at the top has four dots, with the first dot highlighted in blue. The form contains several fields: 'Page Number' with the value '9', 'Page Name' with the value 'Customer Details', 'Page Mode' with a dropdown set to 'Normal', 'Page Group' with a dropdown set to '- Select Page Group -', 'Branch Here on Submit' with a dropdown set to '4', 'Cancel and Go To Page' with a dropdown set to '4', and 'Breadcrumb' with a dropdown set to '- do not add breadcrumb region to page -'. Two red boxes highlight the 'Page Number' and 'Page Name' fields, and another red box highlights the 'Branch Here on Submit' and 'Cancel and Go To Page' fields. A vertical watermark on the right side reads 'For Instructor Use Only. This document should not be distributed.'

- g. Click **Next >**.
- h. Accept the default for Navigation Menu and click **Next >**.

The screenshot shows the 'Create Form' wizard at the 'Navigation Menu' step. A progress bar at the top has four dots, with the second dot highlighted in blue. The form contains a 'Navigation Preference' section with three radio buttons. The first radio button is selected and highlighted with a red box, and it is labeled 'Do not associate this page with a navigation menu entry'. The other two radio buttons are labeled 'Create a new navigation menu entry' and 'Identify an existing navigation menu entry for this page'. A vertical watermark on the right side reads 'For Instructor Use Only. This document should not be distributed.'

- i. For Data Source:
 - **Source Type:** Select **Table**

- **Table/View Name:** Click the  icon and select **OEHR_CUSTOMERS (table)** from the drop-down list.

Data Source

Data Source Local Database REST Enabled SQL Service Web Source ?

* Source Type ☒ **Table** ☐ SQL Query ?

* Table / View Owner OEHR ?

* Table / View Name OEHR_CUSTOMERS (table)  ?

j. Click **Next >**.

k. For Form Attributes:

- **Select Columns:** Select **CUST_FIRST_NAME**, **CUST_LAST_NAME**, **CUST_EMAIL**, and **ACCOUNT_MGR_ID**. Ensure that these columns are in the right column. Use the right arrow (**>**) and left arrow (**<**) to move the columns.
- **Primary Key Type:** Select **Select Primary Key Column(s)**.
- **Primary Key Column:** Select **CUSTOMER_ID (Number)** from the drop-down list.

Create Form - Columns and Primary Key

* Select Columns to be displayed in the form ?

CUSTOMER_ID (Number)

STREET_ADDRESS (Varchar2)

POSTAL_CODE (Varchar2)

CITY (Varchar2)

STATE_PROVINCE (Varchar2)

COUNTRY_ID (Varchar2)

NLS_LANGUAGE (Varchar2)

NLS_TERRITORY (Varchar2)

⊞

>>

>

<

<<

CUST_FIRST_NAME (Varchar2)

CUST_LAST_NAME (Varchar2)

CUST_EMAIL (Varchar2)

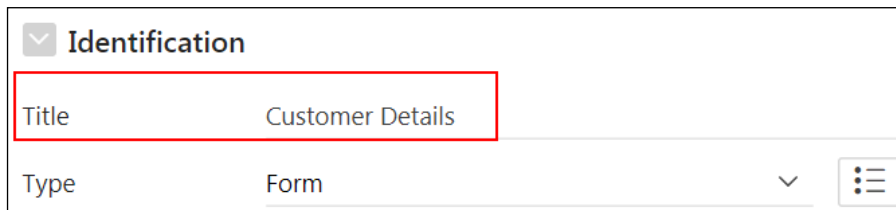
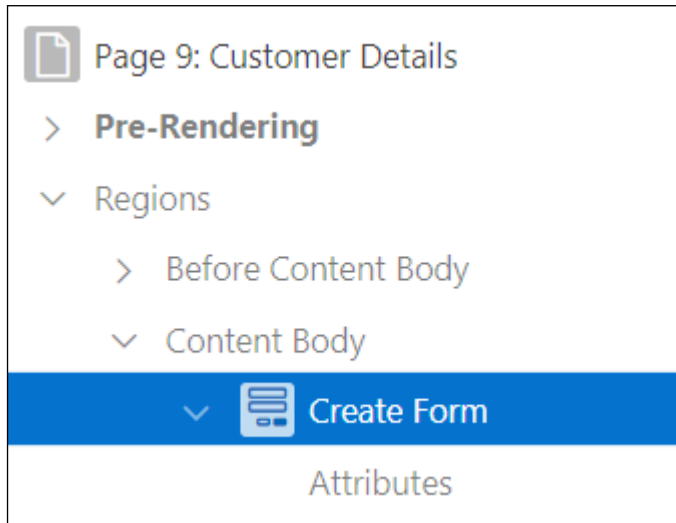
ACCOUNT_MGR_ID (Number)

Primary Key Type ☐ Managed by Database (ROWID) ☒ **Select Primary Key Column(s)**

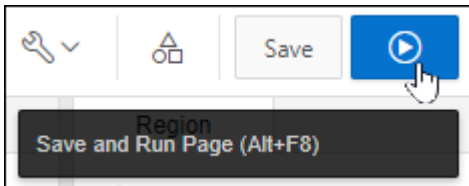
Primary Key Column CUSTOMER_ID (Number)

Secondary Key Column - Select Column -

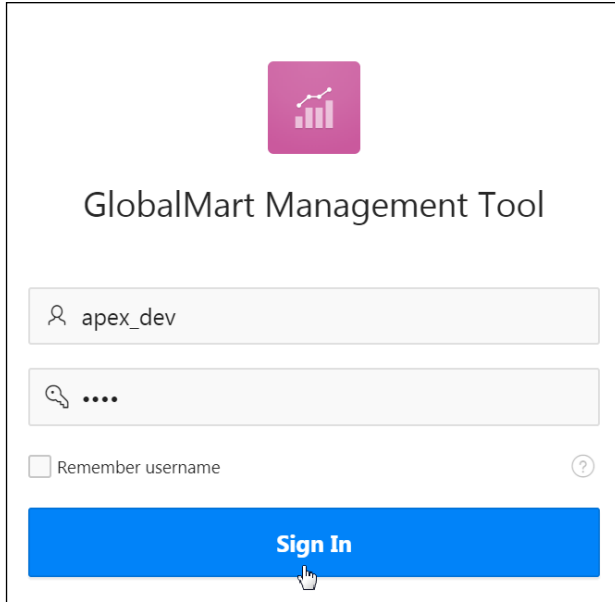
- l. Click **Create**. The Customer Details form page opens in Page Designer view.
- m. Select **Customer Details > Regions > Content Body > Create Form** and in the Property Editor **Identification > Title > Enter Customer Details**.



- n. Click the **Save and Run Page** icon.



- o. If the sign in page appears, enter your credentials and click **Sign in**.



The image shows the sign-in page for the GlobalMart Management Tool. At the top is a purple square icon with a white bar chart. Below the icon is the title "GlobalMart Management Tool". There are two input fields: the first contains the username "apex_dev" and the second contains four dots "....". Below these fields is a checkbox labeled "Remember username" and a question mark icon. At the bottom is a large blue button labeled "Sign In". A mouse cursor is pointing at the "Sign In" button.

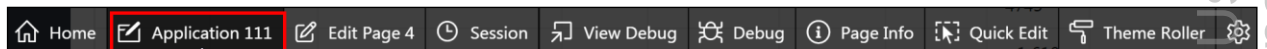
- p. Review the form. Click **Cancel** and the Customer Report page appears.



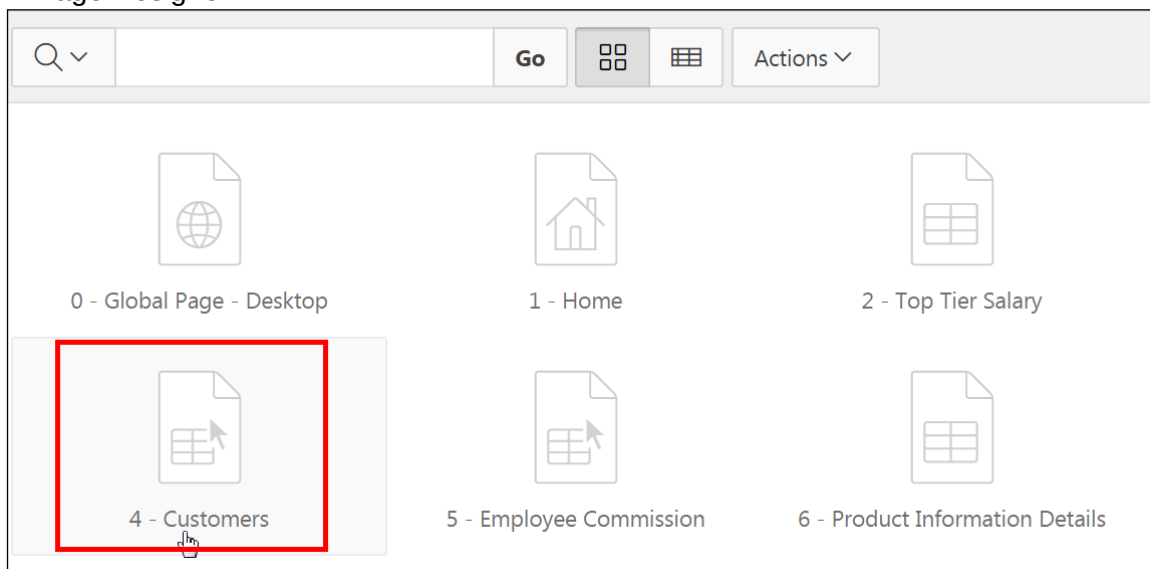
The image shows a form titled "Customer Details" with a red border. It contains four input fields: "Cust First Name", "Cust Last Name", "Cust Email", and "Account Mgr Id". At the bottom left is a "Cancel" button with a red border, and at the bottom right is a blue "Create" button. A vertical watermark on the right side of the form reads "For Instructor Use Only. This document should not be distributed."

Customer Id	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province	Country Id
218	Bryan	Dvrrie	3386 Perrysville Ave	15214	Pittsburgh	PA	US
219	Ajay	Sen	220 Penn Ave # 300	18503	Scranton	PA	US
220	Carol	Jordan	135 S 18Th St # 1	19103	Philadelphia	PA	US
221	Carol	Bradford	522 Swede St	19401	Norristown	PA	US

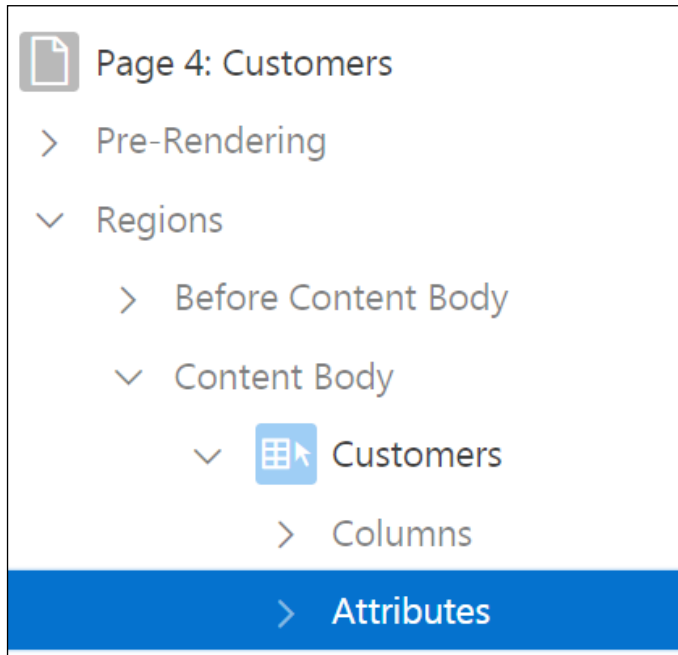
To return to the application home page, click the **Application 111** link on the Runtime Developer toolbar.



2. Link the **Customers** report (page 4) that was created previously to the newly created **Customer Details** form. When the Edit icon in the report is clicked, the user should be redirected to the form and the details populated in the form. After you create the link, run the report and verify the changes.
 - a. On the GMT Application Home page, click **4 - Customers**. The Customers page opens in Page Designer.

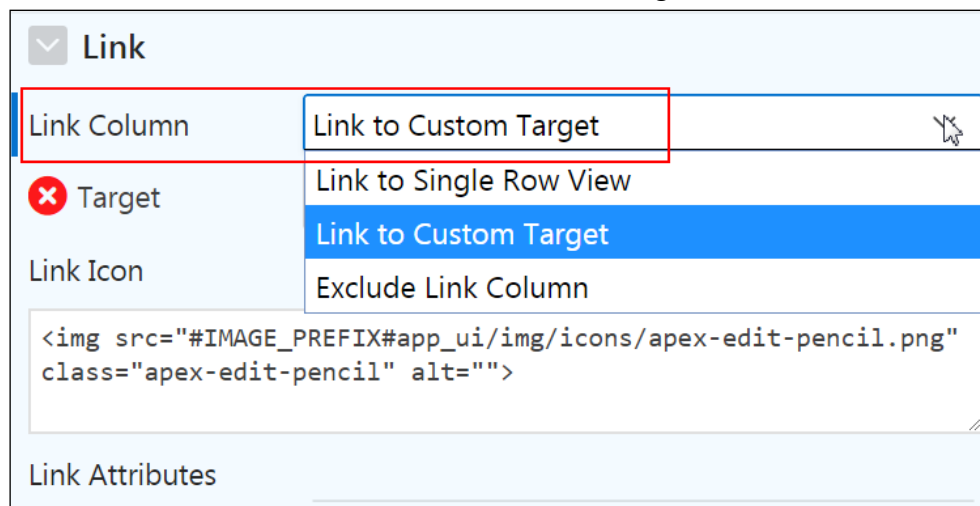


- b. In the Rendering tab, select **Regions > Content Body** and select **Attributes**.



- c. In the Property Editor (the right pane):

- **Link > Link Column:** Select **Link to Custom Target**



- **Link > Target:** Click **No Link Defined**. The **Link Builder –Target** page is displayed.

Link

Link Column

Link to Custom Target

✖ Target

No Link Defined

Link Icon

Link Attributes

Target

Type

Page in this application

Page

Set Items

Name	Value

Clear Session State

Clear Cache

Reset Pagination

Yes

No

Advanced

Cancel

Clear

OK

- **Target > Page:** Click  icon and select **9 - Customer Details** from the drop-down list
- **Set items > Name:** Click  icon and select **P9_CUSTOMER_ID** link from the Name drop-down list
- **Set items > Value:** Click  icon and select **CUSTOMER_ID** link from the value drop-down list

- Click **OK**.

Link Builder - Target

Target

Type Page in this application

Page 9

Set Items

Name P9_CUSTOMER_ID

Value #CUSTOMER_ID#

Clear Session State

Clear Cache

Reset Pagination Yes No

Cancel Clear OK

- d. Click **Save** and then **Save and Run Page**.

Save and Run Page (Alt+F8)

- e. The **Customers** report page opens. Click the **Edit** icon for any one of the rows.

	Customer Id	First Name	Last Name	Street Address	Postal Code	City	State Province	Country
	224	Clara	Krige	101 E Olney Ave	19120	Philadelphia	PA	US
	225	Clara	Ganesan	612 Jefferson Ave	18510	Scranton	PA	US
	226	Ajay	Andrews	223 4Th Ave # 1100	15222	Pittsburgh	PA	US
	227	Kathy	Prashant	36 W 34Th St	16508	Erie	PA	US

- f. Note that you are redirected to the **Customer Details** form and that the values are automatically populated in the form. You can make changes and delete the active record by using this form. To return to the application home page, click the **Application 111** link on the Runtime Developer toolbar.

Customer Details

Cust First Name
Clara

Cust Last Name
Ganesan

Cust Email
Clara.Ganesan@WIGEON.COM

Account Manager
149

Cancel

Delete

Apply Changes

Home

Application 111

Edit Page 9

Session

View Debug

Debug

Page Info

Quick Edit

Theme Roller

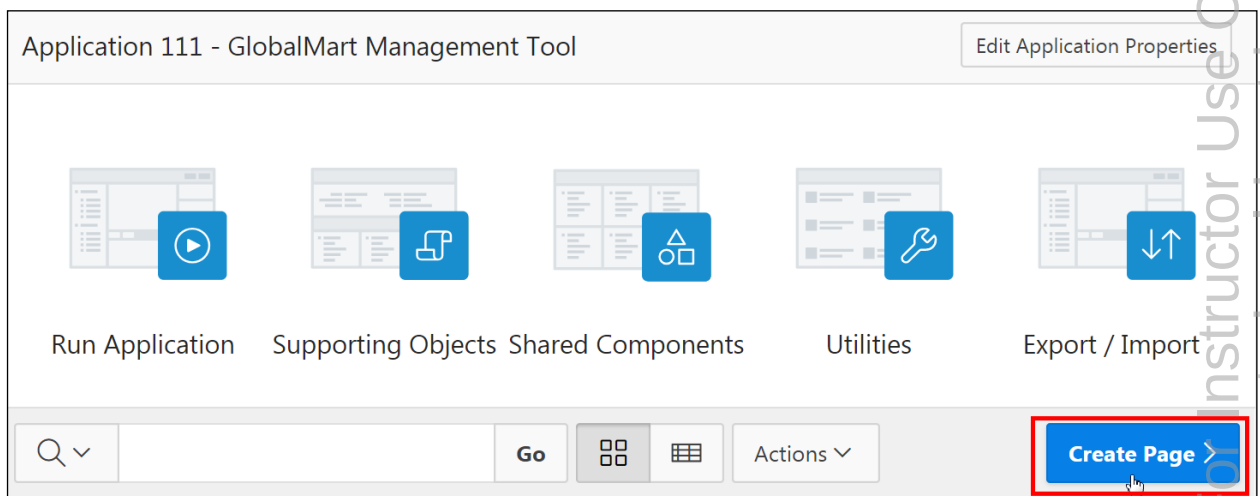
Solution 8-2: Creating a List View with Form

Overview

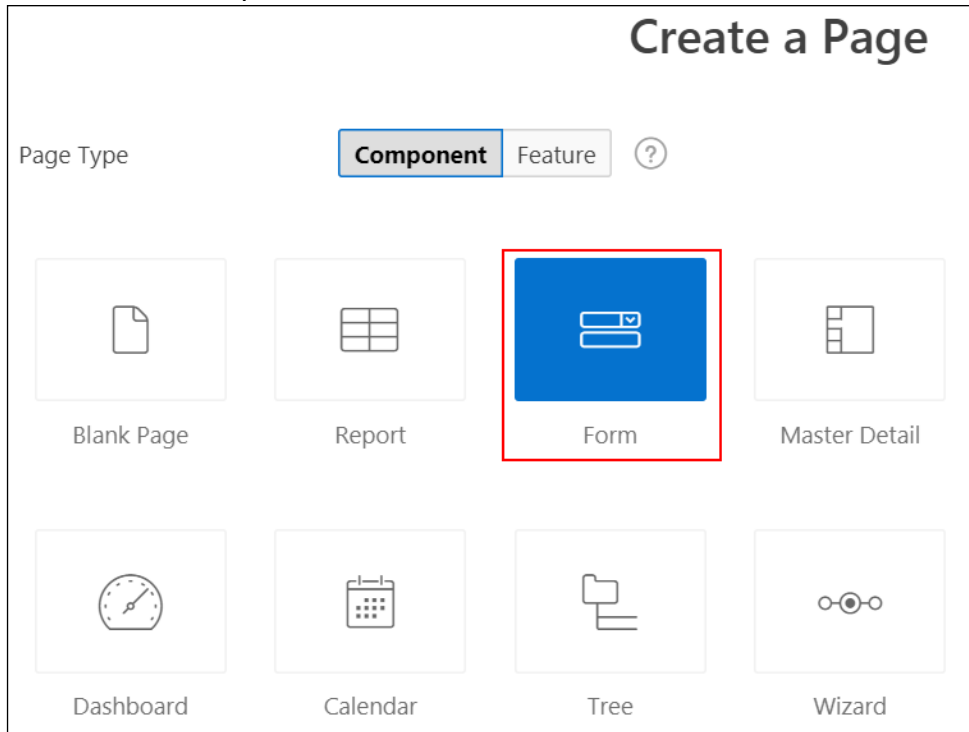
In this practice solution, steps are provided for creating a list view with form for the GMT application.

Steps

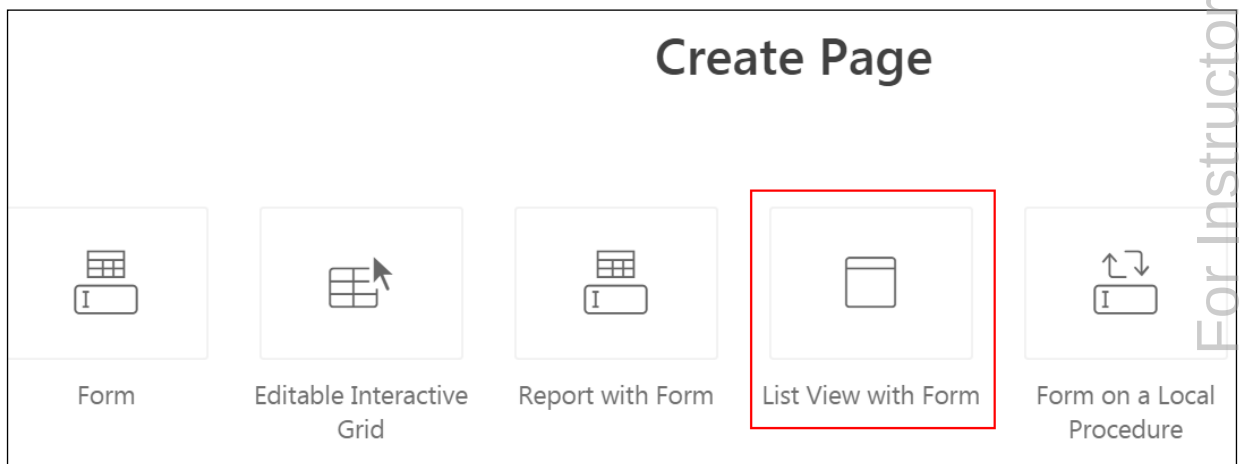
1. Create a list view with a form based on the `Customers` table. Enter 10 as the report page number and 11 as the form page number. The form page should be titled **Customers List View** and contain the Customer's First Name, Last Name, Email, and Manager ID fields. The report page should be titled **Form on Customers** and contain the Customer's First Name, Last Name, Email, and Manager ID fields. The following are the other requirements:
 - Ensure that the primary key has the value `Customer_ID`.
 - The Create, Cancel, Delete, and Apply Changes buttons are required.
 - Run the form for verification.
- a. On the GMT Application Home page, click **Create Page**.



- b. Select the **Form** option.



- c. Click **Next >**.
d. Select **List View with Form**.



- e. For List View Page:
- **Report Page Number:** Enter 10
 - **Report Page Name:** Enter Customers List View
 - **Form Page Number:** Enter 11

- **Form Page Name:** Enter Form on Customers

Create Form and List View

List View Page

* Report Page Number 10 ?

* Report Page Name Customers List View ?

* Form Page Number 11 ?

* Form Page Name Form on Customers ?

Form Page Mode **Normal** Modal Dialog ?

Page Group - Select Page Group - ?

Breadcrumb - do not add breadcrumb region to page - ?

- f. Click **Next >**.

g. For Navigation Menu:

- **Navigation Preference:** Select the **Create a new navigation menu entry**.
- **New Navigation Menu Entry:** Ensure that the text field is set to **Customer List View**.
- **Parent Navigation Menu Entry:** Select **- No parent selected -** (if not selected by default)

The screenshot shows the 'Create Form and List View' wizard with the 'Navigation Menu' step selected. A progress bar at the top has four dots, with the second dot (Navigation Menu) highlighted in blue. The 'Navigation Preference' section has three radio buttons: 'Do not associate this page with a navigation menu entry' (unselected), 'Create a new navigation menu entry' (selected and highlighted with a red box), and 'Identify an existing navigation menu entry for this page' (unselected). Below this, the 'New Navigation Menu Entry' field is highlighted with a red box and contains the text 'Customers List View'. The 'Parent Navigation Menu Entry' field is also highlighted with a red box and shows a dropdown menu with the following options: '- No parent selected -' (selected), 'Home', 'Customers', and 'Products'. Help icons (?) are visible next to the radio buttons and the dropdown menu.

For Instructor Use Only.
This document should not be distributed.

h. For Data Source:

- **Table/View Owner:** Selected by default **OEHR**.
- **Table/View Name:** Select **OEHR_CUSTOMERS** (table) from the drop-down list.
- **Select Columns to be shown in the report:** Select **CUSTOMER_ID**, **CUST_FIRST_NAME**, **CUST_LAST_NAME**, **PHONE_NUMBER**, **CUST_EMAIL**, and **ACCOUNT_MGR_ID**. Ensure these columns are in the right column. Use the right arrow (**>**) and left arrow (**<**) to move the columns.

Data Source

Data Source: **Local Database** REST Enabled SQL Service Web Source

Source Type: **Table** SQL Query

* Table / View Owner: OEHR

* Table / View Name: OEHR_CUSTOMERS (table)

* Select Columns to be shown in the report

Available Columns	Selected Columns
STREET_ADDRESS (Varchar2)	CUSTOMER_ID (Number)
POSTAL_CODE (Varchar2)	CUST_FIRST_NAME (Varchar2)
CITY (Varchar2)	CUST_LAST_NAME (Varchar2)
STATE_PROVINCE (Varchar2)	PHONE_NUMBER (Varchar2)
COUNTRY_ID (Char)	CUST_EMAIL (Varchar2)
NLS_LANGUAGE (Varchar2)	ACCOUNT_MGR_ID (Number)
NLS_TERRITORY (Varchar2)	
CREDIT_LIMIT (Number)	

i. Click **Next >**.

j. On the Form page:

- **Select Column(s) to be included in Form:** Ensure that these columns are in the right column: **CUSTOMER_ID**, **CUST_FIRST_NAME**, **CUST_LAST_NAME**, **PHONE_NUMBER**, **CUST_EMAIL**, and **ACCOUNT_MGR_ID**.
- **Primary Key Column 1:** Select **CUSTOMER_ID** (Number) from the drop-down list.

- k. Click **Create**.

Create Form - Columns and Primary Key

Form Page

* Select Columns to be displayed in the form ?

Available Columns	Selected Columns
STREET_ADDRESS (Varchar2)	CUSTOMER_ID (Number)
POSTAL_CODE (Varchar2)	CUST_FIRST_NAME (Varchar2)
CITY (Varchar2)	CUST_LAST_NAME (Varchar2)
STATE_PROVINCE (Varchar2)	PHONE_NUMBER (Varchar2)
COUNTRY_ID (Varchar2)	CUST_EMAIL (Varchar2)
NLS_LANGUAGE (Varchar2)	ACCOUNT_MGR_ID (Number)
NLS_TERRITORY (Varchar2)	
CREDIT_LIMIT (Number)	

Primary Key Type: ☐ Managed by Database (ROWID) ☒ **Select Primary Key Column(s)**

Primary Key Column: **CUSTOMER_ID (Number)**

Secondary Key Column: - Select Column -

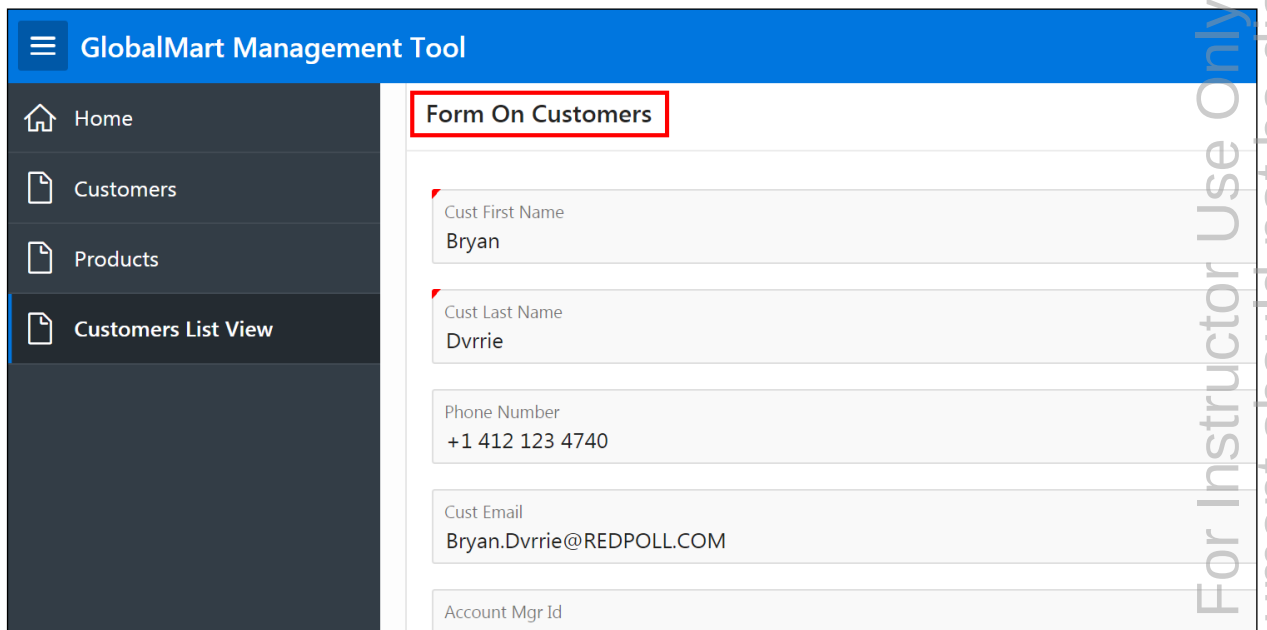
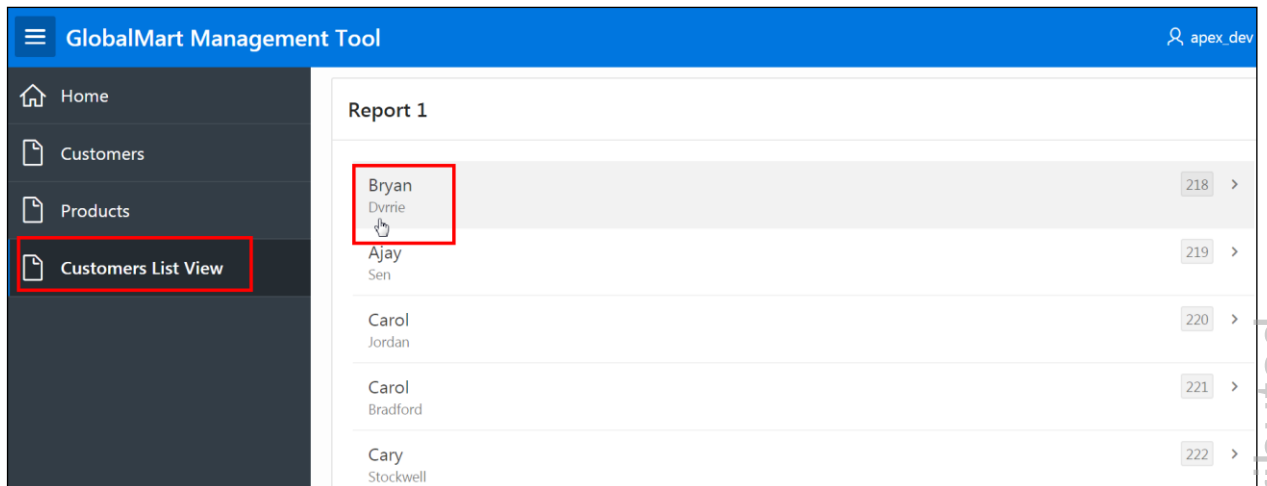
Create

- l. Click **Save and Run Page**.

Save and Run Page (Alt+F8)

- m. If the Sign in page appears, enter your credentials and click **Sign in**.

- n. Review the form. Click on any list view entry. It opens the corresponding **Form on Customers**.



- o. To return to the application home page, click the **Application 111** link on the Runtime Developer toolbar.



Solution 8-3: Creating a Master Detail Form (Side by Side)

Overview

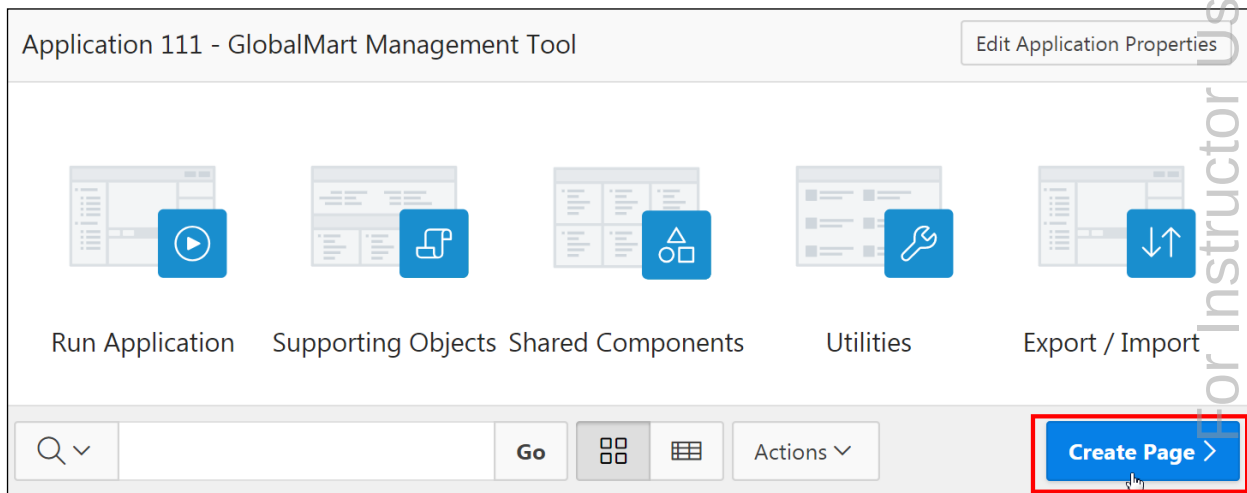
In this practice solution, steps are provided for creating a Master Detail (Side by Side) Form for the GMT application.

Steps

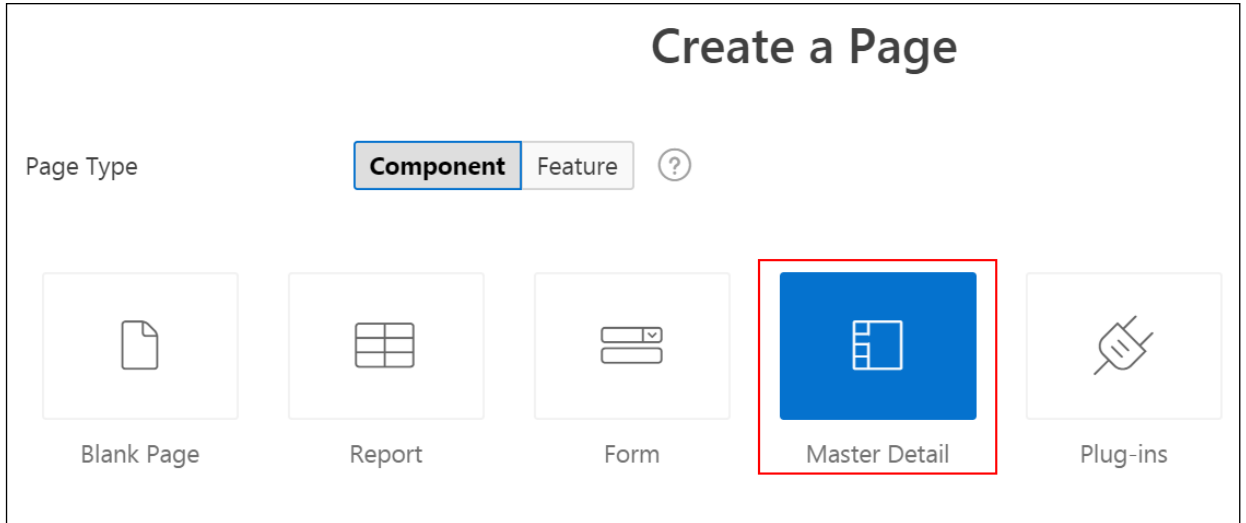
1. On a new page called **Master Detail Form**, create a Master Detail Form (List of Orders page 12) based on the `ORDERS` and `ORDER_ITEMS` tables.

Note: Two additional pages are created while creating the Master Detail Form (Side by Side). You cannot choose the page numbers of these two pages. In this activity, the page numbers are 13 (Oehr Orders) and 14 (Oehr Order Items) . The page numbers may vary with different application IDs.

- Include a report page called **List of Orders** on the master table.
 - Display the master table and detail table forms on a single page.
 - Exclude the `ORDER_ID` column in the `ORDER_ITEMS` table.
 - Include navigation based on the `ORDER_ID` column.
- a. On the GMT Application Home page, click **Create Page**.

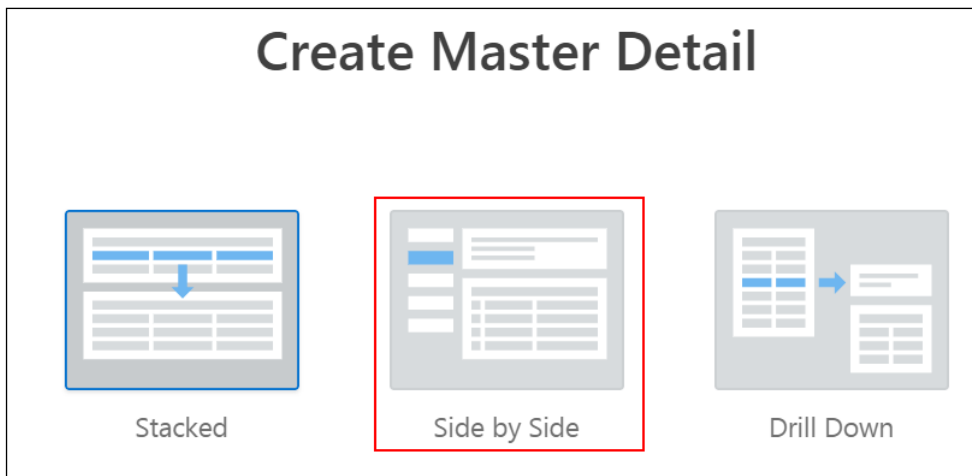


- b. Select **Master Detail**. The Create Master Detail page is displayed.



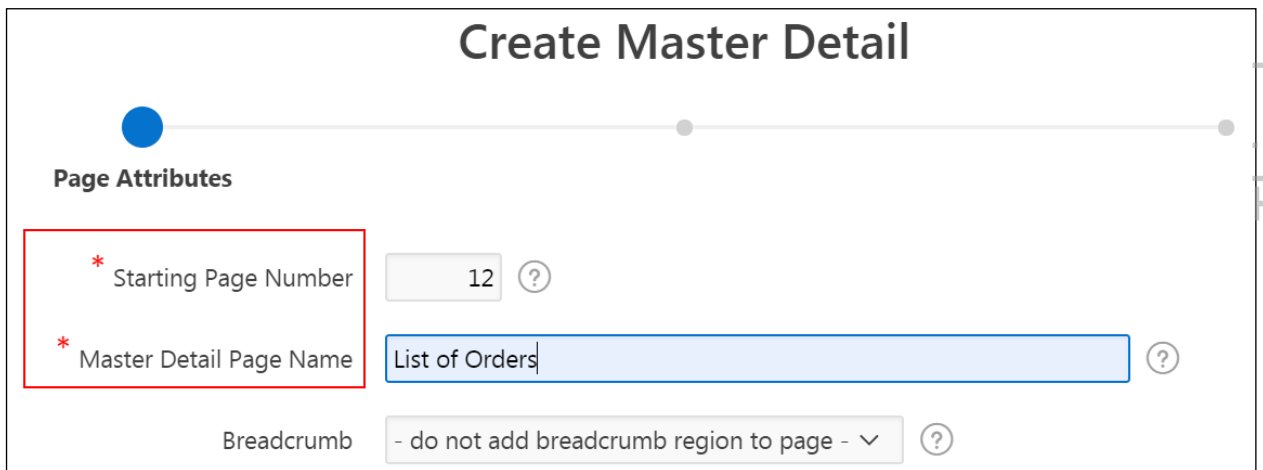
The "Create a Page" dialog box shows the "Page Type" section with two tabs: "Component" (selected) and "Feature". Below the tabs are five icons representing different page types: Blank Page, Report, Form, Master Detail, and Plug-ins. The "Master Detail" icon, which shows a blue square with a white grid, is highlighted with a red border.

- c. Click **Side by Side**.



The "Create Master Detail" dialog box shows three layout options: Stacked, Side by Side, and Drill Down. The "Side by Side" option, which shows two overlapping document icons, is highlighted with a red border.

- d. Enter the Page Attributes and click **Next >**:
- **Starting Page Number:** Enter 12
 - **Master Page Detail Name:** Enter List of Orders.

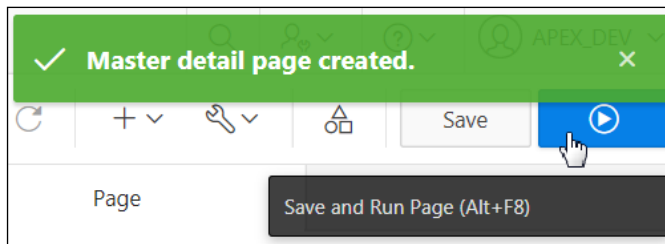


The "Create Master Detail" dialog box shows the "Page Attributes" section. It includes a progress bar at the top with three dots. Below the progress bar, the "Page Attributes" section contains three fields: "Starting Page Number" (with a red asterisk and a value of 12), "Master Detail Page Name" (with a red asterisk and the value "List of Orders"), and "Breadcrumb" (with a dropdown menu set to "- do not add breadcrumb region to page -"). Each field has a help icon (question mark) to its right.

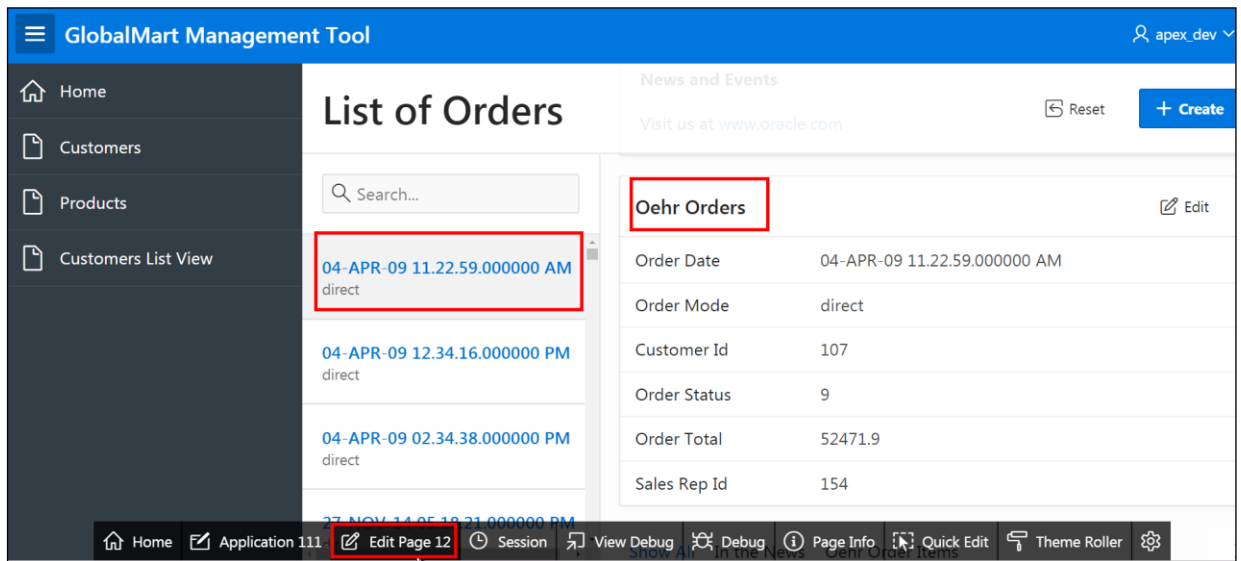
- e. Accept the default for **Navigation Menu** and click **Next >**.

- f. For Master Detail Source:
- **Table/View Owner:** OEHR.
 - **Master Table:** Select OEHR_ORDERS (table) from the drop-down list.
 - **Primary Display Column:** Select ORDER_DATE (Timestamp(6) With Local Time Zone) from the drop-down list.
 - **Secondary Display Column:** Select ORDER_MODE (varchar 2) from the drop-down list.
 - **Detail Table:** Select OEHR_ORDER_ITEMS from drop-down list.
- g. Click **Create**.

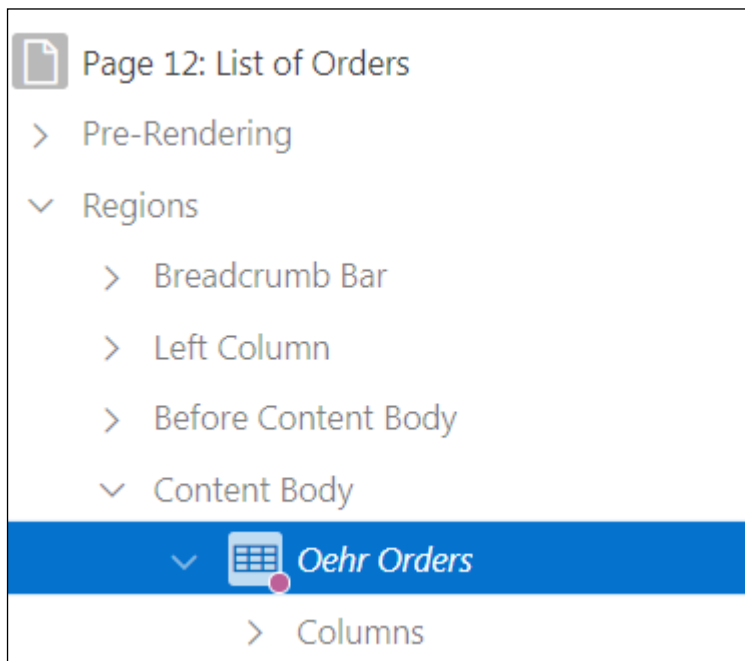
- h. The Master Detail Side by Side page is created. Click **Save and Run Page**, to view the Master Detail Form.



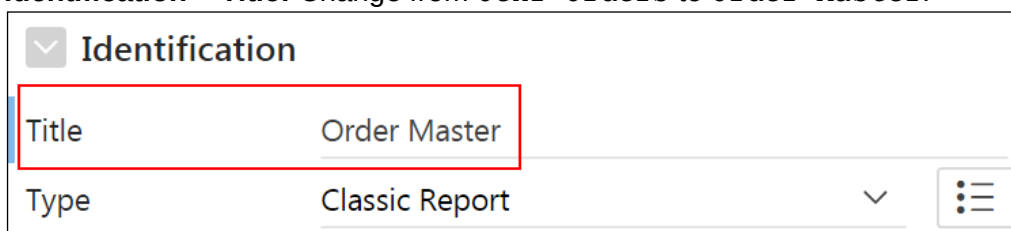
- i. The Master detail Form is displayed. Select any order. The order details are displayed side by side. Review the information. Click **Edit Page 12** on the Runtime Developer toolbar.



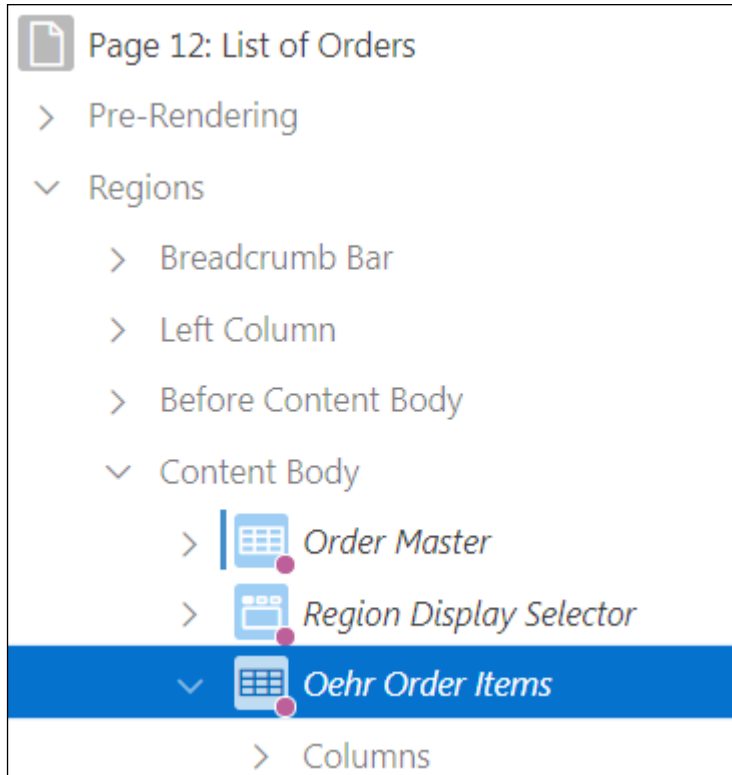
2. On the Master Detail Form page, change the region name for the master region to **Order Master** and change the region name for the detail region to **Order Details**.
 - a. Select **Regions > Content Body > Oehr ORDERS**.



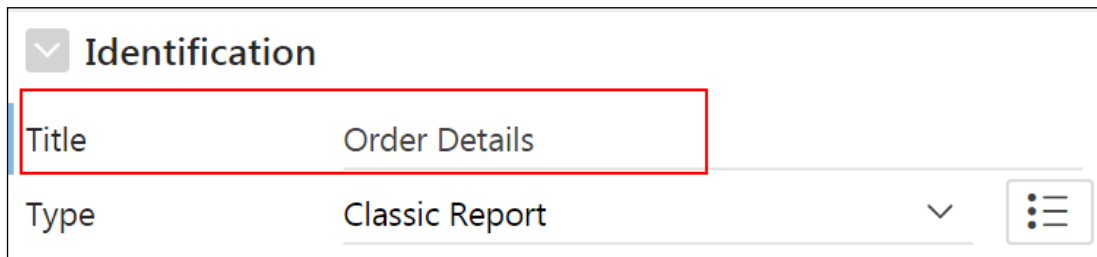
- b. Edit the appropriate attributes in the Property Editor (the right pane):
Identification > Title: Change from Oehr Orders to Order Master.



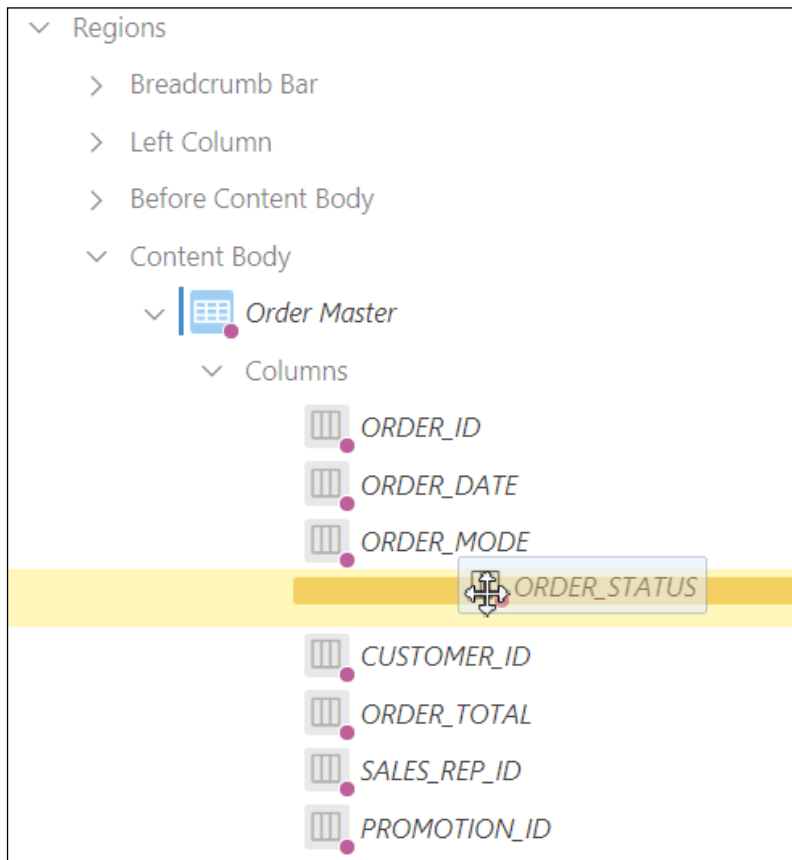
- c. Select **Regions > Content Body > Oehr ORDER_ITEMS**.



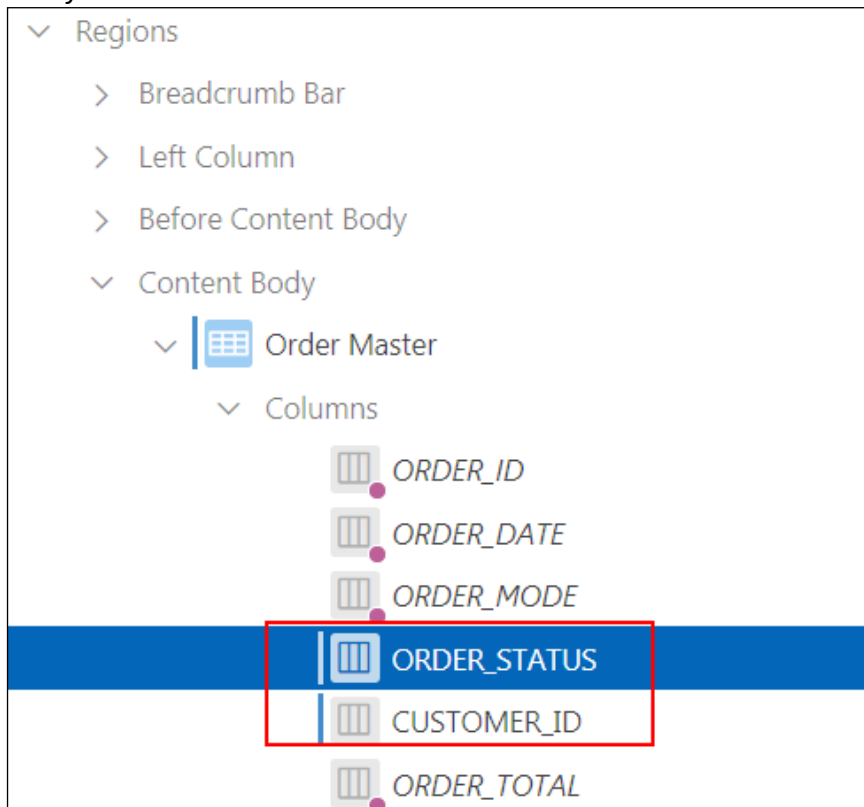
- d. Edit the appropriate attributes in the Property Editor (the right pane):
Identification > Title: Change from Oehr Order items to Order Details.

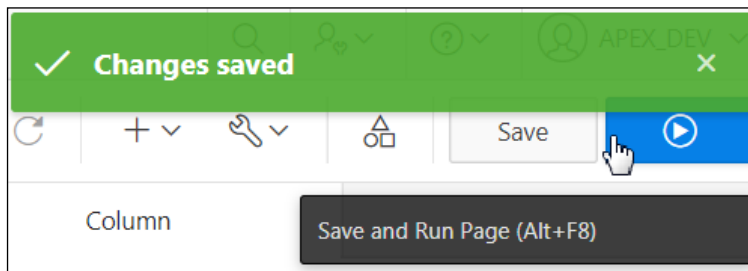


3. On the Master Detail page, move the **Order Status** (under **Order Master > Columns > Order Status**) item to appear before the **Customer ID** in the Order Master region.
- a. You can reorder items very easily by dragging and dropping them. Drag **ORDER_STATUS** to above **CUSTOMER_ID**.



- b. Verify that the item is moved. Click **Save** and then **Save and Run Page**.





- c. Review the information. Click **Edit Page 12** on the Runtime Developer toolbar.

List of Orders

04-APR-09 11.22.59.000000 AM
direct

04-APR-09 12.34.16.000000 PM
direct

04-APR-09 02.34.38.000000 PM
direct

27-NOV-14 05.18.21.000000 PM
direct

06-DEC-15 02.34.50.000000

Visit us at www.oracle.com

Order Master

Order Date: 04-APR-09 11.22.59.000000 AM

Order Mode: direct

Order Status: 9

Customer Id: 107

Order Total: 52471.9

Sales Rep Id: 154

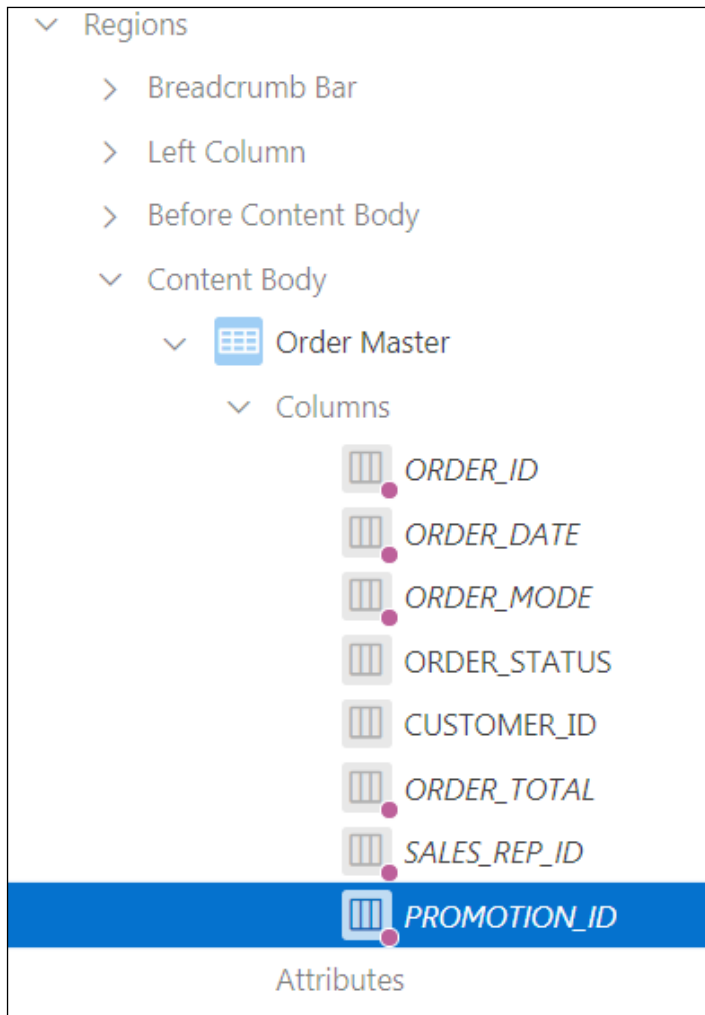
[Show All](#)
[In the News](#)
[Order Details](#)

Order Details

Line Item Id	Product Id	Unit Price	Quantity
1	2402	300	50

Home Application 111 **Edit Page 12** Session View Debug Debug Page Info Quick Edit Theme Roller

4. On the **List of Orders (master report)** page, do not show the `PROMOTION_ID` column. Change the heading of **Sales Rep ID** to **Sales Rep**.
 - a. Select **Regions > Content Body > Order Master > Columns > PROMOTION_ID**.



- b. In the Property Editor (the right pane), under Identification > Type, select **Hidden Column**.

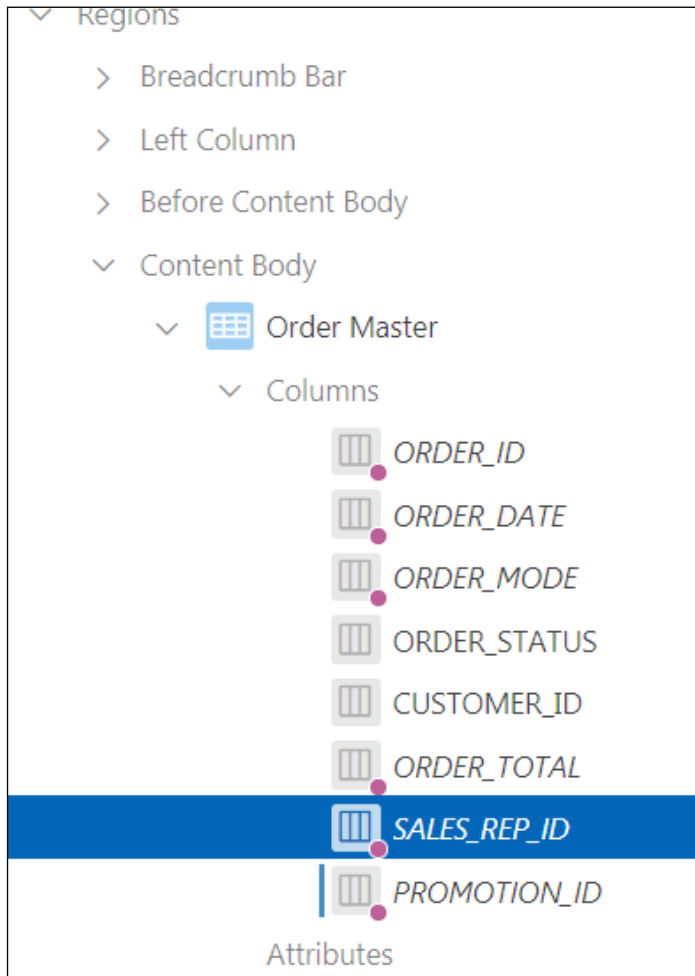
The screenshot shows the Property Editor interface. On the left, there are three expandable sections: 'Identification', 'Layout', and 'UI Defaults Re'. The 'Identification' section is expanded, showing a table with the following properties:

Property	Value
Column Name	PROMOTION_ID
Type	Hidden Column
Sequence	
Table Owner	
Table Name	

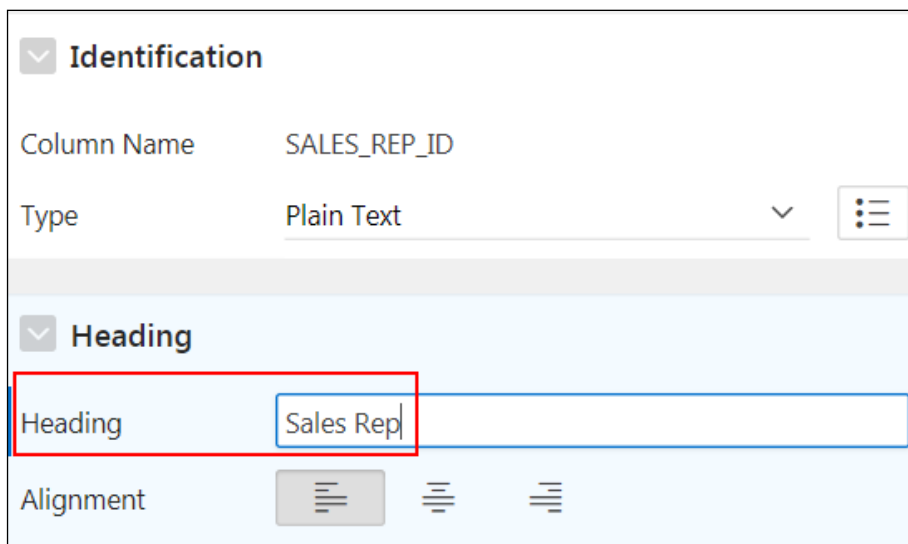
The 'Type' property is highlighted with a red box, and its dropdown menu is open, showing the following options:

- Plain Text
- Plain Text (based on List of Values)
- Link
- Display Image
- Download BLOB
- Percent Graph
- Hidden Column** (highlighted in blue)

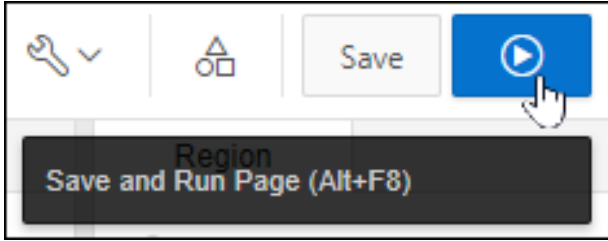
- c. Select **Regions > Content Body > Order Master > Columns > SALES_REP_ID**.



- d. In the Property Editor (the right pane): **Heading > Heading**, change the heading to Sales Rep.



- e. Click **Save** and then **Save and Run Page**.



- f. Review the report. Note that `PROMOTION_ID` is no longer shown and the heading for Sales Rep ID has changed to Sales Rep. Click the **Application 111** link on the Runtime Developer toolbar.

The screenshot displays the GlobalMart Management Tool interface. The top navigation bar is blue with the text 'GlobalMart Management Tool'. The left sidebar is dark grey with links for 'Home', 'Customers', 'Products', and 'Customers List View'. The main content area is titled 'List of Orders' and features a search bar. Below the search bar is a table of orders. The 'Order Master' table on the right shows details for a specific order, with the 'Sales Rep' field highlighted by a red box. The 'Sales Rep' field contains the value '154'. The 'Order Master' table also includes fields for 'Order Date', 'Order Mode', 'Order Status', 'Customer Id', and 'Order Total'. The 'Order Date' is '04-APR-09 11.22.59.000000 AM', 'Order Mode' is 'direct', 'Order Status' is '9', 'Customer Id' is '107', and 'Order Total' is '52471.9'. The 'Sales Rep' field is highlighted with a red box. Below the 'Order Master' table are links for 'Show All', 'In the News', and 'Order Details'. The 'Show All' link is underlined. The bottom toolbar is dark grey and contains icons for 'Home', 'Application 111', 'Edit Page 12', 'Session', 'View Debug', 'Debug', 'Page Info', 'Quick Edit', and 'Theme Roller'. The 'Application 111' link is highlighted with a red box.

Order Master	
Order Date	04-APR-09 11.22.59.000000 AM
Order Mode	direct
Order Status	9
Customer Id	107
Order Total	52471.9
Sales Rep	154

For Instructor Use Only.
This document should not be distributed.

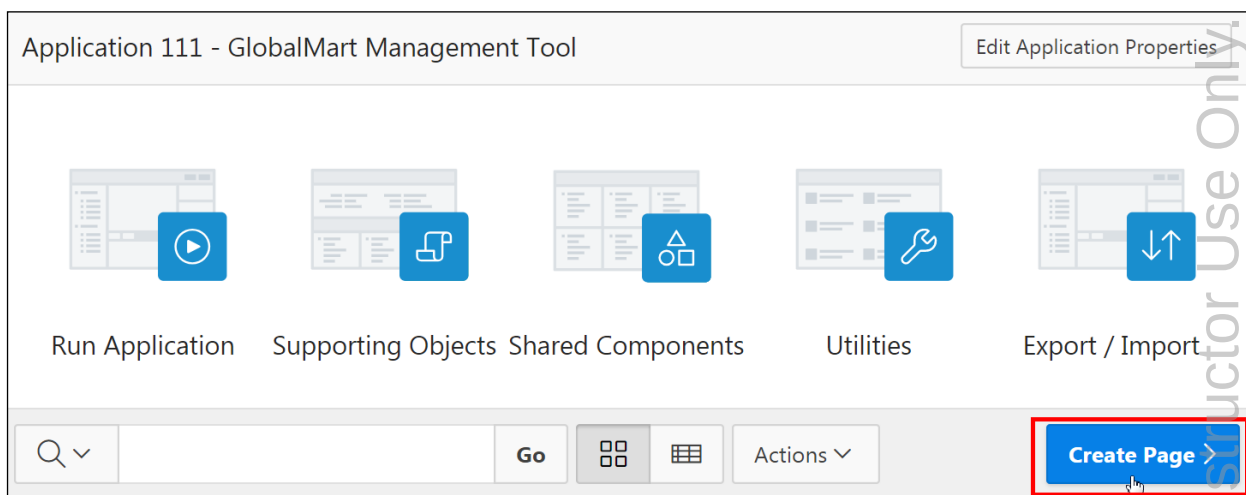
Solution 8-4: Creating a Master Detail Form (Stacked)

Overview

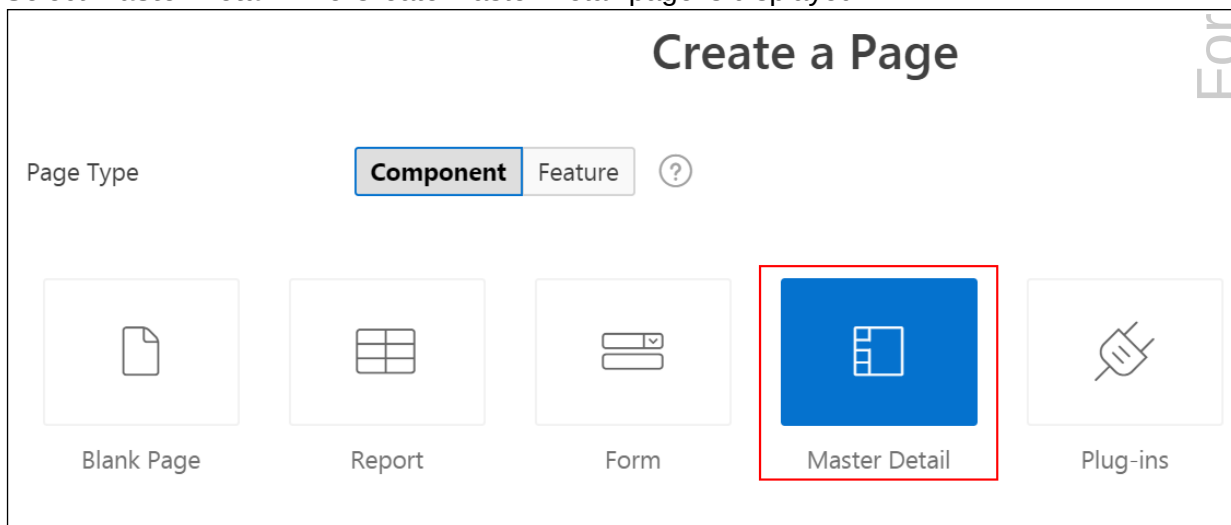
In this practice solution, steps are provided for creating a Master Detail (Stacked) Form for the GMT application.

Steps

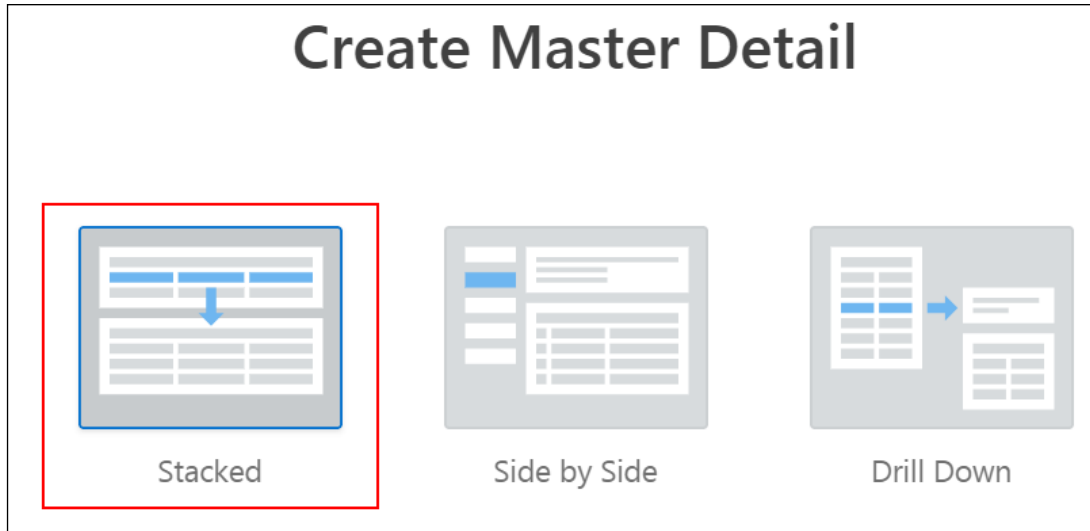
1. On a new page called **Master Detail**, create a Master Detail (Stacked) Form based on the `ORDERS` and `ORDER_ITEMS` tables.
 - Include a report page called **Orders List** on the master table.
 - Display the master table and detail table forms on a single page.
 - Include navigation based on the `ORDER_ID` column.
- a. On the GMT Application Home page, click **Create Page**.



- b. Select **Master Detail**. The Create Master Detail page is displayed.



- c. Click **Stacked**.



- d. Enter the Page Attributes:

- **Page Number:** Enter 15
- **Page Name:** Enter Orders List

The screenshot shows the "Create Master Detail" dialog with the "Page Attributes" section. A red box highlights the "Page Number" and "Page Name" fields. The "Page Number" field contains the value "15" and the "Page Name" field contains the text "Orders List". Below these fields, the "Page Mode" is set to "Normal" and the "Breadcrumb" is set to "- do not use breadcrumbs on page -".

- e. Click **Next >**.

- f. Accept the default for **Navigation Menu** and click **Next >**.

The screenshot shows the "Create Master Detail" dialog with the "Navigation Menu" section. A red box highlights the "Navigation Preference" section, which has a radio button selected next to the text "Do not associate this page with a navigation menu entry". Below this, there are two unselected radio buttons: "Create a new navigation menu entry" and "Identify an existing navigation menu entry for this page".

- g. For Master Source, enter the following and click **Next >**:
- **Master Region Title:** Enter `Orders List`
 - **Table/View Owner:** Selected by default `OEHR`.
 - **Table / View Name:** Click  icon and select `OEHR_ORDERS (table)` from the list.

- **Primary Key Column:** Select ORDER_ID (Number) from the drop-down list.

h. For Detail Source:

- **Detail Region Title:** Enter Item Details
- **Show only Related Tables:** Ensure Yes is selected.
- **Table/ View Name:** Click  icon and select OEHR_ORDER_ITEMS.
- **Primary Key Column:** Select ORDER_ITEM_ID (Number) from the drop-down list.
- **Master Detail Foreign Key:** Select ORDER_ID -> ORDER_ID from the drop-down list.

- i. Click **Create**.

Create Master Detail

Detail Source

* Detail Region Title: Item Details

Show Only Related Tables: ☒ Yes ☐ No

Table / View Owner: OEHR

* Table / View Name: OEHR_ORDER_ITEMS

* Primary Key Column: ORDER_ITEM_ID (Number)

Primary Key Column 2: - Select Column -

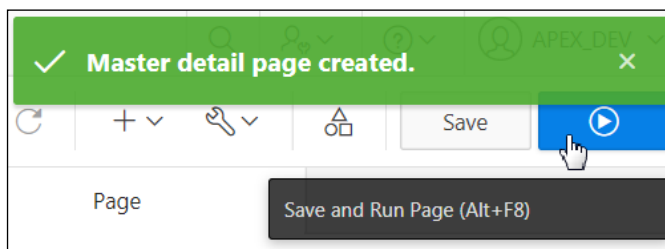
* Master Detail Foreign Key: ORDER_ID -> ORDER_ID

* Select Columns

- ORDER_ID (Number)
- LINE_ITEM_ID (Number)
- PRODUCT_ID (Number)
- UNIT_PRICE (Number)
- QUANTITY (Number)
- ORDER_ITEM_ID (Number)

< Cancel Create

- j. The Master Detail Stacked page is created. Click **Save and Run Page**, to view the Master Detail Form.



- k. The Master detail Form is displayed. Review the information. Click **Application 111** on the Runtime Developer toolbar.

Orders List					
Search: All Text Columns Go Actions Edit Add Row					
☐		Order Date	Order Mode	Customer Id	Order Status
☒		24-APR-18 02.34.12.0...	direct	101	0
☐		28-JUL-18 03.41.54.00...	direct	102	1
☐		10-JUN-18 04.49.34.0...	direct	103	1
☐		23-MAR-19 05.18.23.0...	direct	104	0
☐		16-SEP-18 06.03.12.00...	direct	105	2
☐		21-JAN-19 07.59.08.0...	direct	106	3

Item Details		
Search: All Text Columns Go Actions Edit Add Row		
☐		Line Item Id
☒		6
☐		1
☐		2
☐		3
☐		4
☐		5

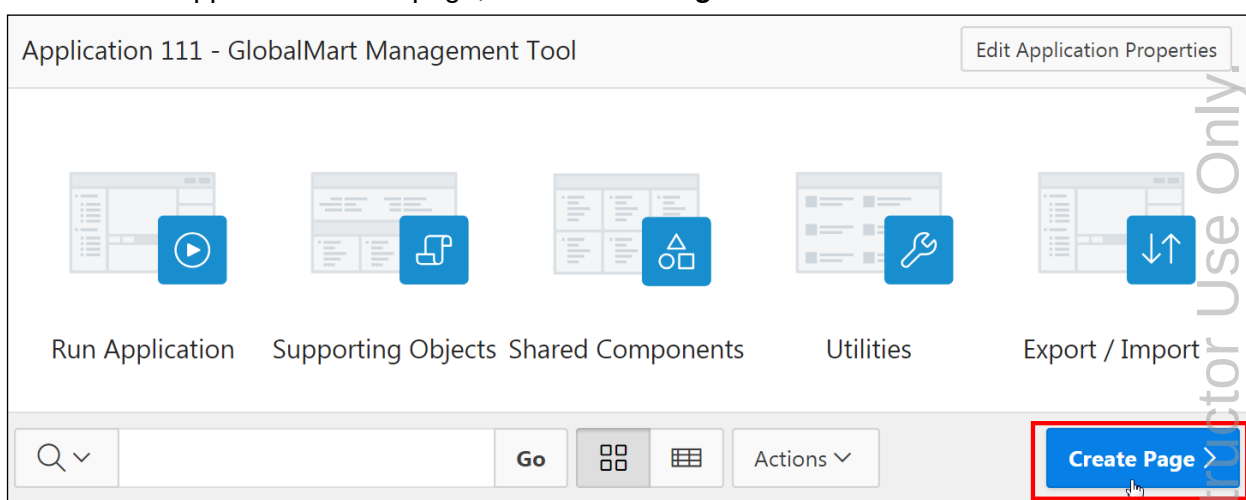
Solution 8-5: Creating a Master Detail Form (Drill Down)

Overview

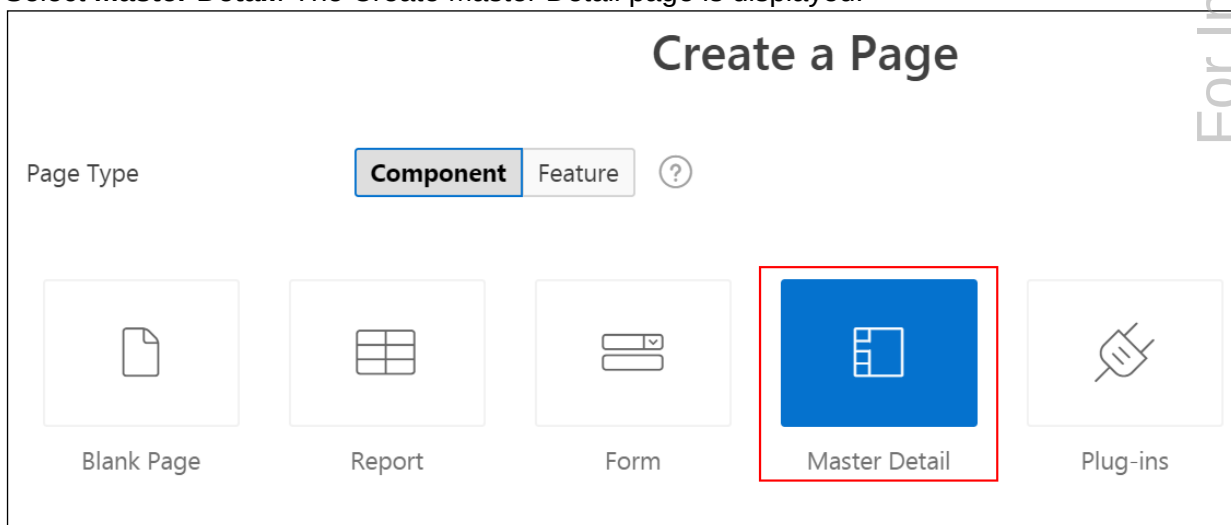
In this practice solution, steps are provided for creating a Master Detail (Drill Down) Form for the GMT application.

Steps

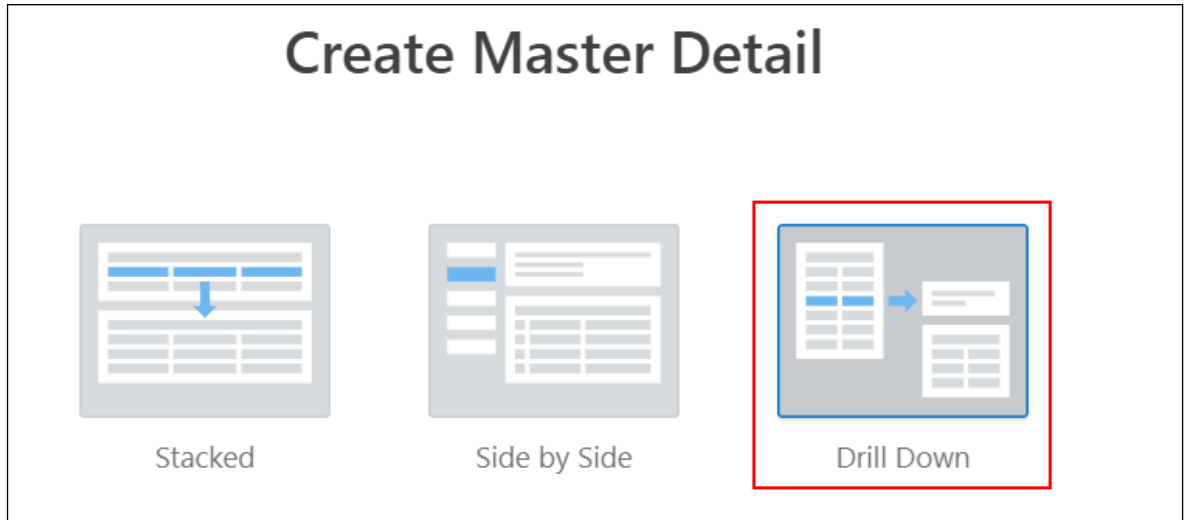
1. On a new page called **Master Detail**, create a Master Detail (Drill Down) Form based on the `OEHR_CUSTOMER` and `OEHR_ORDERS` tables.
 - Include a report page called **Customer Master Information** on the master table.
 - Include navigation based on the `CUST_FIRST_NAME` column.
- a. On the GMT Application Home page, click **Create Page**.



- b. Select **Master Detail**. The Create Master Detail page is displayed.



- c. Click **Drill Down**.



For Instructor Use Only.
This document should not be distributed.

d. Define the Page Attributes and click **Next >**:

- **Master Page Number:** Enter 16
- **Master Page Name:** Enter Customer Master Information
- **Detail Page Number:** Enter 17
- **Detail Page Name:** Enter Customer Order History

The screenshot shows the 'Create Master Detail' wizard with a progress bar at the top. The first step, 'Page Attributes', is highlighted with a blue dot. Below the title, there are four input fields, each with a red asterisk indicating a required field. The first field is 'Master Page Number' with the value '16'. The second field is 'Master Page Name' with the value 'Customer Master Information'. The third field is 'Detail Page Number' with the value '17'. The fourth field is 'Detail Page Name' with the value 'Customer Order History'. At the bottom, there is a 'Breadcrumb' dropdown menu set to '- do not add breadcrumb region to page -'. A red box highlights the four required fields.

e. Accept the default for **Navigation Menu** and click **Next >**.

The screenshot shows the 'Create Master Detail' wizard with a progress bar at the top. The second step, 'Navigation Menu', is highlighted with a blue dot. Below the title, there is a 'Navigation Preference' section. It contains a radio button selected for 'Do not associate this page with a navigation menu entry'. There are two other radio button options: 'Create a new navigation menu entry' and 'Identify an existing navigation menu entry for this page'. A red box highlights the selected radio button.

f. For Master Source:

- **Table/View Owner:** Selected by default OEHR.
- **Table / View Name:** Click  icon and select OEHR_CUSTOMERS (table) from the list.
- **Primary Key Column:** Select CUSTOMER_ID (Number) from the drop-down list.
- **Form navigation Order:** Select CUST_FIRST_NAME (Varchar2) from the drop-down list.

g. Click **Next >**.

Create Master Detail

Master Source

* Table / View Owner: OEHR

* Table / View Name: OEHR_CUSTOMERS (table)

* Primary Key Column: CUSTOMER_ID (Number)

Primary Key Column 2: - Select Column -

* Form Navigation Order: CUST_FIRST_NAME (Varchar2)

* Select Columns

CUSTOMER_ID (Number)
CUST_FIRST_NAME (Varchar2)
CUST_LAST_NAME (Varchar2)
STREET_ADDRESS (Varchar2)
POSTAL_CODE (Varchar2)
CITY (Varchar2)
STATE_PROVINCE (Varchar2)
COUNTRY_ID (Char)

< Cancel Next >

h. For Detail Source:

- **Table/ View Name:** Click  icon and select OEHR_ORDERS.
- **Primary Key Column:** Select ORDER_ID (Number) from the drop-down list.
- **Primary Key Column 2:** Select CUSTOMER_ID (Number) from the drop-down list.
- **Master Detail Foreign Key:** Select CUSTOMER_ID -> CUSTOMER_ID from the drop-down list.

- i. Click **Create**.

Create Master Detail

Detail Source

Show Only Related Tables: ☒ Yes ☐ No ?

Table / View Owner: **OEHR** ?

* Table / View Name: OEHR_ORDERS ?

* Primary Key Column: ORDER_ID (Number) ?

Primary Key Column 2: CUSTOMER_ID (Number) ?

* Master Detail Foreign Key: CUSTOMER_ID -> CUSTOMER_ID ?

* Select Columns ?

ORDER_ID (Number)
ORDER_DATE (Timestamp(6) With Local Time Zone)
ORDER_MODE (Varchar2)
CUSTOMER_ID (Number)
ORDER_STATUS (Number)
ORDER_TOTAL (Number)
SALES_REP_ID (Number)
PROMOTION_ID (Number)

< Cancel Create

- j. The Master Detail Drill Down page is created. Click **Save** and then **Save and Run Page**, to view the Master Detail Form.

✓ Master detail page created.

Save Save and Run Page (Alt+F8)

- k. The Master Detail Form is displayed. Review the information.

GlobalMart Management Tool

Home

Customers

Products

Customers List View

	Customer Id	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province
	218	Bryan	Dvrrie	3386 Perrysville Ave	15214	Pittsburgh	PA
	219	Ajay	Sen	220 Penn Ave # 300	18503	Scranton	PA
	220	Carol	Jordan	135 S 18Th St # 1	19103	Philadelphia	PA

- l. Select a customer, for example, Eddie Bowyer (Customer Id: 159) and click the edit icon.

	Customer Id	First Name	Last Name	Street Address	Postal Code	City	State Province	Country	Phone Number	Nls Language
	159	Eddie	Boyer	411 E Wisconsin Ave # 2550	53202	Milwaukee	WI	US	+1 414 123 4369	us
	160	Eddie	Stern	808 3Rd St # 100	54403	Wausau	WI	US	+1 715 123 4372	us

The Customer Master information and Order Details appear.

Cust Email
Eddie.Boyer@GALLINULE.COM

Account Mgr Id

112 of 321

☒		Order Date	Order Mode	Order Status	Order Total
☒		18-MAR-18 12.34.29.00...	direct	2	69286.4

Click **Application 111** on the Runtime Developer toolbar.



For Instructor Use Only.
This document should not be distributed.

Practices for Lesson 9: Adding Items and Buttons

Practices for Lesson 9: Overview

Overview

There are two practices for this lesson. In these practices, you create a variety of items and buttons and manipulate the way they work.

For Instructor Use Only.
This document should not be distributed.

Practice 9-1: Adding Items and Buttons

Overview

In this practice, you create a Customer Feedback form that contains some items and two buttons. You create a button on the Customers page to link to the feedback form.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Create a blank page called **Customer Feedback**. Enter the page number as 18.
2. Add the following items to the form:

Items	Type
Customer ID	Text Field
Feedback	Text Area
Added On	Date Picker

3. Add a button on the Customers page to redirect to the Customer Feedback page.
4. Add an **Apply** button on the Customer Feedback page.
5. Add a **Cancel** button on the Customer Feedback page to redirect to the Customers page.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 9-1**.

If you haven't completed **Practice 8-5**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.

- c. Click the **Browse** button.
- d. Browse and open the `sol_08_05.sql` file in the `/home/oracle/labs/solutions` directory and click **Next**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

For Instructor Use Only.
This document should not be distributed.

Practice 9-2: Manipulating Items

Overview

In this practice, you add additional item types (select list and cascading LOV) and modify the way the fields in Customer Details forms are displayed.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Modify the Account Manager item to be a select list with the following query (located in /home/oracle/labs/labs/lab_09_02_01.txt):

```
select LAST_NAME||', '||FIRST_NAME display_value, EMPLOYEE_ID
return_value
from OEHR_EMPLOYEES
where DEPARTMENT_ID=80
order by 1
```

2. Add a quick pick list to the Account Manager item with the following labels and values:

Label	Value
Bates, Elizabeth	172
Russell, John	145
Cambrault, Gerald	148

3. Add a select list item called **Country** that contains the following query (located in /home/oracle/labs/labs/lab_09_02_02.txt). This item should allow nulls.

```
select COUNTRY_NAME display_value, COUNTRY_ID return_value
from OEHR_COUNTRIES
order by 1
```

4. Change the Account Manager of an existing customer record.

Note: You must navigate to the customer report and edit one of the records.

5. Create and Modify the Employee Detail form page to take advantage of some of the new HTML5 item types and add a LOV using the following query (located in /home/oracle/labs/labs/lab_09_02_03.txt).

```
select first_name||' '||last_name d, employee_id r
from oehr_employees
```

6. Change the Account Manager of an existing customer record.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 9-2**.

If you haven't completed **Practice 9-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_09_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

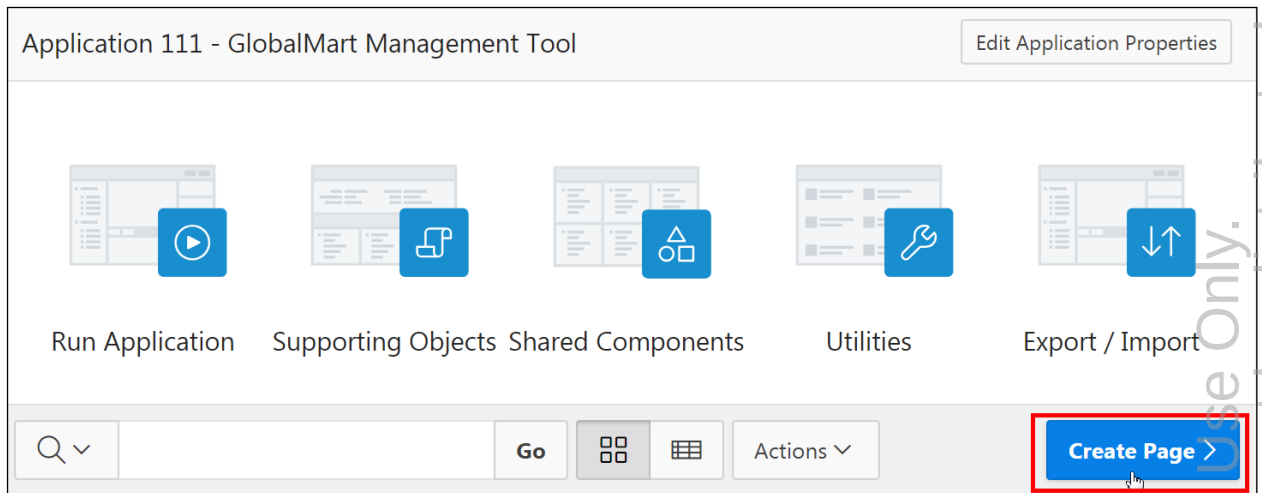
Solution 9-1: Adding Items and Buttons

Overview

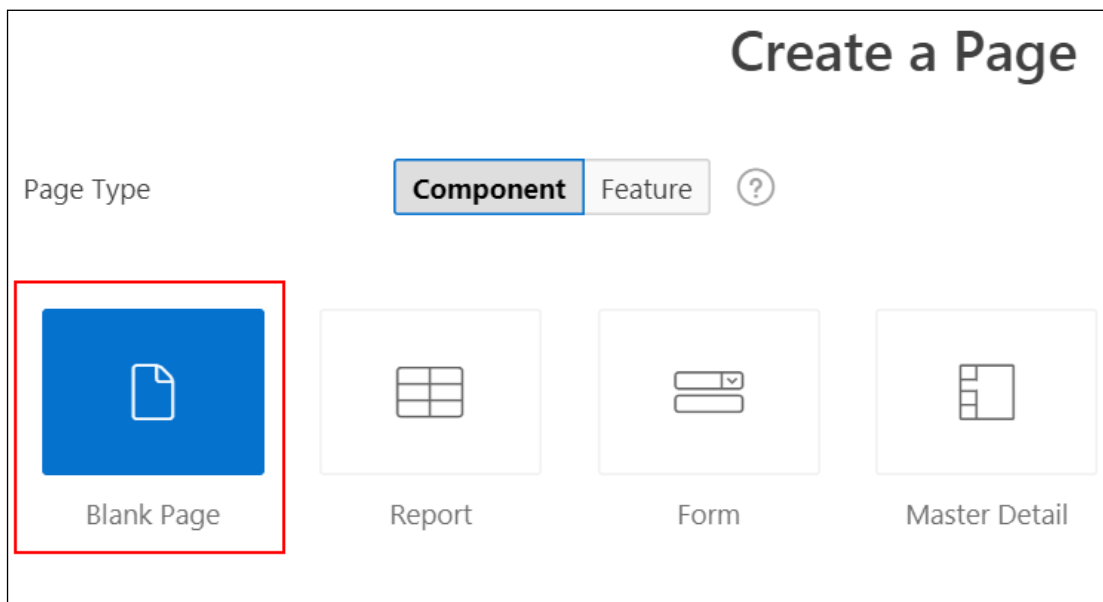
In this practice solution, steps are provided for adding items and buttons to the GMT application.

Steps

1. Create a blank page called **Customer Feedback**. Enter the page number as 18.
 - a. On the GMT Application Home page, click **Create Page**.



- b. Select **Blank Page**.



- c. Enter the Page Attributes:
- **Page Number:** Enter 18
 - **Name:** Enter Customer Feedback

The screenshot shows the 'Create a Blank Page' dialog box with a progress bar at the top. The first step, 'Page Attributes', is active and highlighted with a blue dot. Below the title bar, there are several input fields: 'Page Number' (with a red asterisk and a value of 18), 'Name' (with a red asterisk and the value 'Customer Feedback'), 'Page Mode' (with a red asterisk and options: Normal, Modal Dialog, Non-Modal Dialog), 'Page Group' (a dropdown menu with '- Select Page Group -'), and 'Breadcrumb' (a dropdown menu with '- don't use breadcrumbs on page -'). A red box highlights the 'Page Number' and 'Name' fields. At the bottom, there is a section titled 'Optional Static Content Regions'.

- d. Click **Next >**.
- e. Accept the default for **Navigation Menu** and click **Next >**.

The screenshot shows the 'Create a Blank Page' dialog box with a progress bar at the top. The second step, 'Navigation Menu', is active and highlighted with a blue dot. The first step is completed, indicated by a green checkmark. Below the title bar, there is a section titled 'Navigation Menu'. It contains a 'Navigation Preference' label and three radio button options: 'Do not associate this page with a navigation menu entry' (which is selected and highlighted with a red box), 'Create a new navigation menu entry', and 'Identify an existing navigation menu entry for this page'. A help icon (?) is visible next to the first option.

- f. The Confirm page opens. Click **Finish**.

Create a Blank Page

Confirm

You have requested to create a page with the following attributes. Please confirm your selections.

Application	111
Page	18
Page Name	Customer Feedback
Page Title	Customer Feedback

< Cancel Finish

- g. The **Page 18: Customer Feedback** page opens in Page Designer.

ORACLE App Builder SQL Workshop Team Development App Gallery

✓ Page 18 created.

Application 111 \ Page Designer

18 Go

Rendering Dynamic ... Processing Page Shar...

Page 18: Customer Feedback

- Pre-Rendering
- Regions
 - Before Content Body
 - Content Body
- Post-Rendering

Customer Feedback

- PAGE HEADER
- PAGE NAVIGATION
- BREADCRUMB BAR
- BEFORE CONTENT BODY
- CONTENT BODY

Page

Filter

Identification

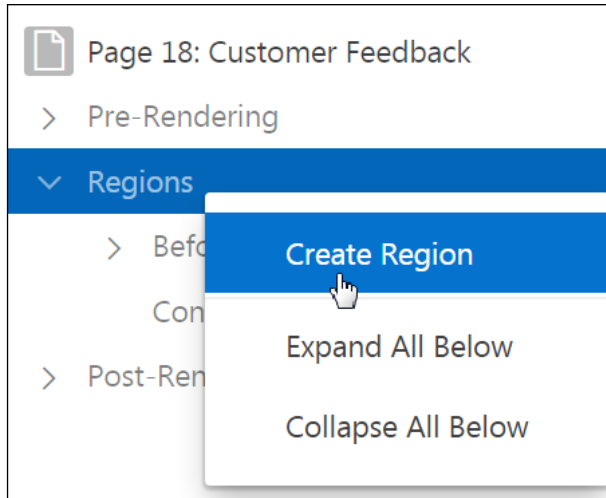
Name Customer Feedback

Page Alias

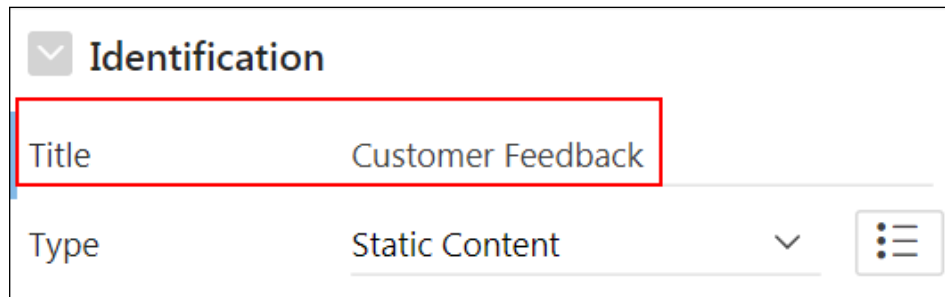
Title Customer Feedback

Page Group - Select -

- h. To create a region on this page, right-click **Regions** and select **Create Region**.



- i. In the Property Editor (the right pane), under **Identification > Title**, enter Customer Feedback.

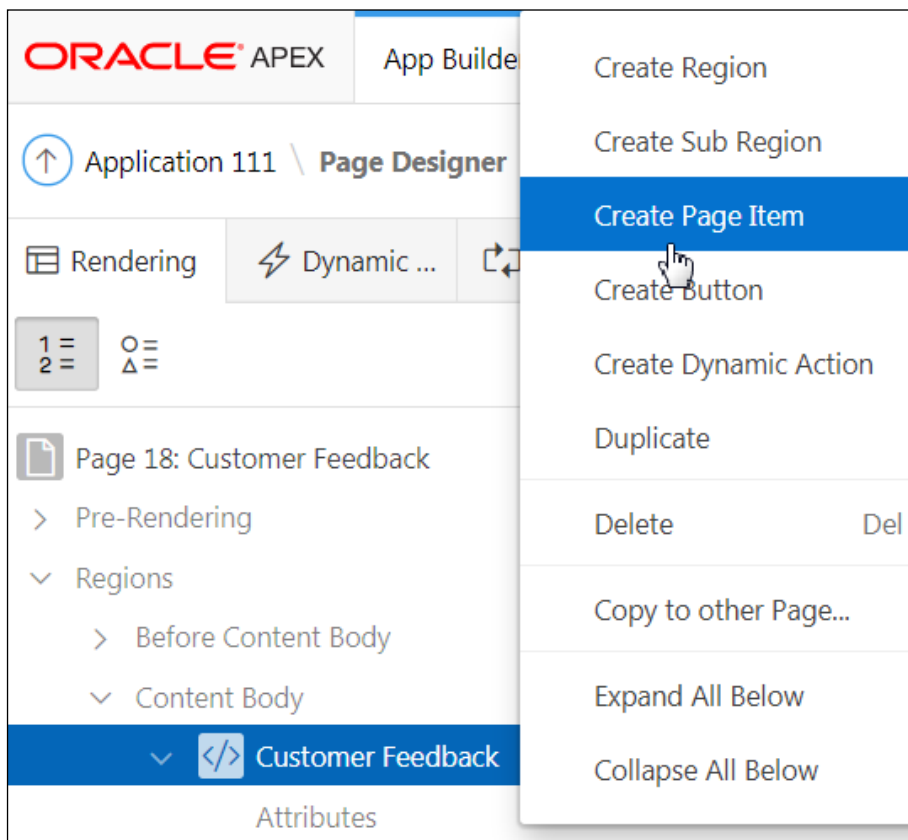


- j. Click **Save**.

2. Add the following page items to the Customer Feedback form:

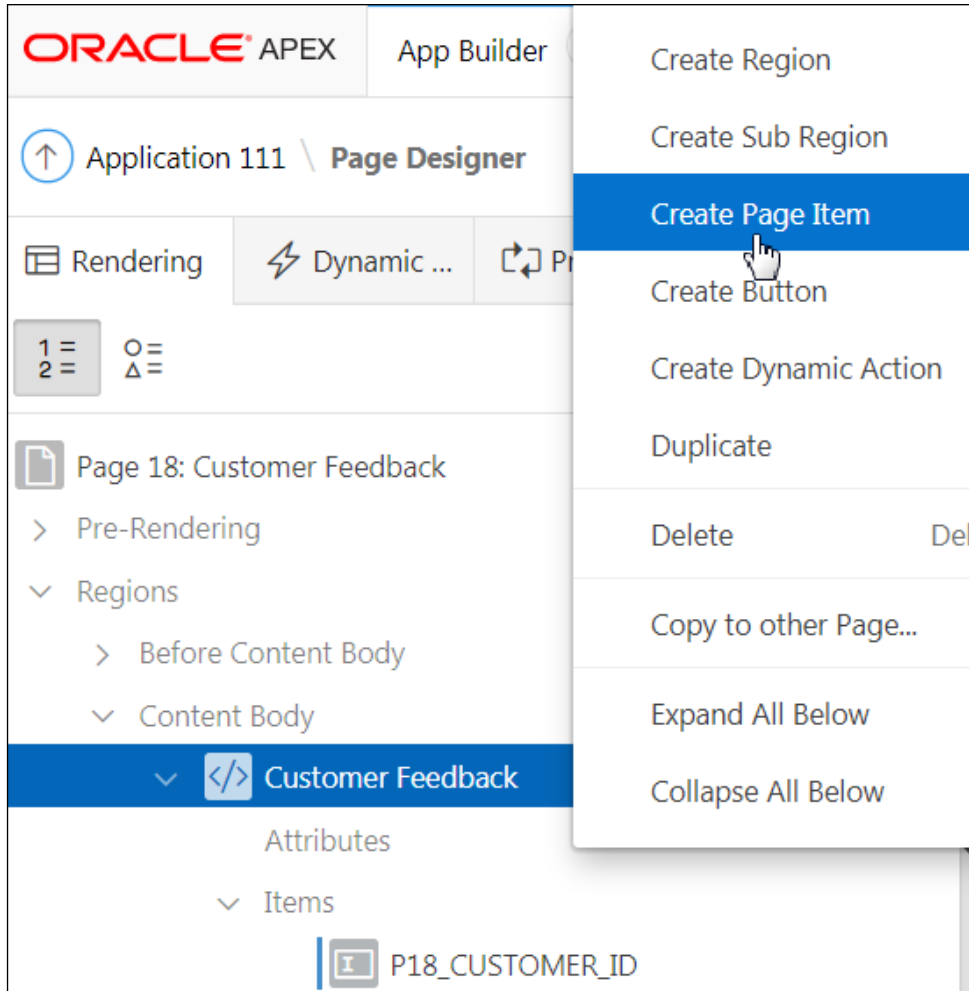
Items	Type
Customer ID	Text Field
Feedback	Text Area
Added On	Date Picker

- a. To create page items in the **Customer Feedback** region, right-click **Customer Feedback** and select **Create Page Item**.

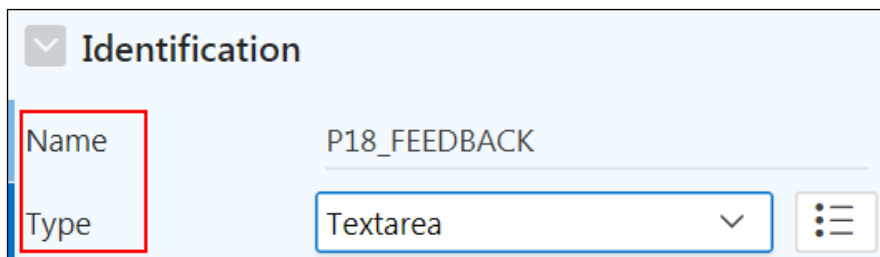


- b. Select the Page Item.
- c. In the Property Editor (the right pane):
 - **Identification > Name:** Enter P18_CUSTOMER_ID
 - **Identification > Type:** Select **Text Field** (if not selected by default)
 - **Source > Type:** Select Database Column

- e. Right-click **Customer Feedback** and select **Create Page Item**.



- f. In the Property Editor (the right pane):
- **Identification > Name:** Enter `P18_FEEDBACK`
 - **Identification > Type:** Select **Textarea**
 - **Source > Type:** Select **Database Column**
 - **Source > Database Column:** Enter `FEEDBACK`



Source

Form Region

- Select -

>

Type

Database Column

>

⋮

Database Column

FEEDBACK

Used

Always, replacing any existing valu

>

Maintain Session State

Per Session (Disk)

>

- g. Right-click **Customer Feedback** and select **Create Page Item**.

ORACLE APEX

App Builder

Application 111

Page Designer

Rendering

Dynamic ...

P

1 =

2 =

O =

Δ =

Page 18: Customer Feedback

> Pre-Rendering

> Regions

> Before Content Body

> Content Body

> **Customer Feedback**

Attributes

> Items

P18_CUSTOMER_ID

P18_FEEDBACK

Create Region

Create Sub Region

Create Page Item

Create Button

Create Dynamic Action

Duplicate

Delete

Copy to other Page...

Expand All Below

Collapse All Below

- h. In the Property Editor (the right pane):
- **Identification > Name:** Enter `P18_ADDED_ON`
 - **Identification > Type:** Select **Date Picker**
 - **Source > Type:** Select **Database Column**

- **Source > Database Column:** Enter ADDED_ON

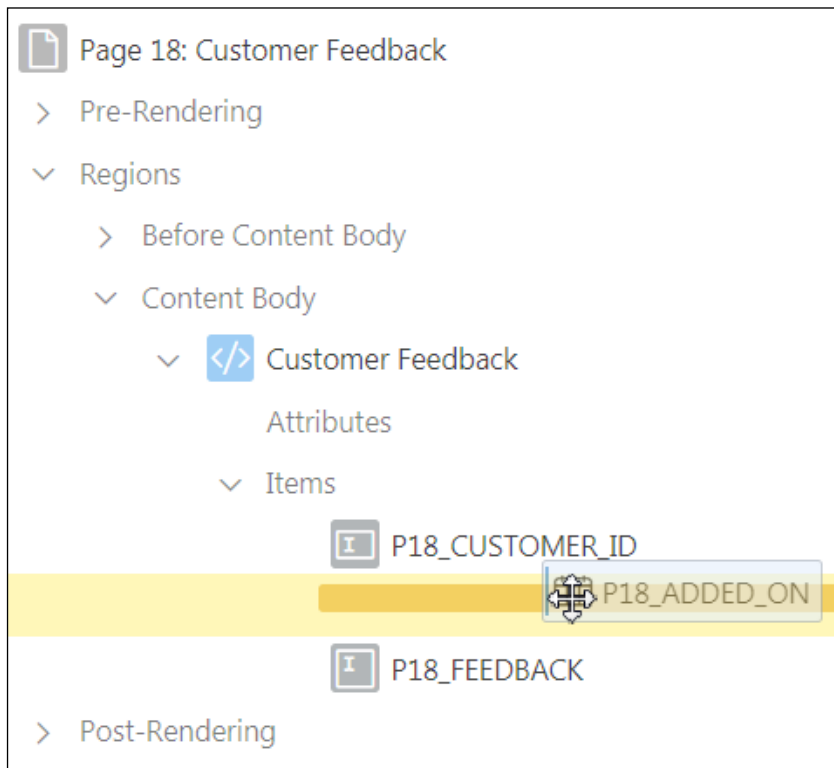
▼ Identification

Name	P18_ADDED_ON	
Type	Date Picker	⋮

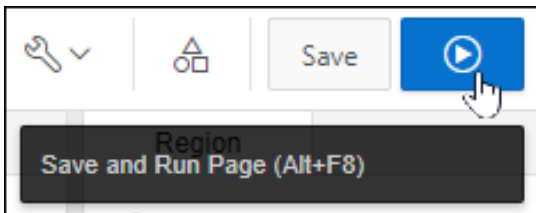
▼ Source

Form Region	- Select -	▼	>
Type	Database Column	▼	⋮
Database Column	ADDED_ON		
Used	Always, replacing any existing valu		▼
Maintain Session State	Per Session (Disk)		▼

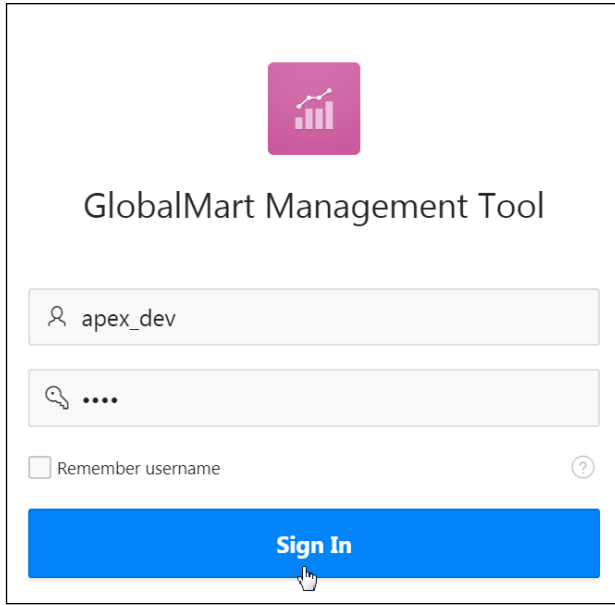
- i. You can drag and drop an item to reposition the item in the list. Drag the P18_ADDED_ON item to the above P18_FEEDBACK item.



- j. Click **Save and Run Page**.

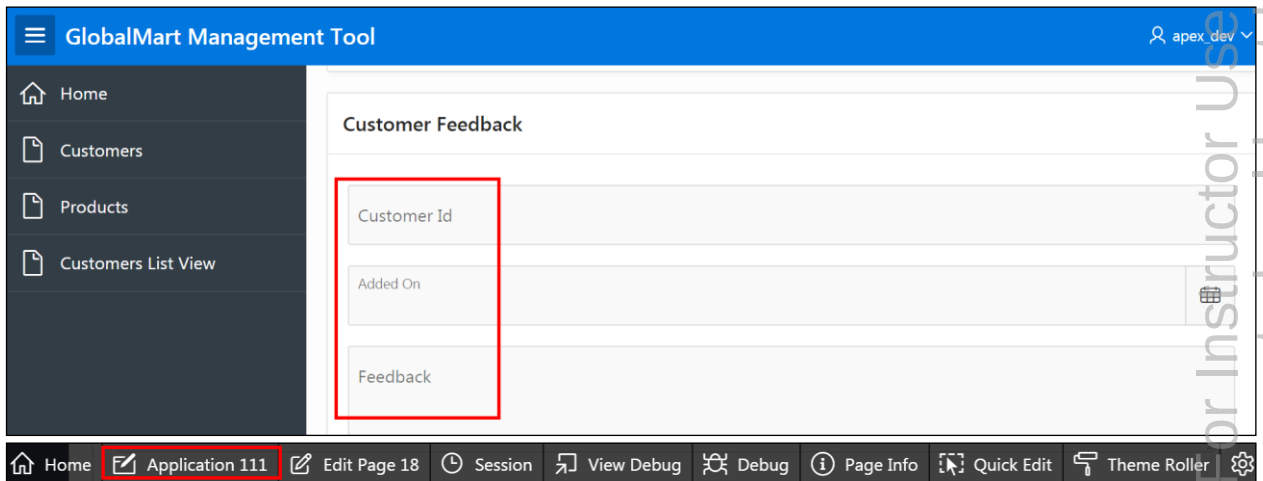


- k. On the sign in page, enter your user credentials if prompted.



The image shows the sign-in page for the GlobalMart Management Tool. At the top is a purple square icon with a white bar chart. Below it is the title "GlobalMart Management Tool". There are two input fields: the first contains the username "apex_dev" and the second contains masked characters "....". Below the input fields is a checkbox labeled "Remember username" and a help icon (?). At the bottom is a large blue button labeled "Sign In". A mouse cursor is pointing at the "Sign In" button.

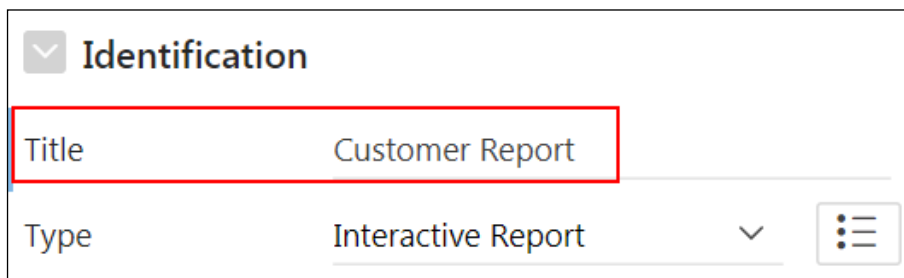
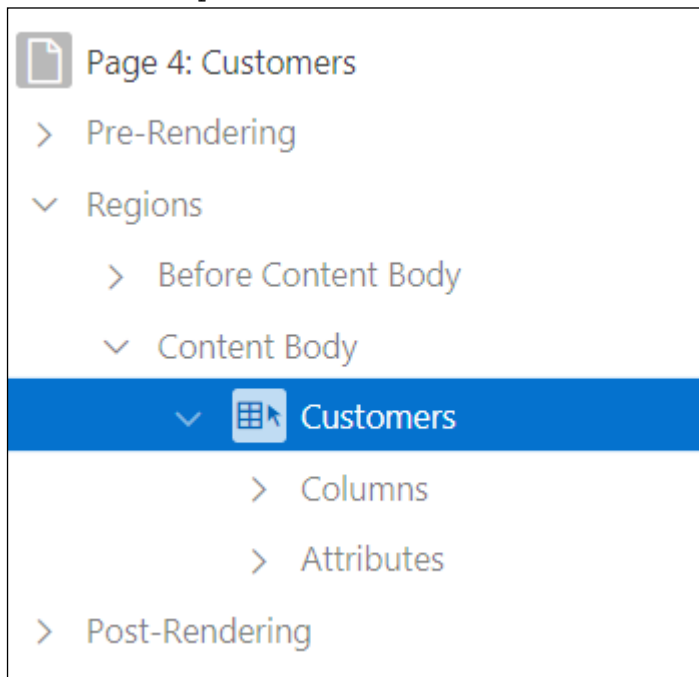
- l. The Customer Feedback page loads with all the fields. Click the **Application 111** link on the Runtime Developer toolbar.



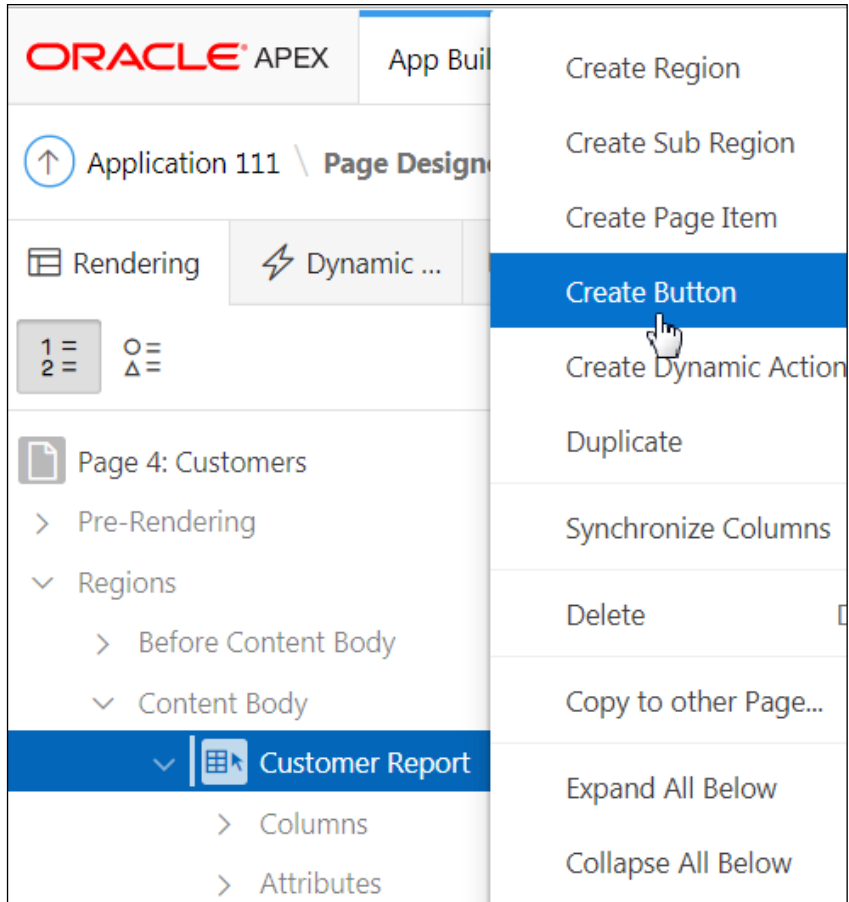
The image shows the "Customer Feedback" page within the GlobalMart Management Tool. The page has a blue header with the title "GlobalMart Management Tool" and a user profile "apex dev". On the left is a dark sidebar with navigation links: Home, Customers, Products, and Customers List View. The main content area is titled "Customer Feedback" and contains three input fields: "Customer Id", "Added On", and "Feedback". A red rectangle highlights these three input fields. At the bottom is a Runtime Developer toolbar with various icons and labels: Home, Application 111 (highlighted with a red box), Edit Page 18, Session, View Debug, Debug, Page Info, Quick Edit, and Theme Roller.

3. Add a button `Customer Feedback` on the Customers page (page 4) to redirect to the Customer Feedback page.
- a. Click Page 4 - **Customers** page.
The Customers page opens in Page Designer.

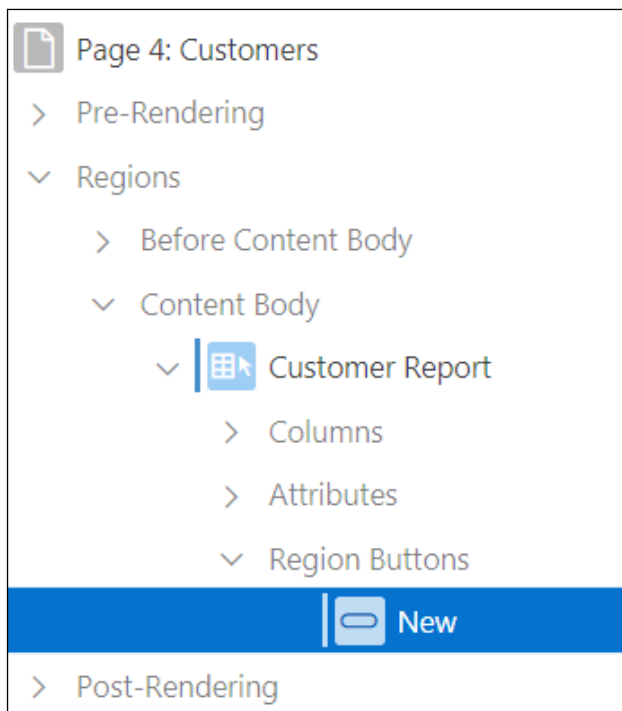
- b. Click **Customers** under Content Body. In **Property Editor > Identification**, enter Customer Report in the **Title** field.



- c. Now, right-click the **Customer Report** region and select **Create Button**.



- d. Select **Customer Report > Region Buttons > New**.



- e. In the Property Editor (the right pane):
- **Identification > Button Name:** Enter `CUSTOMER_FEEDBACK`.
 - **Layout > Button Position:** Select **Right of Interactive Report Search Bar** from the drop-down list.
 - **Appearance > Template Options > Common > Size:** Select **Large** from the drop-down list and click **OK**.
 - **Behavior > Action:** Select **Redirect to Page in this Application** from the drop-down list.
 - **Behavior > Target > Page:** Select **Customer Feedback (Page 18)**.

Identification

Button NameCUSTOMER_FEEDBACK

LabelCustomer Feedback

Layout

Sequence10

RegionCustomer Report

Button PositionRight of Interactive Report Search Bar

Template Options

Common

General

Use Template Defaults

SizeLarge

TypeNormal

StyleDefault

Advanced

Button SetDefault

WidthAuto - Default

Spacing LeftDefault

Spacing RightDefault

Spacing TopDefault

Cancel

OK

▼ Behavior

Action Redirect to Page in this Applica ▼

✖ Target No Link Defined

Warn on Unsaved Changes Do Not Check ▼

Database Action - Select - ▼

Link Builder - Target

▼ Target

Type Page in this application

Page 18

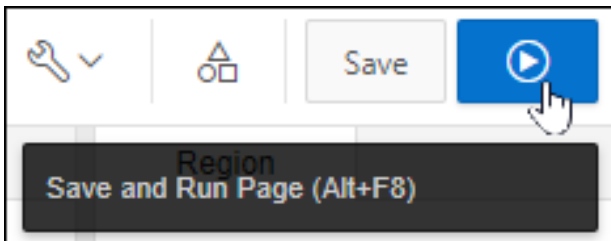
▼ Behavior

Action Redirect to Page in this Application ▼

Target Page 18

Warn on Unsaved Changes Do Not Check ▼

- f. Click **Save and Run Page**.



- g. Click **Customer Feedback**.

GlobalMart Management Tool

apex_dev

Home

Customers

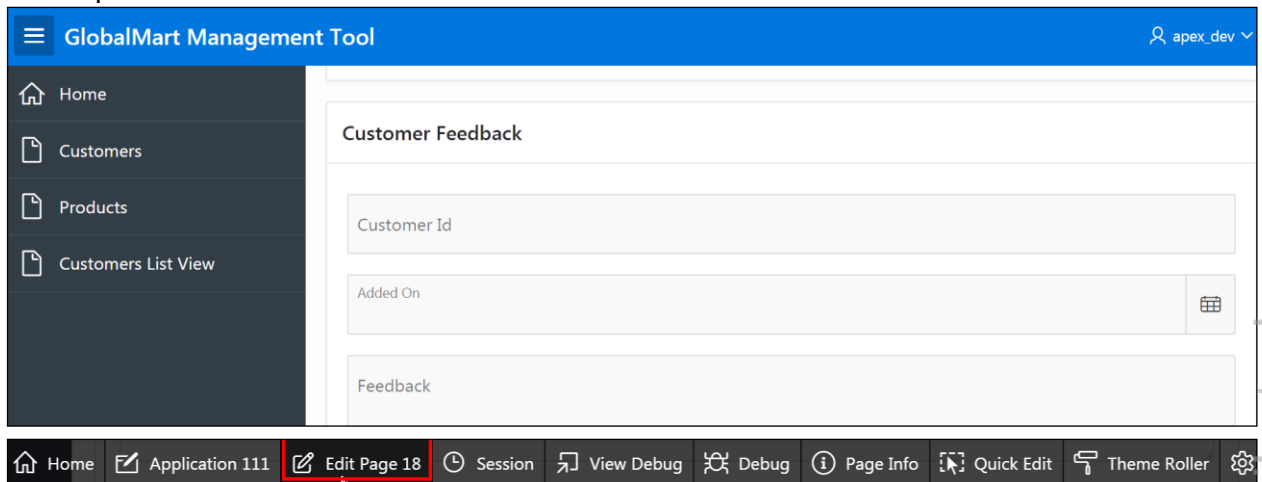
Products

Customers List View

Customer Feedback

	Customer Id	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province	Country Id	Phone Number
	218	Bryan	Dvrie	3386 Perrysville Ave	15214	Pittsburgh	PA	US	+1 412 123 4740
	219	Ajay	Sen	220 Penn Ave # 300	18503	Scranton	PA	US	+1 717 123 4741

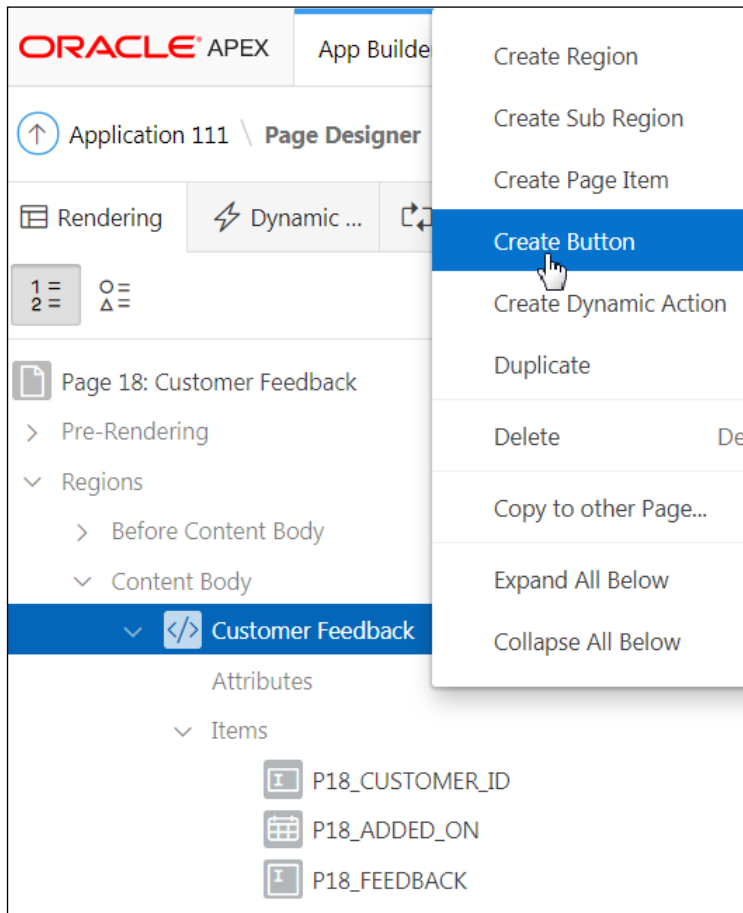
The Customer Feedback page is displayed. Click the **Edit Page 18** link on the Runtime Developer toolbar.



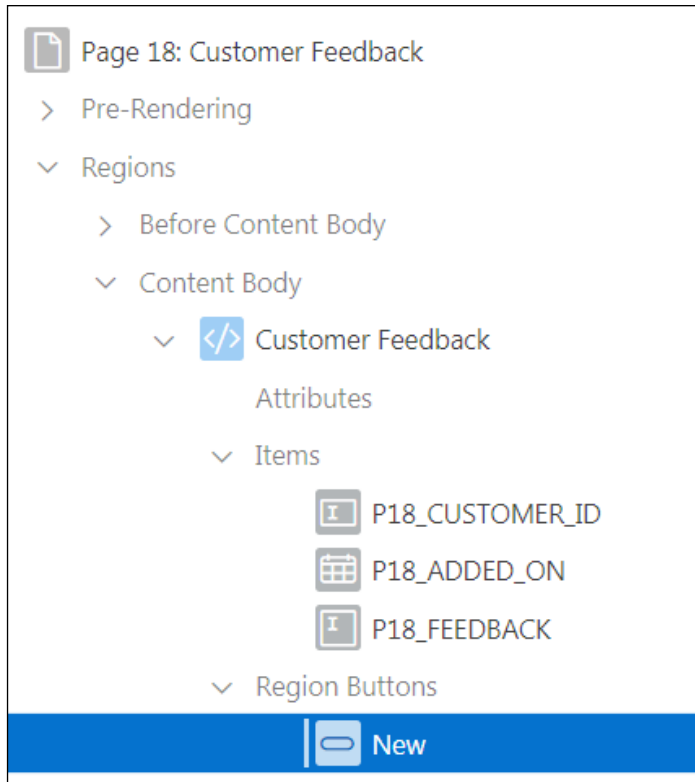
4. Add an **Apply** button on the Customer Feedback page.

For Instructor Use Only.
This document should not be distributed.

- a. The **Page 18: Customer Feedback** opens in Page Designer view. You want to create an **Apply** button on this page.
- b. Right-click the **Customer Feedback** region and select **Create Button**.



- c. Select the **New** button.

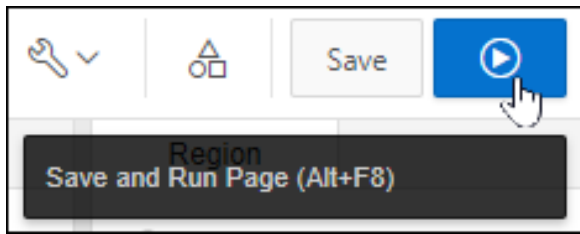


- d. In the Property Editor (the right pane):
- **Identification > Button Name:** Enter `APPLY_CHANGES`
 - **Identification > Label:** Enter `Apply`
 - **Appearance > Hot:** Click **Yes**
 - **Appearance > Template Options > Default Template Options:** In the Template Options dialog, under **Common > Size**, select **Large** from the drop-down list.

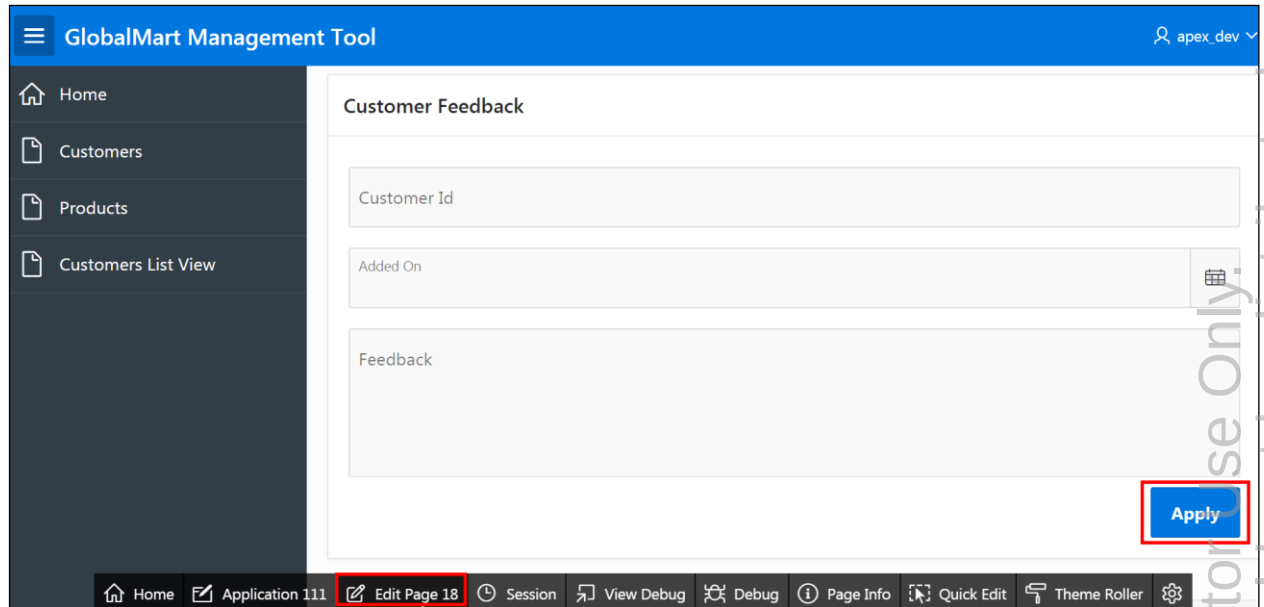
- Click **OK**.

The image displays two screenshots of the ADF component inspector. The top screenshot shows the 'Identification' and 'Appearance' tabs. In the 'Identification' tab, the 'Button Name' is 'APPLY_CHANGES' and the 'Label' is 'Apply'. In the 'Appearance' tab, the 'Button Template' is 'Text', the 'Hot' checkbox is checked, and the 'Template Options' dropdown is set to 'Use Template Defaults'. The bottom screenshot shows the 'Template Options' dialog. In the 'Common' tab, the 'Use Template Defaults' checkbox is checked, and the 'Size' dropdown is set to 'Large'. In the 'Advanced' tab, all settings (Button Set, Width, Spacing Left, Spacing Right, Spacing Top) are at their default values. The 'OK' button is highlighted in the bottom right corner.

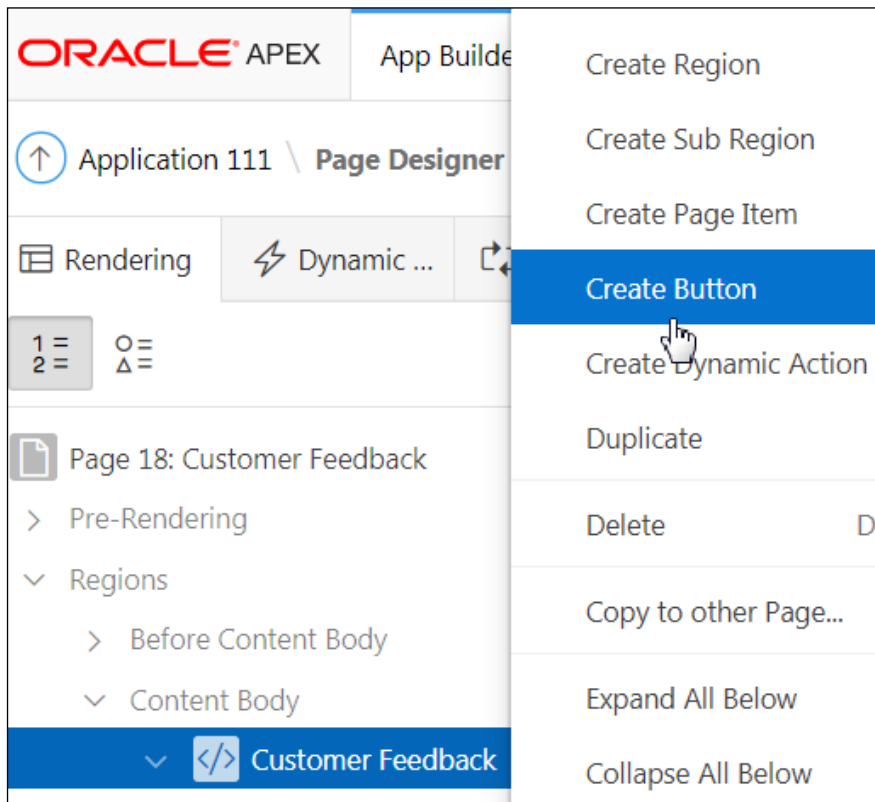
- Click **Save** and then **Save and Run Page**.



- f. Notice that the **Apply** button is created. Click the **Edit Page 18** link on the Runtime Developer toolbar.



5. Add a **Cancel** button on the Customer Feedback page to redirect to the Customers page.
 - a. To redirect to the **Customer Feedback** page when the **Cancel** button is clicked, right-click the **Customer Feedback** region and select **Create Button**.



- b. Select the **New** button.
- c. In the Property Editor (the right pane):
 - **Identification > Button Name:** Enter `CANCEL`.
 - **Identification > Label:** Enter `Cancel` (populates by default).
 - **Appearance > Template Options > Default Template Options:** In the Template Options dialog, under **Common > Size**, select **Large** from the drop-down list.
 - Click **OK**.
 - **Behavior > Action:** Select **Redirect to Page in this Application** from the drop-down list.
 - **Behavior > Target > Page:** Select **Page 4 - Customers**.
 - Click **OK**.

Identification

Button Name	CANCEL
Label	Cancel

Layout

Appearance

Button Template	Text	>
Hot	Yes	No
Template Options	Use Template Defaults	
CSS Classes		

Template Options

Common

General ☒ Use Template Defaults

Size	Large	>
Type	Normal	>
Style	Default	>

Advanced

Button Set	Default	>
Width	Auto - Default	>
Spacing Left	Default	>
Spacing Right	Default	>
Spacing Top	Default	>

Cancel OK

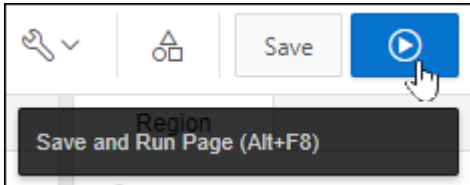
Behavior

Action: Redirect to Page in this Application

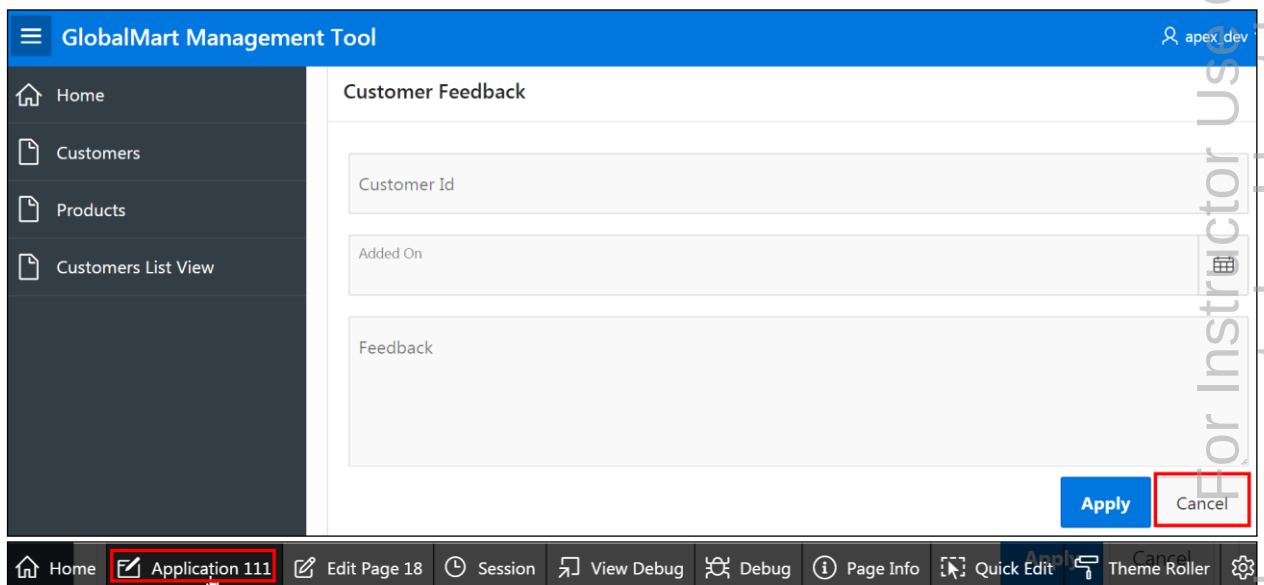
Target: Page 4

Warn on Unsaved Changes: Do Not Check

- d. Click **Save** and then **Save and Run Page**.



- e. The **Cancel** button is created successfully. Observe that when you click the **Cancel** button, it takes you to the Customers report. Click the **Application 111** link on the Runtime Developer toolbar.



Solution 9-2: Manipulating Items

Overview

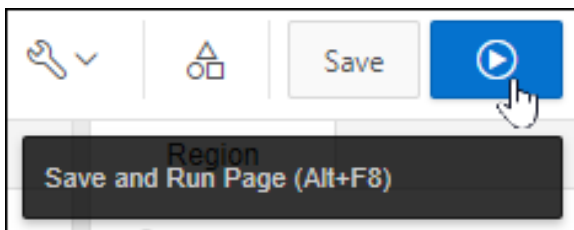
In this practice solution, steps are provided for modifying the items in the GMT application.

Steps

1. Modify the Account Manager item to be a select list with the following query (located in /home/oracle/labs/labs/lab_09_02_01.txt):

```
select LAST_NAME||', '||FIRST_NAME display_value, EMPLOYEE_ID
return_value
from OEHR_EMPLOYEES
where DEPARTMENT_ID=80
order by 1
```

- a. On the GMT Application Home page, click **Page 9 - Customer Details**. **Page 9 - Customer Details** open in Page Designer view.
- b. Click **Save and Run Page**.



- c. You will change the **Account Manager Id** item to a select list and also add a quick pick list. Click the **Edit Page 9** link on the Runtime Developer toolbar,

Customer Details

Cust First Name

Cust Last Name

Cust Email

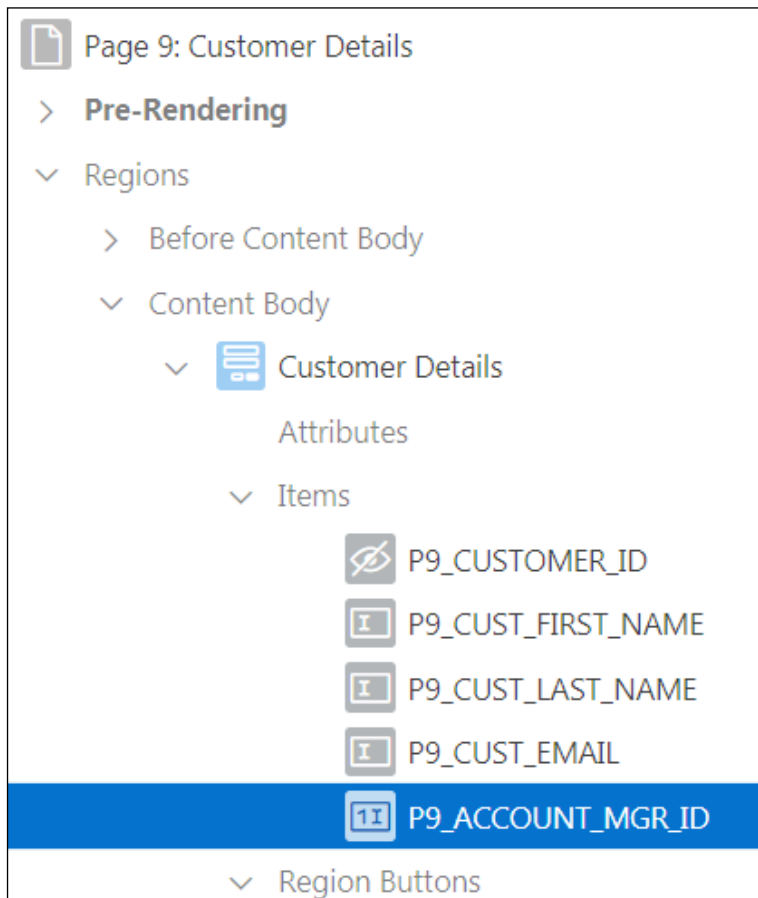
Account Mgr Id

Cancel

Create

- d. **Page 9: Customer Details** opens in Page Designer. Select **Regions > Customer Details > Items > P9_ACCOUNT_MGR_ID**.





e. In the Property Editor (the right pane):

- **Identification > Type:** Select `Select List`
- **Label > Label:** `Account Manager`
- **List of Values > Type:** Select **SQL Query** from the drop-down list.
- **List Of Values > SQL Query:** Enter the following query (located in `/home/oracle/labs/labs/lab_09_02_01.txt`) to display the last name and first name and return `EMPLOYEE_ID`.

```
select LAST_NAME||', '||FIRST_NAME display_value,
EMPLOYEE_ID
return_value
from OEHR_EMPLOYEES
where DEPARTMENT_ID=80
order by 1
```

- **List of Values > Display Null Value:** Click **Yes** (if not selected by default)
- **List of Values > Null Display Value:** Enter `-Select Manager-`

Identification

NameP9_ACCOUNT_MGR_ID

TypeSelect List

Label

LabelAccount Manager

List of Values

TypeSQL Query

SQL Query

```

select LAST_NAME||', '||FIRST_NAME display_value, EMPLOYEE_ID
return_value
from OEHR_EMPLOYEES
where DEPARTMENT_ID=80
order by 1

```

Display Extra Values

Yes

No

Display Null Value

Yes

No

Null Display Value-Select Manager-

Null Return Value

f. Click **Save**.

2. Add a quick pick list to the **Account Manager** item with the following labels and values:

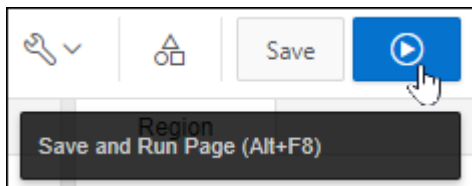
Label	Value
Bates, Elizabeth	172
Russell, John	145
Cambrault, Gerald	148

a. In the Property Editor (the right pane), select P9_ACCOUNT_MGR_ID > **Quick Picks** > **Show Quick Picks**: Select **Yes**.

- b. Enter the following labels and values.

Label	Value
Bates, Elizabeth	172
Russell, John	145
Cambrault, Gerald	148

- c. Click **Save** and then **Save and Run Page**.



- d. Click the **Account Manager** drop-down list to see the list of Account Managers.
e. Click anywhere on the page and then click the **Russell, John** quick pick.

GlobalMart Management Tool

Home

Customers

Products

Customers List View

Customer Details

Cust First Name

Cust Last Name

Cust Email

Account Manager
-Select Manager-
Bates, Elizabeth, Russell, John, Cambrault, Gerald

Cancel

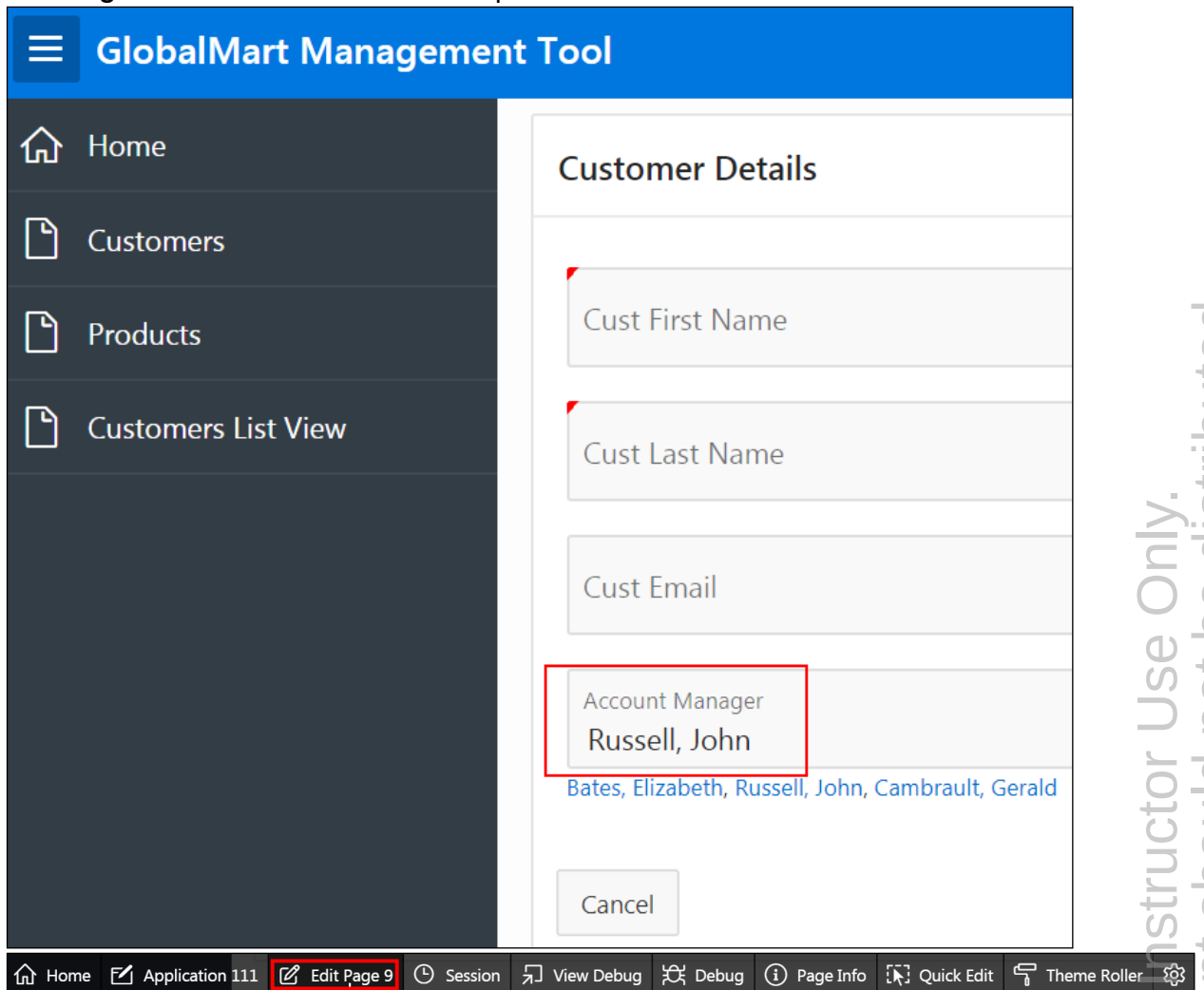
For Instructor Use Only.
This document should not be distributed.

Copyright © 2019, Oracle and/or its affiliates. All rights reserved.

294

Practices for Lesson 9: Adding Items and Buttons

- f. Notice how the name is selected for the **Account Manager** drop-down list. Click the **Edit Page 9** link on the Runtime Developer toolbar.

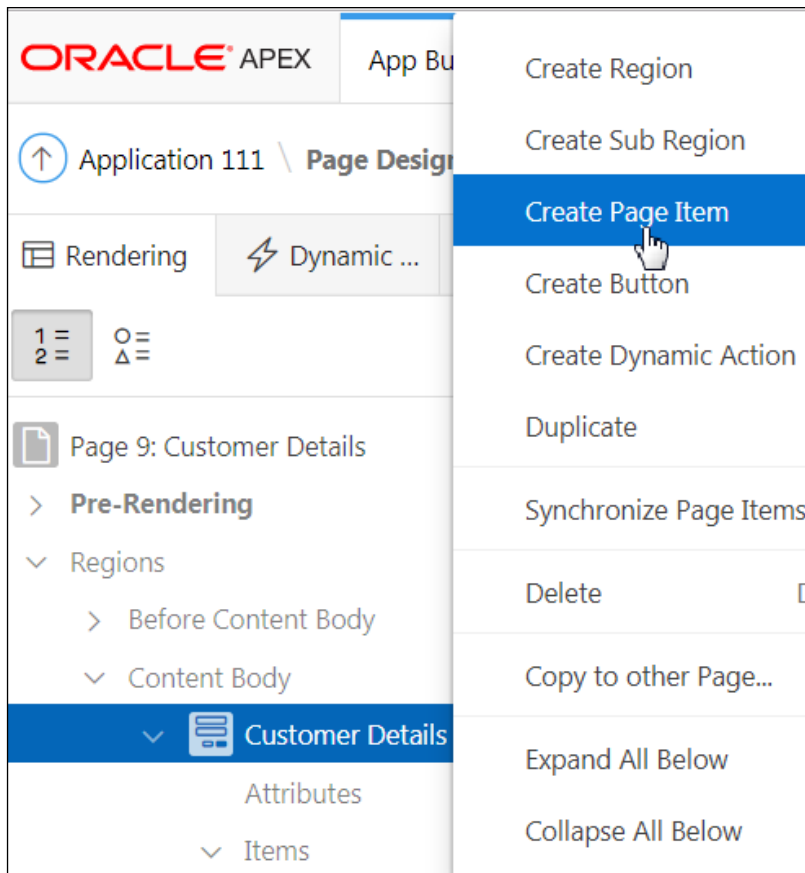


Customer Details page opens in Page Designer.

3. Add a select list item called **Country** that contains the following query (located in /home/oracle/labs/labs/lab_09_02_02.txt). This item should allow nulls.

```
select COUNTRY_NAME display_value, COUNTRY_ID return_value
from OEHR_COUNTRIES
order by 1
```

- a. Open Page 9 in Page Designer and go to **Regions > Customer Details**. Right-click **Customer Details** region and select **Create Page Item**.



- b. Click the new Page Item P9_NEW.
- c. In the Property Editor (the right pane):
 - **Identification > Name:** Enter P9_COUNTRY_ID
 - **Identification > Type:** Select **Select List** from the drop-down list
 - **Label > Label:** Enter Country
 - **List of Values > Type:** Select **SQL Query** from the drop-down list
 - **List of Values > SQL Query:** Enter the following query < located in /home/oracle/labs/labs/lab_09_02_02.txt in this field>


```
select COUNTRY_NAME display_value, COUNTRY_ID return_value
from OEHR_COUNTRIES
order by 1
```
 - **List of Values > Null Display Value:** Enter -Select Country-
 - **Source > Type:** Select **Database Column** from the drop-down list

- **Source > Type > Database Column:** Enter COUNTRY_ID

Identification

NameP9_COUNTRY_ID

TypeSelect List

Label

LabelCountry

List of Values

TypeSQL Query

SQL Query

select COUNTRY_NAME display_value, COUNTRY_ID return_value
from OEHR_COUNTRIES
order by 1

Display Extra Values

YesNo

Display Null Value

YesNo

Null Display Value

-Select Country-

Null Return Value

Source

Form Region

- Select -

Type

Database Column

Database Column

COUNTRY_ID

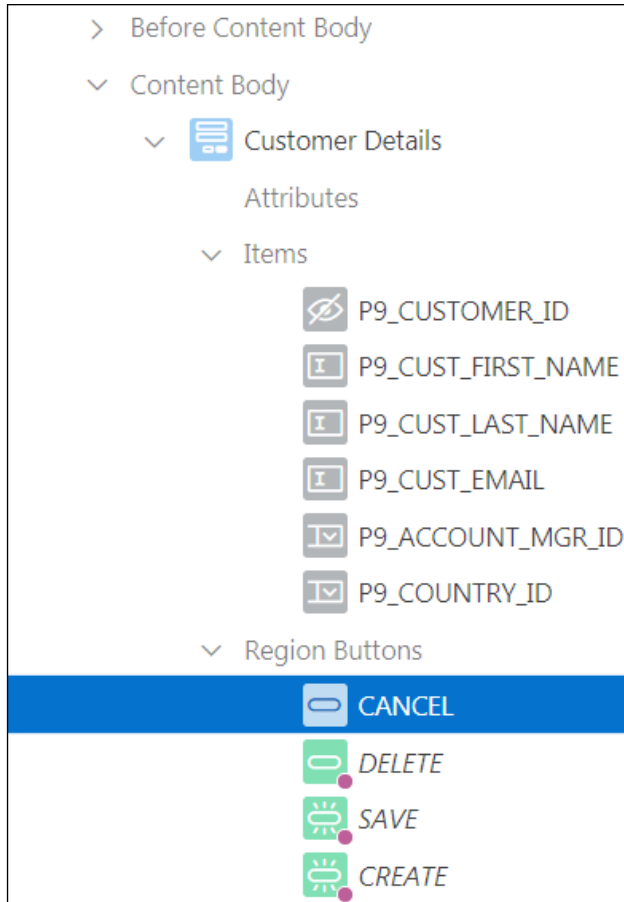
Used

Always, replacing any existing value in session

Maintain Session State

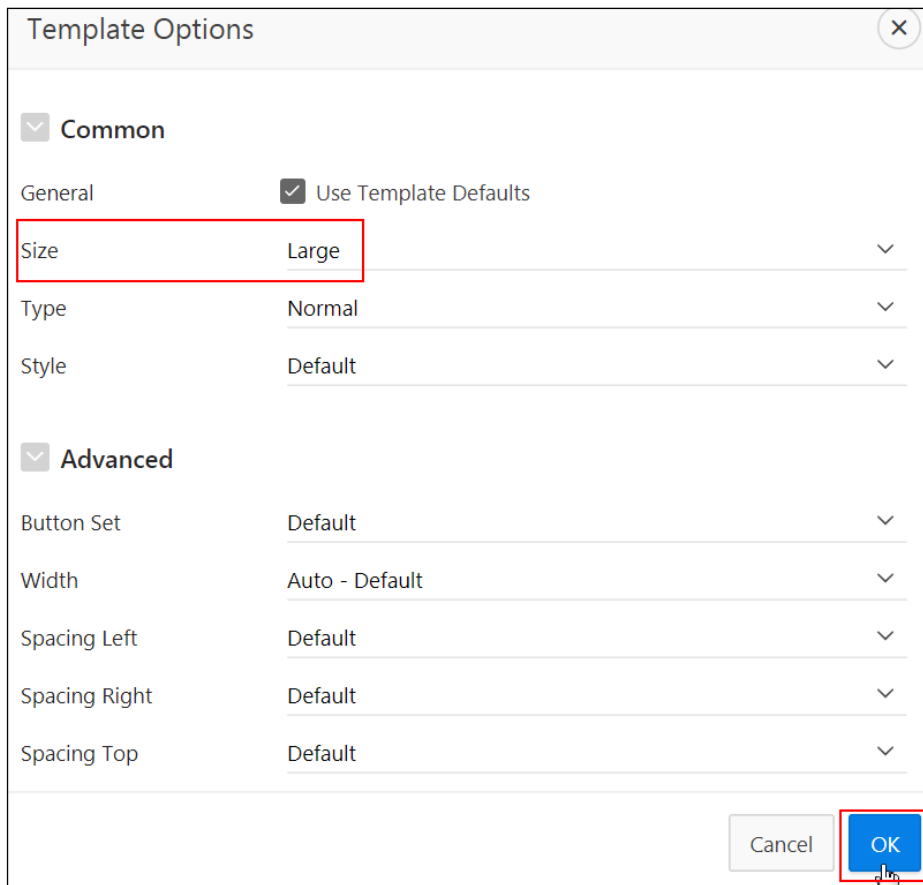
Per Session (Disk)

- d. Select **Region Buttons > Cancel** on the Rendering tab.

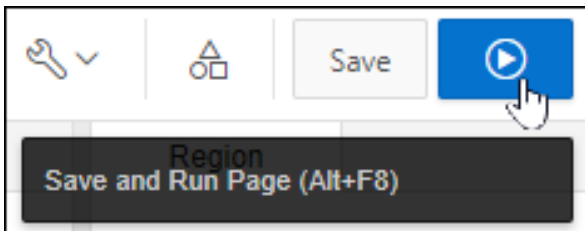


- e. In the Property Editor (the right pane):
- **Appearance > Template Options:** Click **Use Template Default**.

- **Column: Common > Size:** Select **Large** from the drop-down list.
- Click **OK**.



- f. Click **Save** and then **Save and Run Page**.



- g. Notice that the **Country** item shows a select list of country names. Notice also the large size of the **Cancel** button. Click the **Application 111** link on the Runtime Developer toolbar.

Customer Details

Cust First Name

Cust Last Name

Cust Email

Account Manager
-Select Manager-
Bates, Elizabeth, Russell, John, Cambrault, Gerald

Country
-Select Country-

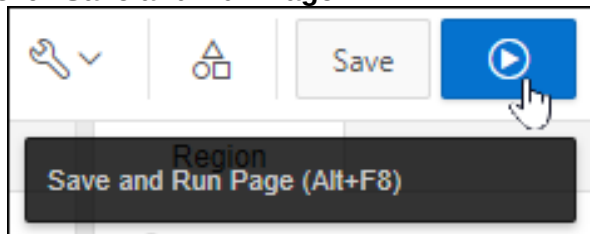
Cancel Create

Home Application 111 Edit Page 9 Session View Debug Debug Page Info Quick Edit Theme Roller

4. Change the Account Manager of an existing customer record.

Note: You must navigate to the customer report (page 4) and edit one of the records.

- a. Open **Page 4: Customers** in Page Designer view.
- b. Click **Save and Run Page**



- c. To modify an existing record, click the **Customers** tab on the left pane.

- d. Click the **Edit** icon for one of the rows. The Customer Details page opens.

	Customer Id	First Name	Last Name	Street Address	Postal Code	City	State Province	Country
	222	Cary	Stockwell	7708 City Ave	19151	Philadelphia	PA	US
	223	Cary	Olin	1801 Lititz Pike	17601	Lancaster	PA	US
	224	Clara	Krige	101 E Olney Ave	19120	Philadelphia	PA	US
	225	Clara	Ganesan	612 Jefferson Ave	18510	Scranton	PA	US

- e. You want to make one more change on the Customer Details page so that the **Apply Changes** button appears larger. Click **Quick Edit** on the Runtime Developer toolbar

and select the  icon next to **Apply Changes** button.

Customer Details

Cust First Name
Clara

Cust Last Name
Krige

Cust Email
Clara.Krige@WHIMBREL.COM

Account Manager
Zlotkey, Eleni
[Bates, Elizabeth](#), [Russell, John](#), [Cambrault, Gerald](#)

Country
-Select Country-

Cancel Delete Apply Changes 

- f. The Live Template Options page is displayed.
In **Live Template Options**, for **Common > Size**, select **Large** from the drop-down list.

Live Template Options [X]

☒ Common

General ☒ Use Template Defaults

Size **Large**

Type Normal

Style Default

☒ Advanced

Button Set Default

Width Auto - Default

Spacing Left Default

Spacing Right Default

Spacing Top Default

Spacing Bottom Default

Cancel Save

- g. Click **Save**.

- h. Notice the change in **Apply Changes** button size.

Customer Details

Cust First Name
Bryan

Cust Last Name
Dvrrie

Cust Email
BRYAN.DVRRIE@REDPOLL.COM

Account Manager
Bates, Elizabeth
Bates, Elizabeth, Russell, John, Cambrault, Gerald

Country
-Select Country-

Cancel Delete **Apply Changes**

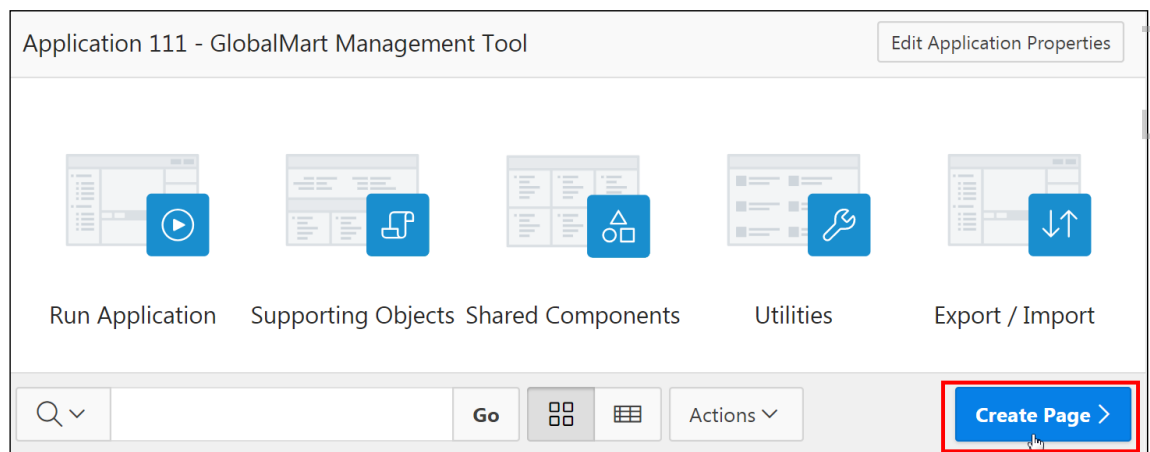
- i. Click the **Application 111** link on the Runtime Developer toolbar.



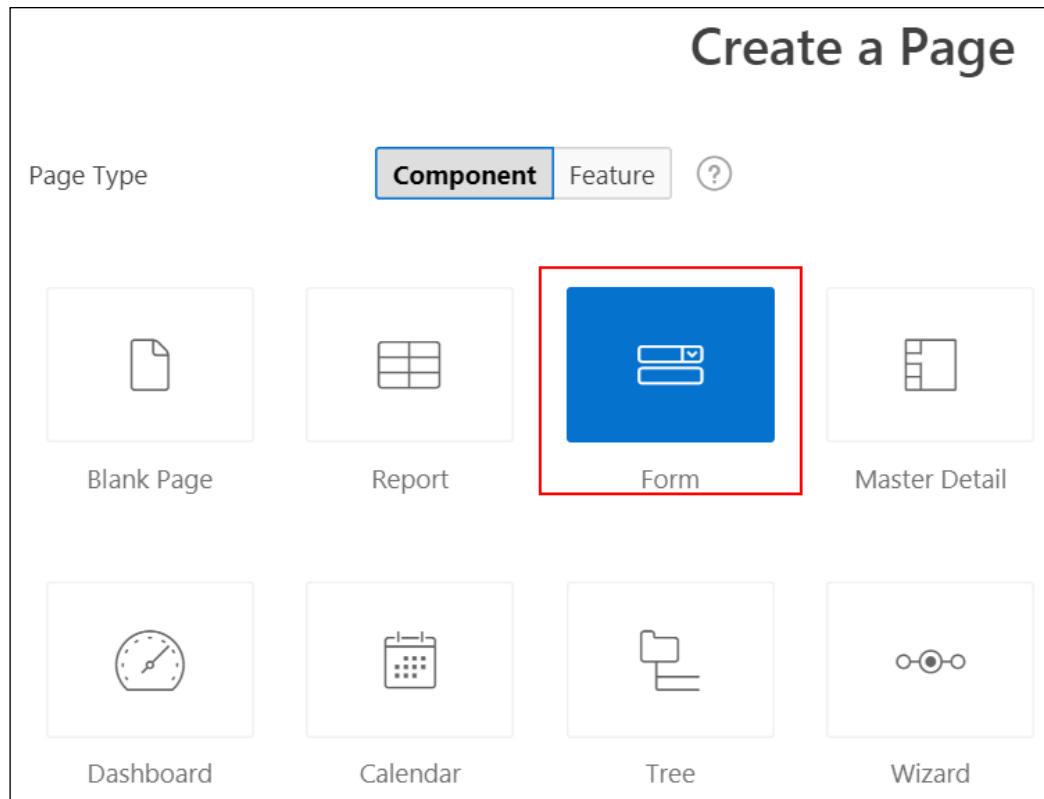
5. Create and modify the **Employee Detail (19 – Employee Detail)** page to take advantage of some of the new HTML5 item types. **Note:** This is mainly for enhancing the user interface. The user creates a subtype for Email, Phone Number, and Manager ID in a select list on a Form page. This ensures that the page uses the appropriate keyboard for email and phone numbers while entering the relevant values in the form.

- a. You will first create a Form Page.

- i. Go to your **Application 111** and click **Create Page**.

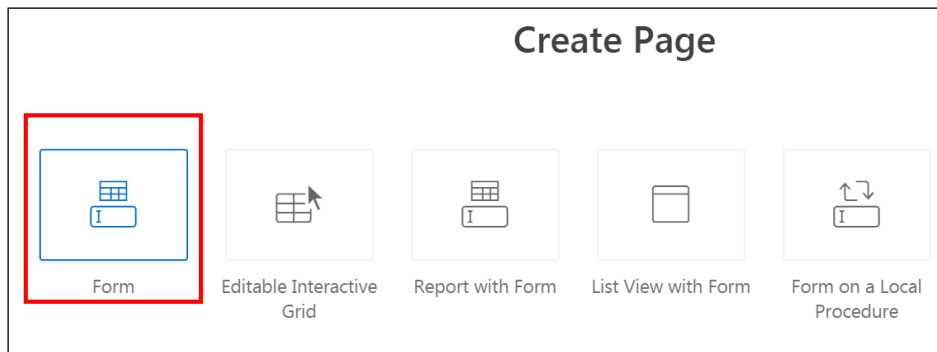


ii. Create **Form**.



iii. Click **Next**.

iv. Select **Form**.



v. Enter the following details and click **Next >**.

- **Page Number:** Enter 19
- **Page Name:** Enter Employee Details
- **Branch Here on Submit:** Select 1
- **Cancel and Go To Page:** Select 1

Create Form

Page Attributes

* Page Number

⋮ ?

* Page Name

?

Page Mode

Normal
Modal Dialog
?

Page Group

- Select Page Group -
▼
?

* Branch Here on Submit

⋮ ?

* Cancel and Go To Page

⋮

Breadcrumb

- do not add breadcrumb region to page -
▼
?

vi. Select default **Navigation Preference** and click **Next >**.

Create Form

Navigation Menu

Navigation Preference

☒
Do not associate this page with a navigation menu entry
?

☐ Create a new navigation menu entry

☐ Identify an existing navigation menu entry for this page

Copyright © 2019, Oracle and/or its affiliates. All rights reserved.

Practices for Lesson 9: Adding Items and Buttons

305

For Instructor Use Only. This document should not be distributed.

- vii. For Table/View Owner, select **OEHR** (if not selected by default). For Table/View Name, select **OEHR_EMPLOYEES (table)** . Click **Next**.

Data Source

Data Source: **Local Database** | REST Enabled SQL Service | Web Source

* Source Type: ☒ **Table** | SQL Query

* Table / View Owner: OEHR

* Table / View Name: OEHR_EMPLOYEES (table)

- viii. Select **Select Primary Key Column(s)** for **Primary Key Type**. For **Primary Key Column**, select **EMPLOYEE_ID (Number)**. Click **Create**.

Create Form - Columns and Primary Key

* Select Columns to be displayed in the form

EMPLOYEE_ID (Number)
FIRST_NAME (Varchar2)
LAST_NAME (Varchar2)
EMAIL (Varchar2)
PHONE_NUMBER (Varchar2)
HIRE_DATE (Date)
JOB_ID (Varchar2)
SALARY (Number)

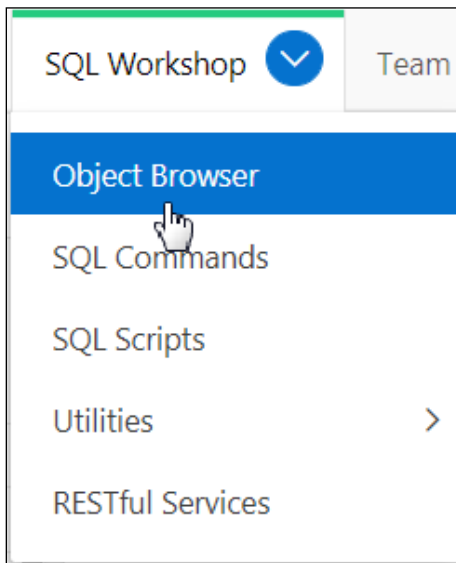
Primary Key Type: ☒ **Select Primary Key Column(s)** | ☐ Managed by Database (ROWID)

Primary Key Column: EMPLOYEE_ID (Number)

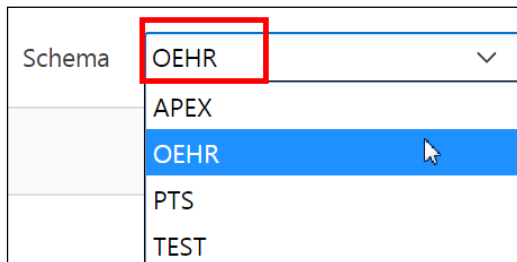
Secondary Key Column: - Select Column -

< Cancel **Create**

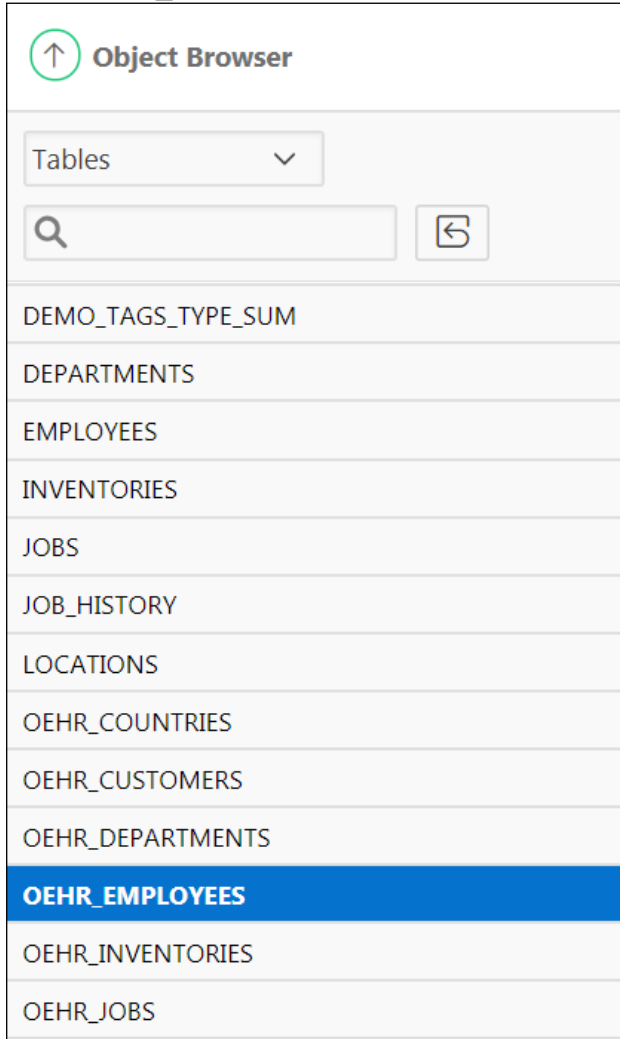
- ix. The form page is created and opens in Page Designer view.
- b. You want to add a new column to the **OEHR_EMPLOYEES** table to store whether an employee is active or not. Select **SQL Workshop > Object Browser**.



- c. Select the appropriate schema. Here you select OEHR.



- d. Select OEHR_EMPLOYEES from the Tables list.



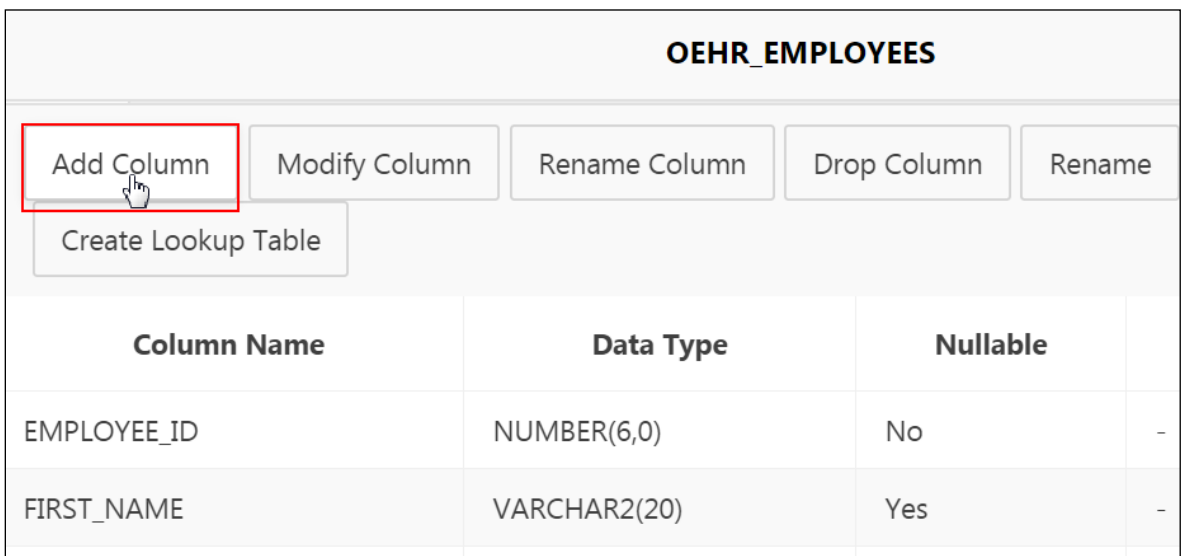
Object Browser

Tables ▾

🔍 ↩

DEMO_TAGS_TYPE_SUM
DEPARTMENTS
EMPLOYEES
INVENTORIES
JOBS
JOB_HISTORY
LOCATIONS
OEHR_COUNTRIES
OEHR_CUSTOMERS
OEHR_DEPARTMENTS
OEHR_EMPLOYEES
OEHR_INVENTORIES
OEHR_JOBS

- e. Click **Add Column**.



OEHR_EMPLOYEES

Add Column Modify Column Rename Column Drop Column Rename

Create Lookup Table

Column Name	Data Type	Nullable	
EMPLOYEE_ID	NUMBER(6,0)	No	-
FIRST_NAME	VARCHAR2(20)	Yes	-

- f. Enter the following:

- **Add Column:** Enter `ACTIVE_YN`
- **Type:** Select **VARCHAR2**
- **Length:** Enter 1

OEHR_EMPLOYEES

Add Column

Schema: **OEHR** ?

Table: **OEHR_EMPLOYEES** ?

* Add Column ?

☐ Preserve Case

Type **VARCHAR2** ?

Length ?

- Click **Next**.

g. Click **Finish**.

OEHR_EMPLOYEES

Confirm your request

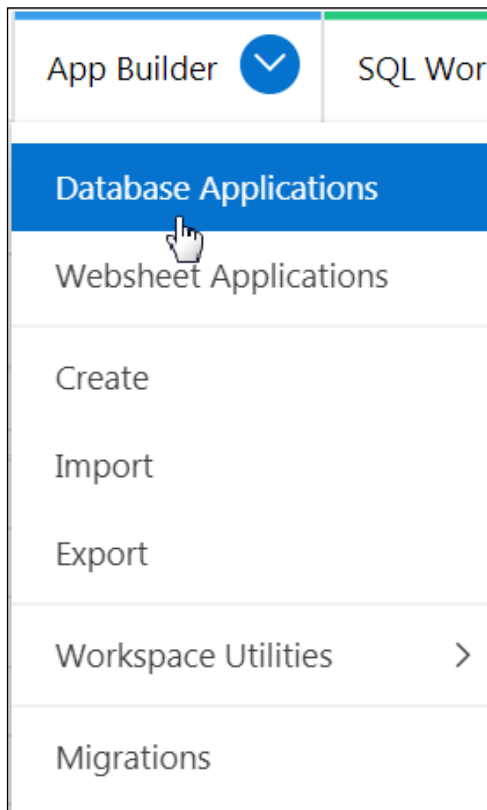
Schema: **OEHR** ?

Object: **OEHR_EMPLOYEES** ?

Action: **Add Column ACTIVE_YN** ?

< Cancel **Finish**

- Now you can modify the **Employee Detail** form page.
 - Click the Application Builder pull-down menu and select **Database Applications**.



- b. Navigate to the GMT application page. Click the **GlobalMart Management Tool** icon.



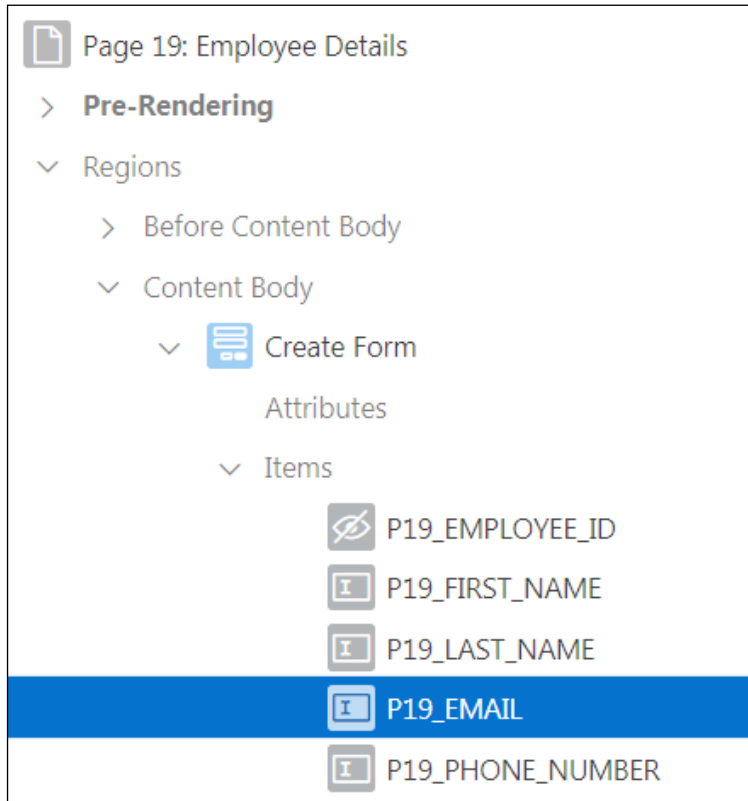
- c. Click the **View Report** icon.



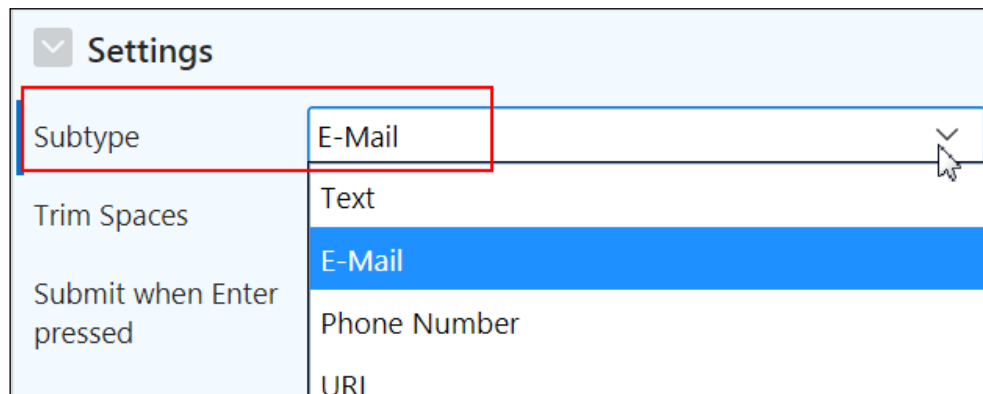
- d. Select **Page 19 - Employee Details**.

Page ↑	Name	Updated	Updated By	Page Type	Group	User Interface	Lock	Run
14	Oehr Orders	19 hours ago	apex_dev	DML Form	Unassigned	Desktop	🔒	
15	Oehr Order Items	19 hours ago	apex_dev	DML Form	Unassigned	Desktop	🔒	
16	Customer Master Information	17 hours ago	apex_dev	Interactive Report	Unassigned	Desktop	🔒	▶
17	Customer Order History	17 hours ago	apex_dev	DML Form	Unassigned	Desktop	🔒	▶
18	Customer Feedback	2 hours ago	apex_dev	Navigation Form	Unassigned	Desktop	🔒	▶
19	Employee Details	14 minutes ago	apex_dev	DML Form	Unassigned	Desktop	🔒	▶
201	Employee List	4 days ago	apex_dev	Static HTML	Unassigned	Desktop	🔒	▶

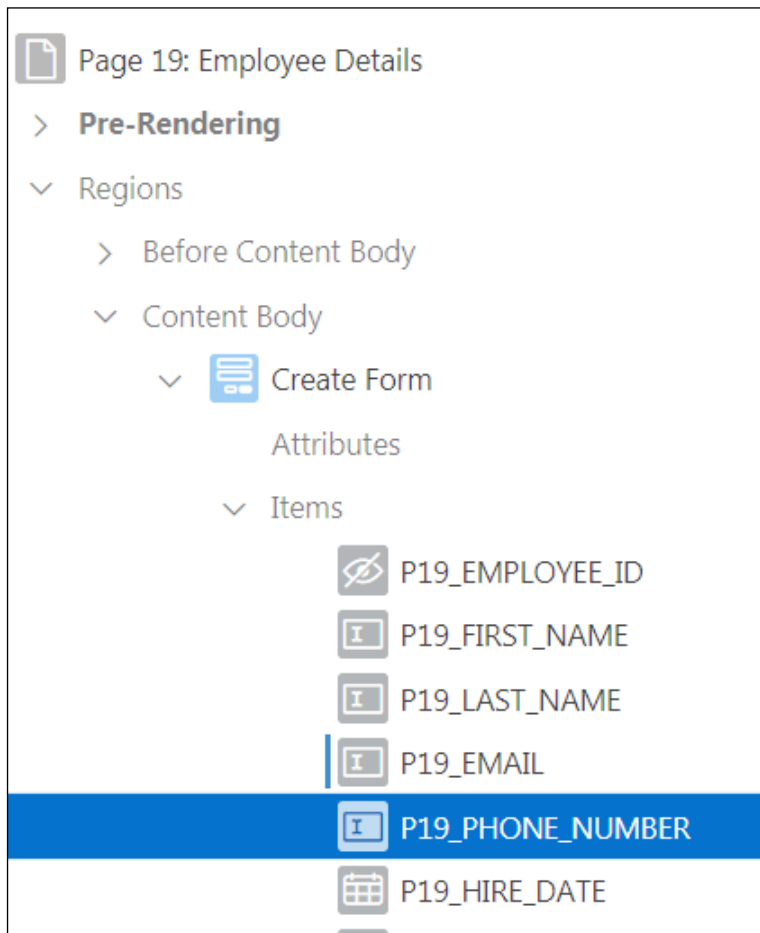
- e. Change the setting for email. In the Rendering tab, select P19_EMAIL.



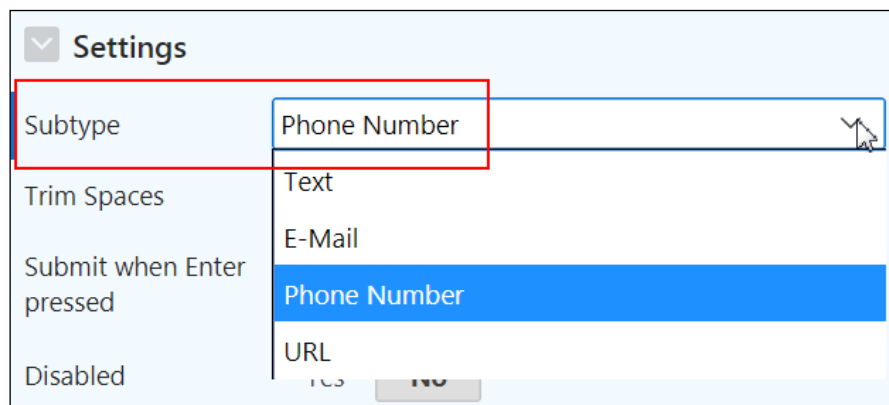
- f. In the Property Editor (the right pane):
- **Settings > Subtype:** Select **E-Mail**



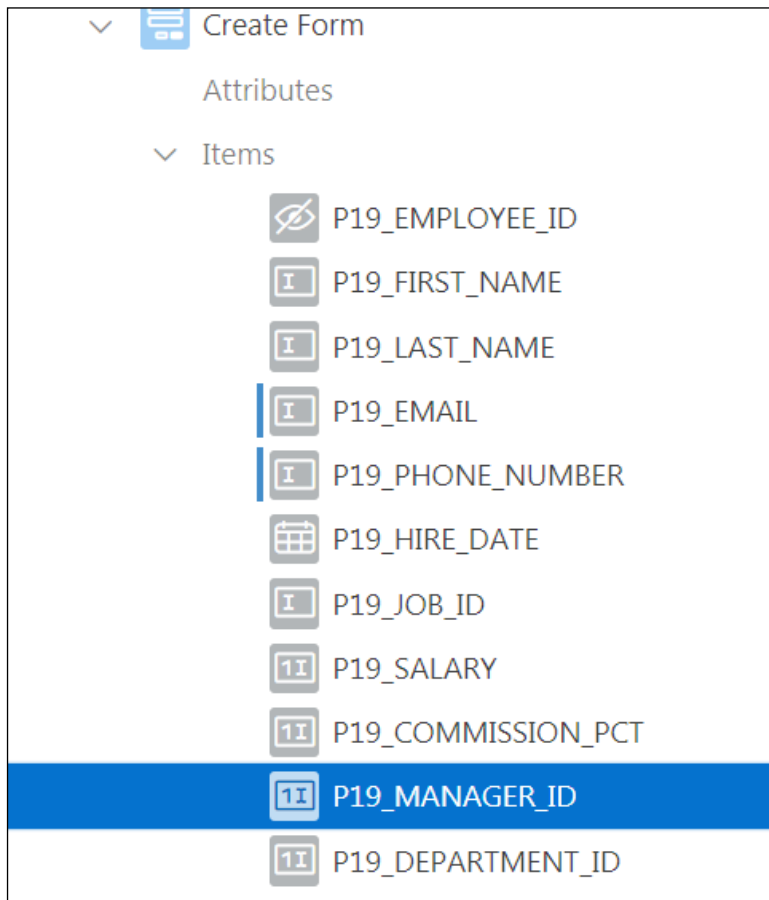
- g. Change the subtype for Phone Number. In the Rendering tab, select P19_PHONE_NUMBER.



- h. In the Property Editor (the right pane):
- **Settings > Subtype:** Select **Phone Number**



- i. Change Manager to show a select list. In the Rendering tab, select P19_MANAGER_ID.



- j. In the Property Editor (the right pane):
- **Identification > Type:** Select **Select List** from the drop-down list
 - **List of Values > Type:** Select **SQL Query** from the drop-down list
 - **List of Values > SQL Query:** Enter the SQL Query <located in /home/oracle/labs/labs/lab_09_02_03.txt in this field>

```
select first_name||' '||last_name d, employee_id r
from oehr_employees
```

▼ Identification

Name P19_MANAGER_ID

Type Select List

▼ List of Values

Type SQL Query

SQL Query

```
select first_name||' '||last_name d, employee_id r
from oehr_employees
```

Display Extra Values No

- k. Create a new page item for the `ACTIVE_YN` column you added to the `EMPLOYEES` table previously.
- i. Select **Create Form** under **Regions > Content Body**, and in the Property Editor **Identification > Title**, enter `Employee Details`.

Page 19: Employee Details

> Pre-Rendering

▼ Regions

> Before Content Body

▼ Content Body

▼  Create Form

Attributes

▼ Items

I. In the Property Editor (the right pane):

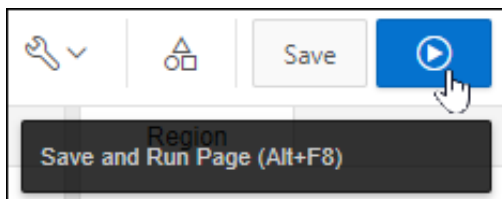
- **Identification > Name:** Enter P19_ACTIVE
- **Identification > Type:** Select **Switch** from the drop-down list
- **Source > Type:** Select **Database Column** from the drop-down list
- **Source > Database Column:** Enter P19_ACTIVE

The screenshot shows the Property Editor with two sections: Identification and Source. In the Identification section, the Name is set to P19_ACTIVE and the Type is set to Switch. In the Source section, the Form Region is set to - Select -, the Type is set to Database Column, and the Database Column is set to P19_ACTIVE. The Used property is set to Always, replacing any existing value in session, and the Maintain Session State property is set to Per Session (Disk).

Identification	
Name	P19_ACTIVE
Type	Switch

Source	
Form Region	- Select -
Type	Database Column
Database Column	P19_ACTIVE
Used	Always, replacing any existing value in session
Maintain Session State	Per Session (Disk)

m. Click **Save** and then **Save and Run Page**.



- n. The Employee Details form displays the Manager Id drop-down list and the **Active** toggle bar. Click the **Application 111** link on the Runtime Developer toolbar.

The screenshot displays the 'GlobalMart Management Tool' interface. On the left is a dark sidebar with navigation links: Home, Customers, Products, and Customers List View. The main content area shows a form for 'Employee Details' with fields for Salary, Commission Pct, Manager Id (a drop-down menu), and Department Id. Below these is an 'Active' toggle bar with 'Yes' and 'No' buttons. At the bottom right of the form is a 'Create' button. The Runtime Developer toolbar at the bottom includes links for Home, Application 111 (highlighted with a red box), Edit Page 19, Session, View Debug, Debug, Page Info, Quick Edit, and Theme Roller.

**Practices for Lesson 10:
Understanding Session State**

Practices for Lesson 10: Overview

Overview

There is one practice for this lesson. In this practice, you examine how session state works in an Application Express application.

For Instructor Use Only.
This document should not be distributed.

Practice 10-1: Understanding Session State

Overview

In this practice, you:

- Modify the Create button on the Customers page of the GlobalMart Management Tool (GMT) application.
- You clear the cache so that when the button is clicked, the Customer Details form is empty.
- You review what is in session URL and then review the change in URL on the Customer Details page, after including a new value.
- Update the title on the Customer Details page to include the customer first name and last name.

Assumptions

You have completed the previous practices.

Note: If you have not completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. On the Customers page in the GlobalMart Management Tool (GMT) application, update the Create button to make it larger by editing Template Options.
2. Save and then run the page and to view the changes.
3. Move your cursor on the Edit option of an existing record and view session URL.
4. Change the Link Column on the Customers report to include the value for the P9_CUST_FIRST_NAME item.
5. On the Customer Details page, change the title of the Customer Details region to use the following (located in /home/oracle/labs/labs/lab_10_01.txt).

`&P9_CUST_FIRST_NAME. &P9_CUST_LAST_NAME. Customer Details`

6. Run the page. What do you observe?

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 10-1**.

If you haven't completed **Practice 9-2**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_09_02.sql` file in the `/home/oracle/labs/solutions` directory and click **Next**.
- e. Click **Next**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

For Instructor Use Only.
This document should not be distributed.

Solution 10-1: Understanding Session State

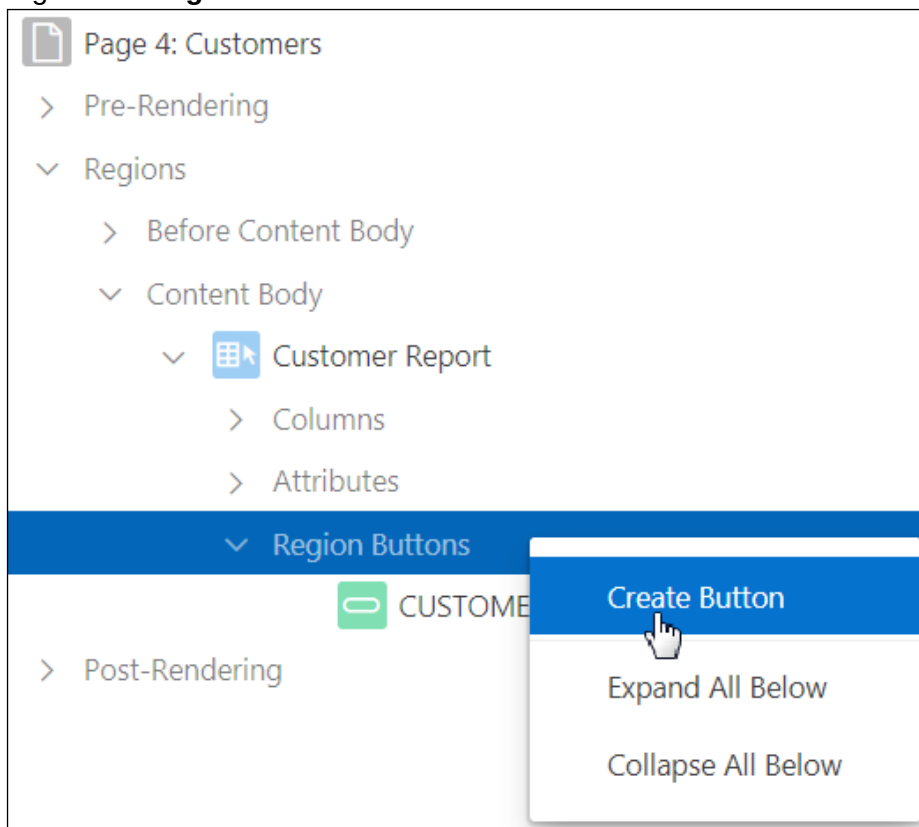
Overview

In this practice solution, steps are provided for:

- Creating and modifying a Create button on the Customers page in the GlobalMart Management Tool (GMT) application
- Reviewing change in session URL on the Customer Details page
- Changing Link Column on the Customers report to include a new value
- Updating the title on the Customer Details page to include the customer first name and last name

Steps

1. On the Customers page in the GMT application, create and edit the **CREATE** button to make it larger.
 - a. On the GMT application homepage, click the **4 - Customers** page. **Page 4 – Customers** opens in Page Designer.
 - b. Right-click **Region Buttons** and click **Create Button**.



For Instructor Use Only.
This document should not be distributed.

c. In Property Editor, define the button as follows:

- **Button Name:** Enter `CREATE`
- **Label:** `Create` (populated by default)
- **Layout > Region:** Select **Customer Report** (populated by default)
- **Layout > Button Position:** Select **Right of Interactive Report Search Bar**

▼ Identification

Button Name	CREATE
Label	Create

▼ Layout

Sequence	20
Region	Customer Report
Button Position	Right of Interactive Report Search Bar

d. Change the size of the Create button by editing Template Options. In the Property Editor under **Appearance > Template Options**, click the **Use Template Defaults** button.

▼ Appearance

Button Template	Text
Hot	Yes No
Template Options	Use Template Defaults
CSS Classes	
Icon	

- e. In the Template Options dialog, for **Common > Size**, select **Large** and click **OK**.

Template Options

Common

General ☒ Use Template Defaults

Size Large

Type Normal

Style Default

Advanced

Button Set Default

Width Auto - Default

Spacing Left Default

Spacing Right Default

Spacing Top Default

Cancel OK

2. Link the button to **Page 9 – Customer Details** and also clear the cache. To do so, select the **Create** button on the Rendering tab and in Property Editor and define the behavior.
- a. In Property Editor, go to **Behavior > Action** and select **Redirect to Page in this Application**.

Behavior

Action Redirect to Page in this Application

Target Submit Page

Warn on Unsaved Changes Redirect to Page in this Application

Database Action Redirect to Page in a different Application

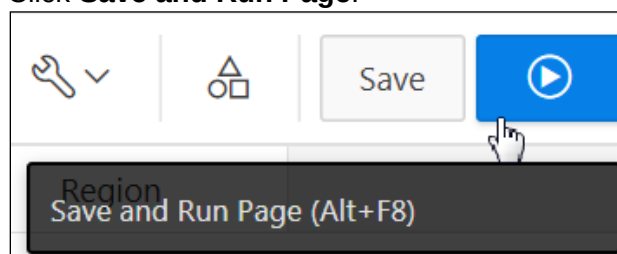
Redirect to URL

Defined by Dynamic Action

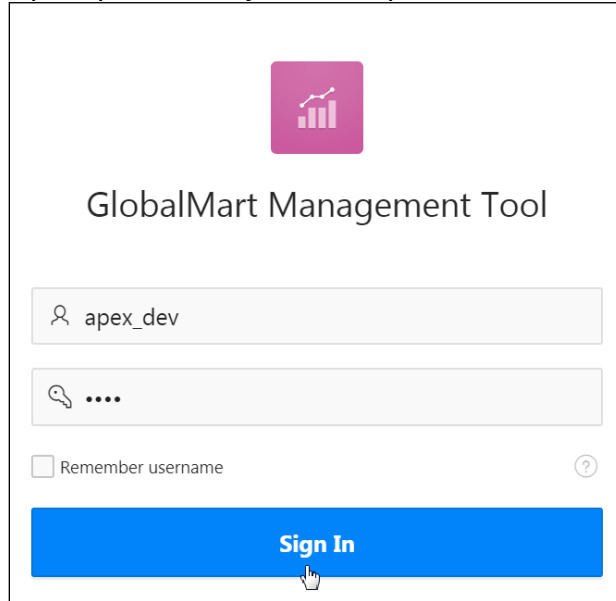
- b. Under **Behavior > Target**, click Link to **Custom Target**. In the **Link Builder - Target** window, for Page select **9**. In the **Clear Cache** field, select page **9**.

The image shows two screenshots from a web application configuration tool. The top screenshot is the 'Behavior' window, where the 'Action' is 'Redirect to Page in this Application' and the 'Target' is 'No Link Defined'. The bottom screenshot is the 'Link Builder - Target' window, where the 'Type' is 'Page' with a value of '9', the 'Clear Session State' is 'Clear Cache' with a value of '9', and the 'Reset Pagination' is set to 'Yes'. The 'OK' button is highlighted in the bottom right corner.

3. Save and then run the page to view the changes.
a. Click **Save and Run Page**.



- b. If prompted, enter your workspace credentials and click **Sign In**.



The image shows a login interface for the 'GlobalMart Management Tool'. At the top is a purple square icon with a white bar chart. Below the icon is the title 'GlobalMart Management Tool'. There are two input fields: the first contains the username 'apex_dev' and the second contains masked characters '....'. Below these fields is a checkbox labeled 'Remember username' and a help icon (a circle with a question mark). At the bottom is a large blue button labeled 'Sign In'. A mouse cursor is pointing at the 'Sign In' button.

- c. The report is displayed. Notice that the **Create** button is now displayed next to the Customer Feedback button. Also, when you click any existing entry, it re-directs you to the **Customer Details** form.

- d. Click **Cancel** and go back to the Customer report. Note that, if you click the **Create** button, it is empty. You do not get any data from the cache. You can create your new records here if you want to.

GlobalMart Management Tool

Home

Customers

Products

Customers List View

Cust First Name

Carol

Cust Last Name

Jordan

Cust Email

CAROL.JORDAN@TURNSTONE.COM

Account Manager

Zlotkey, Eleni

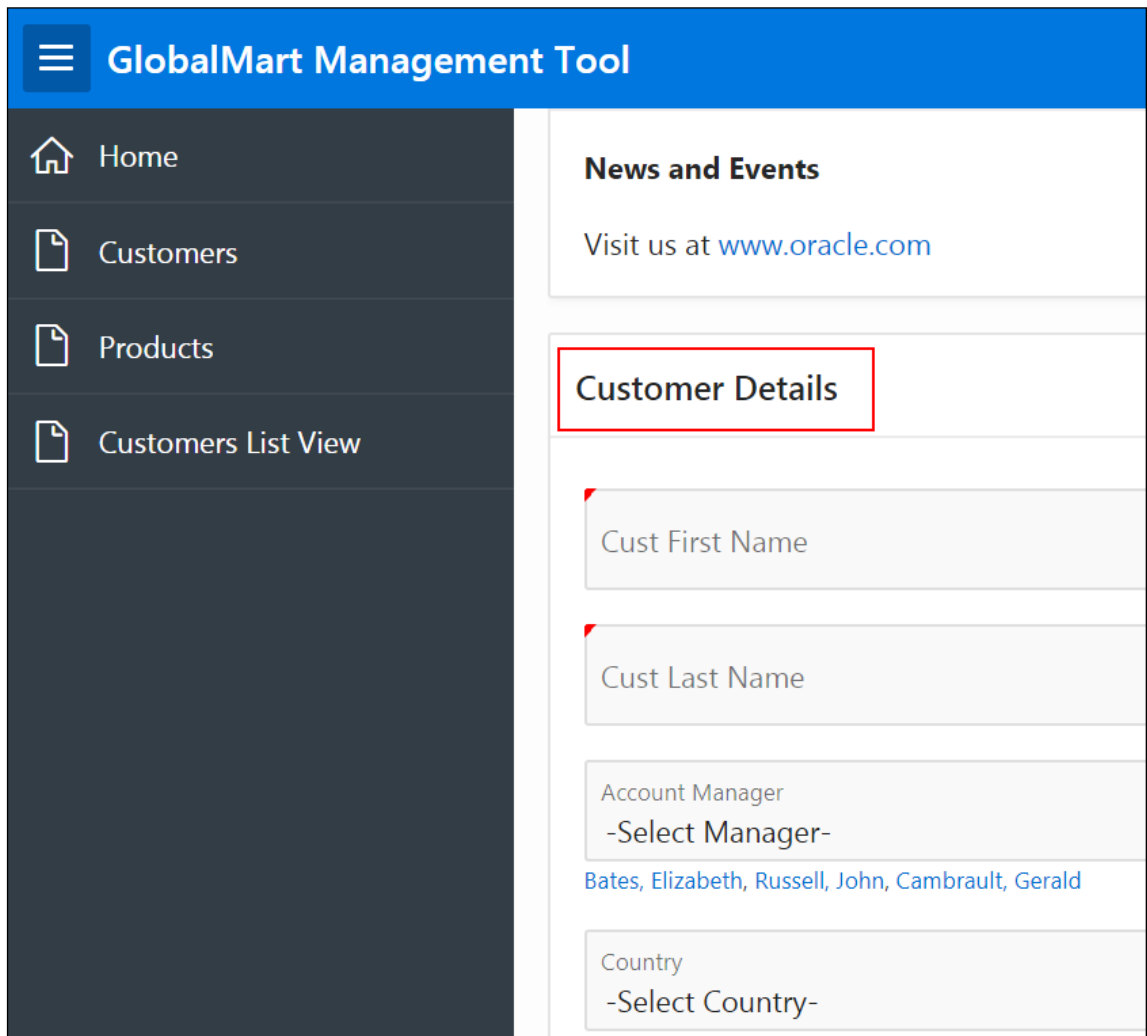
Bates, Elizabeth, Russell, John, Cambrault, Gerald

Country

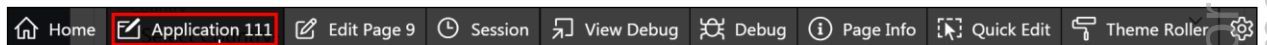
-Select Country-

Cancel

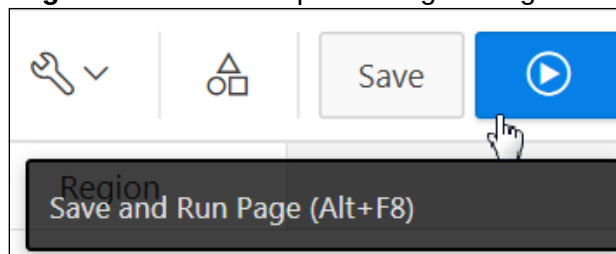
	Customer Id	First Name	Last Name	Street Address	Postal Code	City	State Province	Country	Phone Number
	218	Bryan	Dvrrie	3386 Perryville Ave	15214	Pittsburgh	PA	US	+1 412 123 4740
	219	Ajay	Sen	220 Penn Ave # 300	18503	Scranton	PA	US	+1 717 123 4741
	220	Carol	Jordan	135 S 18Th St # 1	19103	Philadelphia	PA	US	+1 215 123 4743



Click the **Application 111** link on the Runtime Developer toolbar.



4. Move your cursor over the Edit icon of an existing record and view session URL.
 - a. Click **Page 4: Customers**.
 - b. **Page 4: Customers** open in Page Designer view. Click **Save and Run Page**.



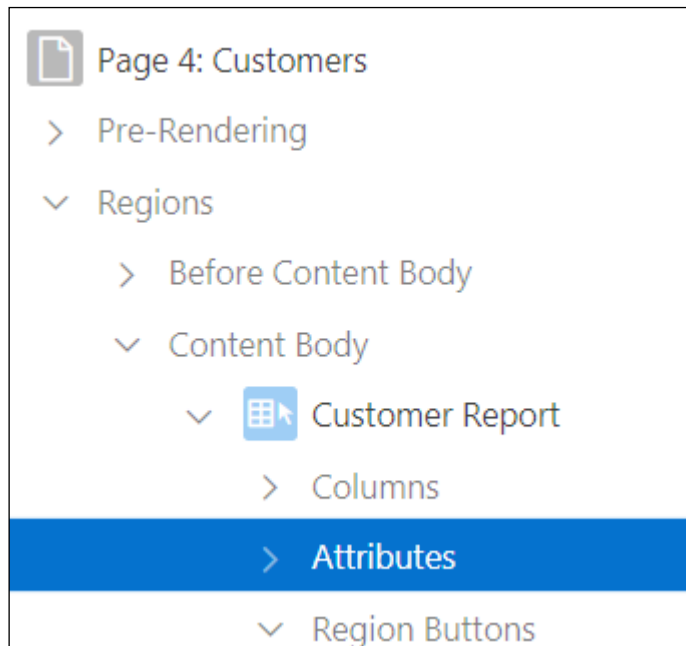
- c. Move your cursor over the **Edit** icon for any row in the report. Notice on the browser's status bar below that **CUSTOMER_ID** are being passed in the session URL.

	Customer Id	First Name	Last Name	Street Address
	218	Bryan	Dvrrie	3386 Perrysville Ave
	219	Ajay	Sen	220 Penn Ave # 300
	220	Carol	Jordan	135 S 18Th St # 1
	221	Carol	Bradford	522 Swede St

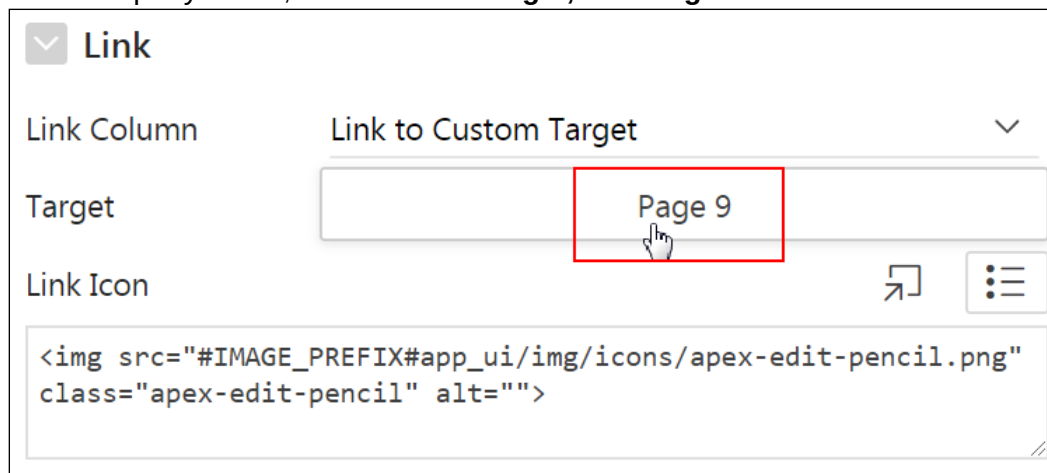
The values on the Customer Details page are based on the value stored in P9_CUSTOMER_ID. The reason the other values are not displayed in the URL is because they have not been submitted to the page.

- d. Click **Edit Page 4** on the Runtime Developer toolbar.

5. Change the Link Column on the Customers Report to include the value for the P9_CUST_FIRST_NAME item.
- a. In the Rendering tab, expand **Regions > Content Body > Customer Report** and then select the **Attributes** node.



- b. In the Property Editor, under **Link > Target**, click **Page 9**.



- c. The **Link Builder – Target** dialog appears. Under Set Items, note that the name for the first item is `P9_CUSTOMER_ID` and the value is `#CUSTOMER_ID#`.

Link Builder - Target

☒ Target

Type: Page in this application

Page: 9

☒ Set Items

Name	Value
P9_CUSTOMER_ID	#CUSTOMER_ID#

- d. Add another item name/value. Under **Value**, click the  icon and select the following for Pick Set Items.
- Name: Select **P9_CUST_FIRST_NAME**
 - Value: Select **#CUST_FIRST_NAME#**
- e. Click **OK**.

Link Builder - Target

Target

Type

Page in this application

Page

9

Set Items

Name		Value		
P9_CUSTOMER_ID		#CUSTOMER_ID#		
P9_CUST_FIRST_NAME		#CUST_FIRST_NAME#		

Clear Session State

Clear Cache

Reset Pagination

Yes

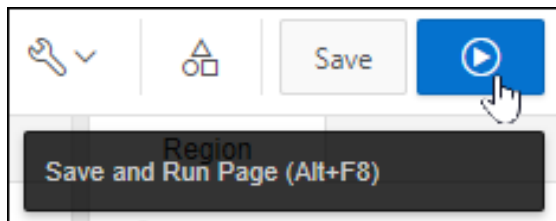
No

Cancel

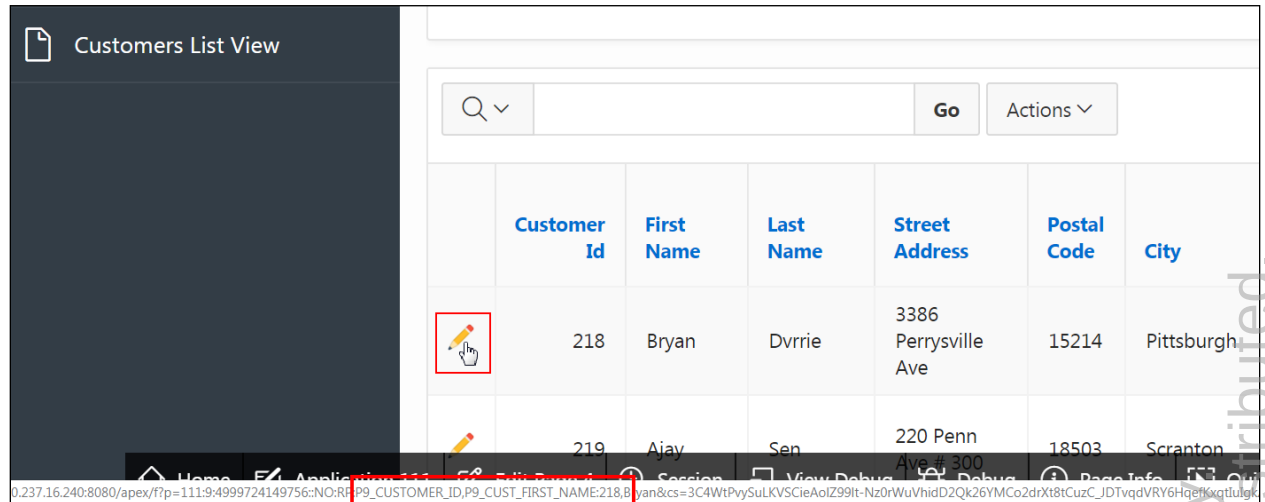
Clear

OK

- f. Click **Save and Run Page**.



- g. Now again move your cursor over the **Edit** icon for any row in the report. Notice on the browser's status bar that `CUSTOMER_ID` and `CUST_FIRST_NAME` are both being passed in the URL.



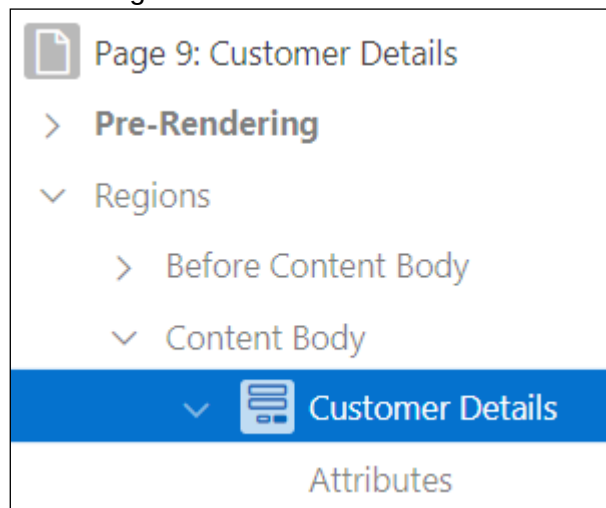
Click **Application 111** on the Runtime Developer toolbar.



6. On **Page 9: Customer Details**, change the title of the Customer Details region to use the following (located in `/home/oracle/labs/labs/lab_10_01.txt`).

`&P9_CUST_FIRST_NAME. &P9_CUST_LAST_NAME. Customer Details`

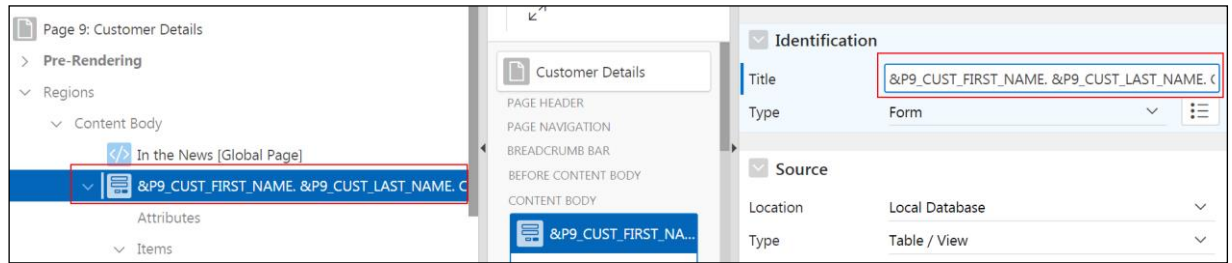
- Open **Page 9: Customer Details** in Page Designer view.
- In the Rendering tab, under **Regions > Content Body** select the **Customer Details** region.



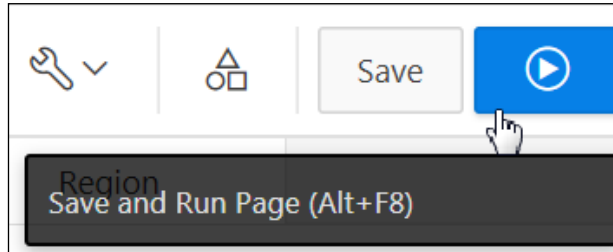
- In the Property Editor, change **Identification > Title** from `Customer Details` to the following (located in `/home/oracle/labs/labs/lab_10_01.txt`).

`&P9_CUST_FIRST_NAME. &P9_CUST_LAST_NAME. Customer Details`

Note: In the preceding text, the period (.) at the end of the item name is required.



- i. Click **Save and Run Page**.



- ii. The Customer Details form opens. Click **Cancel** to go to the Customers report page.

- iii. Click the Edit option to open any existing record. What do you observe? Notice that the Customer Name is included in the title for the displayed record.

GlobalMart Management Tool				
Home				
Customers				
Products				
Customers List View				
	Customer Id	First Name	Last Name	Street Address
	218	Bryan	Dvrrie	3386 Perrysville Ave
	219	Ajay	Sen	220 Penn Ave # 300
	220	Carol	Jordan	135 S 18Th St # 1

GlobalMart Management Tool	
Home	Bryan Dvrrie Customer Details Cust First Name Bryan Cust Last Name Dvrrie Cust Email Bryan.Dvrrie@REDPOLL.COM Account Manager Zlotkey, Eleni Bates, Elizabeth, Russell, John, Cambrault, Gerald
Customers	
Products	
Customers List View	

- d. Return to the App Builder home page. Click Home on the Runtime Developer toolbar.



For Instructor Use Only.
This document should not be distributed.

**Practices for Lesson 11:
Including Page Processing**

For Instructor Use Only.
This document should not be distributed.

Practices for Lesson 11: Overview

Overview

In this practice, you examine how computations, processes, and validations work.

For Instructor Use Only.
This document should not be distributed.

Practice 11-1: Creating and Manipulating Computations, Processes, and Validations

Overview

In this practice, you create the following computations, processes, and validations on multiple pages in the GlobalMart Management Tool application:

- Create a Page Processing computation that changes an email address to uppercase.
- Create a Page Rendering computation to populate a new item on a page.
- Create an On Submit process to populate an Audit table to track changes to order items.
- Examine the built-in Value Required validation for page items.
- Create a validation to verify that a value is greater than zero.
- Create an item string comparison validation to ensure that there are no spaces in a Cust Email field.

Note: If you do not intend to create the table `OEHR_ORDER_ITEMS_AUDIT` that is part of this practice, then you must run the script `create_table_oehr_order_items_audit.txt` located in `/home/oracle/labs/labs/` `create_table_oehr_order_items_audit.txt`

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Page processing computation. Create a page processing computation that changes the email address on the Customer Details page to uppercase after the page is submitted.
2. Page Rendering computation. Create a Page Rendering computation that populates a new item on the Order Details page with the List Price from the `OEHR_PRODUCT_INFORMATION` table.
3. On Submit process. Create an audit table, which tracks changes to order items. To accomplish this, you create a new table by copying the `OEHR_ORDERS_ITEMS` table and adding two new columns, Updated and Updated By. You populate the new Audit table by creating an On Submit process.
4. Examine the **Value Required** validation that exists on some items. Certain items on the Customer Details page are marked as required. This behavior is controlled by the Page Item validation, **Value Required**. Examine the **Value Required** validation that exists on some items on the Customer Details page.
5. Create a validation to verify that a value is greater than zero. On the Order Items page, create a validation on the Quantity field to ensure that it has a value that is greater than zero.

6. Create an item string comparison validation to ensure that there are no spaces in the **Cust Email** field. Create an item string comparison validation to ensure that there are no spaces in the **Cust Email** field on the Customer Details page.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 11-1**.

If you haven't completed **Practice 10-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_10_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

Solution 11-1: Creating and Manipulating Computations, Processes, and Validations

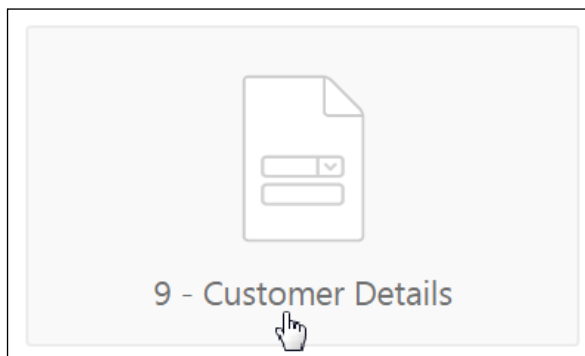
Overview

In this practice solution, steps are provided for creating and manipulating computations, processes, and validations.

Note: If you do not intend to create the table `OEHR_ORDER_ITEMS_AUDIT` that is part of this practice, then you must run the script `create_table_oehr_order_items_audit.txt` located in `/home/oracle/labs/labs/` `create_table_oehr_order_items_audit.txt`

Steps

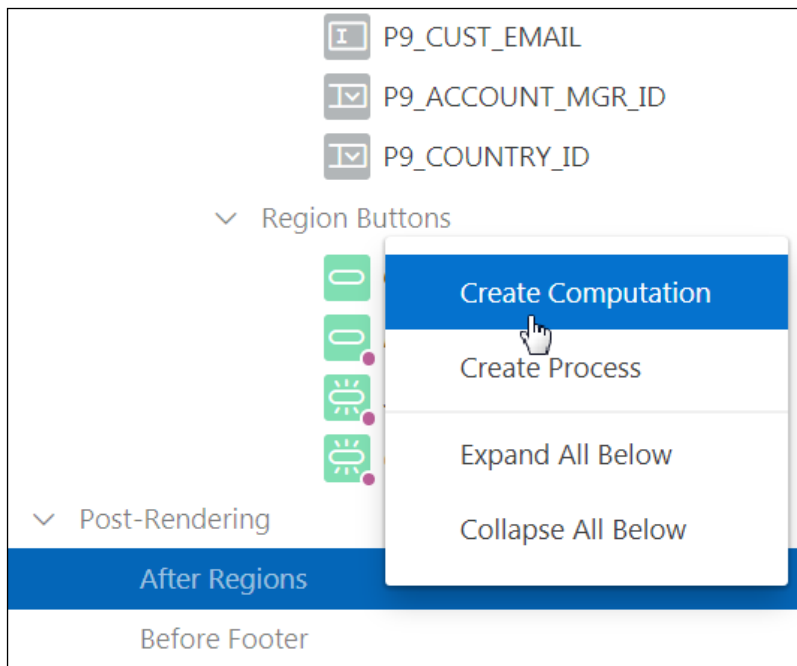
1. Create a page processing computation that changes the email address on the Customer Details page to uppercase after the page is submitted.
 - a. On the GMT Application home page, select page **9 - Customer Details** page.



- b. **Page 9 – Customer Details** opens in Page Designer view. In the left pane of Page Designer, click the Rendering tab.

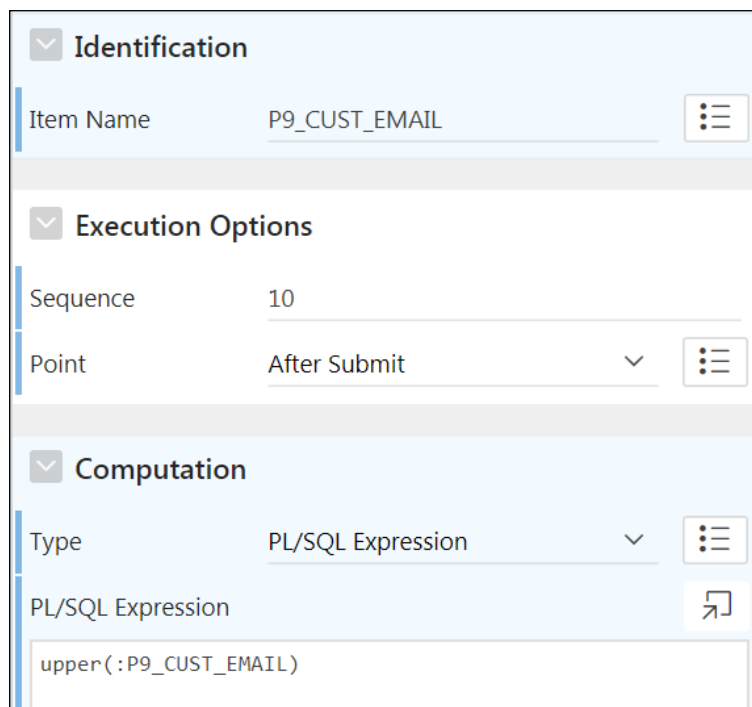
For Instructor Use Only.
This document should not be distributed.

- c. On the Page Rendering tab, scroll down and expand **Post Rendering**. Right-click **After Regions** and select **Create Computation**.

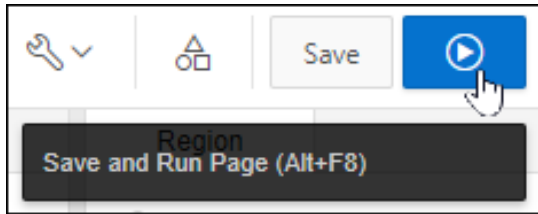


- d. Ensure that the new computation is selected. In the Property Editor, edit the following:

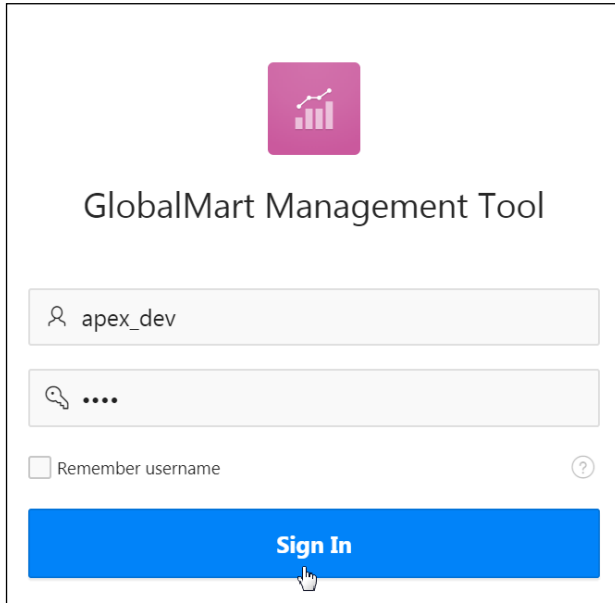
- **Identification > Item Name:** Enter `P9_CUST_EMAIL`.
- **Execution Options > Point:** Select **After Submit**.
- **Computation > Type:** Select **PL/SQL Expression**.
- **Computation > PL/SQL Expression:** Enter `upper (: P9_CUST_EMAIL) .`



- e. Click **Save** and then **Save and Run Page**.



- f. If prompted, enter your workspace user credentials and click **Sign In**.



The image shows a login interface for the 'GlobalMart Management Tool'. At the top center is a purple square icon with a white bar chart. Below the icon is the title 'GlobalMart Management Tool'. There are two input fields: the first contains the username 'apex_dev' and has a person icon on the left; the second is for a password, shown as four dots, and has a key icon on the left. Below these fields is a checkbox labeled 'Remember username' and a small circular help icon with a question mark. At the bottom is a large blue button with the text 'Sign In' in white. A mouse cursor is pointing at the 'Sign In' button.

- g. Click **Cancel** to return to the Customers report.

The screenshot shows the 'GlobalMart Management Tool' interface. On the left is a dark sidebar with navigation links: Home, Customers, Products, and Customers List View. The main content area is white and contains a form with the following fields:

- Cust First Name
- Cust Last Name
- Cust Email
- Account Manager: -Select Manager-
[Bates, Elizabeth, Russell, John, Cambrault, Gerald](#)
- Country: -Select Country-

A red rectangular box highlights the 'Cancel' button at the bottom of the form, with a mouse cursor pointing at it.

For Instructor Use Only.
This document should not be distributed.

- h. On the Customers page, click the **Edit** icon in a row.

GlobalMart Management Tool						
Home		Customer Id	First Name	Last Name	Street Address	Postal Code
Customers		223	Cary	Olin	1801 Lititz Pike	17601
Products		224	Clara	Krige	101 E Olney Ave	19120
Customers List View		225	Clara	Ganesan	612 Jefferson Ave	18510
		226	Ajay	Andrews	223 4Th Ave # 1100	15222

- i. Notice that the email is currently in mixed case format. Click **Apply Changes** to invoke the computation.

Cary Olin Customer Details

Cust First Name
Cary

Cust Last Name
Olin

Cust Email
Cary.Olin@WATERTHRUSH.COM

Account Manager
Zlotkey, Eleni
Bates, Elizabeth, Russell, John, Cambrault, Gerald

Country
-Select Country-

Cancel

Delete

Apply Changes

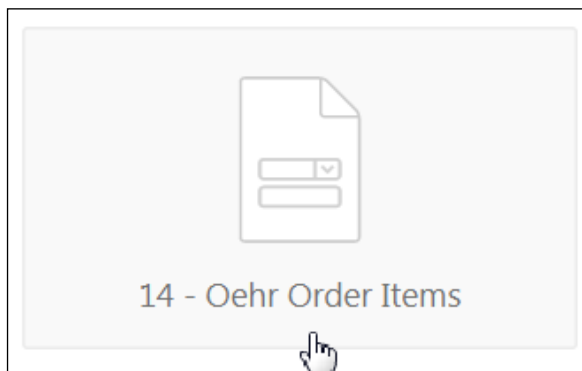
- j. On the Customer Report, scroll to the right and look at the **Cust Email** column. Notice that the email for the edited record (highlighted in red) now appears in uppercase.

Row updated. apex_dev					
Phone Number	Nls Language	Nls Territory	Credit Limit	Cust Email	Account Manager
+1 717 123 4746	us	AMERICA	2300	CARY.OLIN@WATERTHRUSH.COM	149

- k. Click Application 111 on the Runtime Developer toolbar at the bottom of the window.

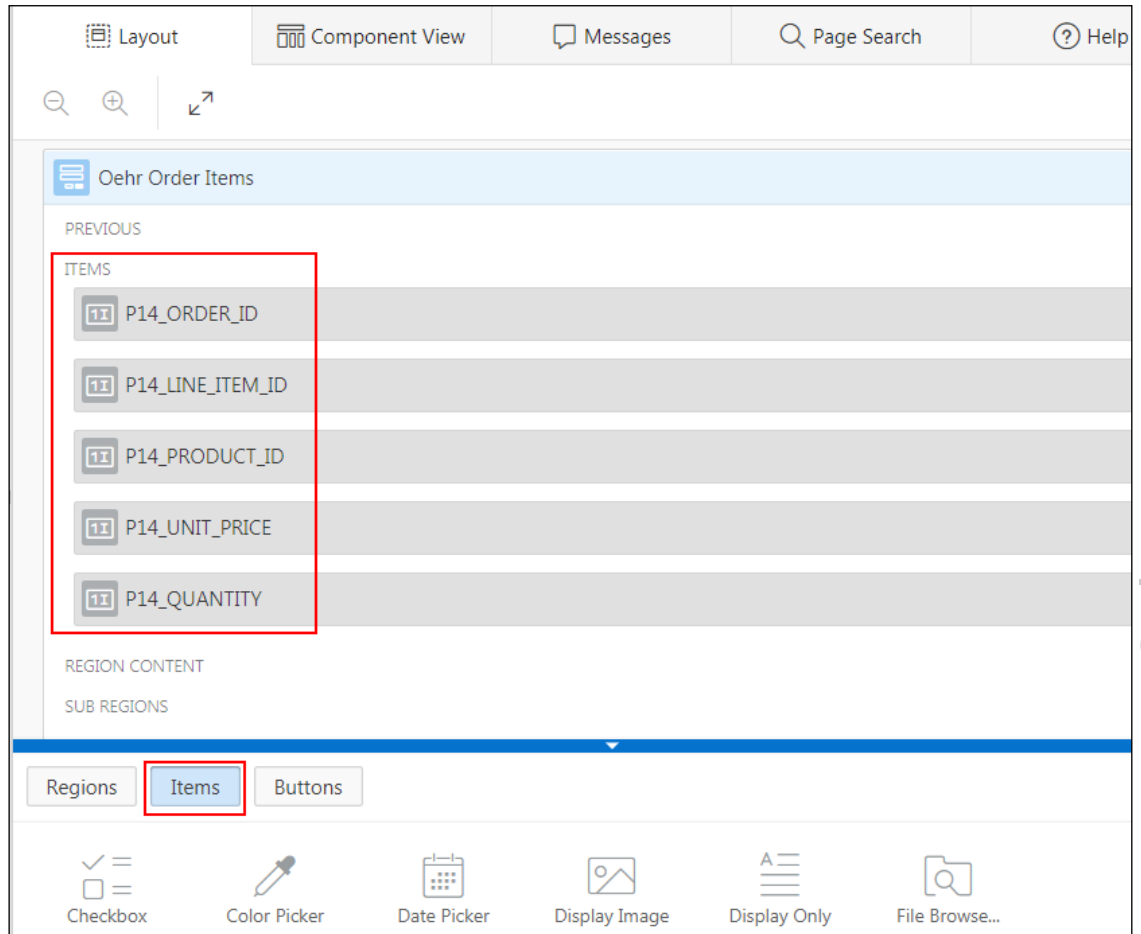
	164	Charlotte	Kazan	2122 Campbell Rd	54601	La Crosse	WI	US	+1 608 123 456
	165	Charlotte	Fonda	3324 N Oakland Ave	53211	Milwaukee	WI	US	+1 414 123 456
	166	Dheera	Alexander	666 22Nd	55418	Minneapolis	MN	US	+1 612 123 456

2. Create a Page Rendering computation that populates a new item on the Order Details page with the List Price from the OEHR_PRODUCT_INFORMATION table.
 - a. On the GMT Application home page, select page **14 – Oehr Order Items**.



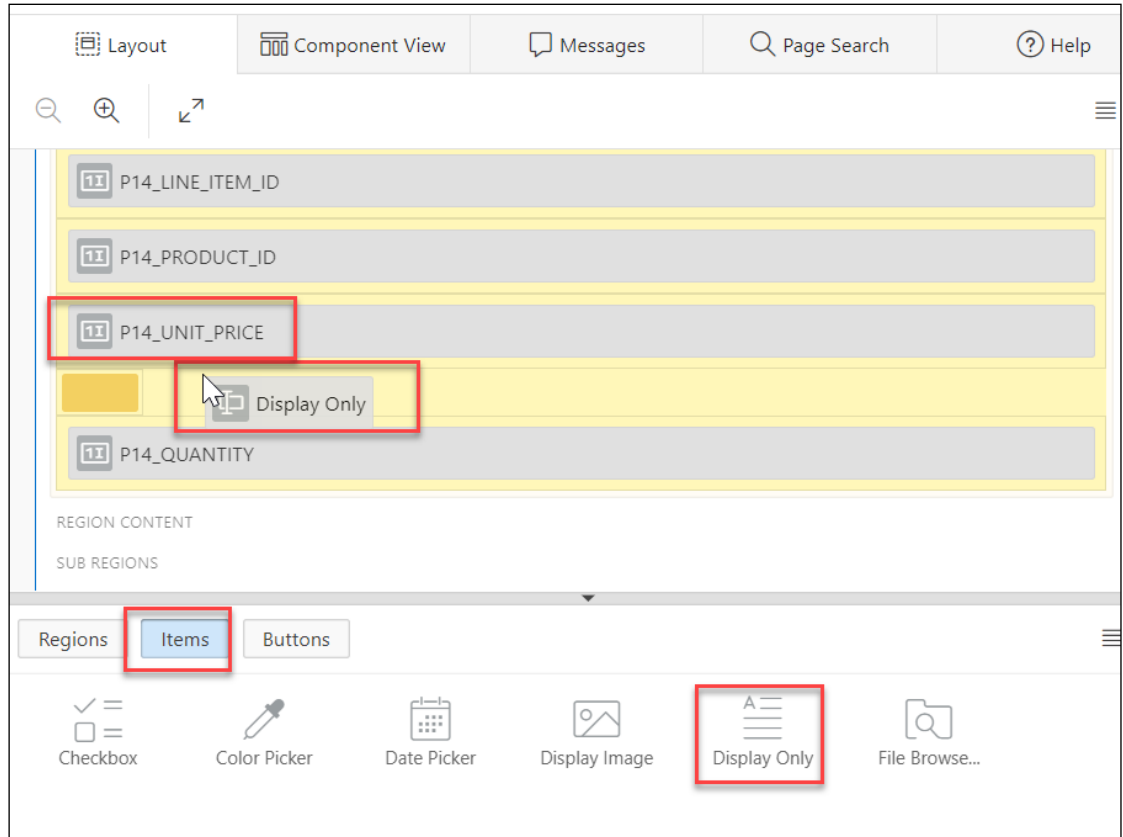
- b. Create a Display Only item:
 - i. On the Layout tab, find the **Oehr Order Items** region.

- ii. From the Gallery at the bottom of the central pane, select the **Items** tab.



For Instructor Use Only.
This document should not be distributed.

- iii. Select **Display Only** and drag it after the P14_UNIT_PRICE.



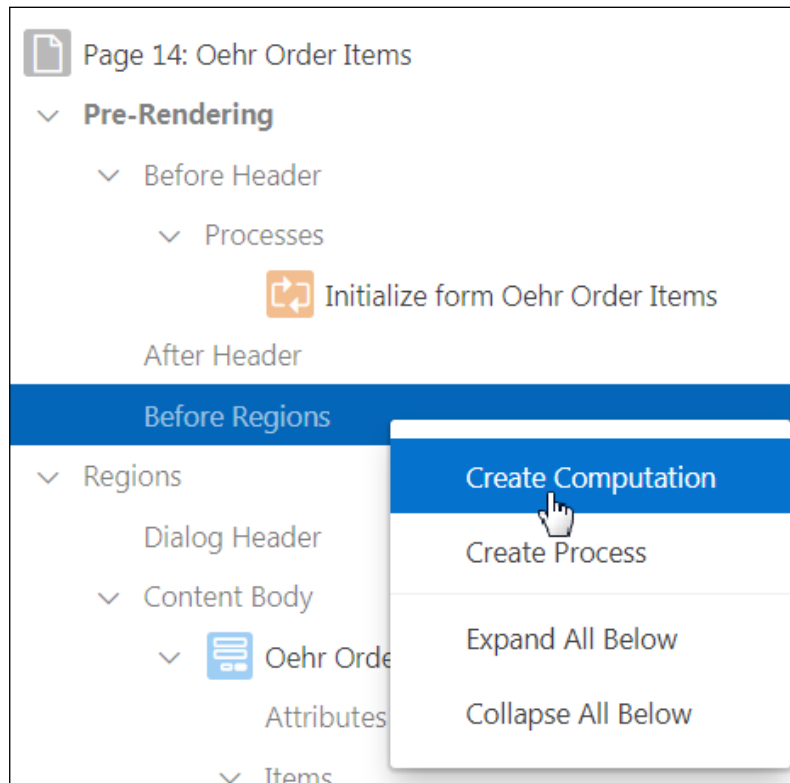
- iv. Select **P14_NEW** and enter the following values:

- **Identification > Title:** Enter P14_LIST_PRICE and press Enter.
- **Source > Type:** Select Null (if not selected by default).

The screenshot shows the Oracle APEX Properties panel for the P14_NEW item. The panel is divided into two sections: Identification and Source. In the Identification section, the Name field is set to P14_LIST_PRICE and the Type is set to Display Only. In the Source section, the Form Region is set to - Select -, the Type is set to Null, the Used checkbox is checked, and the Maintain Session State is set to Per Session (Disk).

c. Create a Pre-Rendering Computation:

- i. On the Rendering tab, expand **Pre-Rendering**.
- ii. Right-click **Before Regions** and select **Create Computation**.



d. In the Property Editor, edit the following:

- i. **Identification > Item Name:** Select P14_LIST_PRICE.
- ii. **Computation > Type:** Select **SQL Query** (return single value), if not selected by default.
- iii. **Source > SQL Query:** Copy and paste the following code located in /home/oracle/labs/labs/lab_11_01_01.txt).

```
select list_price
from oehr_product_information
where product_id = :P14_PRODUCT_ID
```

The screenshot shows the Page Designer configuration window with three sections:

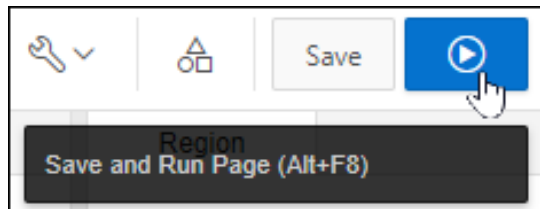
- Identification:** Item Name is set to `P14_LIST_PRICE`.
- Execution Options:** Sequence is `10` and Point is `Before Regions`.
- Computation:** Type is `SQL Query (return single value)`. The SQL Query text area contains:


```
select list_price
from oehr_product_information
where product_id = :P14_PRODUCT_ID
```

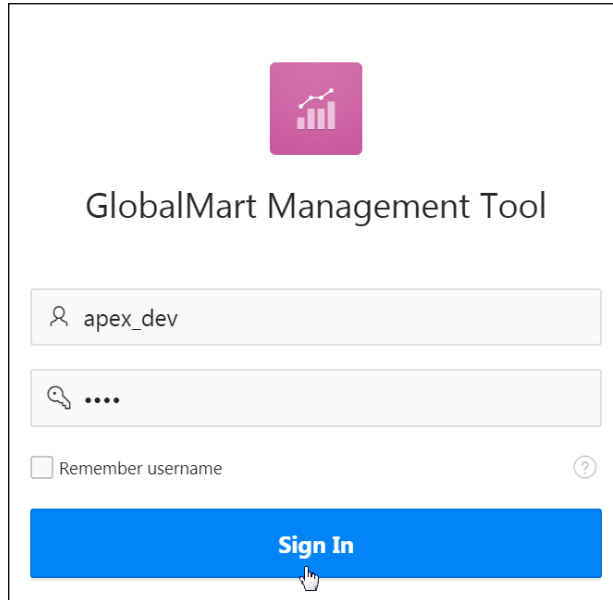
- e. Click **Save**.
- f. Review your changes. Because page **14 – Order Items** is a modal dialog, you cannot run it directly from Page Designer.
 - i. In the Page Selector, in the Page Designer toolbar, enter 12 and click **Go**.



- ii. When page **12: List of Order** appears, click **Save and Run Page**.

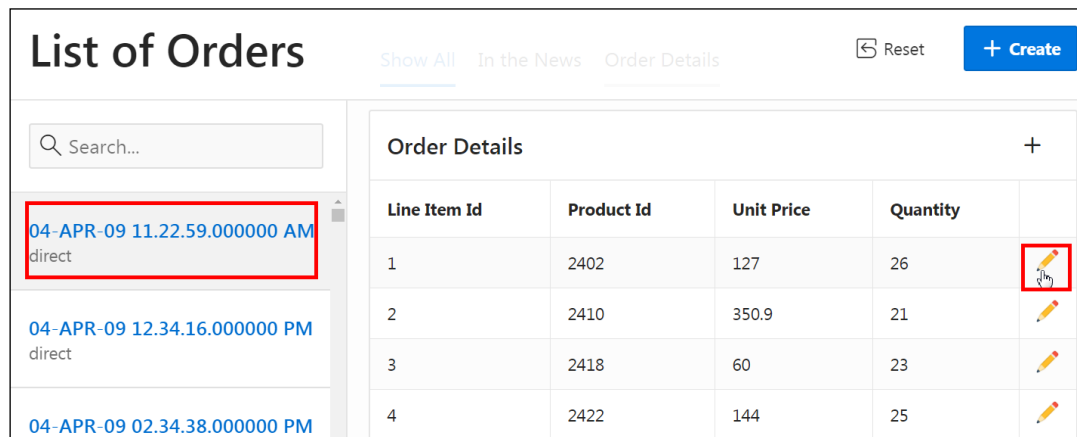


- iii. If prompted, enter your workspace user credentials and click **Sign In**.



The image shows the 'GlobalMart Management Tool' sign-in interface. At the top is a purple square icon with a white bar chart. Below it is the title 'GlobalMart Management Tool'. There are two input fields: the first contains the username 'apex_dev' and the second contains masked characters '....'. Below these fields is a checkbox labeled 'Remember username' and a help icon '?'. At the bottom is a large blue button labeled 'Sign In' with a mouse cursor pointing at it.

- iv. On the page 12 - **List of Orders**, select an order in the left, Master Record region.
- v. Scroll down in the right pane. Under **Order Details**, select Line Item by clicking the Edit icon.



The image shows the 'List of Orders' screen. On the left is a search bar and a list of orders. The first order is highlighted with a red box: '04-APR-09 11.22.59.000000 AM direct'. On the right is the 'Order Details' table. The first row of the table is highlighted with a red box, and the edit icon (pencil) in the last column of that row is also highlighted with a red box.

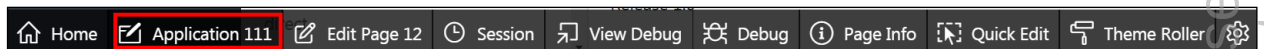
Line Item Id	Product Id	Unit Price	Quantity	
1	2402	127	26	
2	2410	350.9	21	
3	2418	60	23	
4	2422	144	25	

The Order Items dialog box appears. Notice that **List Price** now appears after the **Unit Price**.

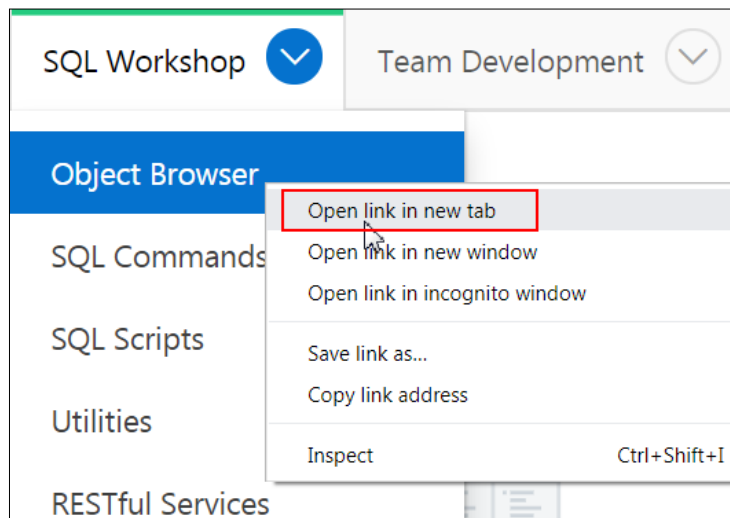
Oehr Order Items ✕

Line Item Id	1
Product Id	2402
Unit Price	127
List Price	127
Quantity	26

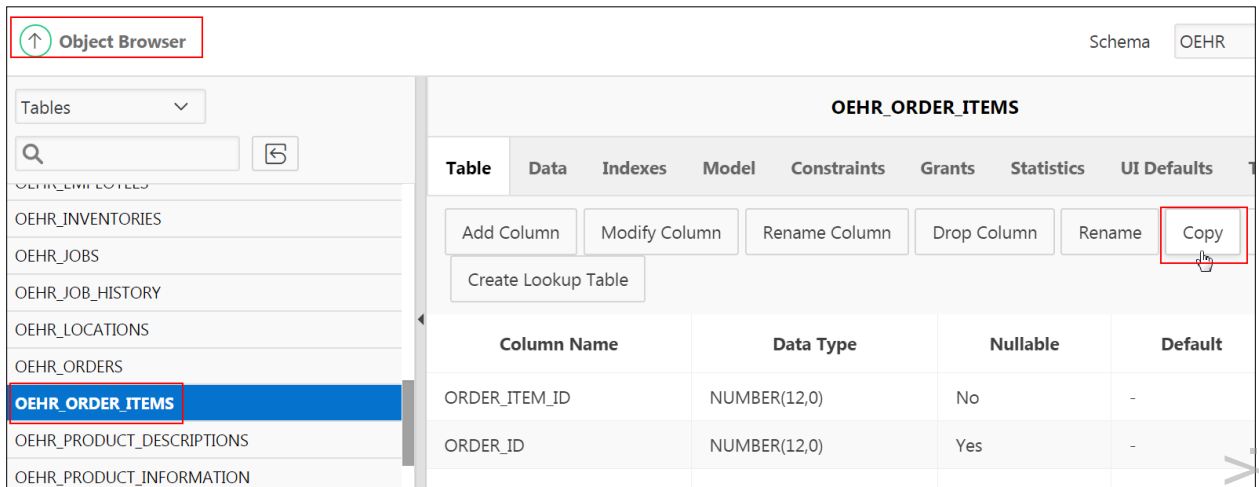
- vi. Click **Cancel** to exit the Order items dialog box.
- g. Click **Application 111** on the Runtime Developer toolbar to return to the Application home page.



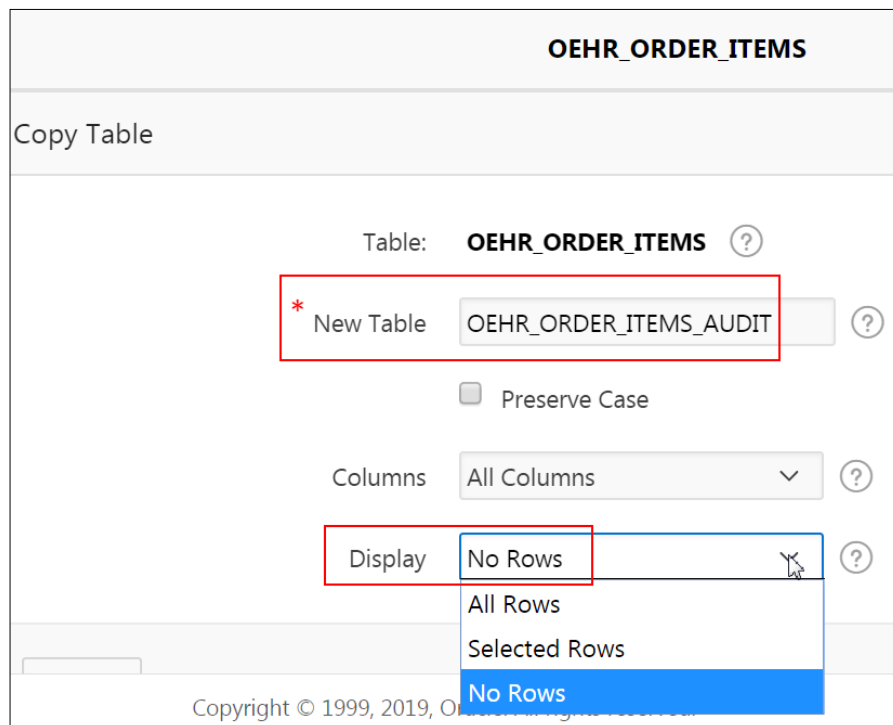
- 3. Create an audit table, which tracks changes to order items. To accomplish this, you create a new table by copying the `OEHR_ORDERS_ITEMS` table and adding two new columns, `UPDATED` and `UPDATED BY`. You populate the new Audit table by creating an On Submit process.
 - a. Open **Object Browser** on a new browser tab.
 - i. On the Application home page, find the **SQL Workshop** menu at the top of the page.
 - ii. Click the down arrow next to **SQL Workshop**.
 - iii. Right-click Object Browser and select **Open Link in New Tab**.



- iv. Click the new browser tab, named Object Browser.
- b. Create a copy of the `OEHR_ORDER_ITEMS` table. In the Object Selection pane, select the `OEHR_ORDER_ITEMS` table and then click the **Copy** tab in the Details pane.



- c. In the Copy Table Wizard:
 - i. New Table: Enter `OEHR_ORDER_ITEMS_AUDIT`.
 - ii. Display: Select **No Rows**.
This option ensures that the wizard only creates the structure but does not copy any columns.



- iii. Click **Next**.

- iv. Click **Finish**.

OEHR_ORDER_ITEMS

Confirm your request

Schema: **OEHR** ?

Object: **OEHR_ORDER_ITEMS** ?

Action: **Copy OEHR_ORDER_ITEMS to table OEHR_ORDER_ITEMS_AUDIT** ?

< Cancel **Finish**

- d. Add a column named updated. In the Object Selection pane, select the **OEHR_ORDERS_ITEMS_AUDIT** table and click the **Add Column** tab in the Details pane.

Object Browser

Tables

OEHR_EMPLOYEES

OEHR_INVENTORIES

OEHR_JOBS

OEHR_JOB_HISTORY

OEHR_LOCATIONS

OEHR_ORDERS

OEHR_ORDER_ITEMS

OEHR_ORDER_ITEMS_AUDIT

OEHR_PRODUCT_DESCRIPTIONS

OEHR_ORDER_ITEMS_AUDIT

Table Data Indexes Model Constraints Grants Statistics

Add Column Modify Column Rename Column Drop Column

Create Lookup Table

Column Name	Data Type	Nullable
ORDER_ID	NUMBER(12,0)	Yes
LINE_ITEM_ID	NUMBER(3,0)	No
PRODUCT ID	NUMBER(6,0)	No

- e. In the Add Column Wizard:
- i. Add Column: Enter **UPDATED**.

- ii. Type: Select **TIMESTAMP WITH LOCAL TIMEZONE**.

OEHR_ORDER_ITEMS_AUDIT

Add Column

Schema: **OEHR** ?

Table: **OEHR_ORDER_ITEMS_AUDIT** ?

* Add Column **UPDATED** ?

☐ Preserve Case

Type **TIMESTAMP WITH LOCAL TIME ZONE** ?

Length ?

- iii. Click **Next**.

- iv. Click **Finish**.

OEHR_ORDER_ITEMS_AUDIT

Confirm your request

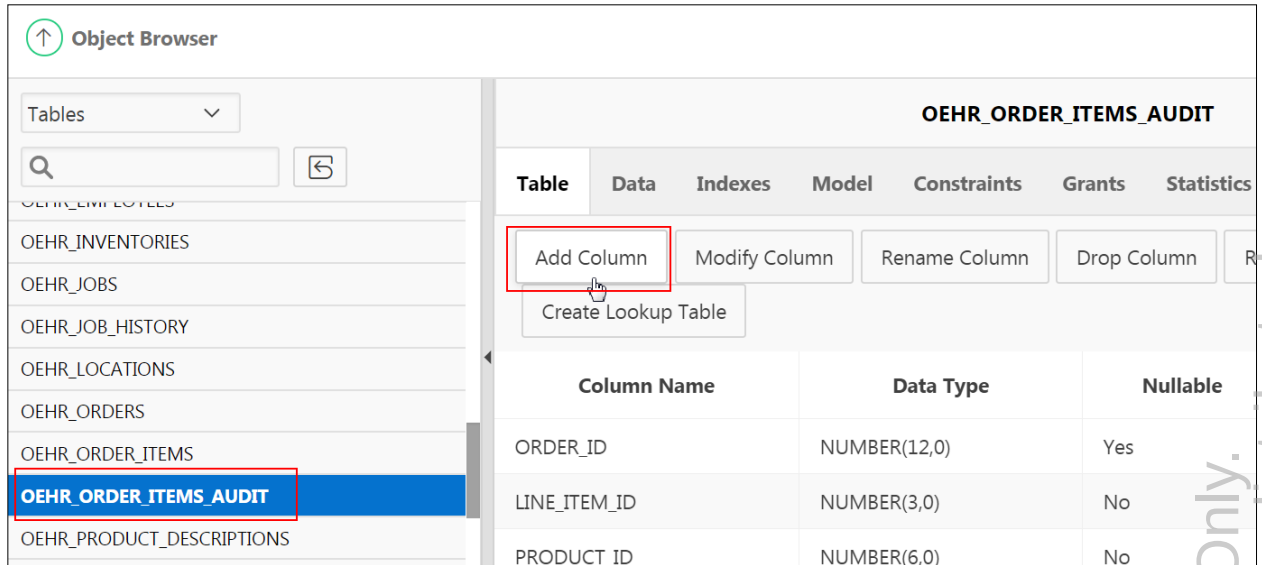
Schema: **OEHR** ?

Object: **OEHR_ORDER_ITEMS_AUDIT** ?

Action: **Add Column UPDATED** ?

< Cancel **Finish**

- f. Add a column named `UPDATED_BY`. In the Object Selection pane, ensure `OEHR_ORDERS_ITEMS_AUDIT` is still selected and click the **Add Column** tab in the Details pane.



- g. In the Add Column Wizard:
- **Add Column:** Enter `UPDATED_BY`.
 - **Type:** Select **VARCHAR2**.
 - **Length:** Enter 255.

The screenshot shows the 'Add Column' wizard for the 'OEHR_ORDER_ITEMS_AUDIT' table. The wizard has the following fields:

- Schema: **OEHR**
- Table: **OEHR_ORDER_ITEMS_AUDIT**
- Add Column: **UPDATED_BY**
- Preserve Case: ☐
- Type: **VARCHAR2**
- Length: **255**

A red box highlights the 'Add Column' field, 'Type', and 'Length' fields.

- Click **Next**.
- Click **Finish**.

OEHR_ORDERS_ITEMS_AUDIT now contains two new columns: UPDATED and UPDATED_BY.

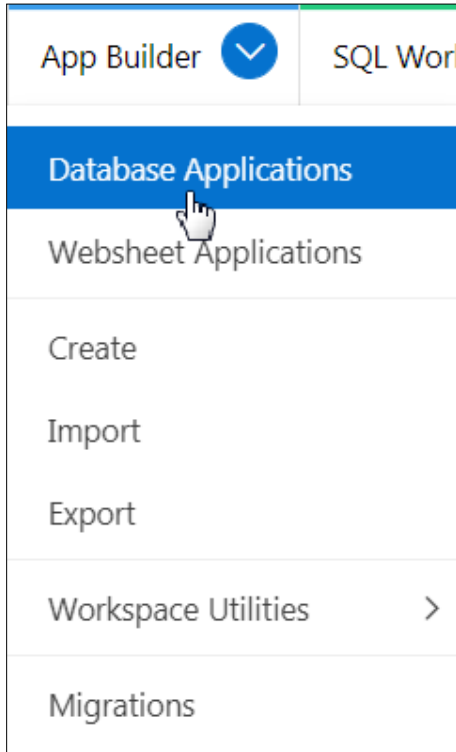
OEHR_ORDER_ITEMS_AUDIT

Table	Data	Indexes	Model	Constraints	Grants	Statistics	UI Defaults	Triggers
Add Column	Modify Column	Rename Column	Drop Column	Rename	Copy	Drop		
Column Name	Data Type							
ORDER_ID	NUMBER(12,0)							
LINE_ITEM_ID	NUMBER(3,0)							
PRODUCT_ID	NUMBER(6,0)							
UNIT_PRICE	NUMBER(8,2)							
QUANTITY	NUMBER(8,0)							
ORDER_ITEM_ID	NUMBER(12,0)							
UPDATED	TIMESTAMP(6) WITH LOCAL TIME ZONE							
UPDATED_BY	VARCHAR2(255)							

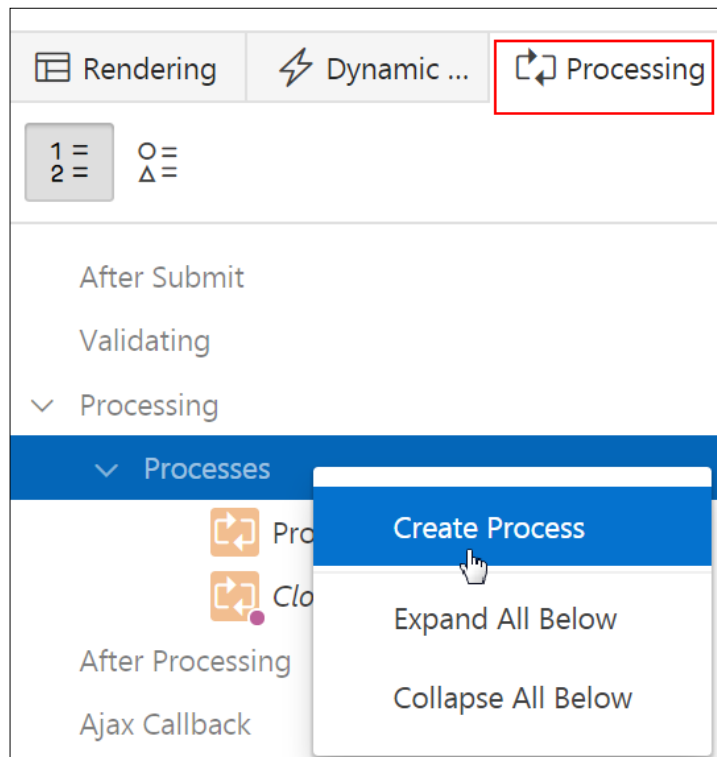
Download | Print

For Instructor Use Only.
This document should not be distributed.

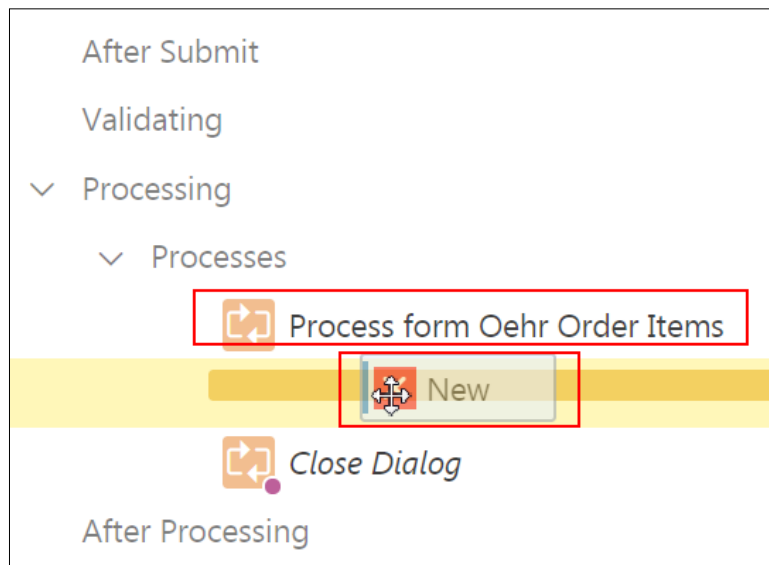
- h. Select **Database Application**; click the GMT application to return to your application.



- i. Click page **14 – Oehr Order Items** to open it in Page Designer.
- j. Create a new process to insert columns into the `OEHR_ORDERS_ITEMS_AUDIT` table:
- In the left pane, click the **Processing** tab .
 - Expand Processing, right-click **Processes**, and select **Create Process**.



- iii. Select the New process and drag it after **Process form OEHR ORDER ITEMS**.



- k. On the Rendering tab, ensure that the **New** process is selected. In the Property Editor, edit the following:
 - i. **Identification > Name:** Enter `Save Audit` and press Enter.
 - ii. **Source > PL/SQL Code:** Click the **Code Editor: PL/SQL Code** icon.

- iii. In the Code Editor, copy and paste the following PL/SQL code (located in /home/oracle/labs/labs/lab_11_01_02.txt) and click **OK**:

```
insert into OEHR_ORDER_ITEMS_AUDIT
( order_id
, order_item_id
, line_item_id
, product_id
, unit_price
, quantity
, updated
, updated_by
)
values
( :P14_ORDER_ID
, :P14_ORDER_ITEM_ID
, :P14_LINE_ITEM_ID
, :P14_PRODUCT_ID
, :P14_UNIT_PRICE
, :P14_QUANTITY
, SYSDATE
, :APP_USER
);
```

Identification

Name: Save Audit

Type: PL/SQL Code

Editable Region: - Select -

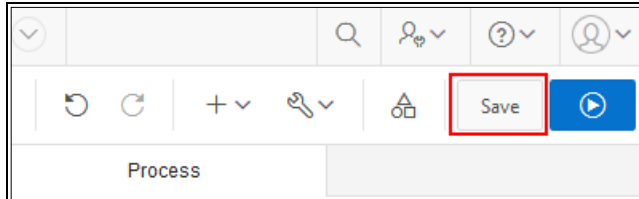
Source

Location: Local Database

PL/SQL Code

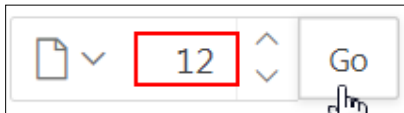
```
insert into OEHR_ORDER_ITEMS_AUDIT
( order_id
, order_item_id
, line_item_id
, product_id
, unit_price
, quantity
, updated
, updated_by
)
values
```

- I. Click **Save**.

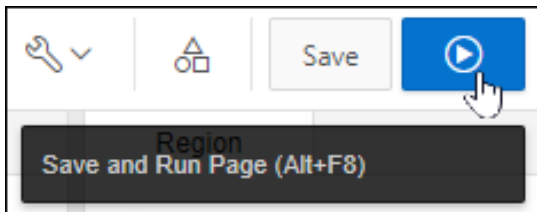


3. Test your changes. Because **Page 15 – Oehr Order Items** is a modal dialog, you cannot run it directly from Page Designer. Navigate to and then run page 12.

- a. In the Page Selector, in the Page Designer toolbar, enter 12 and click **Go**.



- b. When **Page 12: List of Order** appears, click **Save and Run Page**.



- c. If prompted, enter your workspace user credentials and click **Sign In**.

On the **List of Orders** page, select an order in the left, Master Record region.

- d. Scroll down in the right pane. Under **Order Details**, select Line Item by clicking the Edit icon.

A screenshot of the 'List of Orders' page. The page has a header with 'List of Orders' and tabs for 'Show All', 'In the News', and 'Order Details'. A search bar is on the left. The 'Order Details' table is on the right, with columns for 'Line Item Id', 'Product Id', 'Unit Price', and 'Quantity'. The first row of the table is highlighted, and the 'Edit' icon (pencil) is highlighted with a red rectangular box.

Line Item Id	Product Id	Unit Price	Quantity
1	2402	200	70
2	2410	350.9	21
3	2418	60	23
4	2422	144	25
5	2430	173	28

e. Edit the price and quantity for an order in the Order Items dialog box (for Product ID 2402):

- Unit Price: Change to 300.
- Quantity: Change to 50.
- Click **Apply Changes**.

Oehr Order Items

Line Item Id 1

Product Id 2402

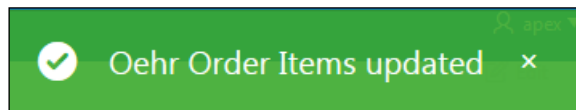
Unit Price 300

List Price 127

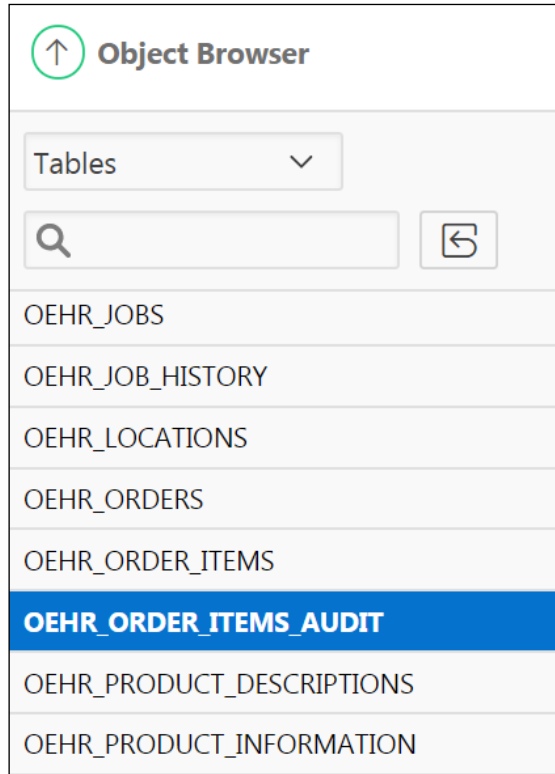
Quantity 50

Cancel Delete Apply Changes

The message “Oehr Order items updated” appears.



- f. View the `OEHR_ORDERS_ITEMS_AUDIT` table in Object Browser and see if the new table records the change.
 - i. Navigate to **Object Browser**.
 - ii. In the Object Selection pane, select the **OEHR_ORDERS_ITEMS_AUDIT** table.



- iii. In the **Details** pane, click the **Data** tab.

OEHR_ORDER_ITEMS_AUDIT							
Data Indexes Model Constraints Grants Statistics UI Defaults Triggers Dependencies SQL REST							
Count Rows		Insert Row					
ORDER_ID	LINE_ITEM_ID	PRODUCT_ID	UNIT_PRICE	QUANTITY	ORDER_ITEM_ID	UPDATED	UPDATED_BY
2442	1	2402	300	50	1089	08-MAY-19 05.15.27.000000 PM	APEX_DEV

Notice that the changes you just made in the Order Details dialog box appear. Also, notice the **Updated** and **Updated By** columns are populated.

- g. Exit Object Browser and return to your running application.
- h. Click **Application 111** on the Runtime Developer toolbar to return to the Application home page.



4. Certain items on the Customer Details page are marked as required. You can enable or disable this behavior by configuring the Page Item validation, **Value Required**. Examine the Value Required validation that exists on some items on the **9 - Customer Details** page.
 - a. Open page **4 - Customers** in Page Designer view.
 - b. Click **Save and Run Page**.
 - c. On the Customers interactive report, click the **Edit** icon to view the Customer Details page.

		Go		Actions ▾		Customer Feedback		
	Customer Id	First Name	Last Name	Street Address	Postal Code	City	State Province	Country
	218	Bryan	Dvrrie	3386 Perrysville Ave	15214	Pittsburgh	PA	US
	219	Ajay	Sen	220 Penn Ave # 300	18503	Scranton	PA	US
	220	Carol	Jordan	135 S 18Th St # 1	19103	Philadelphia	PA	US

- d. Notice that **Cust First Name** and **Cust Last Name** fields are marked with a red triangle in the upper-left corner. This red triangle means that the items must have a value.

Bryan Dvrrie Customer Details

Cust First Name
Bryan

Cust Last Name
Dvrrie

Cust Email
BRYAN.DVRRIE@REDPOLL.COM

Account Manager
Bates, Elizabeth
[Bates, Elizabeth, Russell, John, Cambrault, Gerald](#)

Country
-Select Country-

- e. Delete the value in the **Cust First Name** field and click **Apply Changes**.
- f. An error message appears stating that you must fill out the field. You can enable or disable this validation by editing the attributes for the item.

Cust First Name

Please fill out this field.

Cust Last Name
Dvrrie

Cust Email
BRYAN.DVRRIE@REDPOLL.COM

Account Manager
Bates, Elizabeth
[Bates, Elizabeth, Russell, John, Cambrault, Gerald](#)

Country
-Select Country-

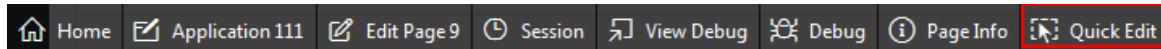
Cancel Delete Apply Changes

Correct errors before saving.

OK

- g. Click **OK** to exit the error message.
- h. Review the attributes that define the **Cust Last Name** and control this validation.

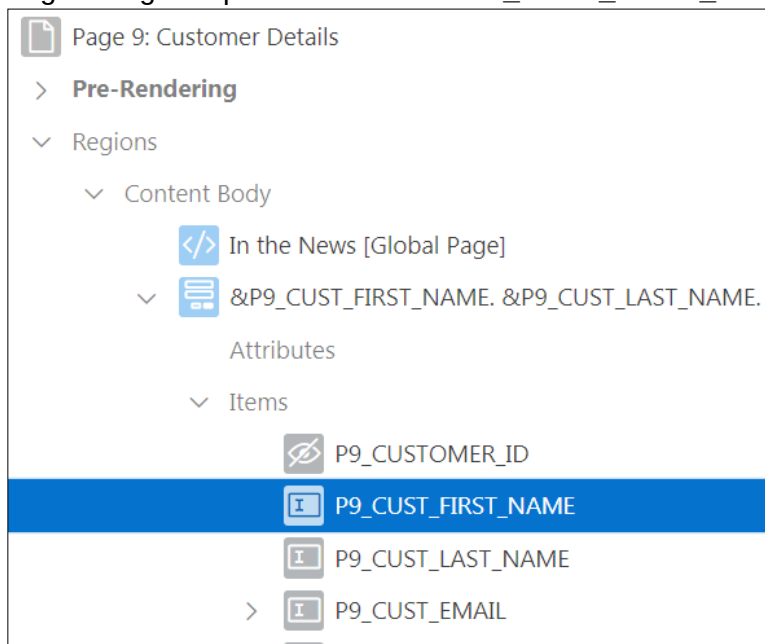
- i. Click the **Quick Edit** link on the Runtime Developer toolbar.



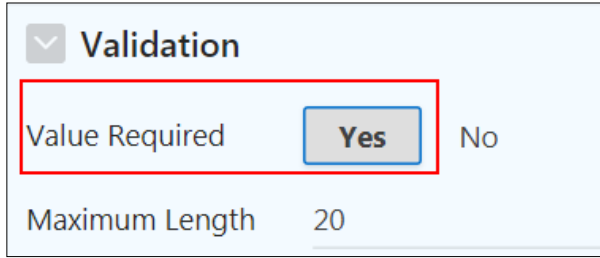
- ii. Select the **Cust First Name** field.

The image shows a web form titled 'Bryan Dvrrie Customer Details'. The form contains several input fields: 'Cust First Name' with the value 'Bryan', 'Cust Last Name' with the value 'Dvrrie', 'Cust Email' with the value 'BRYAN.DVRRIE@REDPOLL.COM', 'Account Manager' with the value 'Bates, Elizabeth', and 'Country' with a dropdown menu showing '-Select Country-'. The 'Cust First Name' field is highlighted with a blue border, and a plus sign (+) is visible to its right.

- i. Page Designer opens with the item **P9_CUST_FIRST_NAME** selected.



- j. In the Property Editor, find the attribute **Validation > Value Required**.



The screenshot shows a 'Validation' section in a property editor. It contains a 'Value Required' attribute with a 'Yes' button selected and a 'No' button. Below it, there is a 'Maximum Length' attribute set to '20'.

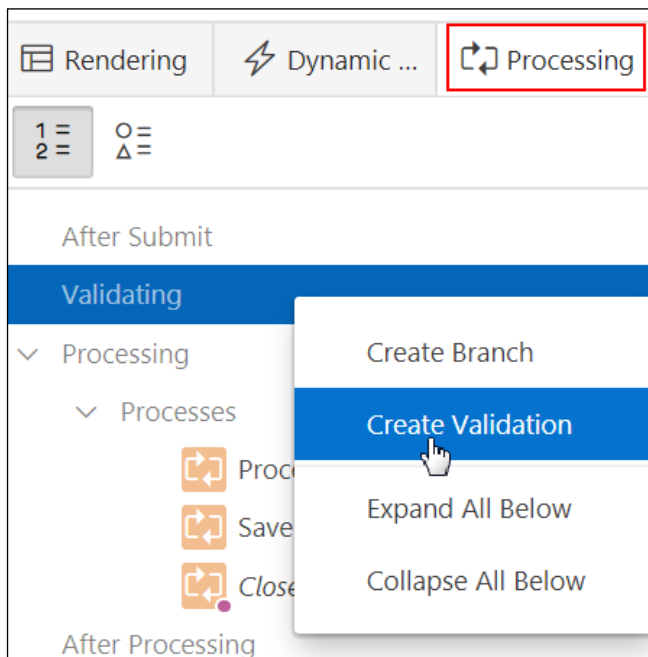
Note that **Value Required** is set to **Yes**.

Note: As an alternative, you could set **Value Required** to **No** and manually create a Not Null validation on this item. The advantage of this approach is that you can then create and custom the error message.

- k. Click the **Application 111** breadcrumb to return to the Application home page.



5. On the Order Items page, create a validation on the Quantity field to ensure that it has a value that is greater than zero.
- On the Application home page, select page **14 – Oehr Order Items**.
 - In the left pane of Page Designer, click the **Processing** tab .
 - Right-click **Validating** and select **Create Validation**.



- d. On the Processing tab, ensure that the newly created validation is selected.

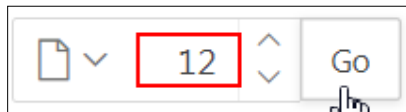
The screenshot shows the APEX interface with the 'Processing' tab selected. Under the 'Validating' section, the 'Validations' subsection is expanded, and a 'New' button is visible.

- e. In the Property Editor, edit the following:

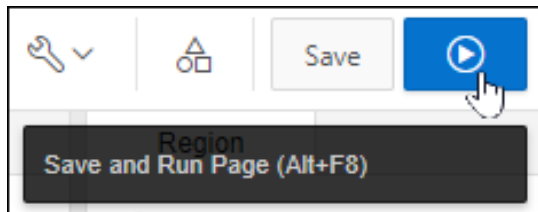
- **Identification > Name:** Enter `Quantity Validation`.
- **Validation > Type:** Select **Item is NOT NULL or zero**.
- **Validation > Item:** Select `P14_QUANTITY`.
- **Error > Error Message:** Enter `Quantity must be greater than 0.`

The screenshot shows the Property Editor for a validation rule. The 'Name' field is set to 'Quantity Validation'. The 'Type' is set to 'Item is NOT NULL or zero' and the 'Item' is set to 'P14_QUANTITY'. The 'Always Execute' checkbox is checked.

- Click **Save**.
- f. See how the validation works by running the page.
 - i. In Page Selector, in the Page Designer toolbar, enter 12 and click **Go**.



- ii. When page 12: List of Order appears, click **Save and Run Page**.



- iii. If prompted, enter your workspace user credentials and click **Sign In**.

- iv. On the **List of Orders** page, select an order in the left, Master Record region.
- v. Scroll down in the right pane. Under Order Details, select a Line Item by clicking the Edit icon.

List of Orders

Order Total: 52471.9 Reset + Create

Sales Rep: 154

Search...

28-JAN-09 11.22.59.000000 AM direct

28-JAN-09 12.34.16.000000 PM direct

28-JAN-09 02.34.38.000000 PM direct

Show All In the News Order Details

Order Details

Line Item Id	Product Id	Unit Price	Quantity	
1	2402	300	50	
2	2410	350.9	21	
3	2418	60	23	
4	2422	144	25	

- vi. In the Quantity field in the Order Items dialog box, enter 0 and click **Apply Changes**.
An error message appears that reads *Quantity must be greater than 0*.

Oehr Order Items

Order Id

Line Item Id

Product Id: 2402

Unit Price: 300

List Price: 127

Quantity: 0

Quantity must be greater than 0.

Cancel Delete Apply Changes

1 error has occurred

- Quantity must be greater than 0.

- vii. Click **Cancel** to exit the Order Items dialog box.
- viii. Click **Application 111** on the Runtime Developer toolbar to return to the Application home page.

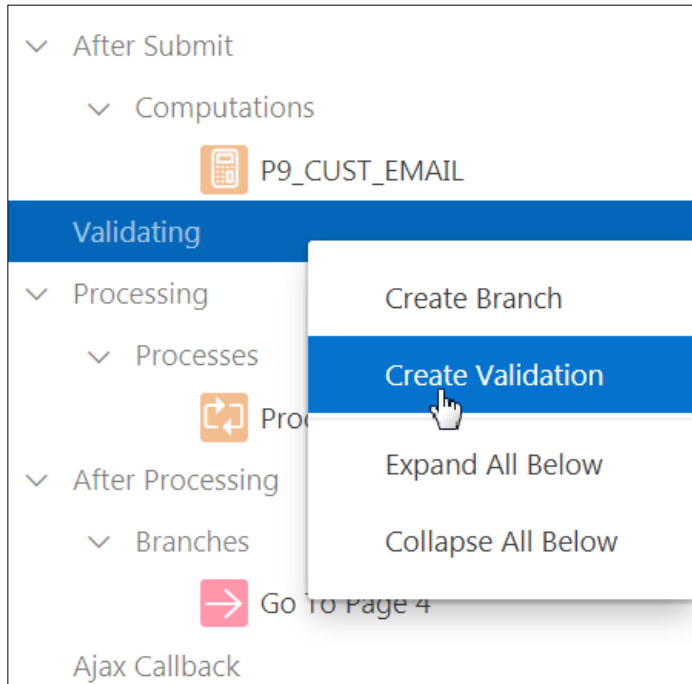
6. Create an item string comparison validation to ensure that there are no spaces in the **Cust Email** field on the Customer Details page.

- a. On the Application home page, select the page, **9 - Customer Details**.

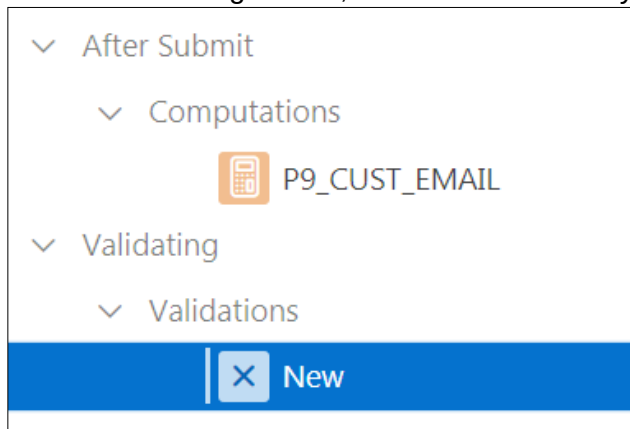
- b. In the left pane of Page Designer, click the **Processing** tab .

Note: The Processing tab should already be selected.

- c. Right-click **Validating** and select **Create Validation**.



- d. On the Processing tab , ensure that the newly created validation is selected.



- e. In the Property Editor, edit the following:

- **Identification > Name:** Enter `Email Validation`.
- **Validation > Type:** Select **Item contains no spaces**.
- **Validation > Item:** Select `P9_CUST_EMAIL`.

- **Error > Error Message:** Enter Email must not contain spaces.
- **Error > Display Location:** Selected by default: **Inline with Field and in Notification**

Identification	
Name	Email Validation
Execution Options	
Sequence	10
Validation	
Editable Region	- Select -
Type	Item contains no spaces
Item	P9_CUST_EMAIL
Always Execute	Yes No
Error	
Error Message	
Email must not contain spaces.	
Display Location	Inline with Field and in Notification
Associated Item	- Select -

- **Server Side Condition > Type:** Select **Item is NOT NULL**.
Note: You want this validation to fire only when the item is not null.

- **Server-side Condition > Item:** Select P9_CUST_EMAIL.

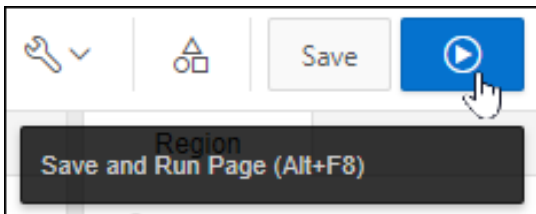
☒ **Server-side Condition**

When Button Pressed: - Select -

Type: Item is NOT NULL

Item: P9_CUST_EMAIL

- f. See how the validation works. Click **Save and Run Page**.



- l. Click Cancel to go to the **4 - Customers** page.
- m. Click the Edit button next to any record.
- n. Edit the Cust Email field by adding a space to the email address and click **Apply Changes**.

GlobalMart Management Tool

Eddie Boyer Customer Details

Cust First Name: Eddie

Cust Last Name: Boyer

Cust Email: EDDIE.BOYER@GALLINULE.COM

Account Manager: Bates, Elizabeth

[Bates, Elizabeth, Russell, John, Cambrault, Gerald](#)

An error message is displayed inline and also at the top of the page.

GlobalMart Management Tool

1 error has occurred

- Email must not contain spaces.

Eddie Boyer Customer Details

Cust First Name
Eddie

Cust Last Name
Boyer

Cust Email
EDDIE. BOYER@GALLINULE.COM

Email must not contain spaces.

Account Manager
Bates, Elizabeth

Bates, Elizabeth, Russell, John, Cambrault, Gerald

- o. Remove the space you added in the Cust Email field and click **Apply Changes** again. The Customers interactive report displays again without the error messages. Note that if you have added Email as Subtype for the Email item, then it also runs this validation check in case you enter an invalid character (for example “space”) while entering the customer email address in the Customer Details form.
- p. Click **Cancel** to exit the Customer Details page.
- q. Click **Application 111** on the Runtime Developer toolbar.



Practices for Lesson 12: Using Dynamic Actions and Plug-Ins

Practices for Lesson 12: Overview

Practices Overview

In these practices, you examine how dynamic actions and plug-ins work.

For Instructor Use Only.
This document should not be distributed.

Practice 12-1: Using Dynamic Actions on Your Page

Overview

In this practice, you create dynamic actions in the GlobalMart Management Tool application.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. **Show Item when another item is Not Null:** On the Customer Details page, show `P9_CUST_EMAIL` only if the `P9_CUST_FIRST_NAME` is not null.
2. **Refresh report data using custom filters:** On the Customers page, create a Quick Filter region with two select list items `P4_CUSTOMER_ID` and `P4_CITY_ID`. Create a dynamic action on the select lists to refresh the report when the values of the select list change.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_admin` user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the `apex_dev` user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 12-1**.

If you haven't completed **Practice 11-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the `apex_dev` user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_11_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).

- h. Wait for the application to successfully install.

For Instructor Use Only.
This document should not be distributed.

Practice 12-2: Importing and Using Plug-Ins on Your Page

Overview

In this practice, you import and use an item type plug-in and a dynamic action plug-in.

Note: The solution script generated at the end of this practice does not contain the **Rating** column.

Assumptions

You have completed the previous practices.

Note: If you haven't completed the previous practices, execute the steps mentioned in the Catch Up section of this practice.

Tasks

1. Add a rating column to the `CUSTOMERS` table.
2. Import the Modern Star Rating Item plug-in.
3. Add the Modern Star Rating plug-in to your Customer Details page.

Catch Up

If you haven't completed **Practice 2-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_admin** user.
- b. Click the arrow next to **SQL Workshop** and select **SQL Scripts**.
- c. Click the **Upload >** button.
- d. Browse and upload the `sol_02_01.sql` file in the `/home/oracle/labs/solutions` directory.
- e. Run the script by clicking the **Run** icon.
- f. Click **Run Now**.
- g. Make sure that the script executed successfully and the **apex_dev** user is created.

If you haven't completed **Practice 2-3: Installing a Sample Dataset**, perform the steps to install a sample dataset before you start with **Practice 12-2**.

If you haven't completed **Practice 12-1**, perform the following steps:

- a. Sign in to the Application Express workspace as the **apex_dev** user.
- b. Click the arrow next to **Application Builder** and select **Import**.
- c. Click the **Browse** button.
- d. Browse and open the `sol_12_01.sql` file in the `/home/oracle/labs/solutions` directory and click **Next >**.
- e. Click **Next >**.
- f. Select **Reuse Application ID 111 From Export File** for the **Install As Application** option and click **Install Application**.
- g. Click **Replace Application** (if prompted).
- h. Wait for the application to successfully install.

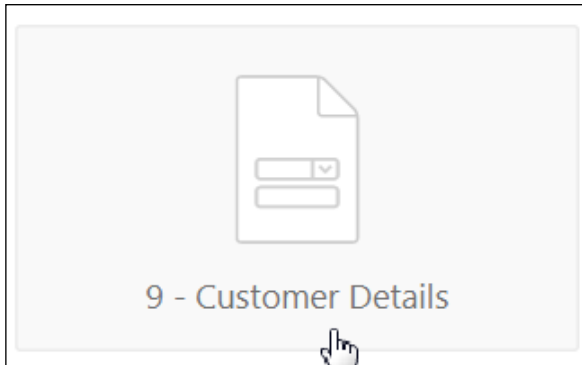
Solution 12-1: Using Dynamic Actions on Your Page

Overview

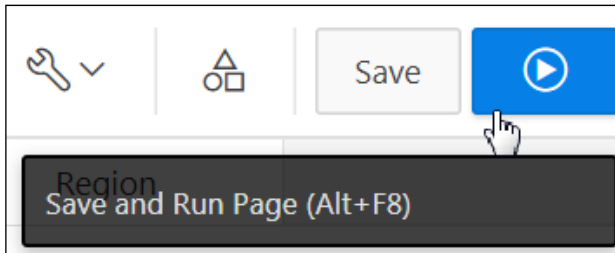
In this practice, you create various dynamic actions on the Customer Details page.

Steps

1. **Show Item when another item is Not Null:** On the Customer Details page, show P9_CUST_EMAIL only if the P9_CUST_FIRST_NAME is not null.
 - a. On the Application home page, select **9 - Customer Details** page.

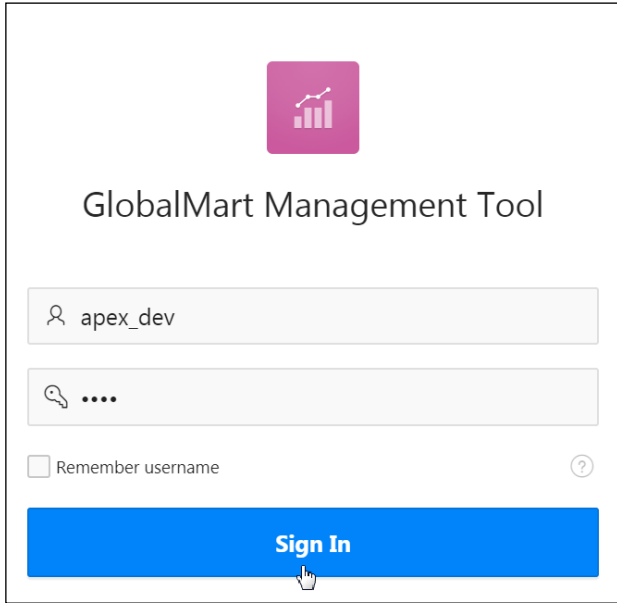


- b. Page Designer appears, displaying **Page 9: Customer Details**. Click **Save and Run Page**.



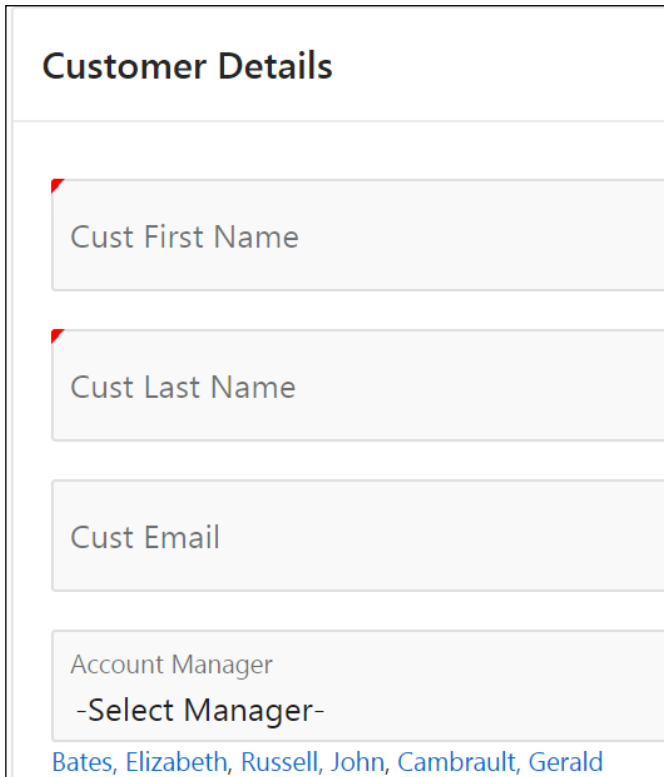
For Instructor Use Only.
This document should not be distributed.

- c. If you are prompted to sign in, enter your workspace credentials and click **Sign In**.



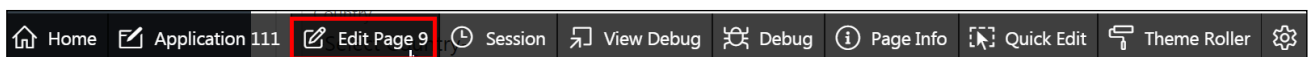
The image shows the sign-in page for the GlobalMart Management Tool. At the top is a purple square icon with a white bar chart. Below the icon is the title "GlobalMart Management Tool". There are two input fields: the first contains the username "apex_dev" and the second contains masked characters "....". Below the input fields is a checkbox labeled "Remember username" and a help icon (?). At the bottom is a large blue button labeled "Sign In". A mouse cursor is pointing at the "Sign In" button.

- d. The Customer Details page appears. Add a dynamic action that shows the **Cust Email** field only if the user enters a value in **Cust First Name**.

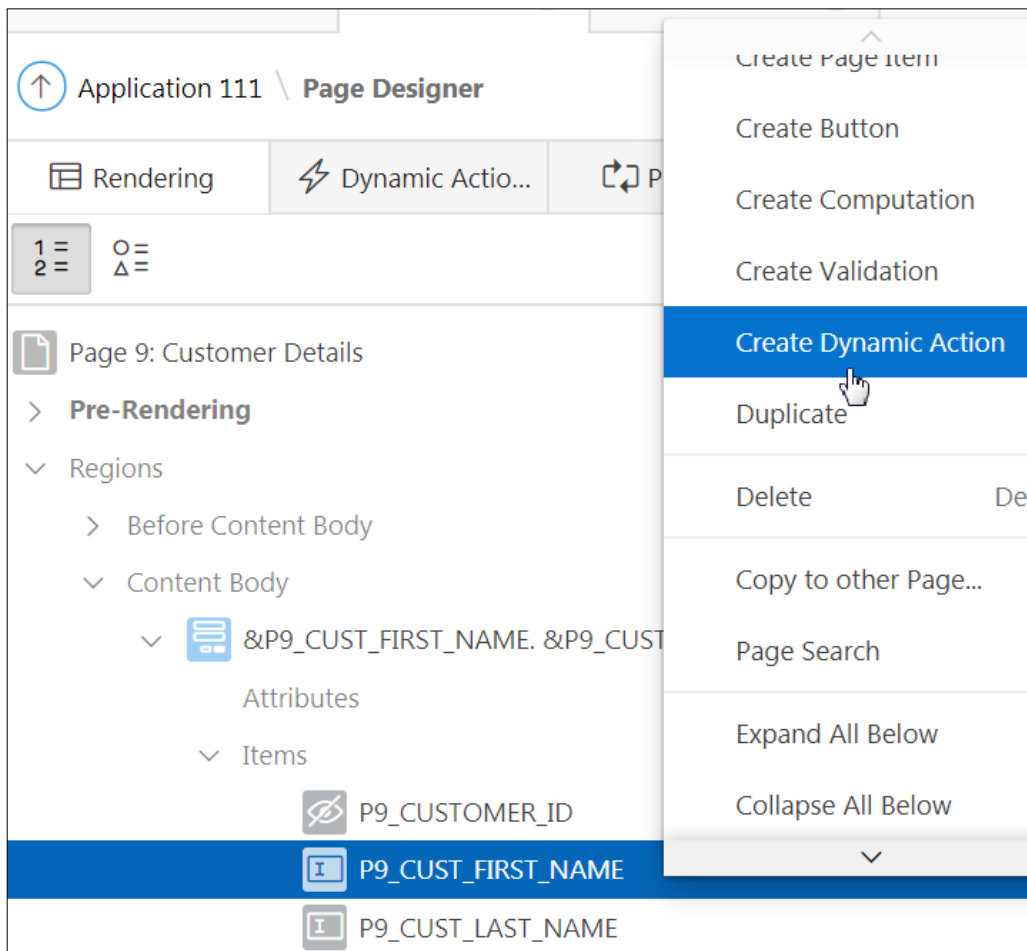


The image shows the "Customer Details" page. It has a title "Customer Details" at the top. Below the title are four input fields: "Cust First Name", "Cust Last Name", "Cust Email", and "Account Manager". The "Account Manager" field has a dropdown menu with the text "-Select Manager-". Below the input fields is a link that reads "Bates, Elizabeth, Russell, John, Cambrault, Gerald".

- e. Click the **Edit Page 9** icon on the Runtime Developer toolbar.



- f. On the Rendering tab, right-click `P9_CUST_FIRST_NAME` and select **Create Dynamic Action**.



A new dynamic action appears on the Rendering tab under `P9_CUST_FIRST_NAME`.

- g. Select the **New** dynamic action. In the Property Editor (right pane), enter the following:
- **Identification > Name:** Enter `Show Email`.
 - **Client Side Condition > Type:** Select **Item is not null**.

The screenshot shows the Property Editor for a new dynamic action. It is divided into four main sections: Identification, Execution Options, When, and Client-side Condition. The Identification section has a 'Name' field set to 'Show Email'. The Execution Options section has a 'Sequence' field set to '10'. The When section has three fields: 'Event' set to 'Change', 'Selection Type' set to 'Item(s)', and 'Item(s)' set to 'P9_CUST_FIRST_NAME'. The Client-side Condition section has two fields: 'Type' set to 'Item is not null' and 'Item' set to 'P9_CUST_FIRST_NAME'. Red boxes highlight the 'Name' field in the Identification section and the 'Type' field in the Client-side Condition section.

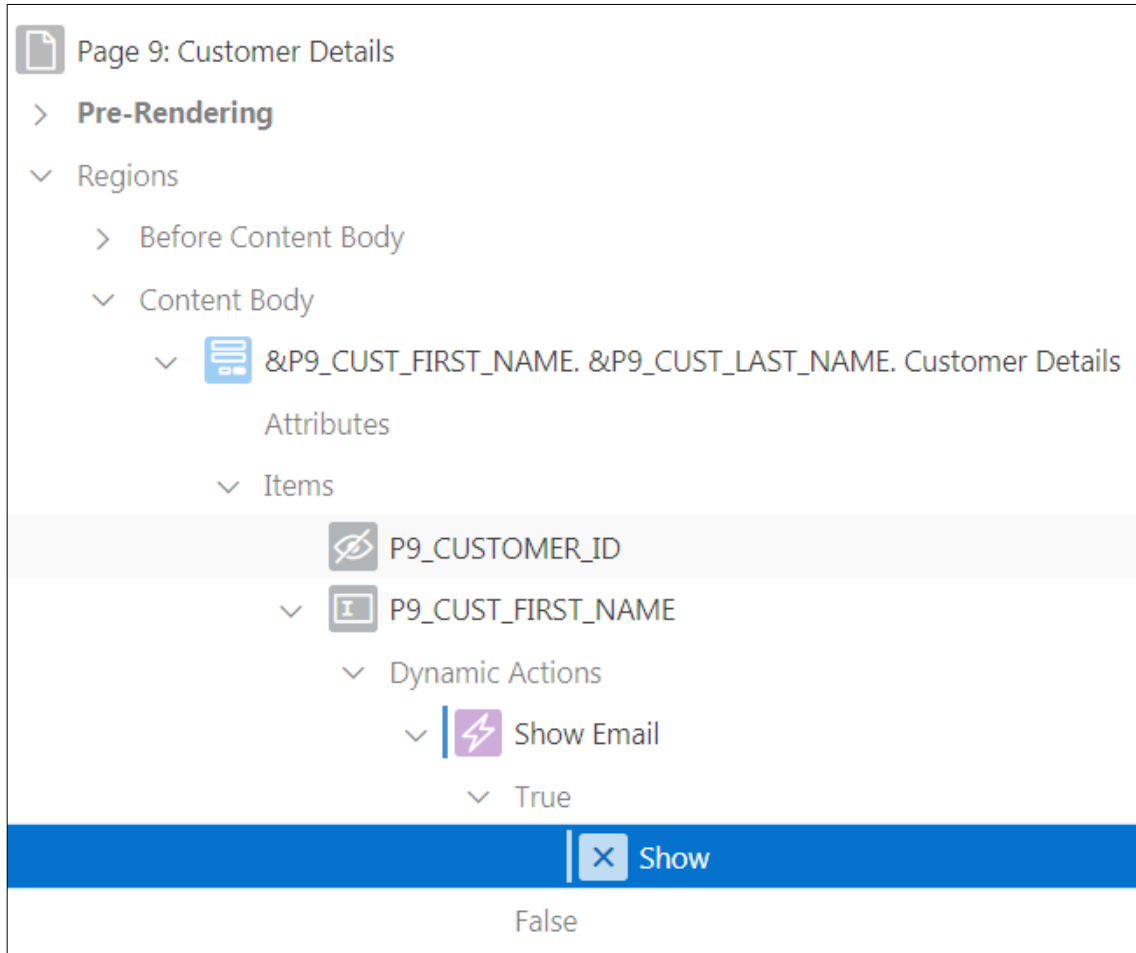
Identification	
Name	Show Email

Execution Options	
Sequence	10

When	
Event	Change
Selection Type	Item(s)
Item(s)	P9_CUST_FIRST_NAME

Client-side Condition	
Type	Item is not null
Item	P9_CUST_FIRST_NAME

- h. On the Rendering tab (left pane), select **Show Email > True > Show**.

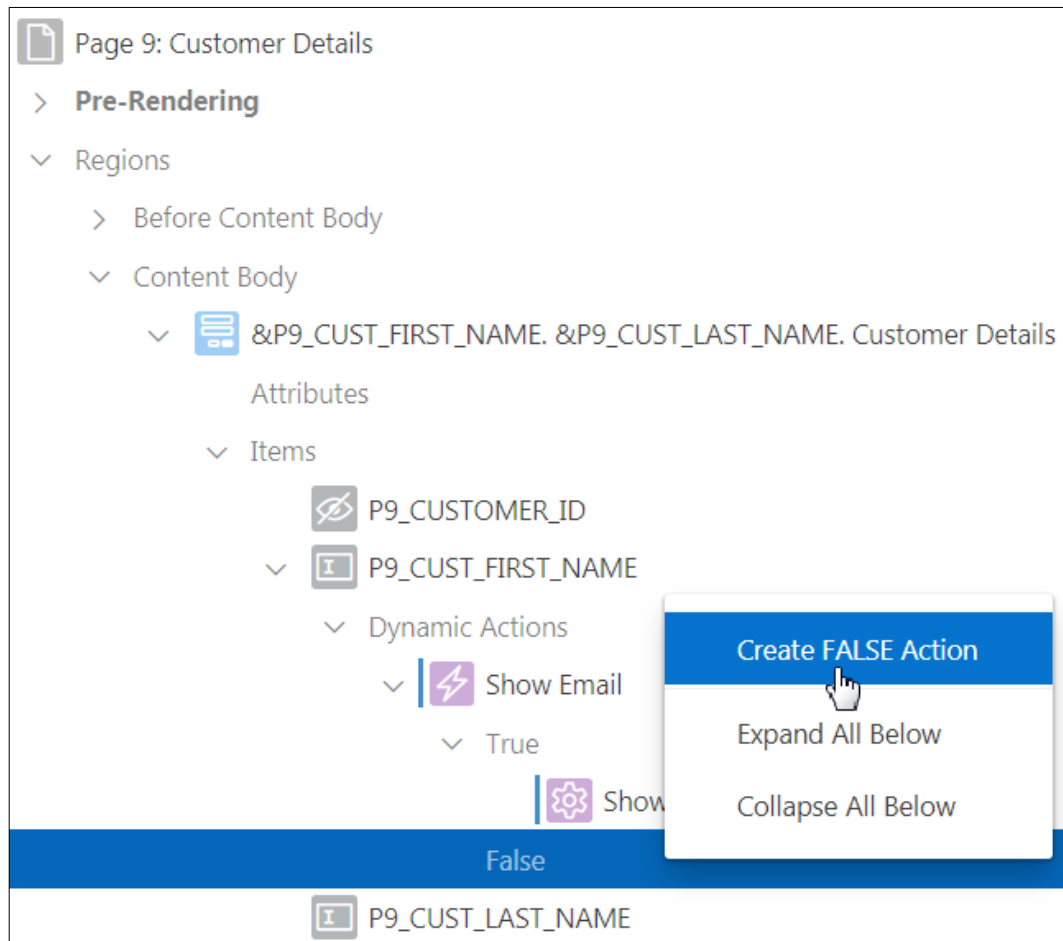


- i. In the Property Editor (right pane), enter the following:
- **Identification > Action:** Select **Show** (selected by default).
 - **Affected Elements > Selection Type:** Select **Item(s)**. (selected by default)
 - **Affected Elements > Item(s):** Select `P9_CUST_EMAIL`.

The screenshot shows the Property Editor interface with three main sections:

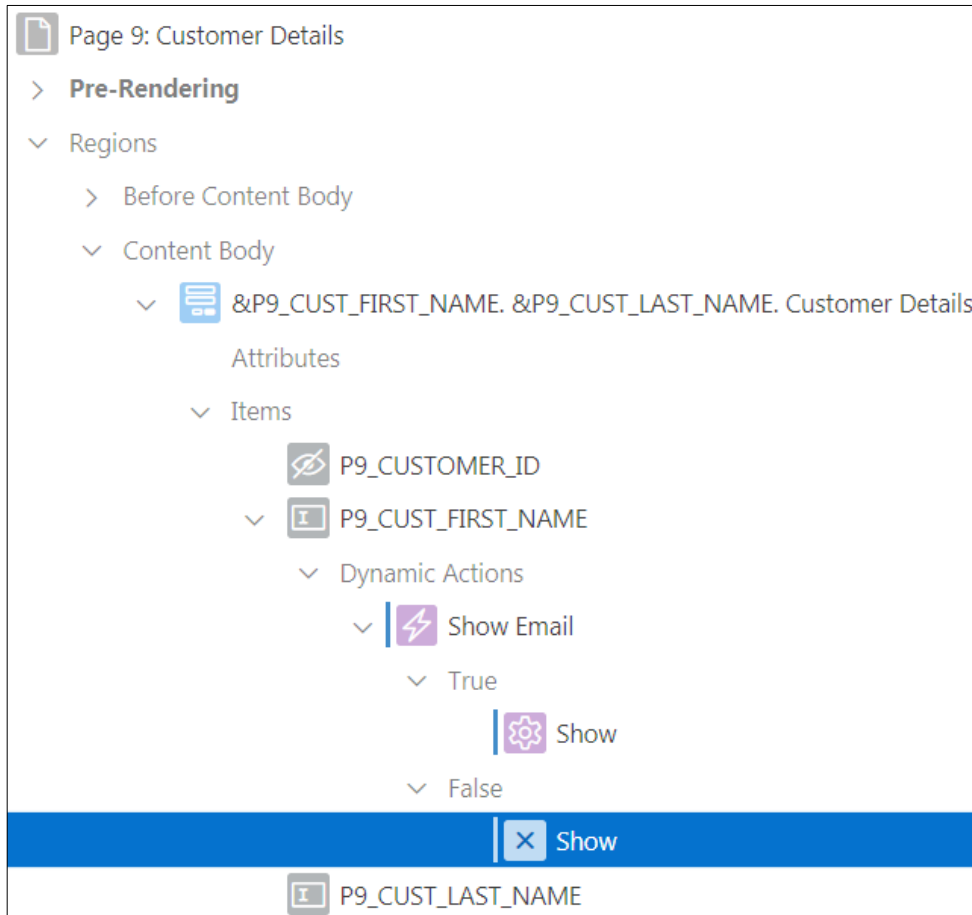
- Identification:** Contains a dropdown menu labeled "Action" with the value "Show" selected. A red box highlights the "Action" label.
- Settings:** Contains a toggle switch for "Show all on the same line" with "Yes" selected and "No" as an alternative option.
- Affected Elements:** Contains two dropdown menus. The first is labeled "Selection Type" with "Item(s)" selected. The second is labeled "Item(s)" with "P9_CUST_EMAIL" selected. Red boxes highlight both dropdown labels.

- j. On the Rendering tab, right-click **Show Email > False** and select **Create FALSE Action**.



For Instructor Use Only.
This document should not be distributed.

k. Ensure that **Show Email > False > Show** is selected on the Rendering tab.

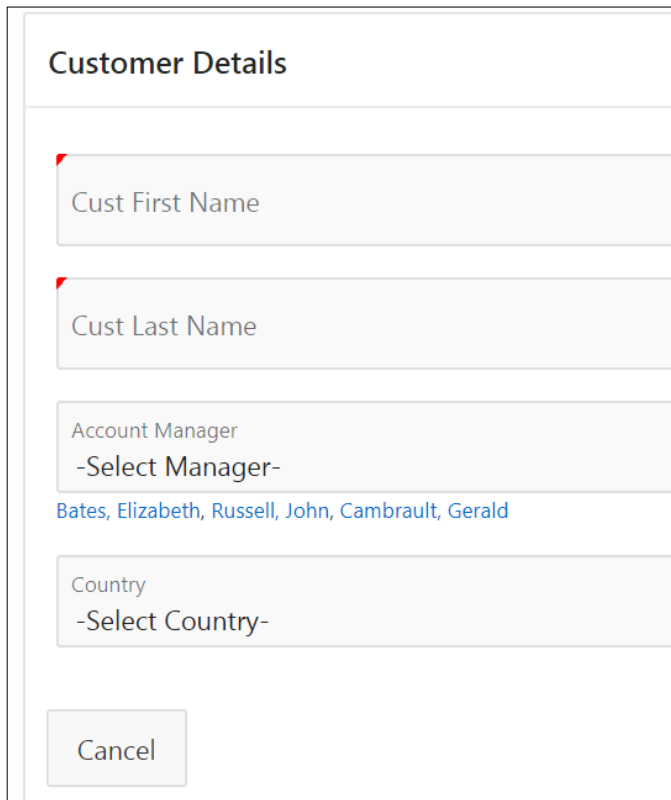


- I. In the Property Editor, enter the following:

- **Identification > Action:** Select **Hide**.
- **Affected Elements > Selection Type:** Select **Item(s)** (selected by default).
- **Affected Elements > Item(s):** Select P9_CUST_EMAIL.

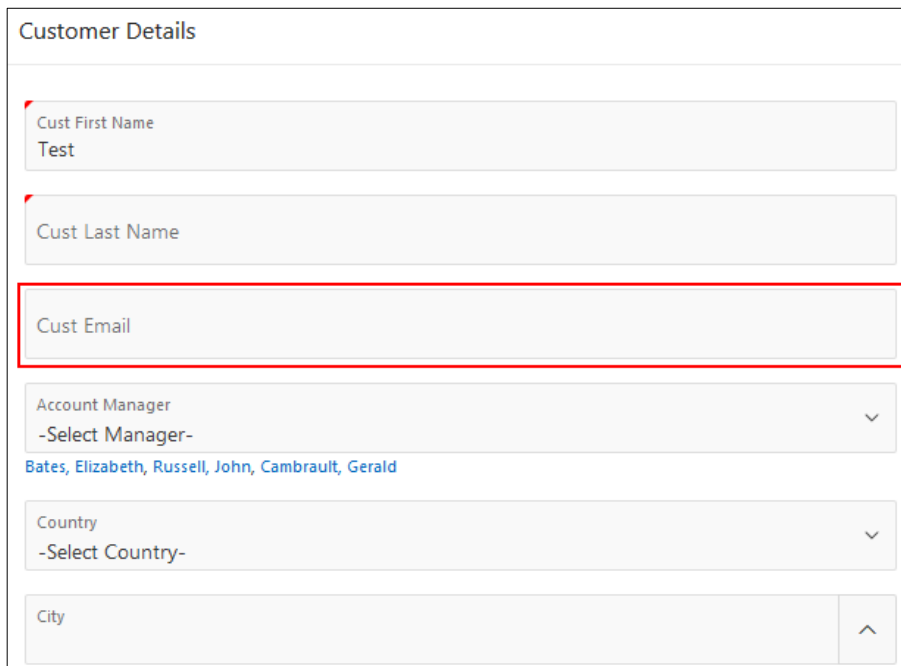
-

Notice that when you first run the page, and there is no value in the **Cust First Name** field, the **Cust Email** field is no longer displayed. This is because of the dynamic action you just created.



The screenshot shows a form titled "Customer Details". It contains several input fields: "Cust First Name", "Cust Last Name", "Account Manager" (a dropdown menu with "-Select Manager-" and a list of names: Bates, Elizabeth, Russell, John, Cambrault, Gerald), and "Country" (a dropdown menu with "-Select Country-"). There is also a "Cancel" button at the bottom left. The "Cust Email" field is not visible in this state.

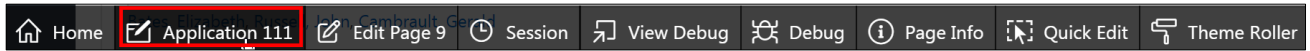
- n. Enter `Test` in the **Cust First Name** field and then click outside the field. Notice that the **Cust Email** field reappears.



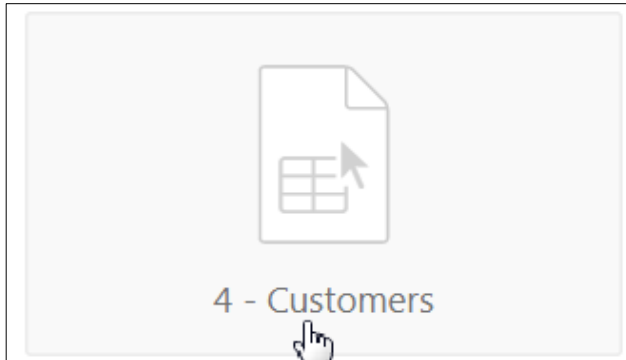
The screenshot shows the same "Customer Details" form, but now the "Cust Email" field is visible and highlighted with a red border. The "Cust First Name" field now contains the text "Test". The other fields and the "Cancel" button remain the same.

Delete your entry in the **Cust First Name** field and click outside the field. Notice that the **Cust Email** field disappears.

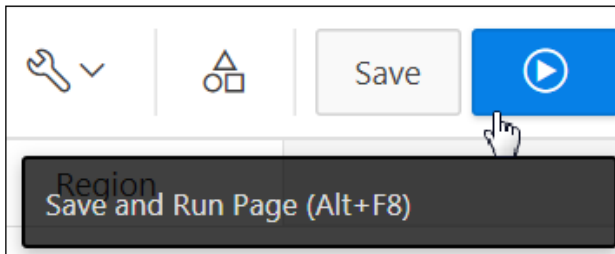
- o. Click the **Application 111** link on the Runtime Developer toolbar.



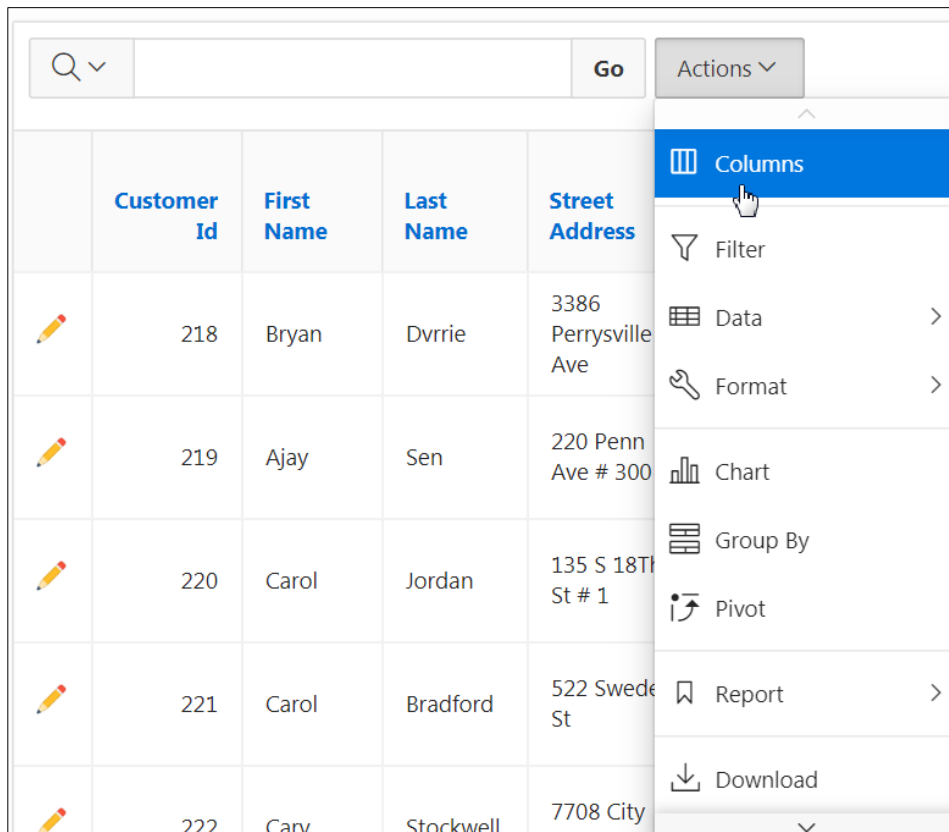
2. **Refresh report data using custom filters:** On the Customers page, create a Quick Filter region with two select list items P4_CUSTOMER_ID and P4_CITY_ID. Create a dynamic action on the select lists to refresh the report when the values of the select list change.
- a. On the Application home page, select the **Customers** page.



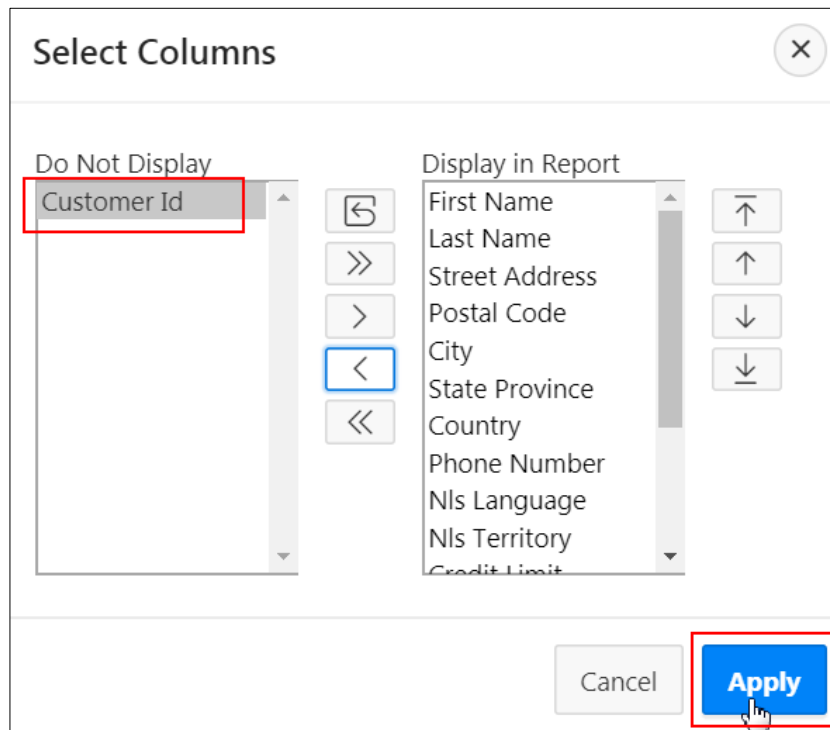
- b. Click the **Save and Run Page**.



- c. On the Customers report, change the columns that display in the report.
- i. Click the **Actions** menu and select **Columns**.

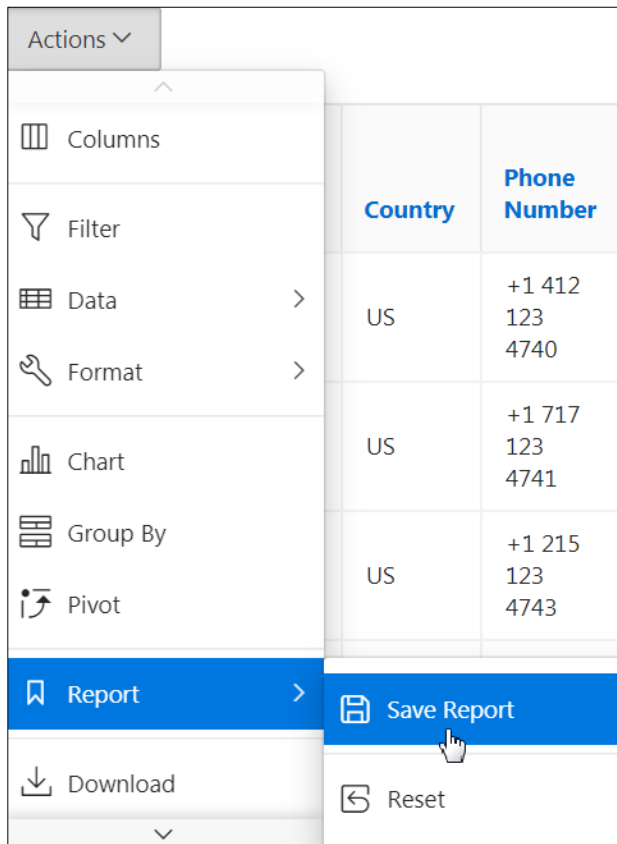


- ii. In the **Select Columns** dialog box, select **Customer Id**, click **Remove** (<), and then click **Apply**.

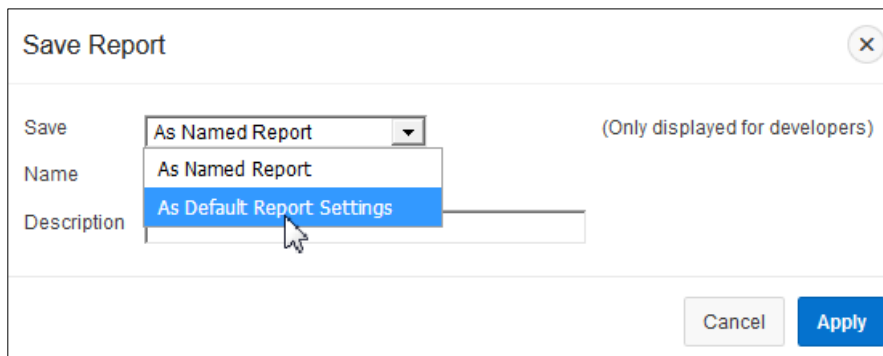


d. Save a new Primary, Default Report.

i. Click the **Actions** menu and then select **Report** and **Save Report**.



ii. The Save Report dialog box appears. From **Save**, select **As Default Report Settings**.



iii. The Save Default Report dialog box appears. Select **Primary** and click **Apply**.

Save Default Report

×

The current report settings will be used as the default for all users.

Default Report Type ☒ Primary ☐ Alternative

Cancel

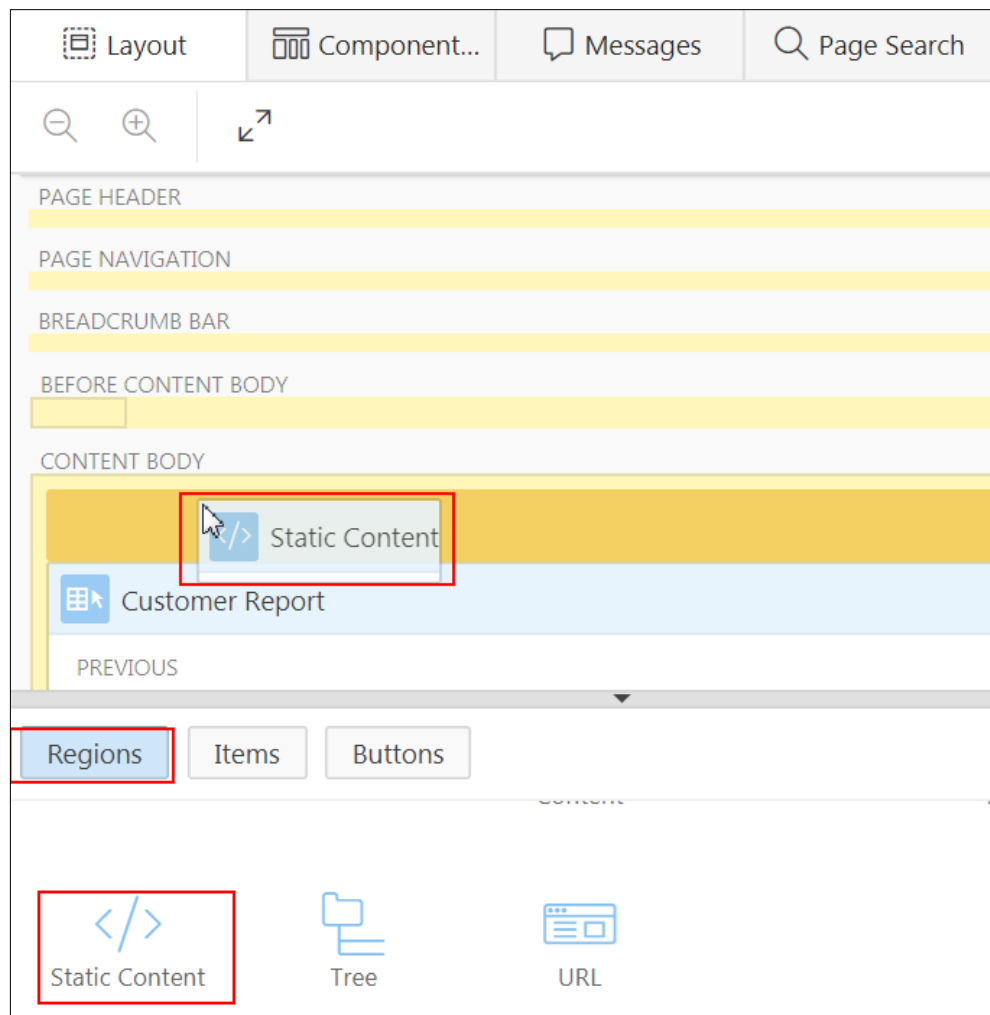
Apply

- e. The report settings are saved. Click the **Edit Page 4** link on the Run Developer toolbar.

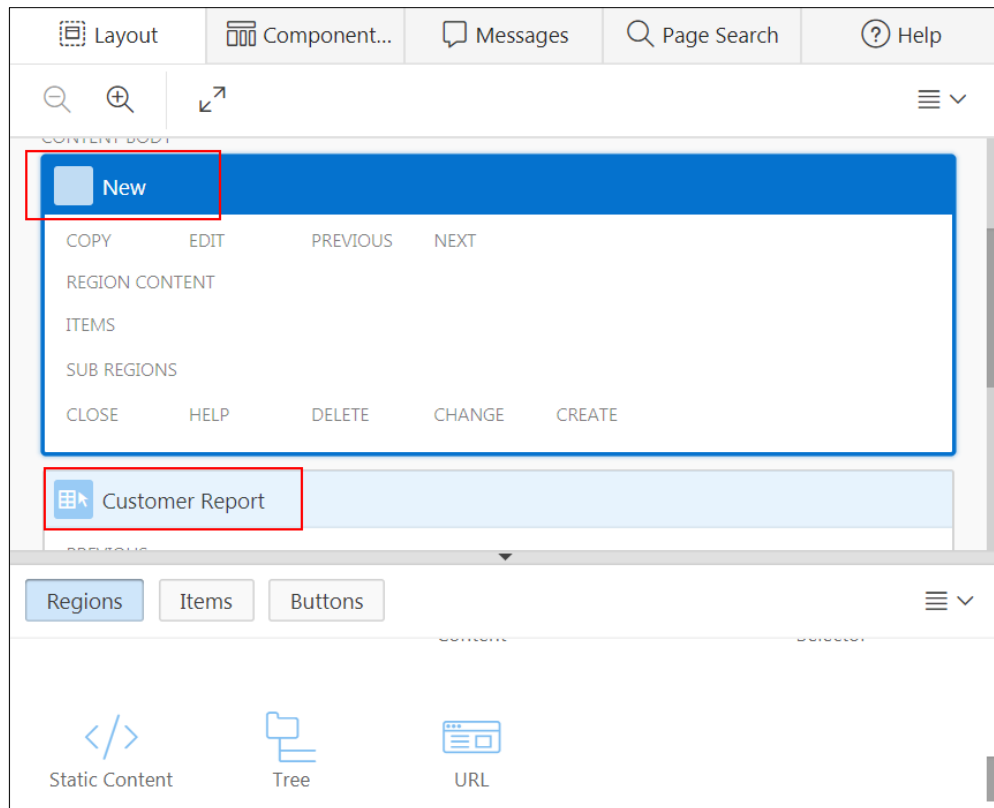
	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province	Country Id	Phone Number	Nls Language
	Eddie	Boyer	411 E Wisconsin Ave # 2550	53202	Seattle	WI	US	+1 414 123 4369	us
	Eddie	Stern	808 3Rd St # 100	54403	Wausau	WI	US	+1 715 123 4372	us
	Ernest	Weaver	300 Crooks St	54301	Green Bay	WI	US	+1 414 123 4373	us
	Ernest	George	122 E Dayton St	53703	Madison	WI	US	+1 608 123 4374	us

Home Application 111 **Edit Page 4** Session View Debug Debug Page Info Quick Edit

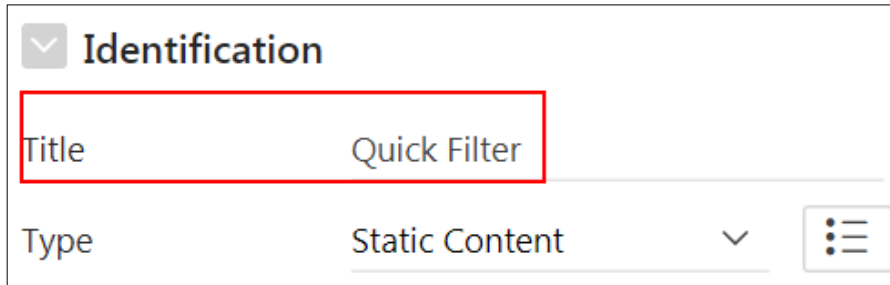
- f. On Page 4: Customers, create the region to contain the filtered item.
- From the Gallery at the bottom of the central pane, select the **Regions** tab.
 - Find the **Static Content** region. Select **Static Content** and drag it above the **Customer Report** region.



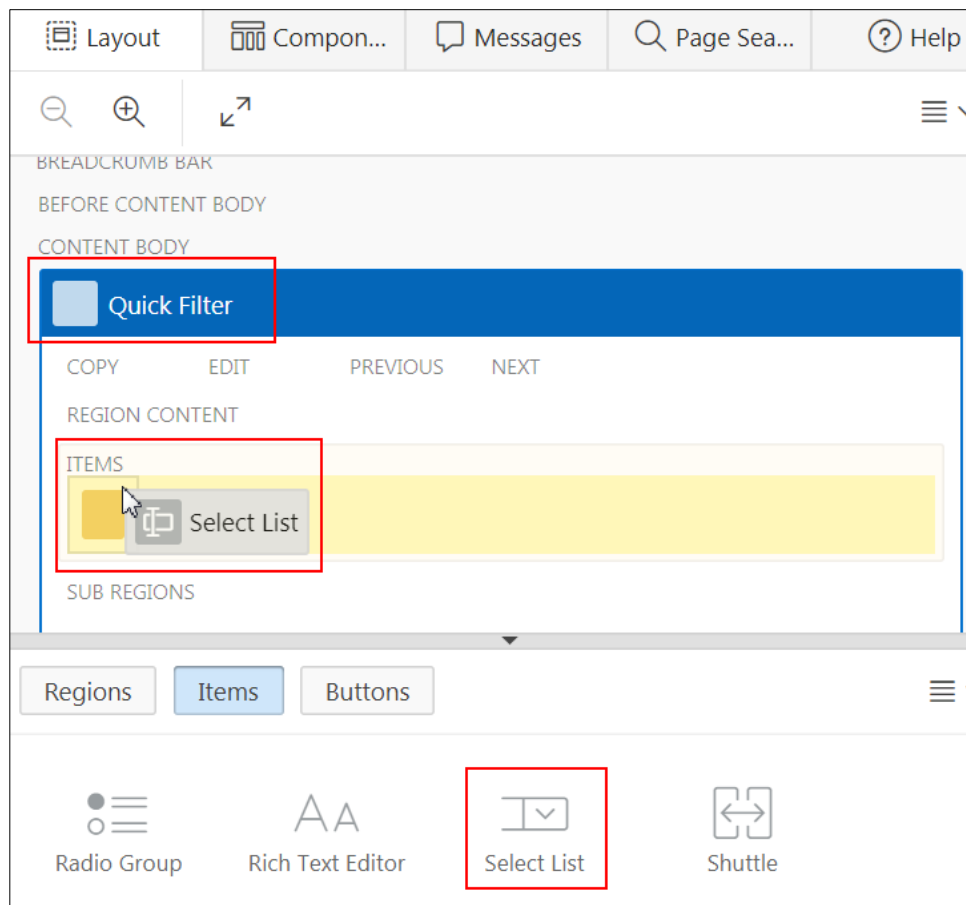
For Instructor Use Only.
This document should not be distributed.



- g. On the Rendering tab, select the new region. In the Property Editor, in **Identification** > **Title** enter Quick Filter.



- h. In the Quick Filter region, create two filter items: P4_CUSTOMER_ID and P4_CITY to filter the report. Create an item for P4_CUSTOMER_ID:
- From the Gallery at the bottom of the central pane, select the **Items** tab.
 - Select **Select List** and drag it to **ITEMS** in the Quick Filter region.



- i. On the Rendering tab, select the new item.
- j. In the Property Editor, edit the following:
 - **Identification > Name:** Enter P4_CUSTOMER_ID.
 - **Label > Label:** Enter Customer.

Identification

Name

P4_CUSTOMER_ID

Type

Select List

Label

Label

Customer

- **List of Values > Type:** Select **SQL Query**.
- **SQL Query:** Enter the query, which is provided in
/home/oracle/labs/labs/lab_12_01_01.txt.

```
select cust_first_name||' '||cust_last_name d, customer_id r
from oehr_customers
order by 1
```

- **List of Values > Display Null Value:** Select **Yes**.
- **List of Values > Null Display Value:** Enter - Show All -.

List of Values

Type: SQL Query

SQL Query:

```
select cust_first_name||' '||cust_last_name d, customer_id r
from oehr_customers
order by 1
```

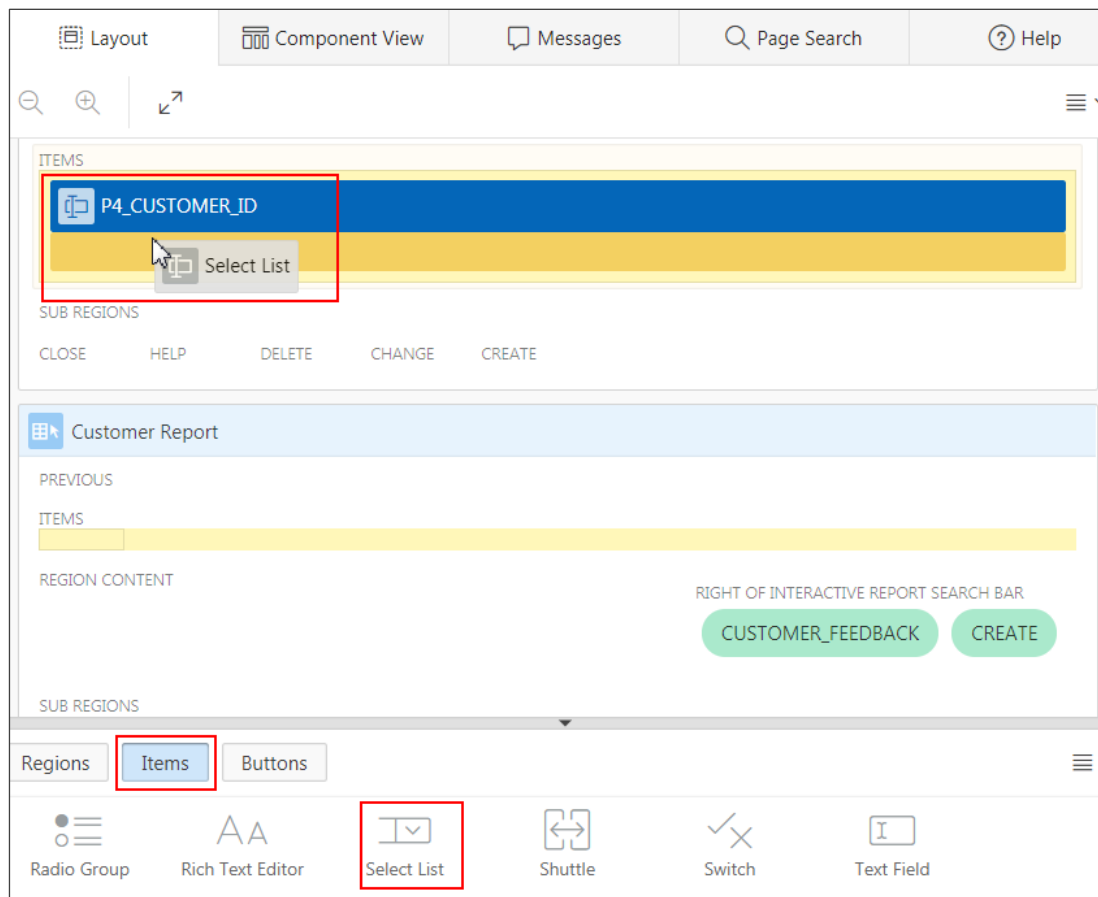
Display Extra Values: **Yes** No

Display Null Value: **Yes** No

Null Display Value: - Show All -

Null Return Value:

- k. Create an item for P4_CITY:
- From the Gallery at the bottom of the central pane, select the **Items** tab.
 - Select **Select List** and drag it to the Quick Filter region below the P4_CUSTOMER_ID item.



For Instructor Use Only.
This document should not be distributed.

- l. On the Rendering tab, select the new item.
- m. In the Property Editor, edit the following:
 - **Identification > Name:** Enter P4_CITY.
 - **Label > Label:** City (populated by default).
 - **Layout > Start New Row:** Select **No** (if not selected by default).
 - **Layout > New Column:** Select **Yes**.

The screenshot shows the Property Editor with three sections: Identification, Label, and Layout. In the Identification section, the 'Name' property is set to 'P4_CITY'. In the Label section, the 'Label' property is set to 'City'. In the Layout section, the 'Start New Row' property is set to 'No' and the 'New Column' property is set to 'Yes'. Red boxes highlight these four properties.

- **List of Values > Type:** Select SQL Query.
- **List of Values > SQL Query:** Enter the query that is provided in /home/oracle/labs/labs/lab_12_01_02.txt.

```
select distinct city d, city r
from oeher_customers
order by 1
```


- **List of Values > Display Null Value:** Select **Yes** (if not selected by default).
- **Null Display Value:** Enter - Show All -.

List of Values

Type: SQL Query

SQL Query: `select distinct city d, city r
from oeher_customers
order by 1`

Display Extra Values: **Yes** No

Display Null Value: **Yes** No

Null Display Value: - Show All -

Null Return Value:

- Click **Save**.
- n. In the left pane, click the **Dynamic Actions** tab.

Rendering **Dynamic Actions** Processing

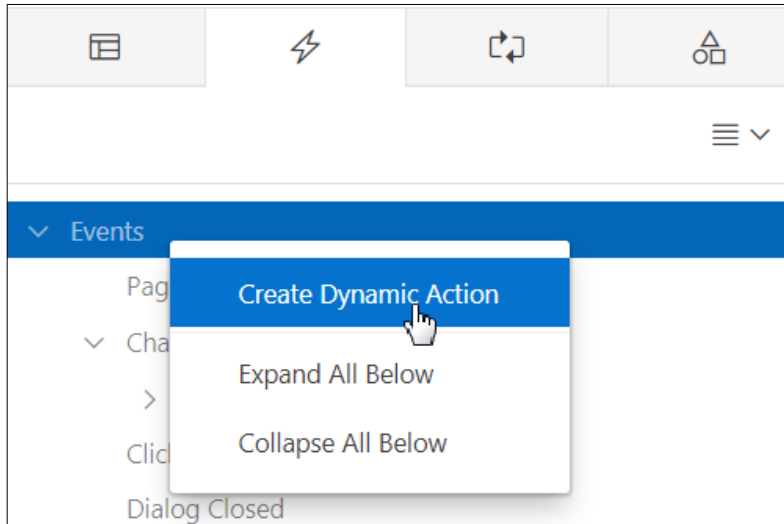
1 = 2 =

Page 4: Customers

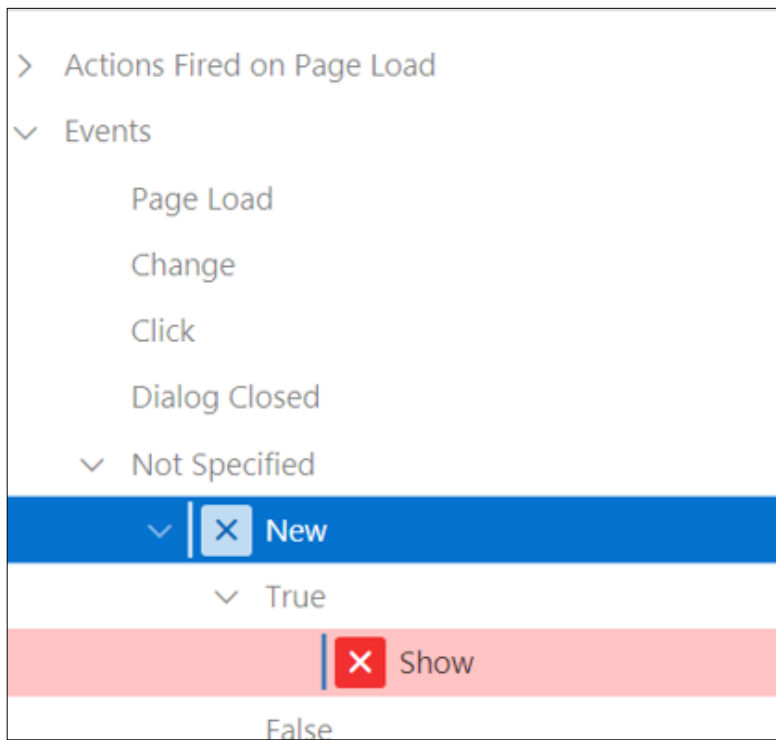
- > Pre-Rendering
- > Regions
 - > Before Content Body
 - > Content Body
 - > Quick Filter
 - Attributes
 - Items
 - P4_CUSTOMER_ID
 - P4_CITY

Customer Report

- o. Right-click **Events** and select **Create Dynamic Action**.



- p. On the Dynamic Actions tab, select the new dynamic action.



- q. In the Property Editor, edit the following:
- **Identification > Name:** Enter `Quick Filter Refresh`.
 - **When > Event:** Select **Change**.
 - **When > Selection Type:** Select **Item(s)**.
 - **When > Item(s):** Select `P4_CUSTOMER_ID, P4_CITY`.

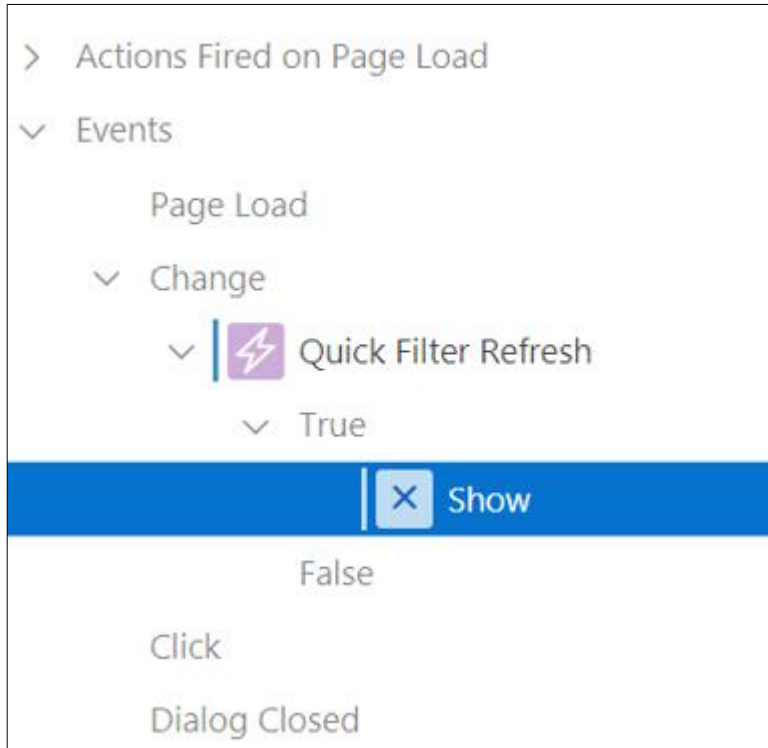
The screenshot shows the Property Editor interface with three main sections: Identification, Execution Options, and When. The 'When' section is expanded and highlighted in light blue. Red boxes highlight the 'Name' field in the Identification section, the 'Event', 'Selection Type', and 'Item(s)' fields in the When section, and the menu icons to the right of these fields.

Identification			
Name	Quick Filter Refresh		

Execution Options			
Sequence	10		

When			
Event	Change	▼	⋮
Selection Type	Item(s)	▼	⋮
Item(s)	P4_CUSTOMER_ID,P4_CITY		⋮

- r. On the Dynamic Actions tab, select **Quick Filter Refresh > True > Show**.



- s. In the Property Editor, enter the following:
- **Identification > Action:** Select **Refresh**.
 - **Affected Elements > Selection Type:** Select **Region**.
 - **Affected Elements > Region:** Select **Customer Report**
 - **Execution Options > Fire on Initialization:** Select **No** (if not selected by default).

Identification

Action

Refresh

Affected Elements

Selection Type

Region

Region

Customer Report

Execution Options

Sequence

10

Event

Quick Filter Refresh

Fire When Event Result Is

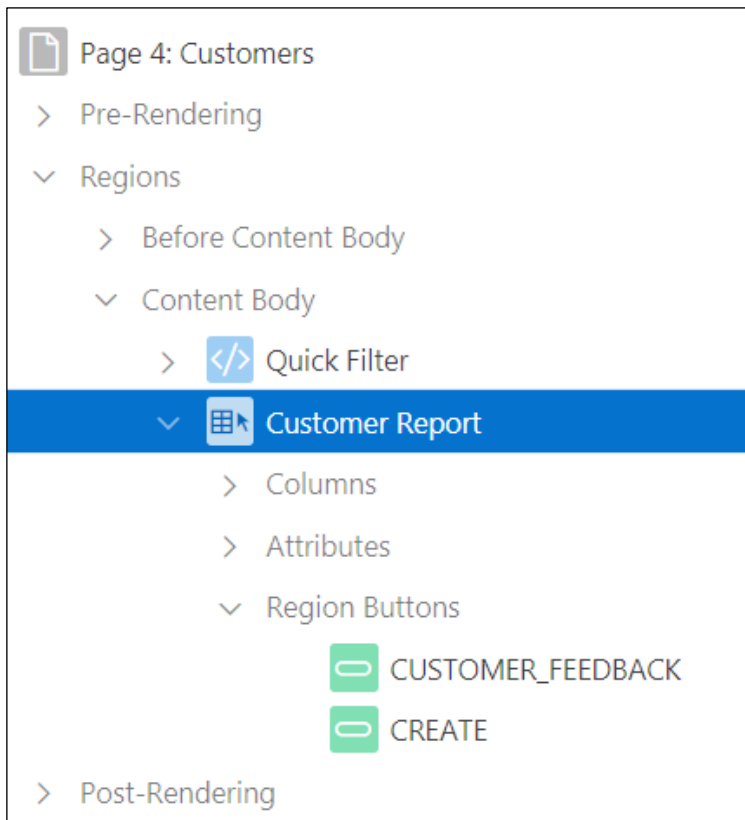
True

Fire on Initialization

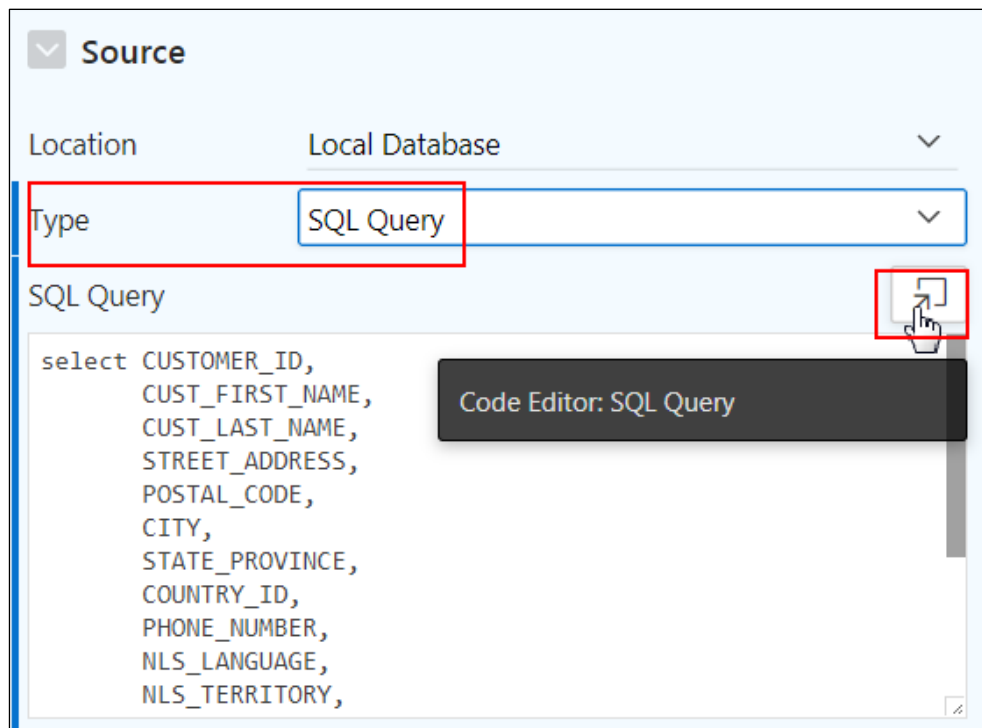
Yes

No

- t. On the Rendering tab, under **Regions > Content Body**, collapse **Quick Filter**. Select **Customer Report**.

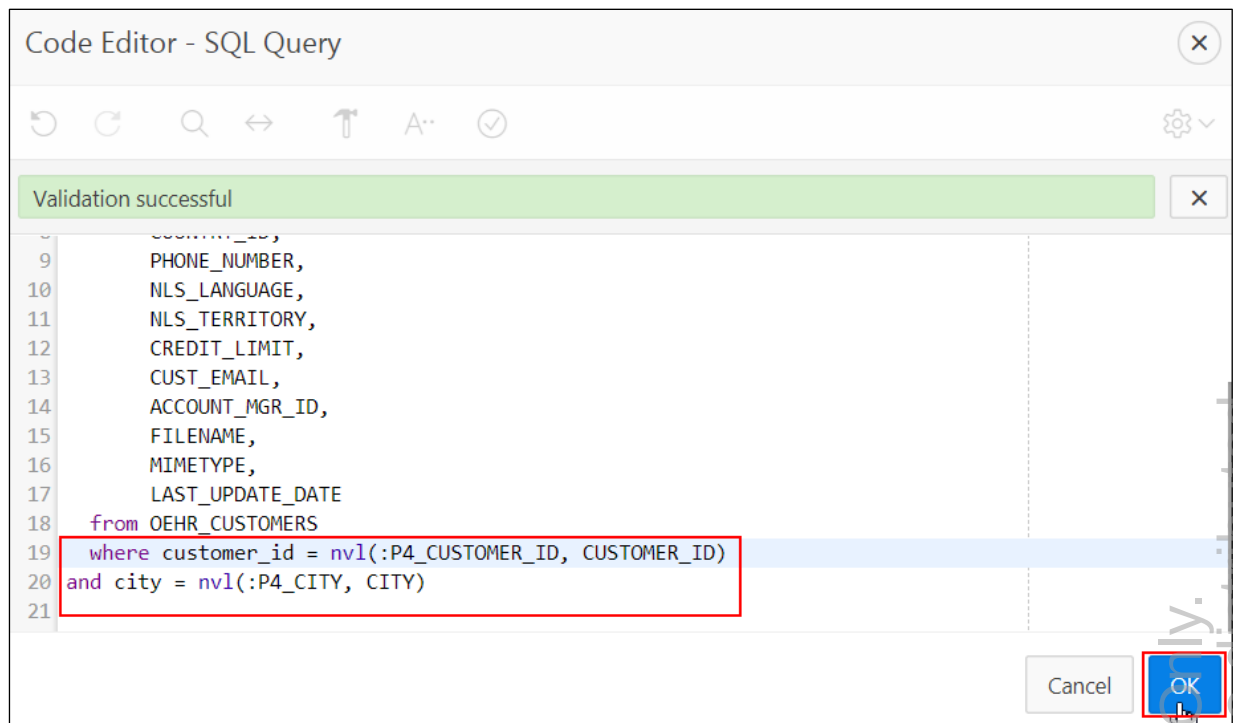


- u. On the Property Editor tab, edit the following:
- **Source > Type: Select SQL Query**
 - **Source > SQL Query:** Click the **Code Editor** icon.

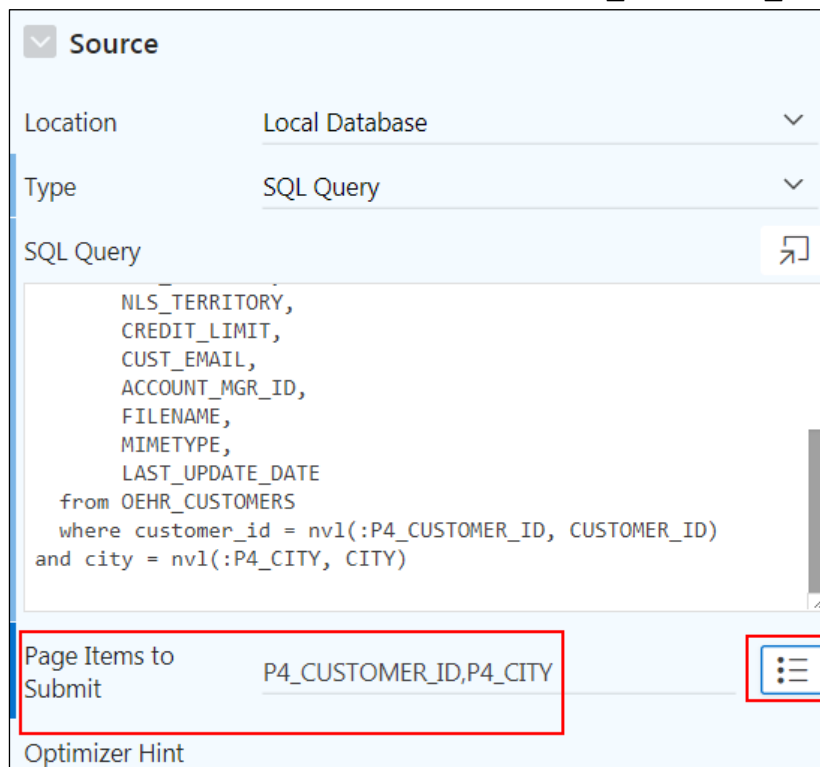


- In the Code Editor, add the following WHERE clause (located in the /home/oracle/labs/labs/lab_12_01_03.txt file) to the end of the SELECT statement and click **OK**.

```
where customer_id = nvl(:P4_CUSTOMER_ID, CUSTOMER_ID)
and city = nvl(:P4_CITY, CITY)
```



- In **Source > Page Items to Submit**, select P4_CUSTOMER_ID, P4_CITY.



- From **Layout > Start New Row**, select **Yes**.

Layout

Sequence: 20

Parent Region: - Select -

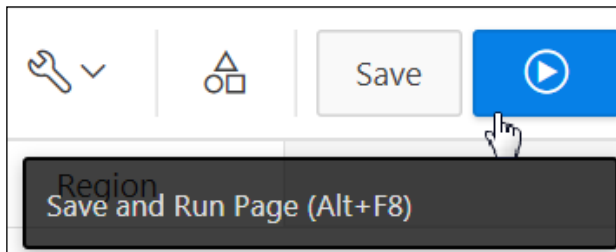
Position: Content Body

Start New Row: **Yes** No

Column: Automatic

Column Span: Automatic

- v. Click the **Save and Run Page**.



- w. If you are prompted to sign in, enter your workspace credentials and click **Sign In**.
- x. The Customers report appears. Review the revised report.
 - i. Note that when both the Customer and City filters are set to **- Show All -** all the records display.

Quick Filter

Customer
- Show All -

City
- Show All -

Q

Go

Actions

Create

Customer Feedback

	Cust First Name	Cust Last Name	Street Address	Postal Code	City	State Province	Country Id	Phone Number	Nls Language
	Eddie	Boyer	411 E Wisconsin Ave # 2550	53202	Seattle	WI	US	+1 414 123 4369	us
	Eddie	Stern	808 3Rd St # 100	54403	Wausau	WI	US	+1 715 123 4372	us
	Ernest	Weaver	300 Crooks St	54301	Green Bay	WI	US	+1 414 123 4373	us

- ii. From the Customer filter, select a customer from the list. What do you observe?

Quick Filter

Customer
- Show All -

- Show All -
Ajay Andrews
Ajay Sen
Alain Barkin
Alain Dreyfuss
Alain Siegel
Alan Hunter
Alan Minnelli
Albert Bel Geddes
Albert Dutt
Albert Spacek
Alec Idle

City
- Show All -

	Country	Phone Number
		+1 412 123 4740
		+1 717 123 4741
		+1 215 123 4742

For Instructor Use Only.
This document should not be distributed.

The revised report appears, displaying the customer.

Quick Filter

Customer

Ajay Andrews

City

- Show All -

Q

Go

Actions

Customer Feedback

	First Name	Last Name	Street Address	Postal Code	City	State Province	Country	Phone Number	Nls Language
	Ajay	Andrews	223 4Th Ave # 1100	15222	Pittsburgh	PA	US	+1 412 123 4763	us

iii. From the Customer filter, select **-Show All-**. From the **City** filter, select **Albany**.

Quick Filter

Customer

- Show All -

City

- Show All -

Albany

Alexandria

Altoona

Ann Arbor

Baden-Daettwil

Baltimore

Bangalore

Bangalore - India

Q

Go

Action

	First Name	Last Name	Street Address	Postal Code	City
	Bryan	Dvrrie	3386 Perrysville	15214	Pitts

The revised report appears, displaying the customers in the selected city.

Customer
- Show All -

City
Albany

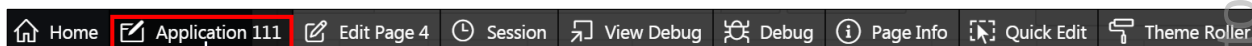
Q

Go

Actions

	First Name	Last Name	Street Address	Postal Code	City	State Province	Country	Phone Number	Nls Language
	Dom	McQueen	8 Automation Ln	12205	Albany	NY	US	+1 518 123 4532	us
	Gerhard	Seignier	49 N Pine Ave	12203	Albany	NY	US	+1 518 123 4459	us
	Harry dean	Forrest	137 Lark St	12210	Albany	NY	US	+1 518 123 4474	us

iv. Click the **Application 111** link on the Runtime Developer toolbar.



Solution 12-2: Importing and Using Plug-Ins on Your Page

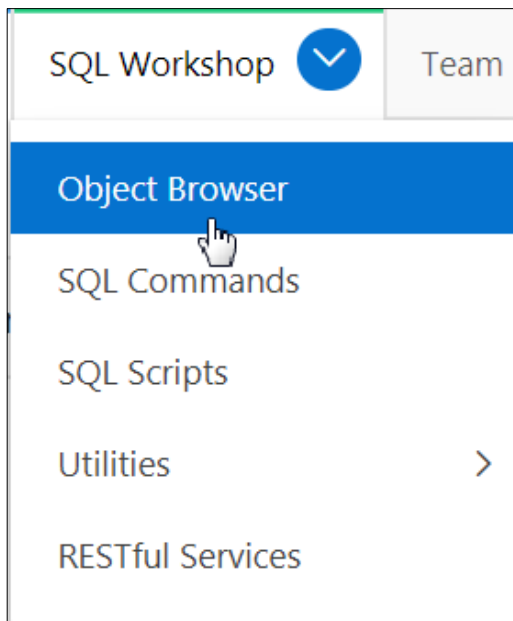
Overview

In this solution, you import and use an item type plug-in and a dynamic action plug-in.

Note: The solution script generated at the end of this practice does not contain the **Rating** column.

Steps

1. Add a rating column to the `CUSTOMERS` table.
 - a. On the Workspace home page, select the **SQL Workshop** menu and then select **Object Browser**.



- b. In Object Browser:
 - i. For **Schema** at the top right of the window, select **OEHR**.
 - ii. In the Object Selection pane (left pane), select `OEHR_CUSTOMERS`.

For Instructor Use Only.
This document should not be distributed.

 **Object Browser**

Tables 





INVENTORIES

JOBS

JOB_HISTORY

LOCATIONS

OEHR_COUNTRIES

OEHR_CUSTOMERS

OEHR_DEPARTMENTS

OEHR_EMPLOYEES

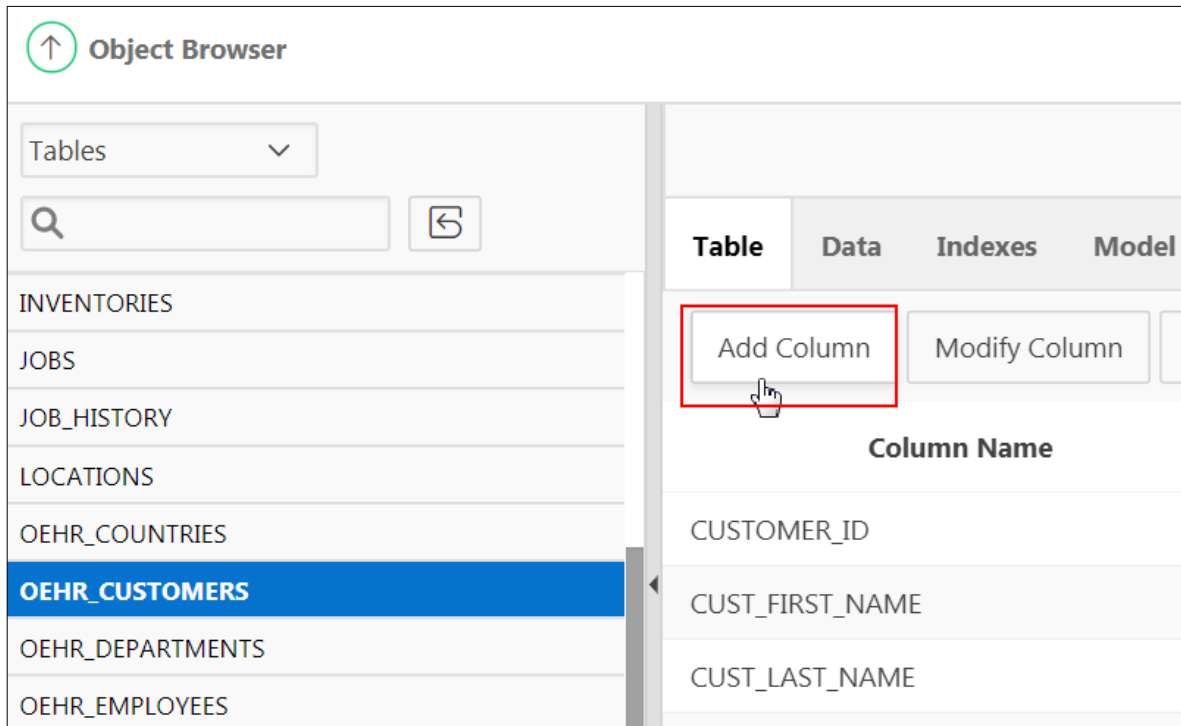
OEHR_INVENTORIES

OEHR_JOBS

OEHR_JOB_HISTORY

OEHR_LOCATIONS

- c. In the Detail pane (right pane), click the **Add Column** button.



- d. On Add Column:
- **Add Column:** Enter `RATING`.
 - **Type:** Select **Number**.
 - **Precision:** Enter `2`.

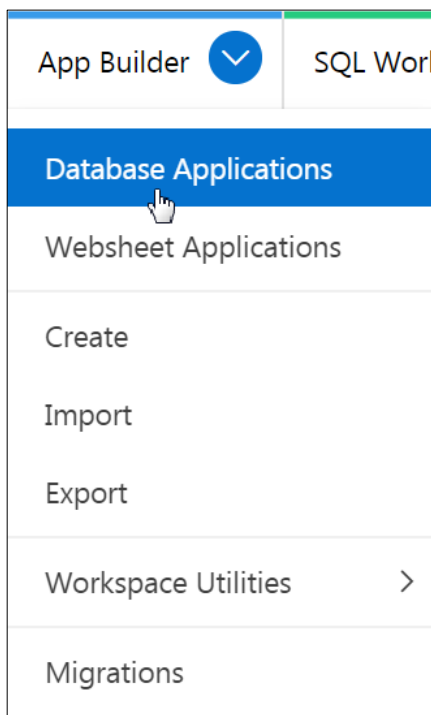
The screenshot shows the 'Add Column' dialog for the 'OEHR_CUSTOMERS' table. The 'Schema' is set to 'OEHR' and the 'Table' is 'OEHR_CUSTOMERS'. The 'Add Column' button is highlighted with a red box. The 'Column Name' field contains 'RATING'. The 'Type' dropdown is set to 'NUMBER'. The 'Precision' field contains '2'. There is also a 'Length' field and a 'Preserve Case' checkbox.

- Click **Next >**.

- Click **Finish**. The column is added.

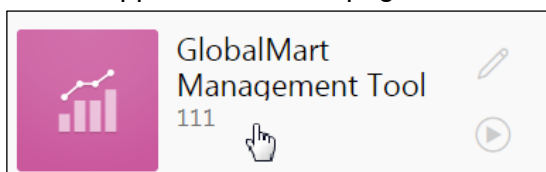
Tables	OEHR_CUSTOMERS		
NLS_TERRITORY	VARCHAR2(30)	Yes	
CREDIT_LIMIT	NUMBER(9,2)	Yes	
CUST_EMAIL	VARCHAR2(30)	Yes	
ACCOUNT_MGR_ID	NUMBER(6,0)	Yes	
FILENAME	VARCHAR2(255)	Yes	
MIMETYPE	VARCHAR2(255)	Yes	
LAST_UPDATE_DATE	DATE	Yes	
RATING	NUMBER(2,0)	Yes	

- Click the **App Builder** menu and select **Database Applications**.

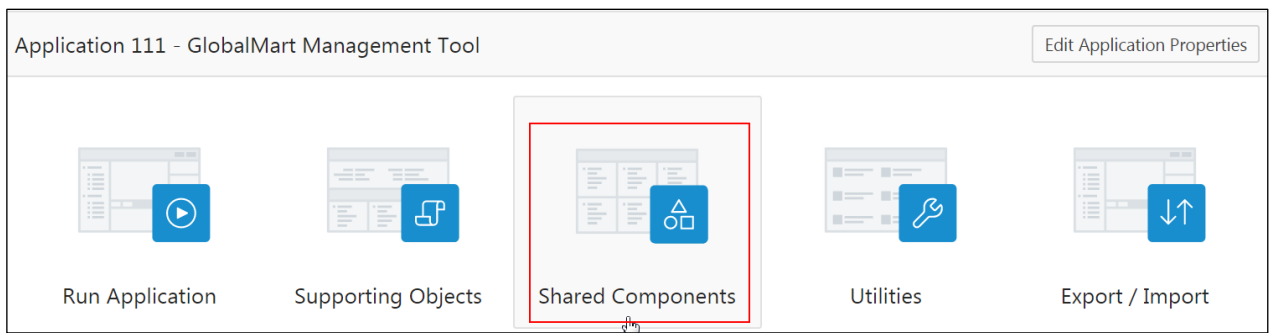


- Import the Modern Star Rating Item plug-in.

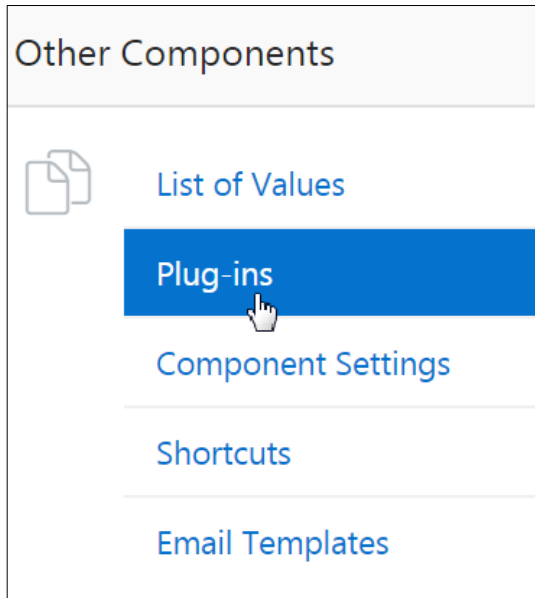
- On the App Builder Home page, select the **GlobalMart Management Tool** application.



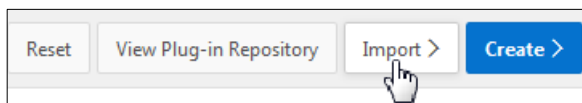
- Click **Shared Components**.



- c. Under **Other Components**, select **Plug-Ins**.



- d. On the Plug-ins page, click **Import >**.



- e. On the Import page, click **Browse** and select `/home/oracle/labs/labs/item_type_plugin_com_oracle_apex_starrating_modern.sql`. On the Import page, click **Next >**.

Import

Select the file you wish to import to the export repository. Once imported, you can install your file.

If the imported file is a productivity or sample application export, the installation wizard will allow you to run the app installation scripts after installing the application definition.

* Import file Choose File item_type_pl..._modern.sql ?

* File Type:

- ☐ Database Application, Page or Component Export ?
- ☐ Worksheet Application Export
- ☒ **Plug-in**
- ☐ Theme Export
- ☐ User Interface Defaults
- ☐ Team Development Feedback
- ☐ CSS Export [Deprecated]
- ☐ Image Export [Deprecated]
- ☐ File Export [Deprecated]

File Character Set Unicode UTF-8 ?

Cancel
Help

Next >

- f. After the file is imported, the File Import Confirmation dialog box appears. Click **Next >** to install it.

↑
Import

Import

✓
●
●
●

File Import Confirmation

The export file has been imported successfully.

If you wish to install now, click the **Next** button. You can also install this file at a later time by navigating to the Export Repository.

> **Tasks**

<
Cancel

Next >

- g. The Install Plug-in window appears. For **Install Into Application**, verify that the GlobalMart Management Tool application is selected and click **Install Plug-in**.

Note that if the plug-in already exists, it will be replaced.

Install Plug-in

When you install a plug-in into the current application, the new plug-in will overwrite an existing plug-in having the same plug-in name. If the installation succeeds, the installation of the plug-in becomes permanent. If any errors are encountered, the actions are rolled back, resulting in no permanent changes.

Export File Version: **2018.05.24**

Name: **Modern Star Rating**

Internal Name: **COM.ORACLE.APEX.STARRATING_MODERN**

Install Into Application: **111 GlobalMart Management Tool**

Action: **New plug-in will be created in application 111.**

Install Plug-in

- h. The plug-in is installed. Accept the default Component Settings and click **Apply Changes**.

ORACLE APEX App Builder SQL Workshop Team Development App Gallery

✓ Plug-in installed.

Application 111 \ Shared Components \ Component Settings \ Edit

Component Settings

Use Component Settings to set application level values for built-in Application Express components and installed plug-ins.

Name: **Modern Star Rating [Plug-in]** ?

Clear Tooltip: ?

Active Star Color: ?

Inactive Star Color: ?

Default icon: ?

Application 111 \ Shared Components \ Component Settings

✓ **Component Settings changed.**

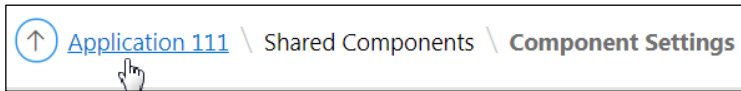
Component Settings History

Q v Go

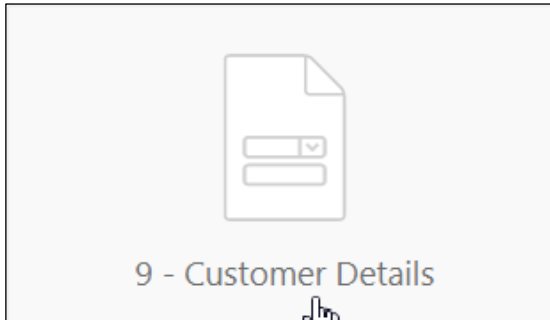
Name ↑≡

Calendar
Color Picker
Interactive Grid
Interactive Report
Modern Star Rating [Plug-in]
Region Display Selector
Rich Text Editor

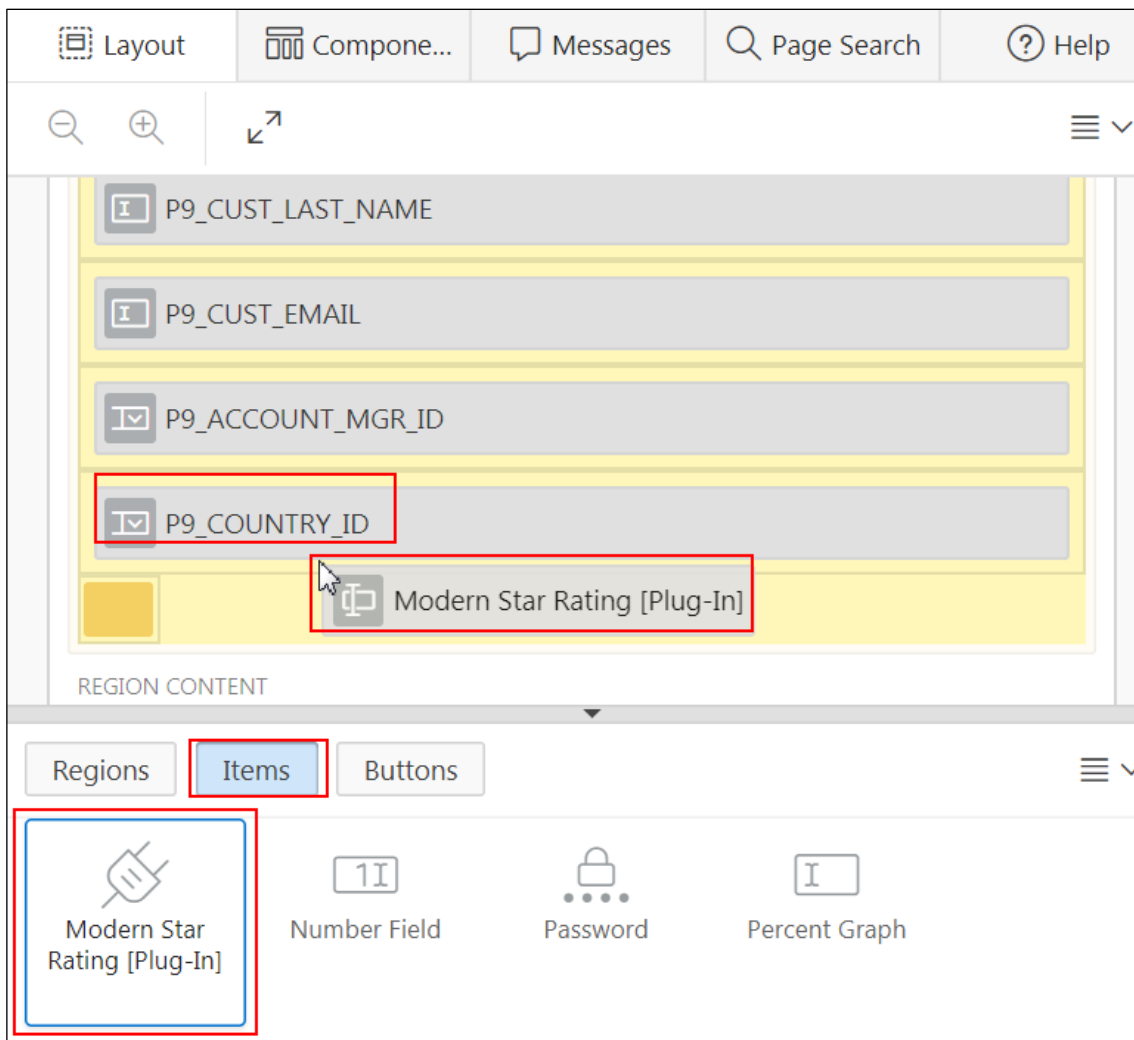
3. Add the Modern Star Rating plug-in to your Customer Details page.
 - a. Click **Application 111** breadcrumb to return to your application.



- b. On the Application home page, select the **9 - Customer Details** page.



- c. Page Designer appears, displaying **Page 9: Customer Details**.
 - d. In the Gallery at the bottom of the central pane, select the **Items** tab and find the **Star Rating [Plug-In]**. Select the **Modern Star Rating [Plug-In]** and then drag and drop it to after P9_COUNTRY_ID in the Layout pane.



For Instructor Use Only.
This document should not be distributed.

- e. On the Rendering tab, verify the newly added plug-in is selected.
- f. In the Property Editor, edit the following:
 - **Identification > Name:** Enter P9_RATING.
 - **Settings > Number of Stars:** Enter 5.

Identification

Name

Type

Label

Label

Settings

Number of Stars

Use Component Defaults ☒ Yes ☐ No

Tooltip Template

- **Appearance > Template:** Select **Optional - Above**.

Appearance

Template

Template Options

CSS Classes

Icon

Format Mask

- **Source > Type:** Select **Database Column**.
- For **Source > Database Column:** Enter RATING.

Source

Form Region

- Select -

>

Type

Database Column

⋮

Database Column

RATING

Used

Always, replacing any exist

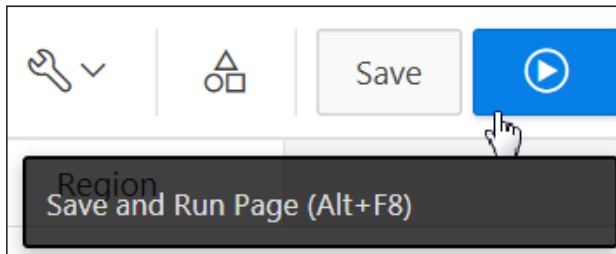
▼

Maintain Session State

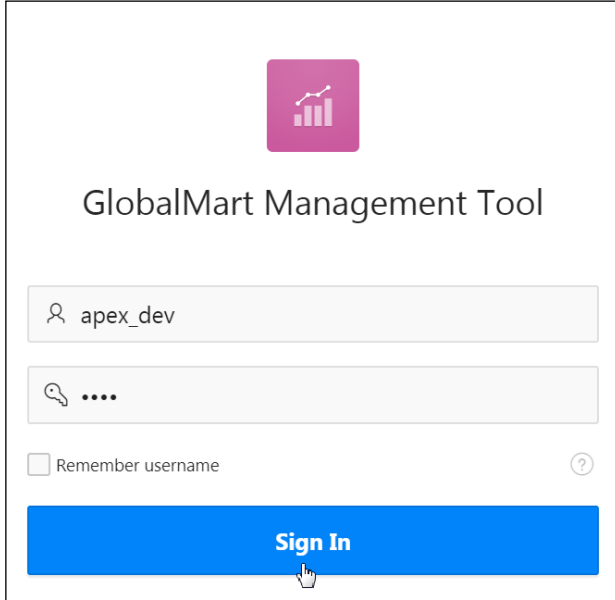
Per Session (Disk)

▼

- g. Click the **Save and Run Page**.



- h. If you are prompted to sign in, enter your workspace credentials and click **Sign In**.



The image shows a login interface for the 'GlobalMart Management Tool'. At the top center is a purple square icon with a white bar chart. Below the icon, the text 'GlobalMart Management Tool' is displayed. There are two input fields: the first contains the username 'apex_dev' and the second contains masked characters '....'. Below these fields is a checkbox labeled 'Remember username' and a small question mark icon. At the bottom is a large blue button with the text 'Sign In' in white. A mouse cursor is pointing at the 'Sign In' button.

- i. Notice that a Rating item now appears at the bottom of the page. Click **Rating**.

The screenshot displays the 'GlobalMart Management Tool' interface. On the left is a dark sidebar with navigation links: Home, Customers, Products, and Customers List View. The main content area is titled 'Customer Details' and contains several form fields: 'Cust First Name', 'Cust Last Name', 'Account Manager' (with a dropdown menu showing '-Select Manager-' and a list of names: Bates, Elizabeth, Russell, John, Cambrault, Gerald), and 'Country' (with a dropdown menu showing '-Select Country-'). At the bottom of the form, there is a 'Rating' section with a radio button icon, which is highlighted by a red square. Below the Rating section is a 'Cancel' button.

For Instructor Use Only.
This document should not be distributed.

- j. Click each rating and note the Tooltips that are displayed.

The screenshot shows the 'GlobalMart Management Tool' interface. On the left is a dark sidebar with navigation links: Home, Customers, Products, and Customers List View. The main content area is titled 'Customer Details' and contains several form fields: 'Cust First Name', 'Cust Last Name', 'Account Manager' (with a dropdown menu showing '-Select Manager-' and a list of names: Bates, Elizabeth, Russell, John, Cambrault, Gerald), and 'Country' (with a dropdown menu showing '-Select Country-'). At the bottom of the form is a 'Rating' section with five stars. The fourth star is selected, and a tooltip displays 'Rating: 4'. A 'Cancel' button is located below the rating stars.

- k. Click the **Application 111** link on the Runtime Developer toolbar.



